

IB SEHS

Sports, Exercise and Health Science

complete bilingual study pack

完整双语学习包

29 topics · 83 sections · English + 中文 throughout

What is in this pack / 本学习包内容

- Every topic, section by section, exactly as the site teaches it.
逐主题、逐小节完整收录，与网站内容一致。
- An everyday picture first, then the idea taken apart.
先用生活化比喻建立直觉，再逐层拆解原理。
- The numbers actually worked through.
所有关键数值均逐步演算。
- The wrong answers students actually arrive with.
针对常见易错点逐条辨析。
- Every key term defined in both languages.
全部核心术语提供中英双语定义。
- Exam-focus questions for each section.
每小节附考核要点与练习问题。

How to use it / 如何使用

- Read the metaphor, close the book, and explain the section to someone else in your own words.
先读比喻，然后合上书，用自己的话把这一节讲给别人听。
- Only then read the detail. The order matters: the picture is the thing you keep.
然后再读详细内容。顺序很重要：真正留在记忆里的是那个画面。
- Work the numbers on paper. Reading arithmetic is not the same as doing it.
数值一定要动笔算一遍。看懂和算出是两回事。
- Read the mistakes before the exam, not the night before.
易错点要提前看，不要考前夜突击。

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Inter-system communication

系统间通讯

Guiding question / 引导问题

How do the nervous and endocrine systems work together to regulate body systems and maintain homeostasis?

神经和内分泌系统如何共同调节身体系统并维持稳态？

The body communicates through fast, precise neural pathways and slower, longer-lasting endocrine signals. Their coordination allows a single change in the environment to trigger rapid protective action and sustained regulation of energy, temperature, fluids and recovery.

身体通过快速而精准的神经通路，以及较慢但持续时间更长的内分泌信号进行通讯。两者的协调使环境变化能够同时引发快速保护反应。

Communication systems · 通讯系统

First, picture it / 先这样想

Your body has two ways to send a message, and they work nothing alike. The nervous system is a phone call: you tap someone on the shoulder and they move before you have finished the sentence. The endocrine system is a noticeboard: you write a note, pin it up, and everyone who reads it over the next hour acts on it. A phone call and a billboard - same information, completely different timing.

身体有两种传递信息的方式，它们的工作方式完全不同。神经系统像一通电话：你轻拍某人的肩膀，他还没等你说完话就已经动了。内分泌系统像一块公告板：你写一张便条钉上去，接下来一小时内看到的人都会照做。同样是传递信息，时间方式却天差地别。

The detail / 详细讲解

A nerve carries an electrical impulse along its length.

神经沿其长度传导电脉冲。

Nerve endings never touch the next cell. The gap is the synapse.

神经末梢从不与下一个细胞直接相接，这道空隙叫突触。

To cross it, the cell releases a neurotransmitter into the synaptic cleft.

为了跨越它，细胞把神经递质释放到突触间隙中。

The transmitter binds a receptor on the far cell, and the signal turns electrical again.

递质与另一侧细胞的受体结合，信号重新变回电信号。

So one nerve signal is electrical -> chemical -> electrical - a relay race with a baton.

因此一个神经信号是电 -> 化学 -> 电，像一场传接力棒的接力赛。

A gland releases hormones straight into the bloodstream.

腺体把激素直接释放到血液中。

Blood carries a hormone to every cell in the body...

血液把激素带到全身每一个细胞.....

...but only a target cell responds to it.

.....但只有靶细胞会对它作出反应。

A cell is a target only if it carries the matching receptor, shaped like a lock.

一个细胞要成为靶细胞，必须带有相应的受体，形状像一把恰好配得上的锁。

Nerve speed: milliseconds. A knee reflex is finished in about 50 ms - before you feel the tap.

神经速度：毫秒级。膝反射约在 50 毫秒内完成--比你自已感觉到敲击还早。

Hormone speed: seconds to days. Insulin acts within minutes; growth hormone works over nights.

激素速度：秒到天。胰岛素在几分钟内起效，生长激素则主要在夜间起作用。

Nerve reach: local and precise - one nerve, one small group of muscles.

神经范围：局部而精准--一根神经，一小群肌肉。

Hormone reach: widespread - bone, skin, liver and muscle all at once.

激素范围：广泛--骨、皮肤、肝脏和肌肉同时受影响。

The flip side: a hormone cannot be aimed, so an unwanted side effect is felt body-wide.

代价在于：激素无法瞄准，所以不想出现的副作用会波及全身。

In a comparison question, speed and reach are the two things that earn the mark.

比较题里，速度和范围就是得分点。

The numbers / 算一算

Line the timings up and the pattern is easy to remember. A reflex arc - stimulus to muscle response - runs about 30 to 50 milliseconds. A hormonal response: adrenaline begins lifting heart rate in roughly 30 seconds to a couple of minutes; insulin works on a scale of minutes to hours; growth hormone and testosterone shape adaptation over weeks to months. A habit worth building for the exam: whenever a question asks you to explain a control mechanism, ask two questions of it - how fast, and how far. Fast and local means neural. Slow and wide means endocrine. That one habit answers most of the comparison questions.

把时间线排在一起，规律就很容易记住。反射弧--从刺激到肌肉反应--大约需要30到50毫秒。内分泌反应：肾上腺素大约在30秒到两三分钟。

Common mistakes / 易错点

"The nervous system is electrical and the endocrine system is chemical." This is only half right and it is a very common loss of a mark. Nerves are electrical along the wire but chemical across every synapse. Endocrine signalling is chemical for the whole journey. Saying "nerves are electrical" with no qualification is the kind of answer that sounds confident and scores partial.

"神经系统是电的，内分泌系统是化学的。"这只对了一半，也是非常常见的丢分点。神经在电线部分是电的，但每跨过一个突触就

"Hormones act everywhere." They travel everywhere, but they only act on target cells carrying the right receptor. Everywhere else is a passenger, not a destination.

"激素作用于全身各处。"它们的确会到达全身，但只对带有相应受体的靶细胞起作用。其他地方只是乘客，不是目的地。

"Adrenaline works instantly because it is fast in sport." It travels fast because the blood moves fast, not because it is a nerve signal. It is still measurably slower than a nerve impulse, and an answer that treats it as instant confuses the two systems.

"肾上腺素在运动中起效很快，所以是瞬间的。"它传得快是因为血流快，不是因为它是神经信号。它仍然可以测出地比神经冲动慢

Why it matters / 为什么重要

A sprint is the cleanest example of both systems running at once. The gun fires. Your nervous system has already asked your leg muscles to contract in well under a second, because the alternative - waiting for adrenaline to arrive - would leave you standing on the blocks. Adrenaline then takes over for the next few minutes, raising heart rate, redirecting blood to the muscles and freeing glucose. Sport does not choose between the two systems. It uses the fast one to start and the slow one to keep going.

短跑是两个系统同时工作的最干净的例子。枪响的那一刻，你的神经系统已经在不到一秒内让腿部肌肉收缩了，因为另一种选择--等肾上腺

Nervous control is rapid, precise, localised and usually short-lived.

神经控制快速、精准、局部化，通常作用时间较短。

Endocrine control is slower, widespread and longer-lasting.

内分泌控制较慢、影响范围广、持续时间较长。

Key terms / 核心术语

Homeostasis 稳态

The maintenance of a stable internal environment within workable limits.

将内部环境维持在可工作的狭窄范围内。

Negative feedback 负反馈

A response that counters a change and returns the variable toward its normal range.

抵消变化，使变量回到正常范围的反馈。

Positive feedback 正反馈

A response that amplifies a change until a specific endpoint.

放大变化直至达到特定终点的反馈。

Compared / 对比

Feature	Nervous system	Endocrine system
Signal	Electrical impulses; neurotransmitters at synapses	Hormones carried in blood
Speed	Milliseconds	Seconds to days
Effect	Fast, precise, localised	Slower, widespread, longer-lasting

中文对照

特征	神经系统	内分泌系统
信号	神经元电脉冲; 突触处神经递质	血液运输的激素
速度	毫秒级	秒至天
作用	快速、精准、局部	较慢、广泛、持久

Worked examples / 实例

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At the start of a sprint, somatic motor output produces rapid skeletal-muscle contraction while adrenaline increases cardiovascular readiness and mobilises energy.

冲刺开始时, 躯体运动输出使骨骼肌快速收缩, 肾上腺素同时提高心血管准备程度并动员能量。

Quick check / 快速检查

1. Name the two main communication systems.
说出两种主要通讯系统。
2. Sequence stimulus, receptor, control centre, effector and response in a negative-feedback loop.
按刺激、受体、控制中心、效应器和反应排列负反馈过程。

Exam focus / 考核要点

Explain how nervous and endocrine control can act at different time scales.

解释神经控制和内分泌控制如何作用于不同时间尺度。

Neural pathways and coordination · 神经通路协调

First, picture it / 先这样想

Think of your body as a two-storey building with a control room on the top floor. The brain and spinal cord are the control room: they receive every report, decide what to do, and send the orders down. Everything else - all the wires running through the building - is the peripheral nervous system. The wires do not decide anything. They carry. A message can go up to the control room, or come down from it, but the decision is always made upstairs.

把身体想成一栋两层楼, 顶层是控制室。脑和脊髓就是控制室: 它们接收所有报告、决定要做什么、再把命令发下去。楼里其余的一切--所有穿行的线路--就是周围神经系统。线路不做任何决定, 它们只负责运送。信息可以送进控制室, 也可以从控制室发下来, 但决定永远是在楼上做出的。

The detail / 详细讲解

The central nervous system - the brain and the spinal cord - is the processing centre.

中枢神经系统--脑和脊髓--是处理中心。

The spinal cord is especially worth understanding on its own terms: it is not just a cable running between the brain and the body.

脊髓尤其值得单独理解: 它不只是一条连接脑和身体的缆线, 它本身就是决策中心。

It is a decision-making centre in its own right.

这正是膝反射在脊髓完好、而大脑并未参与时仍然能工作的原因。

That is why a knee jerk still works when the spinal cord is intact but the brain is not being consulted. The cord can

handle a simple, fast, protective job entirely on its own.

脊髓完全可以独立完成这种简单、快速、保护性的任务。

The peripheral nervous system is everything outside the brain and spinal cord, and it comes in two functionally different halves.

周围神经系统是脑和脊髓之外的一切，它在功能上分成两半。

The somatic system drives skeletal muscle - the muscle you can consciously choose to move, like the muscles of your arm when you decide to jump.

躯体系统驱动骨骼肌--你可以有意识决定让它运动的肌肉，比如你决定起跳时手臂的肌肉。

The autonomic system runs everything you cannot consciously control: glands, smooth muscle in your organs and gut, and the cardiac muscle of your heart.

自主系统负责所有你无法有意识控制的东西：腺体、器官和肠道里的平滑肌，以及心脏的心肌。

You can decide to raise your arm. You cannot decide to raise your heart rate, and you would not want to be able to.

你可以决定抬起手臂，但你无法决定提高心率，而且你也不会希望自己能决定。

The autonomic system then splits again in two, and the two halves pull in opposite directions.

自主系统又分成两半，两半的方向相反。

The sympathetic division prepares you for action or stress.

交感分支为行动或压力做准备：提高心率、扩大气道、把血液重新分配到肌肉、减慢消化。

It raises heart rate, widens the airways, redirects blood to the muscles, and slows digestion.

副交感分支正相反：支持恢复、消化和能量节省。

The parasympathetic division does the opposite: it supports recovery, digestion and saving energy.

教科书的比喻是，这就像同一辆车的一个油门和一个刹车。

The textbook image is that these are two ends of one accelerator and one brake.

一个有用的补充是：它们并不是简单的开和关。

A useful correction to that image: they are not simply on and off. Most of the time both are active, and what changes is the balance between them.

多数时候两者都在工作，变化的是它们之间的平衡。

After exercise the balance shifts.

运动结束后，平衡会移动。

Parasympathetic activity rises, heart rate falls back toward resting levels, and digestion and repair can proceed.

副交感活动上升，心率回落到接近静息水平，消化和修复得以进行。

At the same time the endocrine system is doing slower, longer work: growth hormone and insulin support tissue repair, energy regulation and longer-term adaptation.

与此同时，内分泌系统在做更慢、更持久的工作：生长激素和胰岛素支持组织修复、能量调节和长期适应。

So recovery is not one switch. It is the nervous system winding down while the hormonal system quietly rebuilds.

所以恢复并不是一个开关，而是神经系统慢慢降下来，同时激素系统悄悄地重建。

The numbers / 算一算

Here is a way to make the two halves of the autonomic system stick. At rest, take a resting heart rate of about 60 beats per minute. Sympathetic activation can push it to 180 or more in a maximal effort; the parasympathetic system is what brings it back down afterwards, typically by around 10 to 20 beats per minute once recovery starts. The exam habit: whenever a question gives you a change in heart rate, ask which division is being described and in which direction it is pushing.

这里有一个能让自主系统两半都记住的办法。静息心率大约是每分钟60次。交感激活可以在最大努力时把它推到180次甚至更高；副交感

Common mistakes / 易错点

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"The spinal cord only passes messages to the brain." The single most important correction here. The cord is a processing centre in its own right, which is exactly why reflexes are integrated there and can happen without the brain.

"脊髓只是把信息传给大脑。"这是这一节最重要的纠正。脊髓本身就是处理中心，这正是反射在脊髓中整合、并且不需要大脑参与

"Autonomic means involuntary, so the heart is not muscle." Cardiac muscle is muscle, and it is under autonomic control. Involuntary describes who is in charge, not what the tissue is made of.

"自主就是不受意志控制，所以心脏不是肌肉。"心肌就是肌肉，而且受自主神经支配。不受意志控制说的是"谁在指挥"，不是"这种"

"Sympathetic and parasympathetic are simply on and off." Both are usually active at once; what changes is the balance. Treat them as a dimmer switch, not a light switch.

"交感就是开，副交感就是关。"两者通常同时在工作，变化的是平衡。把它们当作调光开关，而不是普通开关。

Why it matters / 为什么重要

This is the wiring behind every moment of a performance. The sprinter who lifts their arms on the blocks is somatic control. The same athlete's heart rate climbing from 60 to 190 in six seconds is autonomic sympathetic control, and the fact that they do not consciously control it is the point. Being able to name which system is doing the work, and how fast and how wide it acts, is what lets you explain a performance rather than just describe it.

这就是每一次表现背后的线路。起跑器上撑起手臂的短跑运动员，用的是躯体控制；同一位运动员的心率在六秒内从60升到190，用的是

Somatic pathways control voluntary skeletal muscle, such as deciding to jump.

躯体通路控制随意骨骼肌，例如决定起跳。

Autonomic pathways control glands, smooth muscle and cardiac muscle without conscious effort.

自主通路控制腺体、平滑肌和心肌，不需有意识努力。

The sympathetic division prepares the body for action or stress; the parasympathetic division supports recovery, digestion and energy conservation.

交感神经分支为行动或压力做准备；副交感神经分支支持恢复、消化和能量保存。

Key terms / 核心术语

CNS 中枢神经系统

Brain and spinal cord; integration and coordination.

脑和脊髓；整合与协调。

PNS 周围神经系统

Nerves outside the CNS that carry sensory and motor information.

中枢外的神经，传递感觉和运动信息。

Reflex 反射

A rapid automatic protective response integrated mainly at the spinal cord.

主要由脊髓整合的快速自动保护反应。

Formulas / 公式

After exercise, parasympathetic activity can support recovery while endocrine signals such as growth hormone and insulin support repair, energy regulation and longer-term adaptation.

运动后，副交感活动可帮助恢复；生长激素和胰岛素等内分泌信号则支持修复、能量调节和长期适应。

Worked examples / 实例

>

Compare somatic and autonomic control in a sport context.

在运动情境中比较躯体控制和自主控制。

Exam focus / 考核要点

Differentiate sympathetic and parasympathetic activity.

区分交感与副交感活动。

Explain how negative feedback supports homeostasis.

解释负反馈如何支持稳态。

Intra-system communication

系统内通讯

Guiding question / 引导问题

How do organ systems work together within the body to support performance and homeostasis?

身体内的器官系统如何共同支持表现与稳态？

Organ systems are interdependent. Nervous, endocrine, cardiovascular, respiratory, muscular and excretory systems exchange information and constrain one another so that the body can respond to changing demands without losing internal stability.

器官系统相互依存。神经、内分泌、心血管、呼吸、肌肉与排泄系统交换信息并相互制约，使身体能够响应变化需求，同时维持内部稳定。

Systems working together · 系统协同工作

First, picture it / 先这样想

Your organs are not separate machines sitting in the same room. They are one connected system, like the departments of a hospital sharing the same patient. The heart does not just pump; it depends on the lungs to fill it with oxygen and the blood to carry that oxygen to the muscles. The muscles do not just burn fuel; they need the gut to deliver it, the kidneys to manage waste, and the brain to tell them when to fire. Pull any one department out and the patient does not get slightly worse. They get much worse, all at once.

你的器官不是各自独立、放在同一个房间里的机器。它们是一个连在一起的系统，就像同一家医院里各科室共享同一个病人。心脏不只是泵血，它依赖肺给它装满氧气，依赖血液把氧气送到肌肉。肌肉也不只是烧燃料，它们需要肠道把燃料送进来，需要肾脏处理废物，需要大脑告诉它们什么时候收缩。抽掉其中任何一个科室，病人不会只是稍微变差，而是会立刻、同时地大幅变差。

The detail / 详细讲解

Organ systems are interdependent.

各个器官系统是相互依存的。

Nervous, endocrine, cardiovascular, respiratory, muscular and excretory systems all constrain one another, and they all maintain homeostasis together.

神经、内分泌、心血管、呼吸、肌肉和排泄系统彼此制约，并共同维持稳态。

A good way to picture it: homeostasis is not a single thermostat in one room. It is a building full of thermostats - temperature, blood glucose, blood pressure, pH, fluid balance, oxygen and carbon dioxide levels - all holding their own set points at the same time, all being nudged around constantly by exercise, by food and by the weather, and all being corrected by the same two systems you already know: nerves for fast local corrections, hormones for slow wide ones.

一个好的画面是：稳态不是某个房间里的一个恒温器，而是一整栋楼里布满了恒温器—体温、血糖、血压、pH、体液平衡、氧气和二氧化碳。

Why are nerves fast and hormones slow?

为什么神经快、激素慢？

It comes down to how the message travels.

这取决于信息传递的方式。

A nerve impulse is a physical wave of electricity moving down a cable - the speed is limited only by how fast that wave can travel and how far the cable is.

神经冲动是沿着电缆向下传播的一股电能波，速度只受这股波本身传播速度和电缆长度限制。

A hormone has to be released into blood, survive the journey, and diffuse out of the capillary at the far end.

而激素必须先被释放到血液里，在旅途中存活，再从远端的毛细血管扩散出来。

Blood moves at roughly one metre per second at rest, so the delivery time is essentially the distance the blood has to travel.

血液在静息时大约以每秒一米的速度流动，所以送达时间本质上就是血液要走的距离。

That is why a hormone can reach a fingertip in under a second but an organ deep in the abdomen takes longer, and why the whole system is inherently slower than a wire.

这就是为什么激素不到一秒就能到达指尖，而位于腹部深处的器官需要更久，也为什么这套系统在本质上必然比一根电线慢。

Now zoom out to the whole athlete.

再把视角拉远到整位运动员。

Every time a muscle works it needs oxygen delivered and carbon dioxide removed.

每次肌肉工作都需要送来氧气、运走二氧化碳。

The heart and lungs must respond within seconds, and the respiratory system must actually change its own settings - faster and deeper breathing, or slower and shallower - rather than waiting to be told what to do.

心肺必须在几秒内作出反应，呼吸系统必须真正改变自己的设置--呼吸加快加深，或减慢变浅--而不是等着被告知该怎么做。

Meanwhile the digestive system is not busy during a race, which is exactly why the parasympathetic branch quietly pulls back on digestion during effort and lets it resume in recovery.

与此同时，消化系统在比赛期间并不忙碌，这正是副交感分支在用力期间悄悄收回对消化的控制、并在恢复期让它重新工作的原因。

The systems even take turns being the priority.

这些系统甚至会轮流成为"优先事项"。

Different systems are tuned for different jobs, and the body shifts the balance depending on the demand.

不同的系统为不同的工作而调整，身体会根据需求改变平衡。

At rest and in recovery, digestion and repair matter more.

静息和恢复时，消化与修复更重要；

During exercise, delivery of oxygen and removal of waste matter more.

运动时，氧气输送和废物清除更重要。

That re-prioritising is the whole point of inter-system communication: one change in the environment, such as the starter's gun or a sudden rise in body temperature, triggers a rapid protective response and a slower sustained adjustment at the same time.

这种重新排定优先级，正是系统间通讯的意义所在：环境中的一次变化，比如发令枪响或体温突然上升，会同时触发一个快速保护反应

The numbers / 算一算

Numbers to keep in your pocket. Blood flow to skeletal muscle at rest is roughly 20% of cardiac output; during maximal exercise it can rise to around 80-85%. Heart rate at rest is about 60 beats per minute, and 190 or more in a maximal effort. Blood travels at roughly 1 metre per second at rest, which is the basic reason hormone delivery is slower than nerve conduction. Oxygen delivery to the working muscle is the number that explains why training the aerobic system improves the phosphagen and glycolytic systems too - you cannot extract what you never delivered.

需要记在脑子里的数字。静息时骨骼肌的血流约占心输出量的20%，最大运动时可达80-85%。静息心率约60次每分钟，最大努力时190

Common mistakes / 易错点

- "Each system works alone and the brain coordinates them." The brain coordinates, yes, but the systems also constrain and depend on each other chemically and physically all the time. The heart cannot pump what the lungs have not oxygenated.
"每个系统各自工作，由大脑来协调。"大脑确实协调，但系统之间也一直在化学和物理上互相依赖、互相制约。肺没有氧合的东西
- "Nerves are fast and hormones are slow, so hormones are unimportant in sport." Slow is not the same as unimportant. Adrenaline, insulin and growth hormone all matter enormously; they act on a longer timescale and over a wider area.
"神经快、激素慢，所以激素在运动中不重要。"慢不等于不重要。肾上腺素、胰岛素和生长激素都非常重要，只是它们作用的时间
- "Only two systems count: nervous and endocrine." The cardiovascular, respiratory, muscular and excretory systems are the ones being regulated. Treating them as bystanders loses the whole idea of inter-system communication.
"只有两个系统算数：神经和内分泌。"心血管、呼吸、肌肉和排泄系统正是被调控的对象。把它们当成旁观者，就丢掉了"系统间通

Why it matters / 为什么重要

This is where a sports scientist earns their degree. Explaining why a runner needs to taper, why an altitude camp works, why a trained athlete recovers faster, or why a heatwave kills a cyclist all come down to the same idea: two

or more systems pushing the same variable in the same direction until something has to give. When you can name the systems, the direction each one is pushing and the timescale it works on, you can explain almost any performance question the paper sets.

体育科学学位正是在这里考出来的。为什么要减量训练、高原训练为什么有效、训练有素的运动员为什么恢复更快、为什么高温会拖垮一

- The cardiovascular and respiratory systems deliver oxygen and nutrients and remove carbon dioxide and heat.
心血管与呼吸系统输送氧气和营养物质，并移除二氧化碳与热量。
- The muscular system uses oxygen and nutrients to produce force.
肌肉系统利用氧气和营养物质产生力。
- The endocrine system adjusts energy availability, while the excretory system removes waste and excess water.
内分泌系统调节能量可用性，排泄系统清除废物和过量水分。

Key terms / 核心术语

Integration 整合

Coordinated activity across systems that supports performance and survival.
跨系统的协调活动，支持表现与生存。

Formulas / 公式

During exercise, breathing and circulation increase delivery of oxygen and nutrients, muscles use them to produce energy, and endocrine signals help regulate fuel.
运动时，呼吸与循环增加氧气和营养物质输送，肌肉利用这些物质产生能量，内分泌信号帮助调节燃料。

Worked examples / 实例

- > Explain why no organ system works in isolation.
解释为什么器官系统不能孤立工作。

Exam focus / 考核要点

Name two systems involved in thermoregulation.
指出参与体温调节的两个系统。

Feedback and integrated examples · 反馈与综合实例

First, picture it / 先这样想

Your body is full of smoke alarms. Most of them work by negative feedback: the alarm senses smoke, and instead of making more smoke, it switches on the sprinklers and the room goes back to normal. That is negative feedback - the response undoes the change. But a smoke alarm is also wired into a fire drill. Once the alarm is triggered, the building does not calmly return to normal. It escalates, on purpose, until the fire is out. That is positive feedback - the response makes the change bigger, and it is not a fault, it is the design.

你的身体里装满了烟雾报警器。大部分靠负反馈工作：报警器感知到烟雾后，不是制造更多烟，而是打开喷淋，让房间恢复正常。这就是负反馈--反应抵消了变化。但报警器也接着消防演练。一旦报警触发，整栋楼不会平静地回到正常，而是会升级，直到火被扑灭。这就是正反馈--反应让变化变大，这不是故障，而是设计。

The detail / 详细讲解

Negative feedback is the most common control mechanism in the body, and it is the default.
负反馈是身体中最常见的调控机制，也是默认模式。

It counteracts a change and returns the variable toward its normal range.
它抵消变化，把变量拉回正常范围。

When body temperature rises, temperature-sensitive receptors in the skin and the hypothalamus detect it, and the response is sweating and widening of skin blood vessels, which pushes heat out.

当体温上升时，皮肤和下丘脑中的温度感受器检测到变化，反应是出汗和皮肤血管扩张，把热量推出去；

When it falls, the response is shivering and narrowing of skin blood vessels, which keeps heat in.

当体温下降时，反应是发抖和皮肤血管收缩，把热量留住。

Crucially, the response stops as soon as the variable is back in range.

关键在于，一旦变量回到正常范围，反应就会停止。

That switching off is the defining feature of negative feedback, and it is what stops a fever from cooking you.

正是这种“会关掉”的特性构成了负反馈的定义，也正是它让发烧不会把你煮熟。

Positive feedback is rarer, and it has a specific job: it amplifies a change until a particular endpoint is reached.

正反馈较少见，它有一个明确的职责：把变化放大，直到达到某个特定终点。

Blood clotting is the standard example.

凝血就是标准例子。

A small cut damages a vessel wall, platelets gather at the site, and the clotting factors they release attract more platelets, which release more factors.

血管壁有一处小伤口，血小板聚集到损伤处，它们释放的凝血因子又吸引更多血小板，后者释放更多因子。

The loop is self-amplifying.

这个循环是自我放大的。

It runs until the clot seals the damage, and then it stops - but the endpoint is the point.

它一直运行到血凝块封住损伤处才停止--但重点正在于那个终点。

Without positive feedback you would never stop bleeding.

没有正反馈你就永远止不住血。

Childbirth is the other classic example: contractions trigger hormones that strengthen contractions, until the baby is delivered.

分娩是另一个经典例子：宫缩引发激素，激素加强宫缩，直到胎儿娩出。

The integrated examples are where this stops being abstract.

把知识整合起来，抽象概念就落地了。

In thermoregulation during a hot match, temperature receptors detect the rise, the control centre compares it against the set point, the effectors - sweat glands and skin blood vessels - act, and water is conserved through the antidiuretic hormone.

在炎热比赛中的体温调节

：温度感受器检测到上升，控制中心把它与设定点比较，效应器--汗腺和皮肤血管--开始工作，同时抗利尿激素保存水分。

The nervous system handles the immediate correction; the hormonal system handles the slower conservation of fluid.

神经系统处理即时校正，激素系统处理更慢的保水。

One event, two systems, two timescales.

同一件事，两个系统，两种时间尺度。

The fight-or-flight response is the other integrated example.

战斗或逃跑反应是另一个整合例子。

Sympathetic nervous activity and adrenaline together raise heart rate, breathing rate, muscle blood flow and available energy, all at once, in response to one stimulus.

交感神经活动与肾上腺素一起，同时提高心率、呼吸频率、肌肉血流量和可用能量，共同回应同一个刺激。

The body is not choosing between systems here.

身体在这里并没有在系统之间做选择，而是有意同时调用两个，因为威胁既需要即时反应，也需要持续反应。

It is calling both, on purpose, because a threat needs an immediate response and a sustained one. Every part of that sentence is examinable, and the ability to use it on an unfamiliar example is exactly what the command word Apply is asking for.

这句话的每个部分都可以考，而能否把这个框架用在一个陌生的例子上，正是命令词“应用”所要求的能力。

The numbers / 算一算

Work through a real thermoregulation chain with numbers, because that is the shape the exam likes. A runner starts at 37 degC. In a hot race, core temperature climbs by about 1-1.5 degC before the correction catches up. Sweat rate in a hot endurance event can reach 1-2 litres per hour, and the guidance in the workbook is that losing more than about 2% of body mass through sweat is the threshold to act on. For a 60 kg athlete, 2% is 1.2 kg of

fluid. That is why drinking to a plan, not to thirst, is the practical skill this section is really teaching.

用数字走一遍真实的体温调节链，因为考试就喜欢这种形式。跑者起始体温37

degC。在炎热比赛中，核心体温会上升约1到1.5

degC，校正才追得上。高温耐力事件的出汗率可达每小时1到2升，而工作册给出的指导是：因出汗损失超过体重的约2%就是需要行动的

Common mistakes / 易错点

- "Negative feedback means the body stops the change." It means the body opposes the change until the variable returns to range, then switches off. The change itself happened; the response undoes it.
"负反馈就是身体阻止变化。"它的意思是身体对抗变化直到变量回到范围，然后关闭。变化本身已经发生了，反应把它抵消掉。
- "Positive feedback is a fault in the body." It is a designed mechanism with an endpoint. Clotting and childbirth both fail without it.
"正反馈是身体的故障。"它是一个有终点的设计好的机制。没有正反馈，凝血和分娩都会失败。
- "One change in the environment can only produce one response." The whole point of inter-system communication is that one stimulus can trigger a fast response and a slow response at the same time.
Temperature rise gives you sweating and ADH release.
"环境的一次变化只能产生一个反应。"系统间通讯的全部意义正在于，一个刺激可以同时触发一个快反应和一个慢反应。体温上升

Why it matters / 为什么重要

Feedback is what makes an athlete adaptable rather than merely reactive. A body that only reacted once would be helpless the second time a condition changed. Instead, every training session, every acclimatisation and every taper teaches the control systems a slightly better setting, which is the physiological reason training works at all. When a question asks you to justify a training change, the answer is almost always a variable, a receptor, a control centre, an effector and a feedback loop.

反馈机制让运动员变得有适应力，而不只是被动反应。一个只会反应一次的身体，在条件第二次改变时就无能为力了。相反，每一次训练

- Thermoregulation in heat involves sensory detection, nervous control of sweating and vasodilation, and endocrine water conservation through ADH.
高温下的体温调节包括感觉检测、神经控制出汗与血管舒张，以及通过抗利尿激素进行的水分保存。
- After a meal, pancreatic insulin helps cells take up glucose and storage occurs in the liver and muscles.
餐后，胰腺分泌的胰岛素帮助细胞摄取葡萄糖，肝脏和肌肉储存能量。
- Fight or flight uses sympathetic nervous activity and adrenaline to raise heart rate, breathing, blood flow to muscles and available energy.
战斗或逃跑反应利用交感神经活动和肾上腺素，提高心率、呼吸、肌肉血流和可用能量。

Key terms / 核心术语

Thermoregulation 体温调节

Control of temperature through heat loss, heat production and circulation.

通过散热、产热和循环控制体温。

Eustress 积极压力

Useful challenge that energises and focuses performance.

有益的挑战，可激活并集中表现。

Formulas / 公式

A football player in a hot match sweats, vasodilates and shifts blood between the skin, muscles and other tissues.

Explain the nervous and endocrine contributions.

足球运动员在高温比赛中出汗、血管舒张，并在皮肤、肌肉等组织间调整血流。解释其中神经和内分泌的作用。

Worked examples / 实例

>

Compare negative and positive feedback with one example of each.

比较负反馈与正反馈，并各举一个例子。

Exam focus / 考核要点

Why is integration important during exercise?

为什么运动中的系统整合很重要？

Nervous and endocrine control of movement

运动中的神经与内分泌控制

Guiding question / 引导问题

How do the nervous and endocrine systems control movement and physical performance?

神经和内分泌系统如何控制运动与身体表现？

Movement begins with sensory information and a decision in the CNS, continues through motor output to an effector, and is refined by feedback. Hormones modify readiness, energy availability, strength, tissue repair and the physiological response to training.

运动始于感觉信息和中枢作出的决定，经运动输出到达效应器，并由反馈不断修正。激素调节准备状态、能量可用性、力量、组织修复和训练对生理反应的调节。

Voluntary movement and reflexes · 随意运动与反射

First, picture it / 先这样想

Every movement you make on purpose goes through a phone call with the control room. You decide to jump, your brain calls your leg muscles, and they obey. But some things must not wait for a phone call. If something hot touches your hand, waiting for your brain to weigh up the options would be far too slow. So instead a shortcut wire runs straight from the skin to the spinal cord and back to the muscle, with no call to the brain at all. It is a reflex, and the reason it exists is simple: speed beats politeness when you are being burned.

你所有有意识做出的动作，都要先给控制室打一通电话。你决定跳，脑就打电话给腿部肌肉，肌肉服从。但有些事不能等电话。如果有东西很烫碰到你的手，等大脑权衡利弊再决定就太慢了。所以有一条捷径线路直接从皮肤连到脊髓，再回到肌肉，完全不打电话给脑。它就是反射，存在的理由很简单：当你被烫到时，速度比礼貌更重要。

The detail / 详细讲解

Voluntary movement runs through four stages, and examiners like them in this order.

随意运动分四个阶段，考试官喜欢按这个顺序。

First, sensory input: receptors detect something - light, sound, touch or position.

第一，感觉输入：感受器检测到某种东西--光、声、触觉或位置。

Second, central processing: the brain and spinal cord interpret that information.

第二，中枢处理：脑和脊髓解释这些信息。

Third, motor output: motor neurons carry the instruction out to the muscles or glands.

第三，运动输出：运动神经元把指令送到肌肉或腺体。

Fourth, effector response: the muscle or gland actually does something.

第四，效应器反应：肌肉或腺体真正做出动作。

If you can name all four in order for an unfamiliar example, you have answered most of the Trace and Sequence questions in the paper.

如果你能在陌生例子中按顺序说出这四步，卷面上大多数“追踪”和“排序”题就答出来了。

Reaction time is the gap between the stimulus appearing and the movement starting, and it is measurable, which makes it examinable.

反应时间是刺激出现到动作开始之间的间隔，它是可测量的，所以可考。

Visual reaction time is typically around 200-250 milliseconds; auditory is faster, around 140-180 milliseconds, because sound reaches the brain slightly sooner than light does.

视觉反应时间通常约200到250毫秒；

Anticipation is what a trained athlete actually does in competition: they start before the stimulus arrives, which is why sprinters in the blocks are already moving at the gun rather than reacting to it.

听觉更快，约140到180毫秒，因为声音到达大脑略早于光。

Reacting is what untrained players do, and it costs about a fifth of a second.

预判才是训练有素的运动员在比赛中真正做的事：他们在刺激到来之前就启动了，这就是为什么起跑器上的短跑运动员在枪响时已经作出反应的是未经训练的选手，而他们需要付出大约五分之一秒的代价。

Reflexes are fast, involuntary and protective, and the spinal cord is in charge.

反射快速、不受意志控制、具有保护性，并且由脊髓负责。

The knee jerk is the textbook example: the tendon under the kneecap is stretched, the stretch receptors in the muscle fire, the information goes straight to the spinal cord, motor neurons respond, the quadriceps contracts and the lower leg swings forward - all before the brain has registered that anything happened.

膝反射是教科书例子：膝盖下方的肌腱被牵拉，肌肉中的牵张感受器兴奋，信息直接进入脊髓，运动神经元作出反应，股四头肌收缩。

A reflex is not a smaller version of a voluntary movement. It is a different circuit, designed for a different job.

反射不是随意运动的“缩小版”，它是另一套线路，为另一种任务而设计。

The spinal cord is also smart enough to know when not to answer.

脊髓还聪明到知道什么时候不该回应。

If a muscle is already contracted, stretching it further would risk tearing it, so the circuit reduces its own response.

如果一块肌肉已经收缩，再进一步牵拉就有撕裂风险，于是这个回路会降低自己的反应强度。

This is called reciprocal inhibition: the muscle that must relax is deliberately switched off at the same time as the muscle that contracts.

这叫交互抑制：应该放松的那块肌肉，会在收缩肌工作的同时被主动关闭。

It is the reason a knee jerk is a smooth movement rather than a spasm, and it is a detail that separates a strong answer from a correct one.

这正是膝反射是一个平滑动作而不是痉挛的原因，也是把“正确”答案提升为“优秀”答案的一个细节。

Finally, reflexes do not stop at the knee.

最后，反射也不止于膝反射。

Walking, swallowing, breathing, blinking and the withdrawal of a hand from a hot surface are all spinal reflexes running in everyday life.

行走、吞咽、呼吸、眨眼，以及把手从热物体上缩回，都是日常生活中的脊髓反射。

They are not things you decided to do. They are the default wiring that you do not have to think about, which is exactly what makes them reliable under pressure.

它们不是你决定去做的事，而是你不必思考的默认线路--这正是它们在压力下依然可靠的原因。

The numbers / 算一算

The numbers here are small and worth memorising. Visual reaction time is about 200-250 ms; auditory reaction time about 140-180 ms; the total time for a simple reflex arc such as the knee jerk is about 30-50 ms, because it never reaches the brain. Put those three side by side and the design principle becomes obvious: the spinal cord is roughly four times faster than seeing and reacting on purpose. That gap is the entire reason reflexes exist.

这一节的数字很小，值得记住。视觉反应时间约200到250毫秒；听觉反应时间约140到180毫秒；像膝反射这样一条简单反射弧的完整时

Common mistakes / 易错点

-

"A reflex is a very fast voluntary response." It is not voluntary and it is not a response you chose. Involuntary and fast is the whole point; voluntary would make it useless in an emergency.

"反射是一种非常快的随意反应。"它既不随意，也不是你选择的反应。不受意志控制而且快，这正是它的意义；如果变成随意的，

-

"All reflexes are processed in the brain." Most protective reflexes are integrated in the spinal cord. The brain is told afterwards, if at all.

"所有反射都在大脑处理。"大多数保护性反射是在脊髓中整合的。大脑要么之后才被告知，要么根本不会被告知。

-

"Reaction time is how fast you move." Reaction time is the delay before you start moving. Movement speed is a separate thing, and confusing them costs marks.

"反应时间就是你动作有多快。"反应时间是你开始动之前的延迟。动作速度是另一回事，混淆这两者会丢分。

Why it matters / 为什么重要

Reaction time is a real, trainable, measurable number in sport, and it is one of the few physiological things where a beginner and an elite differ by a large, obvious margin. A goalkeeper who reads a penalty earlier by 50 milliseconds has a genuine physical advantage. This is also where the Apply command word earns marks: a

sprinter using a block start, a netballer anticipating a pass, and a goalkeeper setting themselves are all the same principle - using prediction instead of reaction, because prediction is free and reaction always costs you time.

反应时间在运动中是真实、可训练、可测量的数字，也是少数几个新手与顶尖选手差距巨大且明显的生理指标之一。一个能早50毫秒判断

- Reaction time is the interval from the stimulus to the start of movement.
反应时是从刺激出现到运动开始之间的时间。
- A reflex is very fast, involuntary, protective and integrated mainly at the spinal cord. It can protect the body while the brain continues processing other information.
反射快速、不自主、保护性强，主要由脊髓整合。它可以在大脑处理其他信息时保护身体。

Key terms / 核心术语

Motor output 运动输出

Motor-neurone impulses sent from the CNS to an effector.

从中枢传向效应器的运动神经元冲动。

Feedback 反馈

Information about the movement returned to the CNS to adjust the next response.

关于运动的信息返回中枢，以调整下一次反应。

Formulas / 公式

In a knee-jerk response, tapping the patellar tendon stretches the quadriceps muscle spindle, information enters the spinal cord, and the quadriceps contracts so the lower leg moves forward.

在膝反射中，敲击髌腱使股四头肌肌梭被牵张，信息进入脊髓，股四头肌收缩，小腿向前运动。

Worked examples / 实例

>

Distinguish a reflex from a voluntary movement using pathway and speed.

从通路和速度区分反射与随意运动。

Exam focus / 考核要点

Trace a sprint start from stimulus to effector response.

追踪冲刺开始时的刺激到效应器反应。

Hormonal influences and sport applications · 激素影响与运动应用

First, picture it / 先这样想

Nerves are a text message. Hormones are a letter that arrives by post. The text arrives first and is specific: it goes to one phone. The letter travels slowly, reaches everyone in the house, and changes the mood in the whole house. In sport that means a hundredth of a second can come from your nerves, but a change that lasts an hour - how much energy your body is willing to spend today, how much it is willing to repair tonight - comes from hormones.

神经像一条短信，激素像一封走邮寄来的信。短信先到而且精准：它只发到某一部手机。信走得很慢，送到屋子里的每个人那里，改变的是整间屋子的情绪。在运动中这就意味着：百分之一秒的反应来自神经，而持续一小时的改变--你的身体今天愿意投入多少能量、今晚愿意修复多少--来自激素。

The detail / 详细讲解

The nervous system supports timing, precision and correction.

神经系统负责时机、精度和校正：它决定肌肉何时收缩、收缩多大力，并在几毫秒内纠正错误。

It decides when a muscle fires and how hard, and it corrects errors within milliseconds.

内分泌系统做的则不同：它在几分钟、几小时和几天的时间尺度上改变身体的配置。

The hormonal system does something different: it changes how the body is set up over minutes, hours and days.

比赛前，肾上腺素提高心率、把血液重新分配到肌肉并释放葡萄糖，使燃料到位。

Before a race, adrenaline raises heart rate, redirects blood to the muscles and releases glucose so that fuel is

available. None of that is an accident of the moment; it is a deliberate change of the body's settings, made in advance, and it cannot be undone by willpower.

这些都不是偶然发生的，而是对身体设定值有意做出的调整，事先就完成了，而且无法靠意志力撤销。

The major stress hormones are adrenaline and cortisol.

主要的应激激素是肾上腺素和皮质醇。

Adrenaline is the acute one.

肾上腺素是急性的那一个；

Cortisol is the chronic one: it raises available glucose, which is useful, but if it stays high for long enough it promotes protein breakdown, which is exactly the wrong thing for an athlete trying to build or repair tissue.

皮质醇是慢性的：它提高可用血糖，这很有用，但如果长期维持在高水平，就会促进蛋白质分解，而这对一个试图增加或修复组织的运动员来说，恰恰是错误的事情。

This is the clearest example on the whole course of why "more" is not always "better" - a hormone that is essential in a short emergency can be destructive in a long-term situation.

这是整门课程中最清楚地说明“更多不等于更好”的例子--在短暂紧急情况下不可或缺的激素，在长期情境下可能是有害的。

The other hormones you need to know run on much longer timescales.

其他需要了解的激素作用时间要长得多。

Testosterone supports muscle protein synthesis, strength and power, and it is the reason resistance training changes the body structurally rather than just temporarily.

睾酮支持肌肉蛋白质合成、力量和功率，这就是阻力训练能结构性改变身体、而不只是暂时改变的原因。

Growth hormone supports growth and tissue repair, with much of its work happening during sleep.

生长激素支持生长和组织修复，其中很大一部分工作发生在睡眠期间。

Insulin and glucagon between them keep blood glucose in a usable range - insulin moves glucose into cells after a meal, glucagon releases glucose from storage when fuel runs low.

胰岛素和胰高血糖素共同把血糖维持在可用范围--胰岛素在餐后把葡萄糖移入细胞，胰高血糖素在燃料不足时把葡萄糖从储备中释放出来。

Oestrogen is relevant too: it contributes to bone health, and its cyclical change is one reason iron needs differ between people.

雌激素同样相关：它有助于骨骼健康，其周期性变化也是不同人群铁需求存在差异的原因之一。

Notice how the two systems divide the work.

注意两个系统是如何分工的。

Timing, precision and immediate correction are neural.

时机、精度和即时校正属于神经；

Sustained change in capacity - how much power, how much you can repair, how readily you mobilise fuel - is hormonal.

能力的持续改变--力量有多大、能修复多少、能多快调动燃料--属于激素。

Sport is full of moments that need both: the first three seconds of a 100 m sprint are almost entirely nerve, and the adaptation that makes those three seconds faster next month is almost entirely hormone.

运动中充满了同时需要两者的时刻：100米短跑的前三秒几乎完全是神经系统的工作，而让这三秒在下个月变快的适应，则几乎完全是激素。

The boundary matters for exam answers.

边界对考点的答题很重要。

You do not need hormone concentrations, half-lives, receptor subtypes or the precise molecular pathway for any of these.

这些内容你不需要知道激素浓度、半衰期、受体亚型或精确的分子通路。

You need to know which hormone is involved, what broad job it does, which system it belongs to, and give a real sport example where it is the reason something happens.

你需要知道的是：涉及哪种激素、它大致负责什么、属于哪个系统，并给出一个真实的运动例子说明它为什么导致某件事发生。

The numbers / 算一算

Three numbers to keep. A maximal sprint runs about 10 seconds and is powered almost entirely by the phosphagen system, so the adrenaline that arrives after the first few seconds is too late to help the race itself - it helps the ten minutes after it. Bone is the slowest tissue to adapt and the fastest to lose, which is why load and recovery have to be managed over months rather than weeks. And a rough rule of thumb: resistance training produces measurable strength change over about 4 to 8 weeks, and bone density responds over 6 to 12 months.

The fastest thing you can change in sport is nerve and skill; the slowest is bone.

记住三个数字。全力冲刺约10秒，几乎完全由磷酸原系统供能，所以几秒后才到达的肾上腺素对比赛本身来说已经太迟--它帮助的是之后

Common mistakes / 易错点

- "Adrenaline makes you faster at the start of the race." For a maximal sprint it arrives after the decisive seconds. Adrenaline matters most in sustained effort, in the recovery afterwards, and in the seconds-to-minutes window of a 400 m or 800 m.
"肾上腺素让你在比赛开始时更快。"在全力短跑中，它到达时决定性的几秒已经过去了。肾上腺素在持续性项目、之后的恢复期，
- "Growth hormone builds muscle." Its main job is growth and tissue repair, and much of it acts during sleep. For building muscle in a trained adult, resistance training and adequate protein are the levers; growth hormone is a supporting actor, not the star.
"生长激素长肌肉。"它的主要职责是生长和组织修复，而且很大一部分在睡眠中起作用。对训练中的成年人来说，增肌的杠杆是阻
- "Hormones are the slow system, so they do not matter in a fast sport." Slow is not the same as unimportant. The adrenaline that starts your sprint at 0.6 seconds, the growth hormone that repairs tissue overnight and the insulin that decides how fuel is handled after a meal are all slow, and all essential.
"激素是慢的系统，所以在快的运动里不重要。"慢不等于不重要。让冲刺在0.6秒启动的肾上腺素、修复夜间组织的生长激素、以及

Why it matters / 为什么重要

This section is the difference between explaining and guessing. When a question asks why a sprinter needs a good sleep, or why a teenage athlete is growing faster than an adult, or why tapering matters before a competition, hormonal knowledge lets you give the actual mechanism instead of a vague answer about rest. And because the paper is built on command words, the highest-value skill here is not memorising gland names. It is being able to take any hormone, describe its broad role, and link it to a specific moment in a specific sport.

这一节是"解释"和"猜"之间的分界线。当题目问为什么短跑运动员需要充足睡眠、为什么青少年长得比成年人快、为什么赛前减量重要时

- Adrenaline from the adrenal glands increases heart rate, breathing, blood redistribution and glucose mobilisation.
肾上腺分泌的肾上腺素提高心率、呼吸、血流重新分配和葡萄糖动员。
- Testosterone supports muscle protein synthesis, strength and power.
睾酮支持肌肉蛋白质合成、力量和功率。
- Growth hormone supports growth and tissue repair; cortisol increases available glucose but can contribute to protein breakdown when its effect is prolonged.
生长激素支持生长和组织修复；皮质醇提高可用葡萄糖，但长期作用可能促进蛋白质分解。

Formulas / 公式

During a 100 m sprint, the CNS rapidly recruits motor units while adrenaline supports cardiovascular activation. During strength training, recruitment and training adaptations can increase force; during recovery, growth hormone and nutrition support repair.

100米冲刺中，中枢快速募集运动单位，肾上腺素支持心血管激活；力量训练中，募集和训练适应可增加力量；恢复期，生长激素和营养

Worked examples / 实例

- > Name two hormones and connect each to a performance or recovery role.
指出两种激素，并分别联系其表现或恢复作用。

Exam focus / 考核要点

How do nervous and endocrine systems cooperate during a weightlifting event?

力量举过程中，神经和内分泌系统如何合作？

Water and electrolyte balance

水与电解质平衡

Guiding question / 引导问题

How do water and electrolytes maintain health and support performance?

水和电解质如何维持健康并支持表现？

Water and electrolytes are needed for thermoregulation, circulation, metabolism, nerve impulses, muscle contraction and joint function. Fluid balance depends on matching intake with losses from sweat, urine, respiration and faeces.

水与电解质支持体温调节、循环、代谢、神经冲动、肌肉收缩和关节功能。体液平衡取决于摄入与汗液、尿液、呼吸和粪便损失的匹配。

Functions, intake and loss · 功能、摄入与流失

First, picture it / 先这样想

Your body is a swimming pool that is never empty and never still. Two taps fill it: what you drink, and the water inside the food you eat. Four drains empty it: sweat, urine, the water you breathe out, and water leaving in faeces. You are not managing a tank that you fill once. You are managing a pool that is being filled and drained every minute of every day, and health is the level staying inside a fairly narrow band.

你的身体像一个永不枯竭、也从不静止的游泳池。两个进水口负责注入：喝下的水，以及食物中所含的水。四个出水口负责排出：汗液、尿液、呼吸中呼出的水，以及粪便带走的水。你管理的不是一个只需灌满一次的水池，而是一个每一分钟都在注入和排出的水池；健康就是水位始终留在一个相当窄的范围内。

The detail / 详细讲解

About 60% of an adult body is water.

成年人体内约有60%是水。

That sounds like a lot, and it has to be, because water is the medium in which almost everything happens.

这个比例听起来很高，而且必须这么高，因为水几乎是所有事情发生的介质。

Every chemical reaction in the body is a reaction in water.

身体里的每一次化学反应都是在水中进行的；

Every nutrient is transported in water.

每一种营养物质都在水中被运送；

Every joint is lubricated by water.

每一个关节都由水来润滑。

And most importantly for sport, water is the only way the body can carry heat away from working muscle.

而对运动最重要的是：水是身体把热量从工作肌肉带走的唯一途径。

List what water actually does and it is easier to remember: it regulates temperature by sweating, it transports nutrients and waste, it lubricates tissues, it supports blood volume and therefore blood pressure, and it makes chemical reactions possible.

把水实际做的事列出来，就容易记住了：它通过出汗调节体温，运送营养和废物，润滑组织，支持血容量从而维持血压，并使化学反应可能。

Notice that four of those five are about moving or protecting something.

注意这五项中有四项与“运送”或“保护”有关。

Water is less a nutrient and more the medium everything else happens in.

水与其说是一种营养素，不如说是其他一切发生其中的介质。

Electrolytes are the dissolved minerals that make water work.

电解质是溶解在水中的矿物质，正是它们让水发挥作用。

Sodium holds fluid in the right places and lets nerves fire; potassium is what muscle contraction and the heart rhythm depend on; chloride helps balance fluid and acid.

钠把液体维持在正确的位置，并让神经能够放电；钾是肌肉收缩和心律所依赖的东西；

Without them the water is there but the chemistry falls apart, which is why drinking plain water for hours on end

can leave you worse off than not drinking - you dilute the very salts you needed.

氯有助于维持体液和酸碱平衡。

没有它们，水还在，但化学就乱了套--这正是连续数小时只喝白水可能让你状态更差的原因：你把本来需要的盐分稀释掉了。

The key idea is that fluid balance is a matching problem, not a willpower problem.

关键在于：体液平衡是一个"匹配问题"，而不是"意志力问题"。

You stay healthy when intake roughly equals loss, and you fall behind when losses exceed intake.

当摄入大致等于流失时你保持健康，当流失超过摄入时你就开始落后。

In a cool room on a normal day, losses are modest and thirst is a good enough guide.

在凉爽房间里的普通一天，流失不大，口渴本身就是足够好的指引。

In a hot race, losses can be large enough that thirst arrives far too late, which is why drinking to a plan matters more than drinking when you feel like it.

而在炎热比赛中，流失可能大到口渴来得太晚--这就是为什么"按计划补水"比"想喝才喝"更重要。

The numbers / 算一算

Numbers worth having in your head. Total body water is about 60% of adult body mass, so a 70 kg athlete carries roughly 42 litres. That water arrives as drinking and food, and leaves as roughly 1-2 litres of urine a day, a quarter to half a litre of sweat on an easy day, and a smaller amount through breathing and faeces. In the heat, sweat loss rises to about 0.5-2.0 litres per hour with a sodium loss of roughly 300-1000 mg per hour. The exam habit: put the two taps and the four drains on the same page, then explain any scenario by saying which drain opened and whether the tap kept up.

需要记在脑子里的数字。成年人体内水分约占体重60%，所以一位70公斤的运动员体内约有42升水。这些水来自饮水和食物，而每天大约

Common mistakes / 易错点

-

"Water is just a drink you take after exercise." Water is the medium for temperature control, transport and every chemical reaction, and you need it before, during and after - not just after.

"水不就是运动后喝的那杯水吗。"水是体温调节、物质运输和每一次化学反应的介质，你在运动前、进行中和之后都需要它--不只是

-

"Electrolytes only matter in endurance events." Sodium, potassium and chloride are involved in nerve firing, muscle contraction and heart rhythm in every session, including a 100 m sprint.

"电解质只在耐力项目中才重要。"钠、钾和氯参与每一次训练中的神经放电、肌肉收缩和心律，包括100米冲刺。

-

"Thirst tells you when to drink." On a normal day it is a reasonable guide. In a hot long event, fluid loss outruns the thirst signal, which is why plans exist.

"口渴会告诉你什么时候喝水。"普通情况下它是合理指引。在炎热的长距离项目中，流失会跑在口渴信号前面，所以才有"计划"这

Why it matters / 为什么重要

Every performance has a hidden variable, and in the heat it is fluid. Two per cent of body mass lost to sweat is the line the workbook uses, and for a 70 kg athlete that is 1.4 kg. Below it, performance is largely protected. Above it, core temperature climbs, the heart has to beat faster to move the same blood, the gut gets starved of flow, and the central nervous system begins to suffer, which shows up as worse decision-making and earlier fatigue. Drinking is not a chore attached to training. It is one of the variables that decides how the session goes.

每一次表现背后都有一个隐藏变量，而在高温中它就是体液。2%的体重因出汗而下降是工作册给出的界线，对70公斤的运动员来说是1.4

-

Electrolytes include sodium, potassium and chloride. Sodium supports fluid balance and nerve function; potassium supports muscle contraction and heart rhythm; chloride supports fluid and acid-base balance.

电解质包括钠、钾和氯。钠支持体液平衡和神经功能；钾支持肌肉收缩和心律；氯支持体液和酸碱平衡。

Key terms / 核心术语

Dehydration 脱水

Body-water deficit that can increase heart rate, core temperature, fatigue and heat-illness risk.

身体缺水，可增加心率、核心体温、疲劳和热病风险。

Osmolality 渗透压

Concentration of dissolved particles in a solution.

溶液中溶解粒子的浓度。

Formulas / 公式

Exercise in heat can increase sweat loss to about 0.5-2.0 litres per hour and sodium loss to about 300-1000 mg per hour.

高温运动可使汗液损失增加到约每小时0.5-2.0升，钠损失约每小时300-1000毫克。

Worked examples / 实例

>

Explain why fluid replacement must match sweat losses in hot endurance exercise.

解释为什么高温耐力运动中的补液必须与汗液损失相匹配。

Exam focus / 考核要点

Name the main routes of fluid intake and loss.

说出液体摄入和损失的主要途径。

Imbalance and hydration monitoring · 失衡与水合监测

First, picture it / 先这样想

Checking hydration is like checking a car with one instrument on the dashboard. The oil light tells you something, the temperature gauge tells you something else, and neither one is the whole story. A good driver cross-checks. Body-mass change tells you how much fluid left, urine colour tells you roughly how concentrated it is, and urine osmolality tells you how hard the kidneys are working to hold water. Any one of them alone can mislead you.

检查水合状态就像只靠仪表盘上的一个指示灯判断车况。机油灯说明一些事，水温表说明另一些事，而哪一个都不是全部。好的司机会交叉核对：体重变化告诉你流失了多少液体，尿色大致告诉你尿液有多浓，尿液渗透压则告诉你肾脏为了留住水分有多用力。单独依赖其中任何一个都可能误导你。

The detail / 详细讲解

Dehydration is simply a water deficit: you have lost more fluid than you have taken in.

脱水就是水分赤字：你流失的液体超过摄入的液体。

It is the mildest and most common of the three problems here, and it is also the one that a well-designed session should never produce.

它是这里三种问题中最轻、也最常见的一种，同时也是一个设计良好的训练根本不该产生的那种。

A 1-2% deficit sounds trivial, but it reliably raises heart rate at a given workload, raises core temperature, increases perceived effort and multiplies the risk of heat illness.

1%到2%的赤字听起来微不足道，但它会可靠地让同等负荷下的心率上升、核心体温升高、主观用力感增加，并成倍放大热病风险。

Hypernatremia is too much sodium in the blood.

高钠血症是血液中钠过多。

It usually follows heavy sweating with too little fluid replacement, because sweat loses more water than sodium, so replacing the volume with plain water concentrates what is left.

它通常发生在大量出汗而补液不足之后，因为汗液流失的水分多于钠，用白水补回体积只会让剩下的部分更浓。

The picture is a runner who has sweated a great deal, has not drunk enough, and whose blood sodium has climbed.

典型画面是：一位大量出汗、补水不足的跑者，血钠升高。

The fix is fluid with sodium in it, not more plain water.

解决办法是补含钠的液体，而不是更多白水。

Hyponatremia is the opposite: too little sodium in the blood.

低钠血症正相反：血液中钠太少。

It is the one that surprises people, because it is caused by drinking too much rather than too little.

它最让人意外，因为原因是喝得太多而不是太少。

If a runner drinks several litres of plain water over a long hot event and does not replace sodium, the blood sodium

can be diluted below safe levels - and the symptoms, which include nausea, headache and confusion, look like dehydration.

如果一位跑者在炎热的长距离项目中喝了好几升白水却没有补钠，血钠可能被稀释到不安全水平--而症状包括恶心、头痛和意识模糊，

This is the single most useful thing to understand on this page, because the treatment is the opposite of what the patient wants.

这是这一页最值得理解的一点，因为治疗方法和病人想要的恰好相反。

So how do you monitor it?

那么该怎么监测？

The workbook guidance is to keep body-mass loss under 2% during exercise, and to use more than one indicator.

工作册的指导是：运动中体重下降保持在2%以内，并使用一个以上的指标。

Body-mass change is the most direct: weigh yourself before and after, nude and dried, and subtract.

体重变化最直接：运动前后称重，脱去衣物并擦干，然后相减。

Urine colour is a crude but useful hydration marker, and it is a poor stand-in on its own because it is also affected by vitamins, medication and certain foods.

尿色是一个粗糙但有用的水合标志，单独使用并不理想，因为维生素、药物和某些食物也会影响它。

Urine osmolality measures how concentrated the urine is, which reflects how hard the kidneys are working to conserve water.

尿液渗透压测量尿液浓度，反映肾脏为保存水分而工作的程度。

The exam-safe way to answer any monitoring question is to name at least two indicators, say what each one reports, and then say why one alone is not enough.

回答任何监测类问题的稳妥方式是：说出至少两个指标，说明各自反映什么，然后解释为什么单独一个不够。

That last clause is what separates a correct answer from a good one.

最后这一点正是把"正确"答案变成"优秀"答案的关键。

The numbers / 算一算

Do the arithmetic once so it is automatic. A 70 kg athlete who finishes a hot race 1.4 kg lighter has lost exactly 2% of body mass, and 1.4 kg of water is 1.4 litres. For sweat rates, 0.5-2.0 litres per hour over a two-hour race is 1 to 4 litres - which is why a two-hour road event in the heat can empty a normal water bottle several times over. And the sodium side of the same event: at 300-1000 mg per hour, a two-hour race costs 600-2000 mg of sodium, roughly a quarter to three-quarters of a teaspoon of salt. That is the number that justifies electrolyte drinks rather than plain water.

把算式做一遍，以后就自动会了。一位70公斤的运动员在炎热比赛后轻了1.4公斤，正好是体重的2%，而1.4公斤水就是1.4升。出汗率方

Common mistakes / 易错点

"Urine colour tells you everything about hydration." It is one indicator among several, and it is confounded by vitamins, medication and food. Weight change and urine osmolality give complementary evidence.

"尿色能说明水合的全部情况。"它只是若干指标之一，还会受到维生素、药物和食物的干扰。体重变化和尿液渗透压提供的是互补

"Hyponatremia happens when you do not drink enough." It happens when you drink plenty but do not replace sodium. The cause is the opposite of dehydration.

"低钠血症发生在喝得不够时。"它发生在喝得很多但没有补钠时。成因与脱水正好相反。

"Losing 2% of body mass is nothing." It is the point at which heart rate, core temperature, gut blood flow and central nervous system function all start to measurably deteriorate.

"损失2%体重没什么大不了。"这正是心率、核心体温、肠道血流和中枢神经功能开始出现可测量恶化的分界点。

Why it matters / 为什么重要

Hydration monitoring is one of the few places where an amateur and a professional are separated by paperwork rather than talent. Weighing before and after, logging what was drunk, and recording urine colour takes about three minutes and it changes what you can conclude about a session. It is also the safest practical advice in this topic: a plan that tells you to drink to a number, and to include sodium when the session is long and hot, is a plan that has been shown to work repeatedly.

水合监测是少数几个把业余者和职业者区分开的领域之一--靠的是记录，而不是天赋。运动前后称重、记录饮水量、记录尿色，大约花三

Use several indicators rather than urine colour alone: body-mass change, urine colour and urine osmolality give complementary evidence.

应结合体重变化、尿色和尿液渗透压，不要只看尿色。

The workbook guideline is to avoid losing more than 2% of body mass during exercise.

工作册给出的指导是运动中体重下降不要超过2%。

Formulas / 公式

A runner loses body mass in a hot race and has concentrated urine. Explain the pattern without claiming that one indicator is sufficient.

跑者在高温比赛中体重下降且尿液浓缩。解释这一模式，但不要声称单一指标足够。

Worked examples / 实例

Classify dehydration, hypernatremia and hyponatremia by cause and effect.

按原因和影响对脱水、高钠血症和低钠血症进行分类。

Exam focus / 考核要点

Why are multiple hydration measures needed?

为什么需要多种水合指标？

ADH and cardiovascular drift · 抗利尿激素与心率漂移

First, picture it / 先这样想

When your body is short of water it does two things at once, and it is worth keeping them apart. ADH is the water authority turning on a water-saving order: your kidneys stop sending water out in urine and start reabsorbing it. Cardiovascular drift is different - it is what happens to your engine when the coolant level drops, so the same job makes the machine run hotter and the fan has to spin faster.

身体缺水时会同时做两件事，把它们分开理解很有价值。抗利尿激素（ADH）是水务局下达的节水令：肾脏不再把水随尿排出，而是开始重新吸收。而心血管漂移是另一回事--它是冷却液减少时发动机发生的事，同样的工作让机器跑得更热，风扇必须转得更快。

The detail / 详细讲解

The trigger is concentration, not volume.

触发条件是浓度，而不是体积。

Osmoreceptors in the hypothalamus sense the concentration of dissolved particles in the blood, which is osmolality.

下丘脑中的渗透压感受器感知血液中溶解粒子的浓度，也就是渗透压。

When it rises, meaning the blood is more concentrated than it should be, the hypothalamus tells the posterior pituitary to release ADH.

当它升高，也就是血液比应有的更浓时，下丘脑会通知垂体后叶释放抗利尿激素。

ADH then acts on the kidney collecting ducts, increasing water reabsorption, so you produce a smaller volume of more concentrated urine.

ADH随后作用于肾集合管，增加水的重吸收，于是你排出的尿量更少、浓度更高。

The body is not making water; it is keeping the water it already has.

身体并没有制造水，而是不停地留住已有的水。

Sweating is the bigger loss.

出汗才是更大的流失。

In the heat, sweat rate can reach 0.5-2.0 litres per hour, which is far more than most people replace.

在高温下，出汗率可达每小时0.5到2.0升，远比大多数人补充的要多。

Blood volume falls, and blood volume is what fills the heart.

血容量下降，而血容量正是填充心脏的东西。

A heart that is filling less completely has to beat faster to deliver the same amount of oxygen to the working muscle.

充盈不足的心脏必须跳得更快，才能把同样多的氧气送到工作肌肉。

That is the first half of cardiovascular drift: same pace, higher heart rate.

这就是心血管漂移的前半段：配速相同，心率更高。

The second half happens in the skin.

后半段发生在皮肤。

As core temperature rises, more blood is sent to the skin to dump heat, and the skin is far from the working muscle.

随着核心体温上升，更多的血液被送往皮肤散热，而皮肤离工作肌肉很远。

Two things follow: the muscle receives a smaller share of the available blood, and a larger share of the total is going to a non-muscle destination.

于是出现两件事：肌肉从可支配血流中分到的份额减少，而总血流中更大的一份被送往非肌肉部位。

So the working muscle must extract more oxygen from each unit of blood it gets.

因此工作肌肉必须从每单位血液中提取更多氧气。

Both halves point the same way, which is why the numbers all rise together at the end of a long hot effort.

两半原因指向同一个方向，这就是为什么在长时间高温运动的末尾，这些数字会一起上升。

ADH cannot fix the whole picture, and it is worth being clear about that.

ADH无法解决全部问题，这一点值得说清楚。

ADH conserves water; it does not replace the volume you sweated, and it does not bring the sodium back.

ADH只能保存水分，它不能补回你出汗失去的体积，也不能把钠补回来。

So the full recovery picture is three things together: replace the fluid, replace the sodium, and replace the water you need for the next session.

完整的恢复图景是三件事一起做：补回液体、补回钠、补回下一次训练所需的水分。

Athletes often do the first and skip the other two, which is why the same athlete can be both dehydrated and low in sodium.

运动员常常只做了第一件，跳过了另外两件--这正是同一位运动员可能既脱水又低钠的原因。

So a good answer to any drift question names the chain in order: sweat loss, falling blood volume, the heart compensating by beating faster, blood being redistributed to the skin, and the muscle extracting more oxygen from what is left.

因此，任何漂移类问题的好答案都要按顺序说出这条链：出汗流失、血容量下降、心脏代偿性加快跳动、血液重新分配到皮肤、肌肉

Each link causes the next.

每一环都引发下一环。

The exam word for this is a mechanism, and a mechanism is exactly what a list of symptoms is not.

考试里这叫作机制，而机制恰恰不是一串症状的罗列。

The numbers / 算一算

The numbers make the drift concrete. Take a runner in the heat, working at the same speed. At the start, core temperature is near 37 degrees C and heart rate is, say, 150 beats per minute. Over an hour the core temperature may rise by 1-1.5 degrees and the heart rate for the identical pace may climb to 165 or even 175. Nothing about the pace changed. That gap of 15-25 beats per minute is cardiovascular drift, and the cause is on the thermometer and the scale, not in the legs. Sweat rate of 0.5-2.0 L/h and sodium loss of 300-1000 mg/h sit underneath both numbers.

这些数字能让漂移变得具体。想象一位在高温中以同样速度跑步的运动员。开始时核心温度接近37摄氏度，心率比如是每分钟150次。

Common mistakes / 易错点

-

"ADH makes you lose water." It does the opposite. ADH increases water reabsorption in the kidney, so you produce less, more concentrated urine.

"抗利尿激素让你流失水分。"它恰恰相反。ADH 增加肾脏对水的重吸收，所以你的尿量更少、浓度更高。

- "Cardiovascular drift means the athlete is going slower." It means the opposite: at the same pace the heart has to beat faster, so the athlete is working harder, not less.

"心血管漂移意味着运动员变慢了。"它的意思正相反：同样配速下心脏必须跳得更快，所以运动员付出得更多，而不是更少。

- "Drinking more fixes drift." Drinking replaces volume and sodium, which reduces the cause. But drift also comes from heat and from skin blood flow, so fluids alone do not remove it entirely.

"多喝水就能消除漂移。"补液能补回体积和钠，从而减少诱因。但漂移还来自高温和皮肤血流，所以单靠液体并不能完全消除它。

Why it matters / 为什么重要

Cardiovascular drift is the best example on the course of a deceptively simple observation explaining a complicated result. A runner finishes a hot race with a higher heart rate at the same pace and a higher lactate, and it is easy to write that off as being unfit. In reality it is a predictable, well-understood response to a hot environment and an unfinished fluid plan. Recognising it means you can predict the outcome before the event - pacing, hydration and cooling all attack it - and that is the difference between knowing and guessing.

心血管漂移是整门课程中最好的例子：一个看起来简单的观察，解释了一个复杂的结果。跑者在炎热比赛后以同样配速出现更高心率和更

- In prolonged exercise in heat, dehydration reduces plasma volume and more blood is directed to the skin for cooling. Stroke volume can fall, so heart rate rises at the same workload; this is cardiovascular drift.

高温长时间运动中，脱水减少血浆容量，更多血液流向皮肤散热。每搏输出量可能下降，因此相同负荷下心率上升，这就是中心血管

Key terms / 核心术语

Negative feedback 负反馈

ADH release rises with the disturbance, water retention restores the variable, and the response then decreases.

扰动增加时抗利尿激素释放上升，水分保留恢复变量，随后反应减弱。

Formulas / 公式

Plan an endurance session in heat: start hydrated, drink regularly, replace sodium, monitor body mass and adjust the plan.

规划高温耐力训练：开始时水合、规律饮水、补充钠、监测体重并调整方案。

Worked examples / 实例

>

Explain the feedback link between osmolality, ADH, kidney water reabsorption and urine concentration.

解释渗透压、抗利尿激素、肾水分重吸收和尿液浓度之间的反馈联系。

Exam focus / 考核要点

What causes cardiovascular drift and how can it be reduced?

中心血管漂移的原因是什么？如何减轻？

Fuelling for health and performance

健康与表现供能

Guiding question / 引导问题

How can nutrition support health, exercise and recovery?

营养如何支持健康、运动和恢复？

Carbohydrate, protein and lipid have different roles, but all three are needed in suitable amounts. Fuelling should match the activity, duration, environment, body composition, age, sex differences and gastrointestinal tolerance.

碳水化合物、蛋白质和脂质作用不同，但三者都需要适量。供能应匹配活动、时长、环境、身体组成、年龄、性别差异和胃肠耐受。

Macronutrients and individual needs · 宏量营养素与个体需求

First, picture it / 先这样想

Think of a car that runs on three different fluids. Petrol is the fast one: it burns quickly, it gives you power straight away, and if the tank is empty you cannot move. Oil is the slow one: it does not make the car go, but without it the engine wears itself out. And spare parts are the third one: the raw material the body uses to repair and rebuild. Ask what a car cannot do without each one and the answer is always the same: it cannot do anything at all without all three.

把汽车想成靠三种不同液体运行的车辆。汽油是快的那一种：烧得快，立刻给你动力，油箱空了车就动不了。机油是慢的那一种：它不让车跑得更快，但缺了它发动机就会自己磨损。零配件是第三种：身体用来修理和重建的原料。问一台车缺了哪一样会怎样，答案总是同一个：缺了任何一样，它什么都做不了。

The detail / 详细讲解

Carbohydrate is the petrol.

碳水化合物就是汽油。

It is the main fuel for high-intensity exercise, and it earns that place for two reasons: it can be broken down quickly without oxygen, so it delivers ATP fast, and it spares protein, which means the body does not have to break down muscle to fuel itself.

它是高强度运动的主要燃料，理由有两个：它可以在无氧条件下被迅速分解，因此能快速提供ATP；

It also supports immune function and replenishes the glycogen stored in your muscles and liver, which is the reserve you draw on during a hard session.

它还能节省蛋白质，意味着身体不必靠分解肌肉来供能。

它还支持免疫功能，并补充储存在肌肉和肝脏中的糖原--那正是你在高强度训练中动用的储备。

Protein is the spare parts and the building material.

蛋白质既是零配件，也是建筑材料。

It builds and repairs muscle and every other tissue, and it supplies the enzymes that run reactions, the hormones that signal, and the transport proteins that carry things around.

它构建和修复肌肉以及其他所有组织，并提供驱动反应的酶、发出信号的激素，以及运送各种物质的运输蛋白。

This is the macronutrient with the most misunderstood job, because people think of protein as only being about muscle.

这是功用最容易被误解的宏量营养素，因为人们以为蛋白质只和肌肉有关。

It is about constantly replacing worn-out material, and that is happening in every tissue all day.

实际上它关乎持续替换已经磨损的材料，而这在所有组织中、一整天都在发生。

Lipid is the oil, and it does three jobs that carbohydrate cannot.

脂质就是机油，它承担着碳水化合物无法承担的三份工作。

It is far more concentrated as an energy store, so it is the long-duration fuel.

它作为能量储备要集中得多，因此是长时间项目的燃料；

It helps protect organs, forming cushions around the heart and kidneys.

它保护器官，在心脏和肾脏周围形成缓冲层；

And it is required to absorb vitamins A, D, E and K, because those vitamins dissolve in fat rather than water.

它还是吸收维生素A、D、E、K所必需的，因为这四种维生素溶于脂肪而不溶于水。

Remove fat and you do not just lose fuel, you lose the ability to take up four vitamins.

去掉脂肪，你失去的不仅是燃料，还有吸收这四种维生素的能力。

Needs are individual, and this is the part that scores.

需求因人而异，而这正是能得分的部分。

Children and adolescents need energy, protein, calcium and iron for growth.

儿童和青少年需要能量、蛋白质、钙和铁来生长；

Menstruation raises iron requirements, which is why iron is specifically relevant to adolescent and female athletes.

月经会提高铁需求，这也是铁与青少年及女性运动员特别相关的原因。

Beyond that, needs move with body size, lean mass, age and how much activity is actually being done.

除此之外，需求还会随体型、瘦体重、年龄以及实际活动量而变化。

The exam-safe answer to "what should this athlete eat" is never a single universal diet. It is a list of factors, applied to one individual.

当题目问"这位运动员应该吃什么"时，稳妥的答案绝不是一份统一食谱，而是一份需要考虑的因素清单，然后应用到具体个人身上。

There is a second idea hiding here, and it is the one that separates a strong answer: energy availability.

这里还藏着第二个概念，也是把优秀答案区分出来的那个：能量可用性。

It is the energy left over for health and normal life after you have subtracted what training spent.

它指的是扣除训练消耗之后，仍可用于健康和正常生活的能量。

The first question is not "did they eat enough calories" but "was there enough left over to stay well".

第一个问题不是"他们摄入的热量够吗"，而是"剩下的够维持健康吗"。

An athlete can meet their energy needs for performance and still be in deficit for health, and that gap is where injuries, illness and stalled progress come from.

一位运动员可能满足了表现所需的能量，却仍然在健康意义上处于赤字状态，而正是这个缺口带来了伤病、疾病和进步停滞。

The numbers / 算一算

Numbers that let you reason rather than recite. Carbohydrate is the dominant fuel at high intensity, and general guidance is roughly 3-5 grams per kilogram of body mass per day for moderate training, rising to 6-8 g/kg for heavy endurance training. Protein sits around 1.4-2.0 g/kg per day for most athletes. Fat should rarely drop below about 20-30% of energy intake. All three are expressed per kilogram because needs track lean mass and body size, not just appetite. The habit worth building: when a question gives you an athlete, work out the training load first, then the energy the activity spends, then what is left over.

让人能推理而不是背诵的数字。碳水化合物在高强度下是主导燃料，一般指导是中等训练每天每公斤体重约3到5克，重度耐力训练可到6-8克。

Common mistakes / 易错点

- "Protein only builds muscle." It builds and repairs every tissue, and supplies enzymes, hormones and transport proteins. Muscle is the part people notice, not the whole job.
"蛋白质只长肌肉。"它构建和修复所有组织，并提供酶、激素和运输蛋白。肌肉是人们注意到的部分，不是全部工作。
- "Carbohydrate is only for endurance." It is the fastest fuel, so it matters most in short, intense events - the opposite of the popular belief.
"碳水化合物只对耐力项目重要。"它是最快的燃料，所以在短时间高强度项目中才最重要，恰好与普遍看法相反。
- "Low fat is always healthier." Fat carries vitamins A, D, E and K and protects organs. Too little fat is a real problem, and the useful number is a floor, not a ceiling.
"脂肪越少越健康。"脂肪携带维生素A、D、E、K并保护器官。脂肪太少是真实的问题，而有用的数字是下限，不是上限。

Why it matters / 为什么重要

Fuelling is where nutrition most directly becomes performance. A carbohydrate-loaded muscle has bigger glycogen stores and lasts longer before it runs out. Adequate protein and energy availability are what let a muscle be rebuilt rather than merely broken down. And individualisation is not politeness: two athletes on the same training

programme can need different food, and the difference between them is often exactly what decides who adapts and who breaks down.

供能是营养最直接转化为表现的地方。糖原储备充足的肌肉能维持更久才耗尽。充足的蛋白质和能量可用性，让肌肉是被重建，而不只是

- Children and adolescents need energy, protein, calcium and iron for growth and development.
儿童和青少年需要能量、蛋白质、钙和铁以支持生长发育。
- Menstruation increases iron requirements; energy and protein needs also vary with body size, lean mass, age and activity.
月经会增加铁需求；能量和蛋白质需求也随体型、瘦体重、年龄和活动量变化。
- Higher physical activity generally increases energy, carbohydrate and fluid needs.
更高的身体活动量通常会提高能量、碳水和液体需求。

Key terms / 核心术语

Energy availability 能量可用性

The energy available to support health and training after exercise expenditure is accounted for.
扣除运动消耗后，可用于健康和训练的能量。

Formulas / 公式

An endurance athlete needs adequate carbohydrate before and during a long event, plus protein and fluids for recovery and tissue repair.

耐力运动员在长时间项目前后需要充足碳水，恢复和组织修复还需要蛋白质与液体。

Worked examples / 实例

>

Justify a fuelling plan for a two-hour road race in warm conditions.
为高温两小时公路赛设计供能方案并说明理由。

Exam focus / 考核要点

State one role of carbohydrate, protein and lipid.
分别说出碳水化合物、蛋白质和脂质的一项作用。

Pre-exercise, during-exercise and recovery · 运动前、运动中与恢复

First, picture it / 先这样想

Fuelling a training day works like preparing a journey. Before you leave you pack the boot: enough food for the trip, and nothing that will sit heavy and upset you. On the road you top up little and often rather than waiting until the tank is empty, because the engine is running on the way you feel now, not on what you will feel in an hour. Afterwards you do two jobs at once - refill what you spent, and repair what the journey wore out.

为训练日供能，像准备一次出行。出发前你把后备箱装好：足够的食物，以及任何不会压在胃里让你不舒服的东西。在路上你少量多次地补充，而不是等到油箱空了才加，因为发动机消耗的是你此刻的状态，而不是一小时后的状态。结束后你要同时做两件事--补回花掉的，以及修复旅途损耗的。

The detail / 详细讲解

Before exercise, the guidance is 1 to 4 hours beforehand, carbohydrate-focused with some protein, low in fat and fibre, and with fluids taken in the same window.

运动前的指导是：赛前1到4小时进食，以碳水为主并搭配一些蛋白质，减少脂肪和纤维，同时在这个时间窗内补液。

The reason for the timing is digestive: food eaten too close to the start is still sitting in the gut during the session, and gut blood flow is needed by working muscle.

强调时间是为了消化：吃得太靠近开始，训练时食物还留在肠道里，而工作肌肉正需要肠道血流。

The reason for the composition is that carbohydrate is what you will burn, and protein is what you will need afterwards.

强调成分是因为碳水是你马上要烧掉的，而蛋白质是你之后需要的。

During exercise, past about 60 minutes or at high intensity, the workbook recommends roughly 30 to 60 grams of carbohydrate per hour plus regular fluid intake.

运动中，超过约60分钟或强度较高时，工作册建议每小时补充约30到60克碳水，并规律补液。

The mechanism is a supply problem: by the time you feel flat, the problem was created an hour earlier, when the rate of ATP use outpaced what your stores could give.

机制是一个供给问题：等你感觉“掉下去了”，问题其实在一小时前就造成了--ATP消耗速率已经超过了储备所能提供的。

Regular small amounts beat a large amount late, because the gut can only absorb so much at a time and unused carbohydrate in the gut is just weight you carry.

少量多次优于后期大量，因为肠道一次只能吸收有限量，而留在肠道里没被利用的碳水，只是你需要额外背负的重量。

Recovery has two jobs, and people usually only do the first.

恢复有两项工作，而人们通常只做了第一件。

Replacing what you spent means fluid and sodium, and carbohydrate to refill glycogen.

补回花掉的，意味着补液、补钠，并用碳水补充糖原。

Repairing what the session wore out means protein.

修复训练损耗的，意味着蛋白质。

Timing matters most for protein: the muscles are most receptive for a few hours after training, so a meal with protein soon after is more useful than the same protein much later.

时机对蛋白质尤其重要：训练后几个小时肌肉最容易吸收，所以训练后不久就吃一顿含蛋白质的餐，比很晚吃同样多的蛋白质更有用。

Two practical habits sit on top of all of this.

在这一切之上还有两个实用习惯。

First, practise your fuelling in training: the gut adapts to what you give it and when, and nothing is worse than discovering a new gel at kilometre 30 of a race.

第一，在训练中练习你的供能：肠道会适应你给它什么、何时给，而在比赛第30公里才第一次尝试某种能量胶，是再糟糕不过的。

Second, consider sodium for long or heavy-sweat sessions, and match replacement to actual sweat losses and conditions rather than using a fixed rule.

第二，对长时间或大量出汗的训练，考虑补钠，并且让补充量匹配实际汗液损失和条件，而不是套用固定规则。

And keep an eye on energy availability while you are doing all of this.

在做这一切的同时，也要留意能量可用性。

Doing everything above and still not eating enough to cover the training is the route into low energy availability and RED-S.

把上面所有事情都做对了，却仍然没有摄入足以覆盖训练消耗的能量，这就是通往低能量可用性和RED-S的路径。

It is the failure mode that looks like dedication from the outside.

从外面看，这种失败模式看起来反而像自律。

The numbers / 算一算

Do the numbers as a timeline. Pre-exercise: 1-4 hours, focused on carbohydrate, for example a familiar meal of rice, pasta, bread or fruit, with fluid taken in the same window. During: 30-60 g of carbohydrate per hour once the session passes about 60 minutes - a 60-minute race needs very little, a three-hour marathon needs most of that hour after hour. Fluid: replace a meaningful share of sweat loss, and in the heat reckon with 0.5-2.0 L per hour lost.

After: protein within a couple of hours, with carbohydrate to refill glycogen and fluid and sodium to restore what you lost. A simple check that catches most problems: does the energy you took in cover the energy the session spent, plus enough left over to stay well?

把数字当作一条时间线来用。运动前：1到4小时，以碳水为主，比如一顿熟悉的米饭、意面、面包或水果餐，同时在这个时间窗内补液。

Common mistakes / 易错点

-

"The bigger the fuelling session, the better." Food eaten too close to start sits in the gut during the session, and gut blood flow belongs to working muscle. Timing is part of the plan.

"吃得越多越好。"吃得太靠近开始，训练时食物还留在肠道里，而肠道血流属于工作肌肉。时间安排本身就是方案的一部分。

- "Carbohydrate only matters in the marathon." It matters most per minute in short, intense work, where you run out of glycogen fastest. The 400 m runner is more carbohydrate-dependent than the marathon runner, per minute of racing.

"碳水只在马拉松里重要。"按每分钟算，它在短时间高强度项目中最重要，因为那里的糖原消耗最快。每分钟比赛时间，400米跑者比马拉松跑者更依赖碳水。

- "Recovery is just eating more." Recovery has two jobs - replace what was spent and repair what was damaged - and the second one needs protein, on a timescale of a couple of hours.

"恢复就是多吃一点。"恢复有两项工作--补回花掉的，修复损耗的--而第二项需要蛋白质，时间窗口是几个小时。

Why it matters / 为什么重要

Every one of these three windows is a decision you make before you know how the session will go, which is why fuelling belongs to planning rather than improvisation. The athletes who look effortless in a heat are usually the ones whose plan was written the night before and rehearsed in training, not the ones who reacted to thirst and hunger in real time. Practice is what makes the plan executable, and an executable plan is the whole advantage.

这三个时间窗中的每一个决定，都是你在还不知道这次训练会怎样之前做出的--这就是为什么供能属于计划，而不是临场应变。那些在训练中练习供能，让胃肠适应食物和时间安排。

- Practise fuelling in training so the gut becomes familiar with the food and timing.

在训练中练习供能，让胃肠适应食物和时间安排。

- Sodium may be considered for prolonged or heavy-sweat sessions; replacement must match sweat losses and conditions.

长时间或大量出汗的运动可考虑补钠，补充量应与汗液损失和条件相匹配。

Key terms / 核心术语

Low energy availability 低能量可用性

Insufficient energy for health and daily function after exercise energy expenditure.

运动消耗后，能量不足以支持健康和日常功能。

RED-S RED-S

Relative energy deficiency in sport, a health and performance problem linked to low energy availability.

运动中的相对能量缺乏，与低能量可用性相关的健康和表现问题。

Formulas / 公式

A cyclist might use a familiar pre-ride meal, a drink or gel during the ride, and carbohydrate, protein and fluid after finishing.

骑行者可以在骑行前吃熟悉的餐食，骑行中饮用运动饮料或能量胶，完成后补充碳水、蛋白质和液体。

Worked examples / 实例

>

Explain why there is no universal meal plan.

解释为什么不存在适用于所有人的餐单。

Exam focus / 考核要点

How much carbohydrate per hour is suggested during exercise?

运动中建议每小时摄入多少碳水？

Micronutrients, RED-S and microbiome · 微量营养素、RED-S与肠道微生物组

First, picture it / 先这样想

Vitamins and minerals are the punctuation and the letters in a page of writing. Take all the words - carbohydrate, protein, fat, thousands of calories - and remove every full stop, every comma and every single letter, and you end up with a page of nothing but words. You would have all the volume and none of the meaning. That is what a micronutrient deficiency is.

维生素和矿物质，是一页文字里的标点和字母。拿掉所有“词”--碳水、蛋白质、脂肪、几千大卡--再把每一个句号、每一个逗号和每一个字母都拿掉，最后剩下的只是一页没有标点的词。你会有全部的体积，却没有意义。微量营养素缺乏就是这个样子。

The detail / 详细讲解

Start with the four minerals that come up most.

先从最常考到的四种矿物质说起。

Iron builds haemoglobin, the molecule that carries oxygen in the blood, so low iron means less oxygen reaches the muscle and endurance suffers first.

铁构建血红蛋白--血液中携带氧气的分子--所以缺铁意味着到达肌肉的氧气减少，耐力首先受影响。

Calcium builds bone and teeth.

钙构建骨骼和牙齿。

Sodium supports fluid balance, blood volume, nerves and muscle.

钠支持体液平衡、血容量、神经和肌肉。

Potassium supports fluid balance and nerve and muscle function.

钾支持体液平衡以及神经和肌肉功能。

All four are needed every day, in small amounts relative to the macronutrients.

这四样每天都需要，但相对宏量营养素而言量很小。

Then vitamins, which support immunity and the release of energy, with D and K especially relevant to bone health.

然后是维生素，它们支持免疫和能量释放，其中D和K与骨骼健康尤其相关。

The point to hold onto is that vitamin needs are small but not optional.

要抓住的要点是：维生素的需求量小，但并非可有可无。

A deficiency does not make you slightly worse at something; it produces a recognisable clinical picture, which is exactly why these are named and tested rather than being left vague.

缺乏不会让你在某件事上稍差一点，而会产生一个可识别的临床图像--这正是它们被命名、被检测，而不是被含糊处理的原因。

Low energy availability is the condition where intake does not cover training expenditure, so there is not enough left for health.

低能量可用性是指摄入量无法覆盖训练消耗，因而没有足够剩下维持健康的情况。

RED-S, relative energy deficiency in sport, is the name for what that does across systems.

RED-S（运动中的相对能量缺乏）是这种情况在各个系统上造成后果的统称。

It can affect metabolism, menstrual function, bone mineral density, immunity, growth, cardiovascular and psychological health, recovery and performance.

它可以影响代谢、月经功能、骨矿物质密度、免疫、生长、心血管和心理健康、恢复以及表现。

That list is long on purpose: the point is that under-fuelling is not a single problem but a whole-body one.

这个清单故意列得很长：重点在于，供能不足不是单一问题，而是全身性的问题。

The gut microbiome is a third angle, and it is the least examinable, so know its shape.

肠道微生物组是第三个角度，也是最难考的，所以要掌握它的轮廓。

Communities of microorganisms in the gut are influenced by genetics, diet, medication, sleep, stress and activity.

肠道中的微生物群受到遗传、饮食、药物、睡眠、压力和活动的影响。

Diet is the input you control, which is why a varied diet does more than supply nutrients - it feeds a population you did not choose and cannot see.

饮食是你能控制的输入，所以多样化的饮食所起的作用不只是提供营养--它还在喂养一群你既没有选择、也看不见的生物。

Where exam answers usually go wrong here is generalisation.

这一部分考题常错在“笼统作答”。

A safe, strong answer individualises: acknowledge the group where the issue is more common, name the

mechanism that links it to the effect, and be clear that the fix is nutritional adequacy and individual monitoring rather than a supplement prescribed on principle.

稳妥而优秀的答案是个体化的：承认这个问题在哪些人群中更常见，说出把原因连到后果的机制，并明确解决办法是营养充足加个体化。

Iron for adolescent and female athletes is the standard worked example; calcium and potassium come from a varied diet.

青少年和女性运动员的铁是最典型的实例；钙和钾则来自多样化饮食。

The numbers / 算一算

Numbers here are mostly about proportion rather than precision. Vitamins and minerals are needed in milligram or microgram amounts, which is why the daily values on a label look so small next to a carbohydrate figure in grams - and why "take a supplement" is almost never the answer to a dietary shortfall. The one figure worth carrying is the 2% body-mass fluid guideline, because low energy availability and dehydration often arrive together in the same athlete. Practically: a varied diet built on the macronutrient guidance supplies the micronutrients without calculation, and an athlete who is not eating enough to cover training is a candidate for professional assessment, not for a bigger pill.

这一部分的数字更多是比例问题，而不是精确数值。维生素和矿物质的需求以毫克或微克计，所以标签上的每日值和以克计的碳水数字相比。

Common mistakes / 易错点

- "Low energy availability is the same as not eating enough calories." It is not about total calories. It is about what is left after training expenditure, which can be inadequate even when calorie intake looks reasonable.
"低能量可用性就是热量吃得不够。"它不是总热量的问题，而是扣除训练消耗后剩下多少的问题；即使总热量看起来合理，剩余也
- "RED-S is a nutrition deficiency you can fix with iron." It is a whole-body consequence of low energy availability. Iron may be part of the picture, but the underlying problem is the energy deficit.
"RED-S
是补铁就能解决的营养缺乏。"它是低能量可用性造成的全身性后果。铁可能是其中一环，但根本问题是能量赤字。
- "If body mass is normal, energy availability is fine." Body mass can look normal while energy availability is low, which is part of what makes RED-S easy to miss.
"如果体重正常，能量可用性就没问题。"体重可以看起来正常而能量可用性却偏低，这正是 RED-S 容易被忽略的原因之一。

Why it matters / 为什么重要

This is the section that keeps athletes healthy over a career rather than merely fast for a season. The failure mode is not dramatic and it is not visible: an athlete trains harder each year, keeps performing, keeps getting injured, and never connects the two. Bone density, immunity, menstrual function and mood are all on that list, and all of them respond to the same fix - eating enough to cover training plus a margin for health. Recognising that is a professional skill, and it is why energy availability has moved to the centre of sports nutrition research.

这一节决定的是运动员整个职业生涯的健康，而不只是某一个赛季的速度。问题既不戏剧化也不可可见：一位运动员每年训练更重、成绩依然

Formulas / 公式

Define RED-S and name two possible consequences.

定义RED-S并说出两个可能后果。

Worked examples / 实例

>

Name two key sources of iron and calcium.

指出铁和钙的两个主要来源。

Energy systems

能量系统

Guiding question / 引导问题

How do energy systems supply ATP for physical activity?

能量系统如何为身体活动供应ATP?

ATP is the immediate energy currency for contraction, ion pumping, biosynthesis and nerve impulses. The phosphagen, glycolytic and oxidative systems resynthesise ATP continuously; their contribution changes with intensity and duration.

ATP是肌肉收缩、离子泵、合成和神经冲动的即时能量货币。磷酸原、糖酵解和氧化系统持续再合成ATP；它们的贡献随强度和持续时间变化。

ATP and the energy continuum · ATP与能量连续体

First, picture it / 先这样想

ATP is the only currency muscle can actually spend. Everything else is a way of getting more of it. Think of three different workshops all minting the same coin. The first is a small workshop with a fast printing press: it makes coins quickly but can only make a few at a time. The second is a bigger workshop, slower, making more. The third is a real mint, enormous and slow, but capable of supplying a whole city. Your muscles are supplied by all three at once, every minute, and only the proportions change.

ATP 是肌肉真正能花出去的唯一货币，其他一切都是获取更多ATP 的办法。把三个不同的车间想象成都在铸造同一种硬币。第一个是小车间配一台快速印刷机：出币很快，但一次只能造出少量。第二个是更大、更慢的车间，造得更多。第三个是真正的造币厂，庞大而缓慢，但足以供应整座城市。你的肌肉每一分钟都同时接受这三者的供应，变化的只是比例。

The detail / 详细讲解

ATP is the immediate energy currency.

ATP是即时能量货币。

When it is hydrolysed to ADP and inorganic phosphate, energy is released, and that energy is what does the actual work: driving the cross-bridge cycle in muscle, pumping sodium and potassium across membranes, synthesising new molecules and firing nerve impulses.

它水解为ADP和无机磷酸时释放能量，而这些能量才是真正做事的动力：驱动肌肉的横桥循环、把钠和钾泵过细胞膜、合成新分子、发放神经冲动。

ATP is also astonishingly small and short-lived, which is why you cannot simply stockpile it and be done.

ATP还小得惊人、寿命极短，所以不可能靠囤积它来解决问题。

Muscle stores only about 5 mmol per kilogram of wet muscle, which is a few seconds of maximal work.

肌肉每公斤湿肌只储存约5毫摩尔--大约相当于几秒的最大用力。

Because the store is so small, three separate pathways exist purely to keep rebuilding it, and they run at the same time.

因为储备如此之小，存在三条不同的通路专门用来持续重建它，而且它们同时运行。

What changes with intensity is not which one is on, but how much each one contributes.

强度变化时改变的不是哪一条“开”，而是每一条贡献多少。

This is the single most misunderstood idea in exercise science, so it is worth being blunt: there is no switch.

这是运动科学里最常被误解的概念，所以必须直说：没有开关。

A person sprinting up a hill is using all three systems, and so is a person sitting still.

正在上坡冲刺的人同时在使用三个系统，坐着不动的人也一样。

The three systems differ on five axes, and holding those five axes is how you compare them properly: oxygen requirement, fuel, rate of ATP supply, how long they can last, and how quickly they recover.

三个系统在五个维度上不同，握住这五个维度才能正确比较：是否需要氧气、燃料、ATP供应速率、能维持多久、以及恢复有多快。

Power and duration are not opposites of each other. A system can be powerful without being sustainable, and that trade-off is the whole reason sport has events like the 100 m and the marathon rather than just one event.

功率和持续时间并不是彼此的反面--一个系统可以有很高的功率却不持久，而这个权衡正是体育中既有100米又有马拉松、而不是只有

A worked example makes the continuum concrete.

一个实例能让这条连续体变得具体。

A 100 m sprint is dominated by the phosphagen system, a 400 m sprint gains substantial glycolytic contribution, and a 10 km run relies much more on oxidative resynthesis.

100米冲刺以磷酸原系统为主，400米冲刺大幅加入糖酵解贡献，10公里跑则更多依赖氧化再合成。

But in all three cases the others are contributing, and the practical consequence is that training one system changes the others: a runner who trains the oxidative system becomes better at the phosphagen system too, because better delivery and better clearance are what let the fast systems do their job slightly longer.

但三种情况里其他系统都在参与，而实际后果是：训练一个系统会改变其他系统--一个训练了氧化系统的跑者，磷酸原系统也会变好，

Lactate deserves a specific word here, because it is badly misunderstood.

乳酸盐在这里值得单独说一句，因为它被严重误解。

It is a normal product of glycolytic metabolism, not a waste product and not the cause of fatigue by itself.

它是糖酵解代谢的正常产物，不是废物，也不是疲劳的唯一原因。

It is a fuel your heart and other tissues can use, and its concentration rises when production outpaces clearance.

它是一种你的心脏和其他组织可以利用的燃料，而当生成速度超过清除速度时它的浓度会上升。

Fatigue in a hard effort is driven by the whole picture - hydrogen ions, phosphate, depleted stores, heat and central fatigue - not by lactate on its own.

高强度运动中的疲劳由整个图景驱动--氢离子、磷酸盐、储备耗尽、热量和中枢疲劳--而不是单靠乳酸盐。

The numbers / 算一算

Numbers to fix in your head. Stored ATP is about 5 mmol per kilogram of wet muscle; phosphocreatine adds roughly three to four times that, giving the phosphagen system around 6-10 seconds of maximal effort. The glycolytic system dominates from roughly 30 seconds to about 2 minutes. The oxidative system takes over beyond that and can supply ATP for hours. The continuum therefore reads: seconds, then minutes, then hours. And the training consequence: five sets of 20 second maximal efforts with full recovery is the classic way to extend phosphagen capacity, because the adaptation comes from restocking phosphocreatine, not from the effort itself.

要记牢的数字。储存的ATP约为每公斤湿肌5毫摩尔；磷酸肌酸大约是它的三到四倍，因此磷酸原系统大约提供6到10秒的最大用力。糖酵解系统主导从大约30秒到约2分钟。氧化系统接管之后可以供应ATP数小时。因此连续体读作：秒，然后分钟，然后小时。训练后果：五组20秒最大用力，充分恢复是经典方式以扩展磷酸原容量，因为适应来自补充磷酸肌酸，而非努力本身。

Common mistakes / 易错点

- "One energy system switches off and another takes over." All three contribute at every intensity. What changes is the proportion each one contributes.
"一个能量系统关闭，另一个接管。"在任何强度下三个系统都在贡献，变化的是各自贡献的比例。
- "Lactate causes fatigue." Lactate is a normal product and a usable fuel. Fatigue comes from the combination of hydrogen ions, phosphate, depleted stores, heat and central fatigue.
"乳酸盐导致疲劳。"乳酸盐是正常产物，也是可利用的燃料。疲劳来自氢离子、磷酸盐、储备耗尽、热量和中枢疲劳的共同作用。
- "Training one system does not affect the others." It does. Better aerobic delivery and clearance let the fast systems work slightly longer, which is why endurance training also improves sprint capacity.
"训练一个系统不会影响其他系统。"会。更好的有氧输送和清除让快系统能多撑一会儿，这就是耐力训练也会改善冲刺能力的原因。

Why it matters / 为什么重要

This framework is what lets an athlete and a coach make sense of training that otherwise looks contradictory. Why do a sprinter and a marathoner train so differently, when both are running? Why does lifting weights help a distance runner, and why does the weight room come with a sprint programme attached? Why does training make you feel worse before it makes you feel better? All of it falls out of the same five axes once you stop treating the systems as switches. Understanding the continuum is what turns a list of sessions into a plan with a reason behind it.

正是这个框架，让运动员和教练能够理解那些看起来互相矛盾的训练。短跑运动员和长跑运动员都在跑步，为什么练得完全不同？举重训练帮助长跑运动员，为什么举重房会附带短跑项目？训练让你感觉更糟之前它让你感觉更好？这一切都出自相同的五个轴，一旦你停止将系统视为开关。理解连续体是什么将一系列训练课变成一个有理由的计划。

- All three systems contribute at every intensity; the dominant contribution shifts along the energy continuum.
在任何强度下三个系统都会贡献ATP，只是主导贡献沿能量连续体改变。

Key terms / 核心术语

ATP 三磷酸腺苷

Immediate energy currency used by cellular work.

细胞工作的即时能量货币。

Lactate 乳酸盐

A product associated with glycolytic metabolism and fatigue context.

与糖酵解代谢及疲劳情境相关的产物。

Formulas / 公式

A 100 m sprint is dominated by phosphagen contribution, a 400 m sprint gains glycolytic contribution, and a 10 km run relies more on oxidative resynthesis.

100米冲刺主要依赖磷酸原，400米冲刺增加糖酵解贡献，10公里跑更多依赖氧化再合成。

Worked examples / 实例

>

Match a 100 m sprint, 400 m sprint and 10 km run to dominant ATP supply.

将100米、400米和10公里跑步匹配到主要ATP供应。

Exam focus / 考核要点

Why is it inaccurate to say that only one energy system works in a sport?

为什么说运动中只有一个能量系统在工作是错误的？

Comparing the three systems · 三大供能系统比较

First, picture it / 先这样想

Three vehicles, three jobs. A rocket is the phosphagen system: astonishing power, about ten seconds of fuel, and then it is finished. A rally car is the glycolytic system: serious speed that can be sustained for a couple of minutes. A lorry is the oxidative system: slow off the line, but it can carry a load for hours on a small budget. No one argues about which is the best vehicle, because the question is always what the journey is for.

三种交通工具，三种工作。火箭是磷酸原系统：功率惊人，大约十秒的燃料，然后就结束了。拉力赛车是糖酵解系统：速度相当可观，能维持一两分钟。卡车是氧化系统：起步慢，却能用很少的燃料把货运上几个小时。关于哪种车最好，从来没有争论，因为问题永远是：这趟路是干什么的。

The detail / 详细讲解

The phosphagen system uses ATP and phosphocreatine, needs no oxygen, produces very high power from a very small ATP store, and fatigues almost immediately.

磷酸原系统使用ATP和磷酸肌酸，不需要氧气，从很小的ATP储备中产生极高的功率，并且几乎立刻疲劳。

It is the system you are using when you jump out of the blocks, when you lift something heavy once, and when you throw.

它是你冲出起跑器、一次举起重物、或者投掷时所用的系统。

Think of the store as the money in your pocket on day one: it is small, it is immediately spendable, and there is no top-up during the effort.

把这个储备想象成你第一天口袋里的钱：数量不大、立刻可用，而且在这次用力过程中没有补充。

The glycolytic system breaks down carbohydrate without oxygen, supplies a moderate amount of ATP at high intensity, holds that for roughly 30 seconds to 2 minutes, and produces lactate and hydrogen ions as by-products.

糖酵解系统在没有氧气的情况下分解碳水化合物，在高强度下提供中等量的ATP，维持约30秒到2分钟，并产生乳酸盐和氢离子作为副产物。

Those hydrogen ions matter: they shift the pH of the muscle, and the proteins that generate force respond badly to that.

这些氢离子很关键：它们改变肌肉的pH，而产生力量的蛋白质对这种变化反应很差。

This is the system behind 200 to 400 m running, repeated sprints and high-intensity interval training.

这就是200到400米、重复冲刺和高强度间歇训练背后的系统。

The oxidative system requires oxygen, uses carbohydrate, fat and a little protein, supplies a large and steady

amount of ATP, and can run for minutes to hours.

氧化系统需要氧气，使用碳水化合物、脂肪和少量蛋白质，提供大量而稳定的ATP，可以运行数分钟到数小时。

It is the system that supports 1500 m, distance running, cycling and rowing, and it is also the system that makes all the recovery after a session possible.

它支撑着1500米、长跑、骑行和划船，同时也是训练后所有恢复得以进行的基础。

The comparison questions in the paper are always the same five: oxygen use, fuel, ATP supply, duration and recovery.

卷面上的比较题永远就是那五项：氧气使用、燃料、ATP供应、持续时间、恢复。

Answering those five for each system, in that order, will always earn the mark.

把这五项按顺序为每个系统说一遍，标记就到手了。

What loses marks is a paragraph that describes one system fluently and leaves the other two to inference.

丢分的是那种把一个系统讲得头头是道、另外两个却留给读者去猜的段落。

The integrated point is the one that separates a good answer: a 400 m runner still gets aerobic support.

综合性的那一点正是区分好答案的关键：400米跑者仍然得到有氧支持。

There is no switch, so a 400 m effort is phosphagen power with heavy glycolytic contribution and meaningful oxidative contribution underneath.

系统不是开关，所以400米的一次用力是磷酸原的功率、很重的糖酵解贡献，以及底下的实质性氧化贡献。

Pacing and training decide how long each system can keep contributing, which is why a paced 400 m and a raced 400 m are physiologically different despite identical distance and identical time.

配速和训练决定每个系统能维持贡献多久--这就是为什么同样距离、同样时间的400米，配速跑和拼力跑在生理上并不相同。

The numbers / 算一算

A table you can build from memory. Phosphagen: no oxygen, ATP and phosphocreatine, very high power, about 6-10 seconds, fast recovery. Glycolytic: no oxygen, carbohydrate, moderate power, about 30 seconds to 2 minutes, recovery of roughly 10-20 minutes depending on how hard the session was. Oxidative: oxygen required, carbohydrate and fat, lower power, minutes to hours, recovery of hours to a couple of days. The two training consequences that follow: long, easy aerobic work raises the ceiling for how much you can deliver and how fast you clear things, and short, hard phosphagen work raises the ceiling for peak power. In a real programme you need both, and the ratio depends on the event.

一张可以凭记忆搭出来的表。磷酸原：无需氧气，ATP和磷酸肌酸，功率极高，约6到10秒，恢复快。糖酵解：无需氧气，碳水化合物，

Common mistakes / 易错点

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"Phosphagen is the most important system because it gives the highest power." Importance depends on the event. It is the whole answer in a 100 m and almost irrelevant in a marathon.

"磷酸原功率最高，所以最重要。"重要性取决于项目。100米里它是全部答案，在马拉松里几乎无关紧要。

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"Glycolysis produces lactate, so it is the anaerobic waste system." It is not waste. Lactate is a usable fuel for the heart and other tissue.

"糖酵解产生乳酸，所以它是无氧的废物系统。"它不是废物。乳酸盐是心脏和其他组织可以利用的燃料。

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"Each system has its own recovery time, so a 400 m and a 100 m train the same way." They do not. One is a phosphagen event needing high power and long rests, the other a glycolytic event needing a sustained pace.

"每个系统都有自己的恢复时间，所以400米和100米的训练方式一样。"不一样。一个是磷酸原项目，要高功率和长休息；另一个是

Why it matters / 为什么重要

Knowing which system an event lives on tells you what training that event needs before a single session is programmed. A 100 m sprinter lives in the phosphagen system, so their programme is short, maximal and separated by long rests, because the adaptation is phosphocreatine restocking. A 5 km runner lives half in glycolysis and half in oxidative, so their programme is longer, more continuous, and rests are short. A shot putter and a road cyclist are the same distance apart and on opposite ends of the table, which is exactly why one blanket programme fails most athletes.

知道一个项目活在哪个系统里，就能在排第一节课之前知道它需要什么训练。100米短跑运动员活在磷酸原系统里，所以计划是短促、尽

- Phosphagen is strongest for starts, jumps and throws.
磷酸原适合起跑、跳跃和投掷。
- Glycolytic contribution is important for 200-400 m efforts, repeated sprints and HIIT.
糖酵解对200-400米、重复冲刺和高强度间歇训练很重要。
- Oxidative contribution supports 1500 m, distance running, cycling and rowing.
氧化贡献支持1500米、长跑、骑行和划船。

Formulas / 公式

A 400 m runner gains power from anaerobic contribution but still receives aerobic support; pacing and training affect how long each system can contribute.

400米跑者从无氧贡献中获得功率，但仍得到有氧支持；配速和训练影响各系统能维持贡献的时间。

Worked examples / 实例

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Compare oxygen use, fuels, ATP supply, duration and recovery for the three systems.
比较三个系统的氧气使用、燃料、ATP供应、持续时间和恢复。

Exam focus / 考核要点

Compare the three energy systems in a table.
用表格比较三个能量系统。

VO2max, movement economy, LIP and EPOC · VO2max、动作经济性、乳酸拐点与EPOC

First, picture it / 先这样想

Four measurements that decide how far you can go. VO2max is the size of the delivery van: how much oxygen can arrive per minute. Movement economy is how much fuel the courier must burn to carry one parcel, and a good courier barely notices the parcels. The lactate threshold is the point where the van starts to labour on the hill, breathing hard but still coping. EPOC is the engine running hot after you have switched it off.

四个决定你能走多远的测量。最大摄氧量是送货车的大小：每分钟能送来多少氧气。动作经济性是快递员送一个包裹要烧多少油，而好的快递员几乎感觉不到包裹的存在。乳酸阈值是货车在爬坡时开始吃力的那个点--呼吸急促，但仍撑得住。运动后过量氧耗是熄火之后引擎仍在高温运转。

The detail / 详细讲解

VO2max is the maximum rate of oxygen uptake, the ceiling on oxygen delivery.

最大摄氧量是氧摄取的最大速率，也就是氧气输送的上限。

It is the strongest single predictor of endurance performance, and it is the reason an elite distance runner can hold a pace that would leave an untrained person walking.

它是预测耐力表现最强的单一指标，也正是精英长跑运动员能维持一个会让未训练者走不动的原因。

A typical range is about 3-5 litres per minute in trained men and 2-3 in trained women, and it is a ceiling, not a suggestion: if oxygen cannot arrive faster, the pace cannot be faster, no matter how motivated the athlete is.

训练良好的男性典型值约为每分钟3到5升，女性约2到3升。

它是上限而不是建议：如果氧气无法更快到达，配速就不可能更快，无论运动员多有动力。

Movement economy is oxygen cost per kilogram of body mass per minute at a given speed, and it says nothing about fitness.

动作经济性是在给定速度下每公斤体重每分钟消耗的氧气，它与体能无关。

Two athletes with the same VO2max can be separated by economy.

两位最大摄氧量相同的运动员可以被经济性区分开。

Better economy means the same speed at a lower oxygen cost, so there is more oxygen left in the spare capacity.

更好的经济性意味着同样的速度只用更少的氧气，因此还有更多余量。

Mechanism: a trained runner wastes less energy per step through stiffer tendons that return energy like a spring, a smoother stride, less unnecessary movement, and the ability to run at a lower VO₂max than an untrained person at the same pace.

机制在于：训练有素的跑者每一步浪费的能量更少--肌腱更硬、像弹簧一样回弹、动作更流畅、多余动作更少，并且能在与未训练者相比最大摄氧量完成。

The lactate inflection point is where blood lactate begins to rise above baseline.

乳酸拐点是血糖乳酸开始高于基线的位置。

Below it, almost all of the work is aerobic; above it, anaerobic contribution climbs sharply.

在它以下，几乎全部工作都是有氧的；在它以上，无氧贡献急剧上升。

It is not a wall you hit, it is a region you enter, and the practical consequence is that pace selection is a physiological decision, not a preference.

它不是一堵撞上的墙，而是一片进入的区域，其实际后果是：配速选择是一个生理决定，而不是个人偏好。

An athlete who can detect the threshold can sit just below it for a long time; one who cannot will drift above it and pay for it in the last third of the race.

能察觉自己阈值的运动员可以在其下方维持很久；察觉不到的会漂到其上，并在比赛最后三分之一付出代价。

EPOC is the extra oxygen your body keeps consuming after you stop, above what the steady state would need.

运动后过量氧耗是指你停下来之后，身体仍然消耗的、超出稳态所需的额外氧气。

The fast component is roughly the first two minutes as heart rate falls and the most recent lactate is cleared.

快速成分大约是最初两分钟，此时心率下降、最近产生的乳酸被清除。

The slow component then runs on for a while, often hours, because the body is still paying off work: resynthesising phosphocreatine, reloading glycogen, shifting something toward fat, clearing heat and metabolites, and repairing tissue.

之后的慢速成分会持续一段时间，往往长达数小时，因为身体仍在还债：重新合成磷酸肌酸、重新装载糖原、把供能向脂肪转移、清除

It is the physiological reason a hard session is not over when you cross the line.

这就是为什么一次高强度训练在你冲过终点线时并没有结束。

The numbers / 算一算

A few numbers that make these real. VO₂max in trained men is about 3-5 L/min and in trained women about 2-3 L/min. A good movement economy at marathon pace is roughly 5 mL of oxygen per kilogram per minute. The lactate inflection point for an elite endurance athlete sits around 1.6 mmol per litre of blood. And the recovery bill: EPOC after a hard 30 minute effort runs about 2 litres of extra oxygen, and if roughly a third of that goes to restoring glycogen, that is on the order of 90 grams of carbohydrate - which is why a hard session is followed by a carbohydrate-rich meal, not a protein shake and water.

几个让这些概念落地的数字。训练良好的男性最大摄氧量约为每分钟3到5升，女性约2到3升。马拉松配速下良好的动作经济性约为每公斤

Common mistakes / 易错点

- "VO₂max is how hard you are working." It is a capacity ceiling, a measurement of the maximum, not of effort. The same athlete at three different intensities has one VO₂max and three very different heart rates.
"最大摄氧量表示你多努力。"它是一个容量上限，是最大值的测量，不是努力的测量。同一位运动员在三种不同强度下只有一个最
- "VO₂max is a measure of fitness." It measures capacity, not skill or technique. An athlete can have a high VO₂max and still be slow because of movement economy.
"最大摄氧量是体能的度量。"它衡量的是容量，不是技术或动作。运动员可能有很高的最大摄氧量，却因为动作经济性差而仍然很
- "Lactate is waste that has to be cleared before you can run again." It is a usable fuel. What is repaid after a hard effort is phosphocreatine, glycogen, heat clearance and tissue repair, which is why recovery takes hours and not minutes.
"乳酸是废物，必须清空才能再跑。"它是一种可利用的燃料。高强度用力之后真正要还的是磷酸肌酸、糖原、热量清除和组织修复

Why it matters / 为什么重要

These four numbers are what turn a race plan into a physiological argument. An athlete who knows their VO₂max,

their economy, their threshold and their EPOC can answer "why did that race fall apart in the last kilometre" with a mechanism rather than a guess. Coaches use them to set pace targets that are honest about what the athlete can actually deliver on the day, given their fitness, their fatigue and the weather. And they are all trainable, which is the hopeful part: every one of them responds to work.

这四个数字，正是把比赛计划变成生理论证的东西。知道自己最大摄氧量、动作经济性、阈值和过量氧耗的运动员，能用机制而不是猜测

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Intensity below the lactate inflection point is mostly aerobic; above it, anaerobic contribution increases.

低于乳酸拐点的强度主要为有氧；超过后无氧贡献增加。

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EPOC supports restoration of oxygen stores, ATP-PCr, lactate clearance, tissue repair and thermoregulation.

EPOC帮助恢复氧储量、ATP-磷酸肌酸、清除乳酸、组织修复和体温调节。

Key terms / 核心术语

VO2max 最大摄氧量

Maximum oxygen uptake capacity.

最大摄氧能力。

Movement economy 动作经济性

Oxygen cost of a given submaximal movement.

完成给定次最大动作的氧气成本。

EPOC 运动后氧耗

Extra oxygen used after exercise to restore physiological systems.

运动后恢复生理系统所需的额外氧气。

Formulas / 公式

An endurance athlete can improve aerobic capacity and movement economy, while intervals or tempo work can target intensity around the lactate inflection point.

耐力运动员可以提高有氧能力和动作经济性；间歇或节奏训练可针对乳酸拐点附近的强度。

Worked examples / 实例

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Define VO2max, movement economy, LIP and EPOC.

定义VO2max、动作经济性、LIP和EPOC。

Exam focus / 考核要点

Give one training implication for VO2max and one for EPOC.

分别为VO2max和EPOC写出一个训练启示。

Qualities of training

训练品质

Guiding question / 引导问题

How should training be planned to improve performance safely?

训练应如何规划才能安全提高表现？

Effective training is specific, progressively overloaded, varied, recoverable and planned across time. The athlete's current fitness, goal, phase, health and response to load must guide FITT choices and monitoring.

有效训练应具备专项性、渐进超负荷、多样性、恢复和时间规划。运动员当前水平、目标、训练阶段、健康状况和负荷反应应指导FITT选择与监测。

Six qualities and FITT · 六大训练品质与FITT

First, picture it / 先这样想

A training plan is a recipe, and these six words are how you make a dish taste right rather than merely be edible. Specificity is cooking the actual dish, not a substitute. Progressive overload is turning the heat up a notch each week. Recovery is resting the dough. Variety is using different spices. Reversibility is knowing the dish can go back down a notch. Periodisation is deciding what to cook in which season.

训练计划就像一份菜谱，这六个词决定的是这道菜好不好吃，而不只是能不能吃。专项性是真正做那道菜，而不是用替代品；渐进超负荷是每周把火调大一点；恢复是让面团休息；多样性是用不同香料；可逆性是知道这道菜也可以把火调回来；周期化是决定每个季节做哪道菜。

The detail / 详细讲解

Specificity means training the sport, the position and the goal, not a vague version of them.

专项性意味着训练这个项目、这个位置和这个目标，而不是它们的某个笼统版本。

A sprinter who only ever runs long slow miles will get fitter, and will still be slow.

一位只跑长距离慢跑的短跑运动员会变得更健康，也仍然会慢。

The transfer from a training stimulus to a performance is specific to the qualities and movements that matter in the event, which is why a 400 m runner needs speed endurance work and a goalkeeper needs lateral movement, and why a rower gets almost nothing from a squat machine in terms of race-day transfer.

从训练刺激到表现的转化，取决于比赛中真正重要的素质和动作，所以400米跑者需要速度耐力训练，守门员需要横向移动，而赛艇运动员几乎从深蹲器械中获得不了比赛日的转移。

Progressive overload is the engine of adaptation.

渐进超负荷是适应的引擎。

The body adapts to stress placed upon it, and if the stress never increases, adaptation stops.

身体会适应施加在它身上的压力，而如果压力从不增加，适应就停止了。

The practical translation is small and relentless: add a little distance, a little load, a little volume, or reduce a little rest, then hold long enough for the adaptation to happen before increasing again.

实际的做法既小又无情：多加一点距离、多加一点负荷、多加一点训练量，或者少一点休息，然后保持足够长时间让适应发生，再下一次。

The most common failure is jumping too far too early, which produces fatigue and injury rather than adaptation.

最常见的失败是过早跳得太远，那带来的不是适应，而是疲劳和伤病。

Recovery is not a reward for training, it is part of the training.

恢复不是训练的奖励，而是训练的一部分。

Tissue repair, glycogen resynthesis, neural adaptation and the psychological reset all happen while you are not training.

组织修复、糖原重新合成、神经适应和心理上的重置，都发生在你不训练的时候。

Consolidating a session is as much a part of the plan as the session itself.

巩固一次训练和这次训练本身一样，都是计划的一部分。

A week is not seven days of training, it is a week containing training, and the balance between those two is what determines whether the athlete adapts or accumulates debt.

一周不是七天训练，而是包含训练的一周，而这两者之间的平衡，决定了运动员是在适应还是在累积负债。

Variety keeps the athlete and the adaptation alive.

多样性让运动员和适应都保持活力。

Repeating identical work produces a specific adaptation and then stalls; variety raises the stimulus, reduces boredom, reduces overuse risk and, in children in particular, is how you develop general movement skill rather than one narrow strength.

重复完全相同的训练会产生特定的适应然后停滞；

It is also a protection: an athlete who has trained ten different ways is far harder to injure than one who has trained the same way a thousand times.

多样性提高了刺激、减少了枯燥、降低了过度使用风险，而且对孩子来说，它还是发展一般动作能力而不是单一力量的方式。

多样性也是一种保护：用过十种方式训练的运动员，比用同一种方式训练一千次的运动员难得多地会受伤。

Reversibility is the one that gets forgotten and the one that explains most of what people call getting worse.

可逆性是最容易被忘记的一点，也是大多数被称为“退步”的现象的解释。

Detraining works in the opposite direction and just as fast: a few weeks off a trained aerobic athlete measurably reduces VO2max, and strength falls noticeably in a couple of weeks of complete rest.

停训会朝相反方向起作用，而且一样快：一位训练有素的有氧运动员停几周，最大摄氧量就会可测量地下降，完全休息两周力量就会

For a recovering athlete that is the point - resting an injured tissue is not a failure of the programme, it is the programme.

对正在恢复的运动员来说这正是重点--让受伤的组织休息不是计划的失败，而是计划本身。

For a coach it is a warning about the off-season.

对教练来说，这是关于休赛期的一个警告。

FITT is the four dials that turn those six qualities into a session.

FITT是把那六个品质变成一节训练课四个旋钮。

Frequency is how often, intensity is how hard, time is how long, and type is which method.

频率是多频繁，强度是多努力，时长是多久，类型是采用什么方法。

The order matters: you should be able to say why each dial is set where it is, because a session is a set of choices rather than a list of activities.

顺序很重要：你应该能说出每个旋钮为什么设在这个位置，因为一节训练课是一组选择，而不是一份活动清单。

And the honest answer to "why that intensity" is always a reference to something - the phase of the plan, the fitness of the athlete, or the deadline in the calendar.

对“为什么是这个强度”的诚实回答，总是指向某个参照--计划处于哪个阶段、运动员的体能水平，或者日历上的那个截止日。

The numbers / 算一算

Numbers to keep. Across a week, a single demanding session can be built on 2 to 4 hours of quality work, and the volume of easy work around it can be three or four times that. Progressive overload should change one variable at a time and by a small amount - adding roughly 5 to 10% rather than doubling. Recovery between two hard sessions in the same week is what makes the second one productive rather than destructive. And a useful weekly rule: if the athlete is not progressing and not recovering, the answer is usually less, not more. The numbers that matter most are the ones you track week to week, not the ones in a single session.

需要记住的数字。一周之内，一次高强度训练课可以建立在2到4小时的高质量工作之上，而围绕它的低强度训练量可能是它的三到四倍。

Common mistakes / 易错点

-

"Progressive overload means adding more every session." It means adding a small amount and then holding it long enough to adapt. Adding more every session is how you get injured, not how you get stronger.

"渐进超负荷意味着每次训练都加量。"它的意思是加一点量，然后保持足够长时间以适应。每次都加量是受伤的方法，不是变强的

-

"Rest days are wasted days." Tissue repair, glycogen resynthesis and neural adaptation happen while you are not training. A week is not seven training days.

"休息日是浪费掉的日子。"组织修复、糖原重新合成和神经适应都发生在你不训练的时候。一周不是七个训练日。

-

"FITT is four things to do." It is four dials: how often, how hard, how long, which method. Setting them without a

reason is not a plan.

"FITT 是要做的四件事。"它是四个旋钮：多频繁、多努力、多久、什么方法。没有理由地设置它们不叫计划。

-

"More volume is always better." The response to load is individual. Beyond a point, extra volume produces staleness, illness and injury rather than adaptation.

"训练量越多越好。"对负荷的反应因人而异。超过某个点，额外的训练量带来的是停滞、疾病和伤病，而不是适应。

Why it matters / 为什么重要

This section is the practical core of the whole course, because it is the one that decides whether a training programme works. Every other topic on the syllabus is knowledge; this is where knowledge becomes a plan. An athlete who understands the six qualities and FITT can look at any session and say what it is for, what it is building and what it will cost. An athlete who does not will follow a programme that happens to work for somebody else, and will get the injuries that come from training somebody else's plan.

这一节是整门课程的实践核心，因为决定训练计划是否有效的就是它。课程里的其他所有主题都是知识；这一节是知识变成计划的地方。

-

FITT: frequency is how often, intensity is how hard, time is how long and type is what method.

FITT: 频率是训练多频繁，强度是多努力，时间是训练多久，类型是采用什么方法。

-

Progressive overload should be gradual, monitored and compatible with recovery.

渐进超负荷应是渐进的、受监测的，并与恢复相容。

Key terms / 核心术语

Periodization 周期化

Planning training in macro-, meso- and microcycles.

以大周期、中周期和小周期规划训练。

Reversibility 可逆性

The loss of fitness when training stops or decreases.

训练停止或减少时体能下降。

Formulas / 公式

An 800 m runner uses race-specific work, suitable volume and pace, regular recovery and monitoring of split times or heart-rate response.

800米跑者采用专项比赛训练、合适训练量和配速，规律恢复，并监测分段成绩或心率反应。

Worked examples / 实例

>

Turn a performance goal into FITT choices and a monitoring measure.

把表现目标转化为FITT选择和监测指标。

Exam focus / 考核要点

Define specificity, progressive overload, recovery, variety, reversibility and periodization.

定义专项性、渐进超负荷、恢复、多样性、可逆性和周期化。

Periodization, overload and overtraining · 周期化、超负荷与过度训练

First, picture it / 先这样想

Periodisation is planning a journey so that the hard parts come when you are ready for them. You do not sprint on the first day of a mountain climb. The plan works in nested circles: a year or a season on the outside, blocks of several weeks in the middle, and a week containing the daily sessions on the inside. Overtraining is what happens when the plan has no shape - when the same hard week is repeated until the body stops adapting and starts sending you bills.

周期化就是规划一段旅程，让最难阶段出现在你准备好的时刻。你不会在登mountain的第一步就冲刺。这套计划由嵌套的圆组成：最外层是一年或一个赛季，中间是几周为一个的阶段，最里层是包含每天训练的一周。过度训练就是计划失去形状时发生的事--同一个高强度周被反复重复，直到身体停止适应，开始向你收取代价。

The detail / 详细讲解

The three nested timeframes are macrocycle, mesocycle and microcycle.

三个嵌套的时间框架是macrocycle（大周期）、mesocycle（中周期）和microcycle（小周期）。

A macrocycle is a long-term season or year plan.

大周期是长期的赛季或年度计划。

A mesocycle is a phase or block of several weeks, and each one has a purpose: general preparation, specific preparation, competition, or transition.

中周期是若干周为一个的阶段，每个阶段都有目的：一般准备、专项准备、比赛期或过渡期。

A microcycle is the week itself, containing the daily training.

小周期就是这一周本身，包含每天的训练。

The reason this nesting exists is that adaptation takes time and has an order, so a plan that ignores the order gets less out of the same work.

这套嵌套之所以存在，是因为适应需要时间而且有顺序，所以忽略顺序的计划只能用同样的训练量换来更少的结果。

The phases are worth naming because they answer a question coaches are always asked: why are we doing this now.

这些阶段值得点名，因为它们回答了教练总被问到的问题：我们现在为什么做这个。

In the preparation phase you build general fitness and skills.

在准备期你建立一般体能和技能；

In the specific phase you convert that into the exact qualities the event needs, at higher intensity and lower volume.

在专项期你把这些转化成该项目真正需要的素质，强度更高、训练量更低；

In the competition phase you perform and you sharpen.

比赛期你表现并磨尖状态；

In the transition phase you actively recover, which is the one athletes most often waste and the one that most protects the next cycle.

过渡期你主动恢复--这是运动员最常浪费的阶段，也是最能保护下一个周期的一个。

Overtraining is the failure mode of a plan with no shape, and it comes in two forms.

过度训练是没有形状的计划所产生的失败模式，它有两种形式。

Functional overreaching is a planned, short-term increase in load where performance dips temporarily and then comes back, often above where it was, with full recovery afterward.

功能性过训练反应是计划内的、短期增加负荷，表现暂时下降然后恢复，常常还高于之前，之后充分休息。

Overtraining syndrome is the long-term version: load stays too high for too long, performance declines persistently, and you get chronic fatigue, poor sleep, mood problems and a rising risk of illness and injury.

过度训练综合征则是长期版本：负荷长期过高，表现持续下降，并出现慢性疲劳、睡眠不佳、情绪问题以及疾病和伤病风险上升。

The difference is intent and duration, and the treatment is different too - one is a plan, the other is an emergency.

两者的区别在于意图和持续时间，治疗也不同--前者是计划，后者是紧急情况。

Load is not the same as fitness, and confusing them is why tapering feels counter-intuitive.

负荷不等于体能，混淆两者正是减量感觉反直觉的原因。

Fitness is what you have built.

体能是你已经建立起来的东西，负荷是你此刻的要求。

Load is what you are asking for right now.

在体能保持不变的情况下降低负荷，正是让已经完成的适应在表现中显现出来的机制，所以大赛前的减量不是状态的流失，而是状态3

Cutting load while fitness stays is what lets the adaptations you have already made show up in performance, which is why a taper before a big event is not a loss of form, it is the moment the form arrives. Athletes who taper and race well describe it as feeling sharp and strangely light.

减量后比赛成绩好的运动员会形容那种感觉：锋利，而且出奇地轻松。

And a plan is a prediction, which means it has to be allowed to be wrong.

而计划是一个预测，也就是说它必须被允许出错。

A periodised plan is a hypothesis about what will happen if the athlete does these things.

周期化的计划是一个假设：如果运动员做了这些事，会发生什么。

The monitoring loop is how you find out whether it was right, and the honest coach adjusts the plan rather than the athlete.

监测循环就是用来检验这个假设是否成立的，而诚实的教练会去调整计划，而不是调整运动员。

The numbers / 算一算

Numbers that make the structure concrete. A macrocycle might be 9 to 12 months, or a single competitive season. A mesocycle is typically 3 to 6 weeks, and a microcycle is 7 days. So a season of 10 months contains roughly 20 to 30 mesocycles. Over a week, the load distribution matters as much as the total: a classic pattern is one or two high-load days, several moderate ones and genuine rest. The threshold everyone argues about is change - a rule of thumb is that if performance has not moved in 4 to 6 weeks despite consistent training, something in the plan is wrong, and the first thing to examine is the load, not the motivation.

让这套结构变得具体的数字。一个大周期可能是9到12个月，或一个完整的比赛赛季。一个中周期通常是3到6周，一个小周期是7天。所

Common mistakes / 易错点

- "More training always produces more fitness." It produces adaptation only while the load is new and recoverable. Past that point it produces staleness and risk.
"更多训练总是带来更多体能。"只有在负荷是新的、且可以恢复时，它才产生适应。超过这个点，它带来的是停滞和风险。
- "Functional overreaching and overtraining syndrome are the same thing." One is planned, short and followed by supercompensation; the other is chronic, unwanted and needs a real change of load.
"功能性过训练反应和过度训练综合征是一回事。"前者是有计划的、短暂的，之后会有超量恢复；后者是慢性的、非自愿的，需要
- "Tapering means losing fitness." Fitness stays while load falls. The adaptation you already made is what shows up, which is why tapering usually improves rather than reduces performance.
"减量意味着体能下降。"体能保持，负荷下降。已经完成的适应这才显现出来，所以减量通常提升而不是降低表现。
- "A periodised plan is a schedule, so the athlete has to follow it." It is a hypothesis. If monitoring says it is wrong, the plan changes, not the athlete.
"周期化计划就是时间表，所以运动员必须照做。"它是一个假设。如果监测表明它错了，要改的是计划，不是运动员。

Why it matters / 为什么重要

Periodisation is what makes a career instead of a series of good weekends. It protects against the two ways athletes self-destruct: doing too much too soon, and then trying to fix it by resting for a month. A single peak performance is possible without periodisation; a career built of repeated peak performances is not, because every cycle has to begin from a recovered body, and only a plan guarantees that. It also gives a coach a defensible answer to the hardest question in sport, which is why a particular athlete is doing a particular session on a particular day.

周期化让职业生涯成为可能，而不是一连串还不错的周末。它防止了运动员自我毁掉的两种方式：过早做得太多，然后用休息一个月来补

- Functional overreaching is a planned short-term increase in load with temporary performance loss, followed by recovery and possible supercompensation.
功能性过训练反应是计划内短期增加负荷，表现暂时下降，随后恢复并可能出现超量恢复。

Overtraining is an excessive load sustained too long, with persistent decline, chronic fatigue, poor sleep, mood disturbance and higher illness or injury risk.

过度训练是负荷长期过高，表现持续下降、慢性疲劳、睡眠不佳、情绪问题以及疾病或受伤风险增加。

Key terms / 核心术语

Macrocycle 大周期

Long-term plan, usually a season or year.

长期计划，通常是赛季或年度。

Microcycle 小周期

Short-term plan, usually one week.

短期计划，通常是一周。

Formulas / 公式

A coach increases load for one block, schedules recovery and checks whether performance returns; persistent fatigue would prompt a load review.

教练在一个训练块增加负荷，安排恢复并检查表现是否恢复；持续疲劳会促使重新审视负荷。

Worked examples / 实例

>

Distinguish functional overreaching from overtraining.

区分功能性过训练反应与过度训练。

Exam focus / 考核要点

How can monitoring reduce the risk of overtraining?

监测如何降低过度训练风险？

Individualisation and the monitoring loop · 个体化与监测循环

First, picture it / 先这样想

The same training produces a different body in every person who does it. Think of ten athletes all following the identical programme: genetics decide how much of it actually becomes muscle, how they respond to different foods, and how quickly they recover. So the coach cannot read the answer off the plan - the coach has to measure it and adjust. That is what the monitoring loop is: test, record, look at the trend, change one thing, test again.

同一份训练在每个执行者身上产生不同的身体。想象十位运动员执行完全相同的计划：基因决定其中多少真正变成肌肉、他们对不同食物的反应如何、以及他们恢复得多快。所以教练无法从计划里直接读出答案--教练必须测量并调整。这就是监测循环：测试、记录、看趋势、改一个变量、再测试。

The detail / 详细讲解

Training has to respond to the person doing it, and the list of things that differ is long.

训练必须回应执行训练的人，而不同的东西清单很长。

Current fitness comes first, then age, then sex-related factors including menstrual-cycle changes where relevant, then genetics and responder differences, and then the current phase of the macrocycle.

首先是当前体能，然后是年龄，然后是性别相关因素（在相关时包括月经周期变化），然后是遗传和个体反应差异，最后是大周期所

Two athletes on the same programme can genuinely need different things, and the difference is often not visible from the outside - which is precisely why it has to be measured rather than assumed.

执行同一份计划的两位运动员可能真的需要不同的东西，而这个差别往往从外面看不出来--这正是为什么它必须被测量而不是被假设。

The monitoring loop is the mechanism that makes individualisation possible.

监测循环是个体化得以实现的机制。

You establish a baseline, you train, you re-test, you compare, and then you adjust.

你建立一个基线，你训练，你重新测试，你比较，然后你调整。

The critical word is compare: a single test tells you almost nothing, because day-to-day variation in sleep, hydration, caffeine and life stress is large enough to look like a real change.

其中关键的词是“比较”：单次测试几乎说明不了任何事，因为睡眠、水分、咖啡因和生活压力带来的日常波动，大到足以看起来像真实趋势。It is the trend across weeks that carries the information, and an athlete who panics about one bad morning is discarding the only data that mattered.

真正携带信息的是跨数周的趋势，而因为一个糟糕的早上就慌张的运动员，扔掉了唯一重要的数据。

What to track falls into three groups.

该追踪什么分为三组。

Physiological indicators: resting heart rate, heart-rate variability, and to a lesser extent blood markers.

生理指标：静息心率、心率变异性，以及在较小程度上一些血液指标。

Subjective indicators: how the session felt, perceived exertion, soreness, mood, and the athlete own rating of readiness.

主观指标：这节课的感觉、主观用力感、酸痛、情绪，以及运动员自己评的“准备好程度”。

Performance indicators: the numbers that matter to the event, such as a 5 km time or a countermovement jump.

表现指标：对项目真正重要的数字，比如5公里成绩或反向纵跳。

The ones that work best are the ones that are cheap, fast and repeatable, because a measure you cannot take consistently is a measure you cannot use.

最有用的，是那些便宜、快速且可重复的指标，因为一个你无法持续测量的指标就是一个你无法使用的指标。

The loop only works if something actually changes as a result.

这个循环只有在真的会带来改变时才有意义。

The failure mode is measuring everything and adjusting nothing, which produces a very tidy spreadsheet and no progress.

失败的模式是把所有东西都测一遍、却什么都不调整，那会产生一份非常整齐的电子表格和零进步。

Change one variable at a time, keep the rest constant, and you will know which change did the work.

每次只改一个变量，其余保持不变，你才知道是哪一个改变起了作用。

If two things change at once and performance improves, you have learned nothing that you can use next week.

如果同时改了两项而表现提升了，你就学不到下周能用的东西。

And the loop has to be honest about noise.

而且这个循环必须诚实地对待噪声。

A change you cannot explain is not a breakthrough.

你无法解释的改变不是突破。

Before concluding that a programme is working, ask whether sleep, illness, travel, a new job or a broken night could explain it, and be suspicious of any result that appears exactly when you wanted it to appear.

在断定某个计划有效之前，先问一句：睡眠、疾病、出行、一份新工作或者一个破碎的夜晚能不能解释它？

而对任何恰好在你希望它出现的时刻出现的结果，都保持怀疑。

The numbers / 算一算

Numbers that make the loop practical. Resting heart rate is the easiest daily measure: a rise of 5 to 10 beats per minute above your own baseline, persisting for several days, is a meaningful warning signal, provided you are comparing it to yourself and not to a population. Heart-rate variability tends to fall when readiness falls, so the direction matters more than the absolute value. Performance tests should be repeated often enough to see a trend but not so often that fatigue or learning interferes - for most markers, weekly or twice weekly is enough. And a simple rule for the whole loop: if two consecutive weeks show no movement in your key performance marker, change something. Do not wait for month three.

让这个循环变得可操作的数字。静息心率是最容易每天测的指标：比你自己的基线高5到10次每分钟、并且持续数天，就是一个有意义的

Common mistakes / 易错点

- "If the plan says it, they do it." The plan is a hypothesis built for an average athlete. The individual is the one doing the training, so the individual is what the plan adapts to.
"计划怎么说就怎么做。"计划是为一个“平均运动员”建立的假设。而真正在训练的是这个人，所以计划要适应的是这个人。
- "One bad test means the programme has failed." Day-to-day variation in sleep, hydration and life stress is large

enough to look like a real change. Trends across weeks carry the information.

"一次测试不好就说明计划失败了。"睡眠、水分和生活压力的日常波动大到足以看起来像真实变化。真正携带信息的是跨周的趋势

"Heart rate variability is high, so fitness is improving." HRV tends to fall when readiness falls. The direction matters, and so does the context, and it is a supporting measure rather than the decision.

"心率变异性高，说明体能提升了。"心率变异性往往在准备度下降时走低。方向很重要，情境也很重要，而且它是辅助指标而不是

"Measure everything, then decide." Measuring without adjusting produces a tidy spreadsheet and no progress.

Change one variable at a time so you know which one worked.

"把所有东西都测了再决定。"只测不调，会产出一份整齐的电子表格和零进步。每次只改一个变量，你才知道是哪一个起了作用。

Why it matters / 为什么重要

Individualisation is the difference between a coach and someone with a spreadsheet of a training plan. It is also the reason two athletes who look identical on paper can need opposite things, and the reason the best programs in sport keep a written record of every test: not to prove anything, but to be able to notice. A program that never changes because nobody measured anything is a plan that is going to fail slowly and for a long time, and the failure will be blamed on motivation rather than on the decision not to look.

个体化是教练和"手上有一份训练计划电子表格的人"之间的区别。它也是为什么纸面上看起来一样的两位运动员可能需要完全相反的东西

Training data include load, volume, intensity and RPE; wellbeing includes sleep, mood, stress and motivation; performance tests should include relevant sport-specific measures.

训练数据包括负荷、训练量、强度和主观用力程度；健康状态包括睡眠、情绪、压力和动机；表现测试应包含相关专项指标。

Key terms / 核心术语

Responder 反应者

An individual whose response to a training stimulus differs from another person's.

对训练刺激的反应因人而异的人。

Formulas / 公式

Monitor one athlete's training response, adjust one FITT variable and set a new review point rather than applying a universal programme.

监测一名运动员的训练反应，调整一个FITT变量，并设置新的复查点，而不是套用统一计划。

Worked examples / 实例

>

Explain why training must be individualised.

解释为什么训练必须个体化。

Exam focus / 考核要点

What are the stages of a useful monitoring loop?

有用监测循环的阶段是什么？

Benefits to health of being active

身体活动的健康益处

Guiding question / 引导问题

How does physical activity influence health across different populations?

身体活动如何影响不同人群的健康?

Regular activity supports muscular, skeletal, immune, cardiovascular and mental health, and lowers the risk of several non-communicable diseases. Benefits depend on age, starting level, sex differences, progression, health status and whether the activity is appropriate.

规律活动支持肌肉、骨骼、免疫、心血管和心理健康，并降低多种非传染性疾病风险。益处取决于年龄、起始水平、性别差异、渐进

Life stage, sex and energy balance · 生命阶段、性别与能量平衡

First, picture it / 先这样想

Asking for one training plan that suits every age is like asking for one recipe that is right for a growing child, a twenty-year-old athlete and a grandparent. The ingredients are similar, but the amounts are not, and so are the priorities: a child is building the kitchen, a young adult is cooking, and an older adult is maintaining it carefully. Energy balance is the same idea applied to the plate - the body needs enough to grow, to perform and to repair, and each stage puts that requirement in a different place.

要求一份适合所有年龄的训练计划，就像要求一份食谱同时适合正在长身体的孩子、二十岁的运动员和祖父母。食材差不多，但分量不同，重点也不同：孩子在建厨房，年轻成人在做菜，而老年人在小心维护。能量平衡就是把同样的思路用在餐盘上--身体需要足够的能量来生长、表现和修复，而每个阶段把这个需求放在不同的位置上。

The detail / 详细讲解

Children are not small adults.

孩子不是缩小版的大人。

Their first job is fundamental movement skill: running, jumping, throwing, catching, climbing, balancing.

他们的首要任务是基本动作能力：跑、跳、投、接、攀爬、平衡。

Those patterns become the platform everything else is built on, and they cannot be taught, only practised and allowed to develop.

这些模式构成了一切后续能力的平台，而它们不能被教，只能被练习和允许发展。

Children also need energy, protein, calcium and iron for growth, and their bodies are more vulnerable to overuse than adults because their bones, tendons and cartilage are still maturing.

孩子还需要生长所需的能量、蛋白质、钙和铁，而且他们的身体比成人更容易过度使用损伤，因为骨、肌腱和软骨还在成熟。

The mistake to avoid is imposing an adult training load on a child and calling it discipline.

要避免的错误是把成人的训练负荷压到孩子身上，并称之为纪律。

Adolescents are building fitness, body composition and well-being at once, which is a lot of simultaneous projects.

青少年同时在建立体能、身体成分和幸福感，这等于同时在做好几个项目。

This is the period when peak height velocity arrives and the gap between strength and muscle size widens, and it is also when the risk of overtraining and of disordered eating is highest, partly because the body is changing in ways that are visible and therefore commented on.

这个时期会出现身高生长高峰，力量与肌肉体积之间的差距被拉大；

Girls in particular face a social pressure around body change that has nothing to do with sport, and it is a health issue as much as a coaching one.

同时也是过度训练和进食障碍风险最高的时期，部分原因是身体正在以可见、因此会被评论的方式变化。

尤其是女孩，会面临一个与运动无关的、围绕身体变化的社会压力，而这既是健康问题，也是执教问题。

Adults are mostly maintaining function, and older adults gain from a specific and well-evidenced package.

成年人大体上是在维持功能，而老年人从一个有充分证据的特定组合中获益。

After roughly 60, the priority list changes: balance and leg strength become as important as aerobic capacity, because falls rather than cardiovascular events are a leading cause of injury and loss of independence, and

because muscle and bone mass fall unless they are defended.

大约60岁之后，优先清单会改变：平衡能力和腿部力量变得和心肺能力一样重要，因为跌倒而不是心血管事件是伤害和丧失独立性的原因。

Aerobic work still matters enormously for cardiovascular and cognitive health.

有氧训练对心血管和认知健康仍然极其重要。

The plan should be built around what declines fastest and matters most, rather than around what is most intense.

计划应该围绕下降最快且最重要的部分来构建，而不是围绕强度最大的部分。

Sex differences need to be handled carefully and factually.

性别差异需要谨慎而实事求是地处理。

Menstrual cycling does affect energy availability, iron status and perceived exertion, which is why iron is specifically relevant to female athletes, and why monitoring rather than assuming is the right response.

月经周期确实会影响能量可用性、铁状态和主观用力感，这也是铁与女性运动员特别相关的原因，也是应该监测而不是假设的原因。

Bone is a related point: oestrogen contributes to bone health, and any prolonged low energy availability affects bone mineral density in all athletes regardless of sex.

骨是相关的一点：雌激素有助于骨骼健康，而任何长期能量可用性偏低都会影响所有运动员（不论性别）的骨矿物质密度。

The useful framing is not "men and women are different" but "there are known physiological variables that an individual programme can monitor", which is a much stronger position to argue from.

更有用的框架不是“男性和女性不同”，而是“存在一些已知的生理变量，可以纳入个体化计划进行监测”--这是一个强得多的论证立场。

Energy balance ties it together, because it is the requirement behind every one of these stages.

能量平衡把这些串起来，因为它是所有这些阶段背后的要求。

Growth needs an energy surplus.

生长需要能量盈余；

Training needs enough to cover the expenditure.

训练需要足以覆盖消耗；

Health needs a margin left over.

健康需要留下余量。

An athlete at any stage who is not covering expenditure is not managing a small problem; the consequences run through bone, immunity, endocrine function and mood, and they arrive slowly enough to be missed.

任何阶段的运动员如果没有覆盖消耗，那都不是在处理一个小问题：后果会经由骨、免疫、内分泌功能和情绪表现出来，而且它们出现得很慢。

The numbers / 算一算

Numbers to keep. Children and adolescents are generally recommended around 60 minutes of moderate-to-vigorous daily activity. Adults accumulate about 150 minutes of moderate activity per week, and older adults add balance and strength work two or more days a week - often 2 to 3 sessions focused on the legs and on balance. Iron requirement differs between menstruating and non-menstruating athletes, and the direction of that difference is what makes monitoring worthwhile rather than optional. On the energy side, the practical test is the simplest one: growth, training and health all need to be paid for, and a plan that only pays for training has already lost.

需要记住的数字。儿童和青少年一般建议每天约60分钟中高强度活动。成年人每周累计约150分钟中等强度活动，老年人则每周增加两次。

Common mistakes / 易错点

-

"Children just need a scaled-down adult programme." Children need fundamental movement skill development, which is a different job from adult conditioning, and their tissues are still maturing.

"孩子只需要把成人计划按比例缩小就行。"孩子需要的是基本动作能力的发展，这与成人体能训练是两份不同的工作，而且他们的组织仍在成熟。

-

"Older adults should train like younger athletes." After about 60 the priority shifts toward balance, leg strength and maintaining muscle and bone, because falls are a leading cause of lost independence.

"老年人应该像年轻运动员那样训练。"60岁以后，重点会转向平衡、腿部力量以及维持肌肉和骨，因为跌倒是丧失独立性的主要原因。

-

"One plan for everyone, adjusted only for ability." Age, sex-related variables, growth, energy availability and current phase all change what the plan should be, not just how hard it is.

"所有人一份计划，只按能力调整。"年龄、性别相关变量、生长、能量可用性和当前阶段都会改变计划的内容，而不只是它的强度。

"Energy balance only matters for weight." It decides whether growth, training and health are all funded. An athlete who only covers training expenditure has already begun to lose the other two.

"能量平衡只关系到体重。"它决定生长、训练和健康是否都被支付。只覆盖训练消耗的运动员，已经开始失去另外两项了。

Why it matters / 为什么重要

Every room in a sports setting is full of people of different ages and stages, and the people who train them well are the ones who notice the difference. A coach who gives a fifteen-year-old girl a man plan is not being strict, they are setting her up for a bone stress injury. A coach who puts a seventy-year-old on a rowing machine only is not being cautious, they are ignoring the thing most likely to put them in a care home. Population guidance is not a softer version of elite guidance; it is a different set of answers to a different set of questions.

每一个运动场所里都挤满了不同年龄、不同阶段的人，而善于训练他们的人正是那些注意到差别的人。给十五岁女孩套上成年男性的计划

Energy intake is compared with expenditure from basal metabolism, physical activity and the thermic effect of food. A deficit can reduce weight; balance supports stable weight and function; a surplus stores energy.

能量摄入要与基础代谢、身体活动和食物热效应产生的消耗比较。赤字可减重；平衡有助于体重稳定和功能；过剩会储存能量。

Females may have a higher risk of iron deficiency, and hormonal cycles can affect energy and performance.

女性缺铁风险可能更高，激素周期可影响能量和表现。

Key terms / 核心术语

Energy balance 能量平衡

The relationship between energy intake and energy expenditure.

能量摄入与能量消耗之间的关系。

Formulas / 公式

A programme should start from a person's current activity level and progress according to health, age, ability and response.

训练计划应从个人当前活动水平开始，并按健康、年龄、能力和反应逐步进阶。

Worked examples / 实例

>

Why should activity recommendations consider age and sex differences?

活动建议为何要考虑年龄和性别差异？

Exam focus / 考核要点

Explain energy balance and one possible outcome.

解释能量平衡及一个可能结果。

System benefits, chronic disease and progression · 系统益处、慢性病与进阶

First, picture it / 先这样想

Exercise is a bit like servicing a car that is still being driven. Oil changes feel wasteful until you remember what the engine sounds like afterwards. The benefits of activity are not dramatic and they are not instant, and that is exactly why they get devalued by anyone looking for a quick result. They are cumulative: small amounts, kept up, produce a body that is more robust, more tolerant and more capable - and a car that is still running in twenty years.

运动有点像给一辆还在开的车做保养。换机油在感觉上很浪费，直到你记起之后发动机听起来是什么样。活动的益处既不戏剧化也不即时，而这恰恰是那些想看快速结果的人最不看重它的原因。它们是累积的：小量、持续，最终造出一个更结实、更耐受、更能干的body--以及二十年后还在跑的车。

The detail / 详细讲解

Start by listing what activity does, because the list is longer than most people expect.

先把活动能做什么列出来，因为这份清单比多数人预期的长。

It increases muscular strength and endurance, improves bone density, supports immune function, reduces stress

and anxiety, improves mood and sleep, increases confidence and resilience, supports cardiovascular and metabolic health, and helps maintain function and independence into old age.

它能增加肌肉力量和耐力、改善骨密度、支持免疫功能、减轻压力和焦虑、改善情绪和睡眠、提升自信和韧性、支持心血管和代谢健康。

Almost every one of those is on an official list somewhere as a separate recommendation, which is a reasonable hint about how well evidenced the field is.

其中几乎每一项都出现在某份官方清单里、还作为单独的一条建议--这本身就是这个领域证据有多充分的合理提示。

Those benefits divide into two groups that behave differently.

这些益处分成两组，表现方式不同。

Early responses happen within minutes to hours: mood improves, anxiety drops, and the effect is real and repeatable.

早期反应发生在几分钟到几小时内：情绪改善、焦虑下降，效果真实且可重复。

Longer adaptations take weeks to years: bone density, aerobic capacity, insulin sensitivity and strength.

较长的适应需要数周到数年：骨密度、有氧能力、胰岛素敏感性和力量。

This matters because the early effects are what keep people doing it, and the long effects are what actually protect health decades later.

这一点很重要，因为早期效果是让人坚持下来的东西，而长期效果才是几十年后真正保护健康的東西。

Programmes that only sell the first lose people, and programmes that only claim the second cannot recruit anyone.

只卖前者的计划会让人流失，只宣称后者的计划则根本招不到人。

Chronic disease risk is where this becomes concrete.

慢性病风险是这件事变得具体的地方。

Regular physical activity lowers the risk of cardiovascular disease, type 2 diabetes, osteoporosis and some cancers, and it does so through mechanisms rather than magic.

规律的身体活动能降低心血管疾病、2型糖尿病、骨质疏松和某些癌症的风险，而且它是通过机制起作用的，而不是魔法。

More movement means better glucose handling, better blood pressure control, better lipid profiles, better body composition, lower inflammation, and a heart and circulation that are conditioned rather than merely un-damaged.

更多活动意味着更好的血糖处理、更好的血压控制、更好的血脂谱、更好的身体成分、更低的炎症，以及一颗被训练过而不仅仅是没

The single biggest lever is usually not intensity, it is volume and consistency: the people who benefit most are the ones who go from nothing to something, not the ones who go from a lot to slightly more.

最大的杠杆通常不是强度，而是总量和一致性：获益最多的人是从零到有，而不是从很多到更多一点。

Progression is the part that decides whether any of it sticks.

进阶决定这一切能否留住。

Starting too hard, or too much, produces more soreness, more missed sessions and eventual dropout, and the most common way activity programmes fail is not that people cannot see the benefits but that the first four weeks are unpleasant.

开始得太猛或太多，会带来更多酸痛、更多缺席，以及最终的退出；

Progress from a baseline, add a small amount at a time, and allow recovery to be part of the plan rather than an interruption to it.

活动计划最常见的失败方式不是人们看不到益处，而是最初四周很难受。

The people who stay active for decades are not the most disciplined people, they are the ones whose programmes were easy to keep doing.

从一个基线开始，每次加一点，并且把恢复当作计划的一部分而不是对计划的打断。

能坚持几十年的人并不是最自律的人，而是那些计划容易持续下去的人。

And overdoing it has its own cost.

而做过头也有自己的代价。

Excess volume or intensity without progression produces overuse injury, burnout and loss of motivation, and the resulting break is far more costly than a slower start would have been.

没有进阶的过量训练量或强度会带来过度使用损伤、倦怠和动力丧失，由此造成的停顿远比慢慢开始要代价高。

The goal is a routine that can be kept for decades, not the biggest possible week, and an answer that only works for four weeks is not an answer.

目标是能维持几十年的常规，而不是最大可能的那一周；一个只在头四周有效的方案不算方案。

The numbers / 算一算

Numbers to keep. Adults are recommended about 150 minutes of moderate or 75 minutes of vigorous activity per week, and the guidance is deliberately modest because the biggest gains come from moving from inactive to active rather than from pushing hard. Bone responds over months, not weeks. Aerobic capacity over 8 to 12 weeks of consistent training, strength over 4 to 8. Sedentary behaviour is its own risk factor, so breaking up long sitting matters independently of whether you train. And for progression, the useful rule is small and boring: increase one thing by a small amount, then wait long enough for it to work before increasing again.

需要记住的数字。成年人建议每周约150分钟中等强度或75分钟高强度活动，而这个指导刻意定得不高，因为最大的收益来自从不活动到

Common mistakes / 易错点

- "If it is not hard, it is not working." Intensity is one variable. The people who gain most are usually the ones who go from inactive to active, and the biggest risk is dropping out rather than under-training.
"不够难就没有效果。"强度只是其中一个变量。获益最多的往往是那些从不动到动起来的人，而最大的风险是退出而不是练得不够。
- "All the benefits show up in two weeks." Some do. Bone density, aerobic capacity and insulin sensitivity take weeks to months, and that delay is exactly why adherence drops.
"所有益处两周就出现。"有一部分会。骨密度、有氧能力和胰岛素敏感性需要数周到数月，而这个延迟正是坚持率下降的原因。
- "Sore and stiff means it worked." Soreness is a cost, not a target. Progressing faster than recovery produces overuse injury, not a better programme.
"酸痛僵硬说明练到了。"酸痛是代价，不是目标。进阶快于恢复会产出过度使用损伤，而不是更好的计划。
- "More intensity is always more benefit." For inactive beginners, volume and consistency beat intensity almost every time.
"强度越大收益越大。"对不活动的初学者来说，训练量和一致性几乎总是胜过强度。

Why it matters / 为什么重要

This is the section that lets a sport professional make a claim about health rather than about performance, and that is a different and more valuable kind of knowledge. It is what separates a programme that helps an athlete perform from a programme that helps a person stay well across a lifetime. And the strongest version of the argument is the mechanistic one: naming the pathway through which activity changes a risk factor, rather than asserting that it does. Anyone can say exercise is good. The mark is in the mechanism.

这一节让运动专业人员能够对健康作出论断，而不仅仅是对表现--这是另一种更有价值的知识。它区分了"帮助运动员表现"的计划和"帮助

- Regular activity can reduce risk of osteoporosis, obesity, hypertension, cardiovascular disease and type 2 diabetes.
规律活动可降低骨质疏松、肥胖、高血压、心血管疾病和2型糖尿病风险。

Key terms / 核心术语

Progressive intensity 渐进强度

Gradual increase in workload with recovery, monitoring and avoidance of sudden large changes.

在恢复、监测和避免突然大幅增加的前提下逐步提高负荷。

Formulas / 公式

For an older adult returning after inactivity, begin with appropriate work, progress gradually, include strength and balance, and modify for health and injury risk.

久坐老年人恢复活动时，应从适当工作量开始，逐步进阶，加入力量和平衡，并按健康及受伤风险调整。

Worked examples / 实例

- > Prescribe a progression for a population and justify one safety modification.
为某人群设计进阶并说明一项安全调整。

Exam focus / 考核要点

Name two positive and two negative effects of activity if overdone.

指出活动适量时的两个正面效果和过量时的两个负面效果。

HL exercise prescription for populations · HL人群运动处方

First, picture it / 先这样想

Prescribing exercise for a population is writing a plan for a room full of different people rather than for one of them. A single sheet with one line per person is impossible, so you write a recipe with ranges: a direction, a minimum, a maximum, and a note about who to be careful with. That sounds vaguer than a single number, and it is more accurate. "Some aerobic activity most days" is genuinely more useful to a grandparent than "150 minutes", because the grandparent can act on it.

为人群开运动处方，就是给一屋子不同的人写计划，而不是给其中一个人。对每个人一行的一张表是不可能的，于是你写一份带范围的食谱：方向、下限、上限，以及关于谁需要小心的说明。这听上去比一个具体数字更含糊，但它更准确。"多数日子做一些有氧活动"对祖父母来说比"150分钟"更有用，因为祖父母能照着做。

The detail / 详细讲解

The workbook gives population-focused guidance rather than individual numbers, and the reason is that the numbers differ.

工作册给出的是面向人群的指导而不是个人数字，原因是数字本就不同。

Children and adolescents need moderate-to-vigorous activity daily, and what they need alongside it is variety, fun and skill development, because the point at their age is a broad base rather than a specific adaptation.

儿童和青少年需要每天中高强度活动，而在这些之外还需要多样性、趣味和技能发展，因为在他们那个年龄，重点是宽广的基础而不是

Older adults need moderate aerobic activity plus balance and muscle and bone strengthening, and the reason balance appears at all is falls.

老年人需要中等强度有氧活动，加上平衡以及肌肉和骨骼的强化训练，而平衡之所以出现，纯粹是因为跌倒。

Sex, and pregnancy in particular, change the prescription rather than the principle.

性别，尤其是怀孕，会改变处方而不是改变原则。

Moderate intensity activity is appropriate during pregnancy, including pelvic floor and flexibility work, with adjustments for hydration and overheating, and individual medical advice where there is any history.

孕期进行中等强度活动是适宜的，其中包括盆底肌和柔韧性练习，同时要根据水合和避免过热做出调整；

The safe principle is gentle progressive loading, avoiding overheating, maintaining hydration, and seeking professional advice where the medical history warrants it.

有病史时则应寻求专业建议。

There is no reason to be less active, and every reason to be thoughtfully active.

安全的原则是温和的渐进负荷、避免过热、保持水合，并在病史提示需要时寻求专业意见。

没有任何理由减少活动，但有充分理由去"有心地"活动。

Every prescription needs a progression, or it is just a target.

每一份处方都需要进阶，否则它只是一个目标。

A starting dose has to be below the athlete current capacity and then climb.

起始剂量必须低于当前能力，然后逐步爬升。

That is true of populations as much as individuals: someone who has been inactive for years should not begin at the guideline, they should begin at a fraction of it, and the guideline is a ceiling to work towards rather than a starting prescription.

这点对人群和对个人一样：多年不活动的人不应该从指南值开始，而应该从它的一个零头开始；

Designing the wrong starting point produces injuries and dropouts, and it is the most common failure in population programmes.

指南是努力靠近的上限，而不是起始处方。起始点设计错了会产出伤病和退出，这也是人群项目中最常见的失败。

And a prescription has to be built around what the person will actually keep doing.

而一份处方必须围绕这个人真正会持续做的事来构建。

Adherence is not a personality trait, it is a property of the plan.

依从性不是性格特质，而是计划的属性。

A programme built around activities the person already enjoys, at a dose they can fit into an existing routine, in a setting they can get to, will beat a theoretically superior programme they will abandon in a month, every single time.

围绕对方本来就喜欢的活动、他们能塞进现有生活节奏的剂量、以及他们能到达的场地所构建的项目，每一次都会打败那份理论上更

The numbers / 算一算

Numbers that keep the ranges honest. Children and adolescents: about 60 minutes of moderate-to-vigorous activity daily, plus free play and sport. Adults: about 150 minutes of moderate or 75 minutes of vigorous activity weekly, plus two sessions of strengthening. Older adults: the same aerobic target with balance and strength work two or more days a week. And for an inactive beginner, the realistic starting dose is a fraction of that - often 10 to 20 minutes of easy activity a day, walking - with the guideline as the destination rather than the start. The "talk test" is useful at this level: able to talk but not sing indicates moderate intensity, which is the target for most population prescriptions.

让范围保持诚实的数字。儿童和青少年：每天约60分钟中高强度活动，外加自由玩耍和运动。成年人：每周约150分钟中等强度或75分钟

Common mistakes / 易错点

- "Give everyone the guideline." That is the target, not the starting dose. Starting an inactive person at the guideline produces soreness, injury and dropout.
"给每个人指南值。"那是目标，不是起始剂量。让不活动的人从指南值开始，会产生酸痛、伤病和退出。
- "Prescription is one size fits all." It is a direction, a range and a progression, with notes about who needs care. Individual numbers come later, from monitoring.
"处方就是一份通用的。"它是一个方向、一个范围和一条进阶路径，外加关于谁需要小心的说明。具体数字要靠监测以后再定。
- "Only vigorous activity counts." Moderate activity is where nearly all the health benefit is, and it is far more achievable for most people, so it produces better adherence.
"只有高强度活动才算。"中等强度活动几乎承载了全部健康益处，而且对多数人现实得多，因此依从性更好。
- "Pregnancy is a reason to stop training." It is a reason to train thoughtfully: gentle progressive loading, no overheating, good hydration, and medical advice where the history warrants it.
"怀孕是停止训练的理由。"它是去有心地训练的理由：温和渐进负荷、不过热、充分水合，并在病史提示需要时寻求医疗建议。

Why it matters / 为什么重要

Population prescription is the HL extension, and it is the part of sports science that has the largest effect on the most people. The 100 m sprinter population is small and already well served. The inactive adult who is told something they can actually do, at a dose they can actually start, changes a risk factor for the rest of their life. That is a bigger contribution to public health than almost anything else in this course, and it is entirely a communication problem as much as a physiological one: the science is easy compared with getting a person to start.

人群处方是HL的扩展内容，也是运动科学中对最多人影响最大的部分。百米短跑运动员这个群体很小，而且已经得到了很好的服务。而

- Prescription must be individualised and adjusted as needs change.
运动处方必须个体化，并随需求变化调整。

Key terms / 核心术语

Safety modification 安全调整

A change to activity or environment that lowers risk for a person or population.
为个人或人群降低风险而对活动或环境作出的调整。

Formulas / 公式

For pregnancy, progress gently, avoid overheating, maintain hydration and seek professional advice when the individual's history requires it.

孕期应温和渐进、避免过热、保持水合；个人病史需要时应寻求专业建议。

Worked examples / 实例

>

How should exercise prescription differ for children, older adults and pregnancy?

儿童、老年人和孕期运动处方应有何不同？

Exam focus / 考核要点

Name two chronic-disease risks reduced by regular activity.

说出规律活动可降低的两种慢性病风险。

Fatigue and recovery

疲劳与恢复

Guiding question / 引导问题

Why does fatigue develop and how can recovery be supported?

疲劳为何产生，恢复如何得到支持？

Fatigue can be central, arising from reduced brain drive or motivation, or peripheral, arising in muscle or at the neuromuscular junction. Recovery is multidimensional: physiological, symptom-based, psychological and practical indicators should be monitored together.

疲劳可分为中枢性疲劳，即脑部驱动或动机下降，以及外周性疲劳，即肌肉或神经肌肉接点的问题。恢复是多维度的，应同时监测生理、症状、心理和实际指标。

Central and peripheral fatigue · 中枢性与外周性疲劳

First, picture it / 先这样想

Fatigue is a message, and the important question is where it was sent from. Think of a car on a long drive: if the fuel gauge is the problem, the car is empty of something. But if the driver is the problem, the car is full and the driver is simply no longer reading the road. Sport sits exactly in that gap. Some fatigue is a real shortage in the working muscle, and some fatigue is the brain deciding the effort is not worth continuing - and treating the second as though it were the first is how people injure themselves.

疲劳是一条信息，重要的是它从哪里发出。想象一辆长途行驶的车：如果问题出在油量表上，那车是缺了某种东西；但如果问题出在司机身上，那车是满的，而司机只是不再看路了。运动恰好落在这个缝隙里。有些疲劳是工作肌肉真的缺东西，有些疲劳是大脑判定这个努力不值得继续--而把后者当成前者来处理，正是人们弄伤自己的方式。

The detail / 详细讲解

Central fatigue is fatigue generated in the brain, not in the muscle.

中枢性疲劳是产生在大脑而非肌肉的疲劳。

It shows up as reduced drive, lower motivation, and a greater perception of effort for exactly the same task.

它表现为动力下降、积极性降低，以及对完全相同任务的用力感增加。

A runner who feels terrible in week four and then feels fine after two good nights of sleep has demonstrated **central fatigue very clearly.**

一位跑者在第四周感觉糟透了、然后睡了两个好觉就感觉好了--这就是非常清楚的中枢性疲劳。

The practical consequence is that central fatigue is managed by things other than fuel: sleep, motivation, perceived progress, and confidence.

实际推论是，中枢性疲劳靠燃料以外的东西来管理：睡眠、动力、可见的进步和自信。

Peripheral fatigue is fatigue in the working muscle itself, and it has identifiable causes.

外周性疲劳是工作肌肉自身的疲劳，它有可指认的原因。

Metabolic by-products accumulate.

代谢副产物积累；

Potassium shifts out of the muscle and sodium shifts in, which changes the excitability of the muscle.

钾移出肌肉、钠移入，改变肌肉的兴奋性；

Glycogen falls.

糖原下降；

Creatine and phosphocreatine fall.

肌酸和磷酸肌酸下降；

Heat rises.

热量上升。

In the short term these are the direct limiters of performance, and the reason a 400 m runner hits a wall at 300 m is mechanical rather than motivational.

短期内这些是表现的直接限制因素，也正是400米跑者在300米处撞上墙的原因是机械性的，而不是动力性的。

The two interact, and that is the part that makes it complicated.

两者会相互作用，这也是它变得复杂的原因。

A hard session depletes peripherally, which raises the cost of the next effort, and the athlete perceives that cost as central effort.

一次高强度训练造成外周耗竭，从而提高下一次努力的成本，而运动员把这份成本感知为中枢上的用力。

So a caffeine-free week, a poor week of sleep, or a hard block of training can all produce the same feeling in the same athlete for entirely different reasons.

于是断咖啡因的一周、睡眠不好的一周、或者一个高强度训练块，都可能在同一位运动员身上产生完全相同的感受，而原因却完全不同。

This is why the two-way model matters: the treatment only works if you know which one you are dealing with.

这就是双向模型重要的原因：只有知道自己在处理哪一种，治疗才有效。

Athlete monitoring, in a broad sense, exists to tell the two apart.

广义上的运动员监测，目的就是把两者区分开。

Subjective measures - how the session felt, perceived exertion, willingness to train - capture the central side.

主观指标--这节课的感受、主观用力感、训练意愿--捕捉的是中枢那一侧；

Performance measures and physiological markers capture the peripheral side.

表现指标和生理指标捕捉的是外周那一侧。

When the subjective measures deteriorate while performance holds, think central.

当主观指标变差而表现还保持时，考虑中枢；

When performance drops while the athlete still feels fine, think peripheral.

当表现下滑而运动员自己感觉还不错时，考虑外周。

Neither pattern on its own is diagnostic, and chasing the wrong one wastes weeks.

单独任何一种模式都不足以诊断，而追错方向会浪费好几周。

The most useful thing most athletes never do is separate the two on paper.

而多数运动员最有用、却从未做过的一件事，就是把两者在纸面上分开。

Rate session quality out of ten every day, record perceived exertion, and note performance.

每天给训练质量打十分，记录主观用力感，并记下表现。

Two lines of data, taken for a few weeks, will tell an athlete far more about their own training than any single test, because it turns "I feel tired" into a question with an answer.

几周的这样两条数据线，会告诉一位运动员关于他自己训练的事情，比任何单项测试都多--因为它把"我感觉累"变成了一个有答案的问题。

The numbers / 算一算

Numbers that keep the two sides separate. Peripherally, muscle glycogen is the limiter in most events lasting 30 seconds to about 2 hours, and starting a session with low stores is one of the most common and most fixable causes of a flat performance. Phosphocreatine falls essentially to zero within the first 30 seconds of maximal effort.

Centrally, the useful marker is often simply consistency: if session quality ratings drop for four or five consecutive days while nothing has changed in the plan, that is a central signal and it should be answered with sleep and reduced load, not with more conditioning. Practice, performance, fatigue and recovery are four different things and they are routinely confused.

把两侧区分开的数字。在外周方面，肌糖原是大多数30秒到约2小时项目的限制因素，而在储备不足的状态下开始训练，是表现平淡最常见的原因。

Common mistakes / 易错点

-

"Fatigue is always a physical shortage." Sometimes it is, and sometimes the body is adequately fuelled and the brain has simply reduced its drive. Treating all fatigue as a fuel problem means resting a body that needed rest, and training a brain that needed a break.

"疲劳总是身体缺东西。"有时是，有时身体燃料充足而大脑只是降低了动力。把所有疲劳都当成燃料问题，就意味着让一个本该休息的身体去训练，而大脑需要的是休息。

-

"Feeling flat in training means you are getting fitter." It can mean the opposite. Adaptation shows up in performance, not in how the session felt.

"训练时感觉平淡说明你正在变强。"也可能正相反。适应体现在表现上，而不是这节课的感受上。

- "Perceived exertion is the objective measure." Perception is a central report. It is valuable, and it is not the same thing as what the muscle is doing.

"主观用力感是客观指标。"感知是中枢的报告。它很有价值，但它和肌肉实际在做的事不是一回事。

- "Eat more and push through." Pushing through central fatigue is exactly the mechanism that produces overuse injury. Central fatigue is a reason to change the plan.

"多吃点硬扛过去。"硬扛中枢性疲劳正是产生过度使用损伤的机制。中枢性疲劳是应该调整计划的理由。

Why it matters / 为什么重要

Knowing the difference between central and peripheral fatigue is what separates a coach who adjusts a training plan from one who simply tells an athlete to try harder. It changes what you do next: peripheral fatigue is a question of load and fuel, and it responds to a lighter week. Central fatigue is a question of sleep, stress and motivation, and it responds to almost nothing you can do in the gym. Getting these the wrong way round is how a temporary dip becomes an injury, a lost season, or an athlete who decides the sport was not for them.

知道中枢性与外周性疲劳的区别，是区分"会调整训练计划的教练"和"只会叫运动员再努力一点的人"的关键。它决定了你下一步做什么：

- Both central and peripheral factors can interact and limit performance.

中枢和外周因素可能相互作用并限制表现。

- Recovery is not judged only by feeling less tired.

恢复不能只用"感觉不累"来判断。

Key terms / 核心术语

Central fatigue 中枢性疲劳

Reduced central drive, motivation or perceived effort.

中枢驱动、动机或用力感下降。

Peripheral fatigue 外周性疲劳

Fatigue arising in muscle or at the neuromuscular junction.

发生在肌肉或神经肌肉接点的疲劳。

Formulas / 公式

After a hard session, an athlete may feel motivated but still have low glycogen, fluid loss or impaired ion handling, so subjective freshness is insufficient.

高强度训练后，运动员可能感觉有动力，但糖原低、体液流失或离子处理受损，因此主观精神好并不足够。

Worked examples / 实例

>

Distinguish central from peripheral fatigue.

区分中枢性与外周性疲劳。

Exam focus / 考核要点

Give three peripheral contributors and their effects.

给出三个外周因素及其影响。

Recovery nutrition and methods · 恢复营养与方法

First, picture it / 先这样想

Recovery is three deliveries, and they travel to three different addresses. Carbohydrate is fuel sent to the muscle to refill the tank. Protein is a building crew sent to repair what the session damaged. Water and sodium are the restocking of a shop that got emptied. Add the fourth item - creatine monohydrate - and you have a repair shop for the energy currency itself, because phosphocreatine is what the first ten seconds of a sprint is made of. Polyphenols and antioxidants are the tidy-up crew, useful but never the headline.

恢复是三批送到三个不同地址的货。碳水是送往肌肉的燃料，用来把油箱加满；蛋白质是派去修补训练损耗的施工队；水和钠是给被搬空的商店补货。再加上第四样--肌酸一水合物--你就有了为能量货币本身服务的维修站，因为磷酸肌酸正是短跑前十秒的原料。多酚和抗氧化剂是收拾班组，有用，但从来不是主角。

The detail / 详细讲解

Carbohydrate replenishes muscle glycogen, and the detail that surprises people is how specifically it does it.

碳水化合物补充肌肉糖原，而让人意外的是它补得有多“专一”。

Muscle glycogen is stored in the muscles that will use it, so topping up after a leg session fills the legs, and topping up after a hard week benefits the muscles that were actually depleted.

肌糖原储存在将要使用它的那些肌肉里，所以腿部训练后补充主要补的是腿，而一周高强度训练后补充则惠及真正被耗竭的肌肉。

The window is reasonably generous - somewhere in the region of the few hours after training - so the practical instruction is simple: a carbohydrate-containing meal soon after training, and a normal balanced diet thereafter.

时间窗其实相当宽--大致是训练后的几个小时--所以实际指令很简单：训练后不久吃一顿含碳水的餐，之后正常均衡饮食。

Protein provides the amino acids for repair and adaptation, and here timing matters more than most people think.

蛋白质提供用于修复和适应的氨基酸，而这里的时机比多数人以为的更重要。

Muscles are most receptive for a few hours after training, so a meal with protein reasonably soon afterwards does more than the same protein much later.

训练后几个小时肌肉最容易吸收，所以之后不久吃一顿含蛋白质的餐，效果好过很晚吃同样多的蛋白质。

The total across the day matters too: around 1.4 to 2.0 grams per kilogram of body mass, spread across several meals rather than concentrated in one enormous evening serving, because the body cannot use a large surplus in one sitting and simply oxidises it for fuel.

全天的总量同样重要：大约每公斤体重1.4到2.0克，分散在几餐里，而不是集中在晚餐一大盘里--因为身体无法在一次进食中利用大量

Water rehydrates, and the detail is that you cannot simply drink your way back from a deficit.

水负责补回脱水和流失，而细节在于你不能只靠多喝水来补回赤字。

Rehydration needs both volume and sodium, because without sodium the fluid you drink does not stay where it should - it is lost to the kidneys faster.

再水合既需要容量也需要钠，因为没有钠，你喝进去的液体不会留在该留的地方--它会被肾脏更快地排掉。

A practical consequence: after a big sweat session, an isotonic drink or salty food with the fluid is measurably better than plain water, and the difference is not subtle in the heat.

实际推论是：一次大量出汗的训练之后，含电解质的饮料或配着盐一起摄入的液体，明显优于白水，而在高温中这个差别并不细微。

Creatine monohydrate is the fourth item, and it is the one that needs a reason.

肌酸一水合物是第四样，而且是需要理由的那一样。

Creatine and phosphocreatine are stored in muscle, and plain dietary meat simply does not supply enough to saturate them, so a loading phase raises the store above what food alone achieves.

肌酸和磷酸肌酸储存在肌肉里，而普通饮食中的肉类根本无法让它们达到饱和，因此一个负荷期可以把储备提高到单靠食物达不到的

The effect is a modest increase in repeated-sprint and high-intensity capacity, recovered through a further benefit: more muscle mass.

它的效果是重复冲刺和高强度能力的适度提升，并通过一个额外益处得到回报：更多肌肉量。

It is not a general performance aid, it is a specific one, and the honest framing is that it earns its place in a programme that already trains the phosphagen system.

它不是全面的表现增强剂，而是有针对性的；诚实的说法是，它在一份本来就在训练磷酸原系统的计划里才站得住脚。

And then the tidy-up crew.

然后是收拾班组。

Polyphenol-rich foods and antioxidants are widely described as helping recovery, and the honest position is that the effect is smaller and less certain than the marketing suggests, while a diet that is simply varied and adequate will supply them without any special effort.

富含多酚的食物和抗氧化剂被广泛描述为有助于恢复，而诚实的立场是：其作用比营销所说的更小、也更不确定；

Antioxidant supplements taken in large doses have their own concerns.

与此同时，一份单纯多样且充足的饮食就会提供它们，不需要任何特别努力。

The version of the answer that scores is the version that separates what is well evidenced from what is merely marketed.

大剂量服用抗氧化补充剂本身也有顾虑。能得分的答案，是把“有充分证据的”和“只是被宣传的”分开的那种答案。

The numbers / 算一算

Numbers to keep. Carbohydrate for recovery runs around 1.0 to 1.2 grams per kilogram per hour in the first few hours, which for a 70 kg athlete is about 70 to 85 grams an hour - a large, real number, and the reason carbohydrate after a hard session matters more than a protein shake. Protein 1.4 to 2.0 g/kg across the day, split across three or four meals. Fluid: aim to replace most of what you lost, and a useful test is that your urine should be pale rather than dark, though colour is never a complete guide. Creatine monohydrate at around 3 to 5 grams per day, with an optional loading phase of about 20 grams per day split into four for five to seven days. The four-to-one ratio worth remembering is that carbohydrate restoration after a session is worth several times more than any single supplement.

需要记住的数字。恢复期的碳水大约是最初几小时内每小时每公斤1.0到1.2克，对一位70公斤的运动员来说是每小时约70到85克--这是一

Common mistakes / 易错点

- "Drink as much water as you can after training." Excess plain water without sodium dilutes the blood and can cause hyponatremia. Volume and sodium travel together.
"训练后喝越多水越好。"过量的白水而没有钠会稀释血液，并可能导致低钠血症。容量和钠必须一起走。
- "Protein is only needed after training." It is needed every day, and the day after a session is simply the day it is easiest to get right. Spread it across the day.
"蛋白质只在训练后需要。"它每天都需要，而训练后那一天只是最容易做对的那天。要分散到全天。
- "Supplements replace recovery nutrition." Carbohydrate and protein at the right dose do more than any supplement, and the effect size for most supplements is much smaller than the marketing claims.
"补充剂可以代替恢复营养。"剂量正确的碳水和蛋白质作用大于任何补充剂，而大多数补充剂的效果量远小于宣传。
- "Antioxidants prevent soreness and speed recovery." The evidence is much weaker than the marketing. A varied, adequate diet supplies what is needed without supplements.
"抗氧化剂能防止酸痛并加快恢复。"证据比宣传弱得多。一份多样且充足的饮食就能提供所需，不需要补充剂。

Why it matters / 为什么重要

Recovery is where the training is actually absorbed, and getting it wrong is a quiet, expensive way to fail. An athlete who trains hard and recovers badly does not plateau at their best; they accumulate debt, get ill more often, and eventually injure. The order matters too, because the first ten seconds of a sprint, the last kilometre of a marathon and the tissue repair that happens at 2 am are all paid for in specific nutrients at specific times. A coach who asks an athlete what they ate, drank and slept after the session has asked the question that decides the next four weeks.

恢复是训练真正被吸收的地方，而恢复做错是一种安静而昂贵的失败方式。训练刻苦却恢复不好的运动员不会停在他们的最好水平：他们

- The recovery timeline is 0-30 minutes to refuel and rehydrate, 30-120 minutes to repair and restock, 2-4 hours to restore and refuel, and daily nutrition to support adaptation.

恢复时间线：0-30分钟补充能量和水分，30-120分钟修复和补充，2-4小时恢复和补充，之后依靠每日营养支持适应。

Key terms / 核心术语

Recovery indicator 恢复指标

A physiological, symptom, psychological or practical sign used to judge readiness.

用于判断准备状态的生理、症状、心理或实际迹象。

Formulas / 公式

After a long race, prioritise carbohydrate, protein and fluid, then use sleep and practical routines to support the next session.

长距离比赛后优先补充碳水、蛋白质和液体，再通过睡眠和实际安排支持下一次训练。

Worked examples / 实例

>

Evaluate recovery with more than one indicator.

用不止一个指标评估恢复。

Exam focus / 考核要点

Name three key recovery nutrients and their roles.

说出三种关键恢复营养素及其作用。

Recovery indicators, sleep and travel · 恢复指标、睡眠与旅行

First, picture it / 先这样想

Knowing you are recovered is a bit like knowing your car is roadworthy. The engine will start either way. But the oil light, the temperature gauge and how the steering feels tell you whether the thing underneath is coping or is quietly running on fumes. Recovery indicators are those gauges, and the mistake is waiting for the car to break down instead of reading them. Sleep is the one part of the whole recovery picture that no powder, drink or gadget can replace.

判断自己是否恢复，有点像判断一辆车是否适合上路。发动机反正都能启动，但机油灯、水温表和方向盘的手感，会告诉你底下的东西是在应付，还是正在悄悄靠残喘运转。恢复指标就是那些仪表，而错误在于等车抛锚而不是去读它们。睡眠是整个恢复图景中唯一没有任何粉末、饮料或电子设备能替代的部分。

The detail / 详细讲解

Recovery indicators come in three groups, and the strongest programmes use all three.

恢复指标分为三组，而最强的计划三组都用。

Physiological: resting heart rate returning toward your baseline, improved heart-rate variability, less soreness, reduced inflammatory markers, better sleep quality.

生理指标：静息心率回到基线附近、心率变异性改善、酸痛减轻、炎症标志物下降、睡眠质量变好。

Symptomatic: how the session felt, how the muscles feel now, mood, and the athlete own sense of readiness.

主观指标：这节课的感觉、现在肌肉的感觉、情绪，以及运动员自己的准备感。

Performance: the numbers that matter to your event, tested on a schedule.

表现指标：对项目重要的那些数字，按固定时间测试。

The physiological measures are the most objective, the symptomatic ones are the most sensitive, and the performance ones are the ones that actually tell you whether the work is being converted.

生理指标最客观，主观指标最敏感，而表现指标才是真正告诉你训练是否被转化的那一组。

Sleep is the one everybody knows about and almost nobody optimises, and it is arguably the biggest single lever in recovery.

睡眠是大家都知道、却几乎没人真正优化的一项，而且它可以说是恢复中最大的单一杠杆。

It is when growth hormone does much of its repair work, when the body consolidates motor learning, and when the autonomic nervous system resets.

生长激素的大部分修复工作、动作学习的巩固以及自主神经系统的重置，都发生在睡眠时。

The two things that matter most are duration and consistency: roughly 7 to 9 hours for most adults, at similar times.

最重要的两件事是时长和规律性：多数成年人约7到9小时，时间点相对固定。

The measures people obsess over - sleep trackers, temperature, heart-rate variability during sleep - are far less useful than simply going to bed at the same time and getting enough of it.

人们最着迷的那些测量--睡眠追踪器、体温、睡眠中的心率变异性--远不如“按时上床并睡够”有用。

The day before competition is a genuine trap, because the instinct is to rest hard the day before.

赛前一天是一个真正的陷阱，因为本能是前一天拼命休息。

Total inactivity produces a feeling of freshness that is actually detraining: your aerobic system begins to lose, muscles feel different under load, and the first day of competition arrives slightly flatter than it needed to be.

完全静止会产生一种“新鲜”的感觉，而那其实是在退化：有氧系统开始下降，肌肉在负荷下感觉不同，比赛第一天到来时状态比需要的

The better approach is an active taper - reduced volume, retained intensity, familiar and enjoyable - which keeps the adaptations you have built available while reducing accumulated fatigue.

更好的做法是主动减量--降低训练量、保留强度、内容熟悉且愉快--这样既让你已经建立的适应保持可用，又减少累积疲劳。

Travel is a genuine recovery problem that is usually treated as an inconvenience.

旅行是一个真正的恢复问题，却通常只被当作不便。

Crossing time zones moves your circadian rhythm, and the body takes roughly one day per time zone to adjust, so a three-zone flight westward arrives with your body clock roughly three days out.

跨时区会移动你的昼夜节律

，而身体大约每个时区需要一天来适应，所以向西飞行三个时区，抵达时你的生物钟大约还差三天。

Practical countermeasures: shift sleep and meal times by an hour a day in the direction of travel, use light exposure at the destination to re-anchor, keep caffeine to the first half of the local day, and arrive a day early if the event matters.

实用的应对办法：按一小时一天的幅度朝目的地方向调整睡眠和进餐时间，用目的地的时间晒光重新锚定，把咖啡因限制在当地上午。

Hydration and eating on schedule matter more than most travellers expect.

补水和按时进食比多数旅行者预期的更重要。

And the honest summary is that recovery is not a day off, it is a process with a rate.

而诚实的总结是：恢复不是一天的休息，它是一个有速率的过程。

The adaptations from a session are built over hours to days, they are paid for in nutrients and sleep, and they are checked by indicators rather than by how you feel on day two.

一次训练的适应在数小时到数天内被建立，由营养和睡眠支付，再由指标而不是由你第二天的感觉来检验。

Athletes who track this properly recover faster from illness, miss fewer sessions, and - the part that matters at the top level - arrive at competition able to express what they built rather than merely carrying it.

认真追踪这一点的运动员恢复更快、缺席更少--而在最高水平上最重要的是--他们能在比赛时表现出所建立的东西，而不只是带着它。

The numbers / 算一算

Numbers to keep. Sleep 7 to 9 hours for adults, with a consistent time. Taper: reduce volume substantially in the final week while keeping some intensity, and keep the content familiar. Travel: about one day of adjustment per time zone crossed, which is why arriving a day early is worth it for an important event. Monitoring: resting heart rate meaningful as a deviation of roughly 5 to 10 beats per minute from your own baseline sustained over several days, and the same caution as ever - a trend across days means something, a single reading means almost nothing. The number that predicts most about competition readiness is not on any device: it is whether the last three nights of sleep were normal.

需要记住的数字。成年人睡眠7到9小时，时间点保持规律。减量：最后一周显著降低训练量，同时保留一些强度，内容保持熟悉。旅行：

Common mistakes / 易错点

- "I can tell I am recovered because I feel fine." Feeling fine is a subjective report and it lags. The indicators that change first are physiological, and the ones that matter most are performance-based.
"我感觉不错就说明我恢复了。"感觉不错是主观报告，而且会滞后。最先变化的指标是生理指标，而最重要的是表现类指标。
- "Total rest the day before is the best preparation." Inactivity begins to detraining you. An active taper keeps the adaptations available while reducing fatigue.
"赛前一天完全休息是最好的准备。"静止会开始让你退化。主动减量能在降低疲劳的同时让适应保持可用。
- "Sleep tracking tells me if I slept well." A tracker is far less useful than a consistent bedtime and enough hours. The device measures; it does not fix.
"睡眠追踪能告诉我睡得好不好。"追踪器远不如固定的就寝时间和足够的时长有用。设备只负责测量，不负责改善。

-
"Jet lag is cured by arrival." The body adjusts at roughly one day per time zone. Shift sleep and meals before you fly, use light at the destination, and arrive early if it matters.

"抵达就克服了时差。"身体大约每个时区需要一天适应。飞行前就调整睡眠和进餐，用目的地的时间晒光，重要的话提前抵达。

Why it matters / 为什么重要

Recovery is the part of a programme that nobody measures, so it is the part that quietly goes wrong, and the cost falls on the athlete as illness, injury and stalled progress. Building the habit of tracking a small set of indicators over weeks is what turns recovery from a mood into a process, and it is also the cheapest intervention in the whole course - it requires no equipment, no expense and no new theory, only the willingness to write down what would otherwise be guessed. The athletes who stay healthy for years are usually not the ones with the best recovery science; they are the ones who checked.

恢复是计划中没人测量的部分，所以它也是悄悄出问题的部分，而代价由运动员承担：疾病、伤病和进步停滞。用几周时间追踪一小组指

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Formulas / 公式

How do sleep and travel disruption affect recovery?

睡眠和旅行干扰如何影响恢复？

Worked examples / 实例

>

Name three physiological indicators of recovery.

说出三个恢复的生理指标。

Anatomical position, planes and movement

解剖位置、平面与运动

Guiding question / 引导问题

How can precise anatomical language describe human movement?

精确的解剖语言如何描述人体运动？

Anatomical position is the standard reference position. Planes divide the body, axes run perpendicular to planes, and movement terms describe actions at joints. Precise language supports analysis, coaching, equipment design and injury prevention.

解剖位置是标准参考位置。平面分割身体，轴与平面垂直，运动术语描述关节处的动作。精确语言支持动作分析、教练、装备设计和预防。

Skeleton, position and directional terms · 骨骼、方位与方向术语

First, picture it / 先这样想

The skeleton is two buildings with two different jobs. The axial skeleton is the tower in the middle: a strong central column with the vital services running through it, built mainly to be hit by things without failing. The appendicular skeleton is everything that sticks out - the arms and legs - built to move. That split is the whole logic of it: what needs protecting goes in the middle, and what needs to move goes on the outside.

骨骼是两栋功能不同的建筑。中轴骨骼是中间那座塔：一根坚固的中央柱，重要“管线”从中穿过，主要为了被撞倒而不失效而建造。外周骨骼是伸出去的一切--手臂和腿--为了运动而建造。这个划分就是全部逻辑：需要保护的放在中间，需要运动的放在外面。

The detail / 详细讲解

The axial skeleton is the skull, the vertebral column, the rib cage and the sternum.

中轴骨骼包括颅骨、脊柱、胸廓和胸骨。

Its job is protection, and it does it by arrangement rather than by thickness: the vertebrae form a stacked column that carries load and absorbs shock progressively, the ribs form a cage around the heart and lungs, and the skull wraps the brain.

它的任务是保护，而它靠的是排列而不是厚度来实现：椎骨堆成一根柱子，承受负荷并逐级吸收震荡；肋骨围成笼子保护心脏和肺；

It is also the framework that everything else hangs off, so an axial injury is serious in a way that a limb injury often is not.

颅骨包裹大脑。它同时也是其他一切悬挂其上的框架，所以中轴部位的损伤远比四肢损伤严重。

The appendicular skeleton is the shoulder and pelvic girdles plus the upper and lower limbs.

外周骨骼包括肩带、骨盆带以及上下肢。

Its job is movement, and it is built for leverage rather than for armour: long bones, joints placed between segments, and muscles attaching across joints so that shortening a muscle moves the bone beyond it.

它的任务是运动，而且它是为杠杆效率而不是为盔甲而建造的：长骨、置于骨段之间的关节，以及跨关节附着的肌肉，使肌肉缩短能带动骨骼运动。

The pelvic girdle is worth singling out because it is the junction - it transfers the weight of the trunk onto the legs, and in doing so it is both the strongest region of the hip and, in childbirth, a functional compromise.

骨盆带值得单独提出，因为它是衔接处：它把躯干的重量转移到双腿上，在这样做的同时既是髋部最坚固的区域，也（在分娩时）是薄弱环节。

Directional terms exist because there is only one correct way to describe where something is, and that way must be relative to the person rather than to the observer.

方向术语之所以存在，是因为描述“在哪里”只有一种正确方式，而这种方式必须相对于人本身而不是观察者。

Standing and lying describe position.

站立位和卧位描述姿势。

Terms such as superior and inferior, anterior and posterior, medial and lateral, proximal and distal, and superficial and deep describe location, and each of them means the same thing on every person because they are all defined from the anatomical position - standing, facing forward, arms at the sides, palms forward.

诸如上与下、前与后、内侧与外侧、近端与远端、浅与深等术语描述位置，而它们之所以对每个人含义相同，是因为全部以解剖学姿势为基准。

The pair that confuses people most is anterior and posterior, because the body has no "front" in the abstract.

最让人困惑的一对是前与后，因为身体并没有抽象意义上的“前面”。

Anterior means toward the front of the body, posterior toward the back, medial toward the midline and lateral away from it, and the mistake people make is using them as if they meant left and right.

前是指朝身体前面，后是指朝背面，内侧是指向中线、外侧是指离中线；

Proximal and distal are the ones specific to limbs: proximal is nearer the trunk, distal is further from it, so the knee is proximal to the ankle and distal to the hip.

人们犯的错是当它们表示“左”和“右”来用。

近端与远端是四肢专有的：近端更靠近躯干，远端更远离躯干，所以膝盖在踝的上方端、在髋的下方端。

A precise vocabulary is not pedantry here.

在这里，精确的词汇不是吹毛求疵。

When a clinician says "pain at the medial aspect of the knee", every person who hears it pictures the same structure.

当临床医生说“膝关节内侧疼痛”时，每个听到的人脑海中都是同一个结构。

When a description says "it hurts somewhere in my knee at the side where the bone sticks out", it might be the lateral collateral ligament, the iliotibial band, the fibula, or the joint line, and nobody can act on it.

而当描述是“我的膝盖侧边、骨头突出的地方有点疼”时，那可能是外侧副韧带、髂胫束、腓骨或关节线，谁都无法据此行动。

Language is the compression that makes the rest of the diagnosis possible.

语言是一种压缩，正是它让诊断的其余部分成为可能。

The numbers / 算一算

Numbers worth knowing. An adult skeleton is about 206 bones, and that number varies with age because some bones fuse during development - which is itself a useful idea, because a newborn has around 270 separate bones and the difference is mostly fusion rather than loss. Stature is one of the most reproducible anthropometric measures there is, and arm span is very close to equal to it, within a few centimetres in most adults, which is why height is so often estimated from span. Vertebral count is the standard reference for describing spinal level: 7 cervical, 12 thoracic, 5 lumbar, 5 sacral and 4 coccygeal, and it is the numbering that lets everyone discuss the same vertebra.

需要知道的数字。成年人骨骼约有206块骨头，而这个数字随年龄变化，因为有些骨在发育过程中会融合--这本身是个有用的概念：新生

Common mistakes / 易错点

- "Anterior and posterior mean left and right." They mean toward the front and toward the back of the body. Left and right are lateral and medial in the sense of which side, and you should say the side directly.
"前和后表示左和右。"它们表示朝身体前面和朝背面。左右指的是哪一侧，而你应该直接说哪一侧。
- "A newborn has more bones so they can grow easier." The count is higher because bones are separate; growth happens at the growth plates and by remodelling, not by keeping extra bones.
"新生儿骨头多所以更容易长。"数量多是因为骨头是分开的；生长发生在生长板并通过重塑进行，而不是靠保留多余的骨头。
- "Directional terms depend on how you are standing." They are defined from the anatomical position, which is why they mean the same thing whichever way the person is facing or lying.
"方向术语取决于你怎么站。"它们是从解剖学姿势定义的，所以无论人面朝哪个方向或躺着，含义都相同。
- "Precise anatomical language is being difficult for its own sake." It is the compression that makes a diagnosis actionable. Vague description means nobody can act on it.
"精确的解剖学语言是在故意为难人。"它正是让诊断可执行的压缩手段。含糊的描述意味着谁都无法据此行动。

Why it matters / 为什么重要

This is the foundation topic, and the reason to care about it is precision rather than memory. Almost every injury question in the paper names a location, and the location is what determines the mechanism, the likely tissue and the management. An athlete who can name the medial side of the knee, the proximal part of the hamstring and the anterior aspect of the ankle has given a clinician most of what they need before any test is done, and has also

given themselves a much better chance of describing a problem that is actually getting worse.

这是基础主题，重视它的理由是精确而不是记忆。卷面上几乎每一道损伤题都会指出一个位置，而位置决定了机制、可能受累的组织以及

Anatomical position: standing upright and facing forward, head level, eyes forward, arms at the sides with palms forward, feet flat and toes forward.

解剖位置：直立并面向前方，头部水平、眼睛向前、手臂位于身体两侧且掌心向前、双脚平放、脚尖向前。

Superior/inferior describe above/below; medial/lateral describe toward/away from the midline; proximal/distal describe nearer/farther from the point of origin; anterior/posterior describe front/back.

上/下表示更靠头部/足部；内/外表示靠近/远离中线；近端/远端表示距离起源较近/较远；前/后表示身体前面/后面。

Key terms / 核心术语

Anatomical position 解剖位置

The standard reference standing position used to describe the body.

描述身体的标准站立参考位置。

Formulas / 公式

"The sole turns inward" is inversion; "the sole turns outward" is eversion. Explain each using directional terms.

"脚底向内"是内翻；"脚底向外"是外翻。用方向术语解释。

Worked examples / 实例

>

Name the plane, axis and movement terms for a squat or biceps curl.

为深蹲或弯举说出平面、轴和运动术语。

Exam focus / 考核要点

Define proximal and distal.

定义近端和远端。

Planes, axes and movement · 平面、轴与运动

First, picture it / 先这样想

Planes are imaginary sheets of glass and axes are imaginary nails. Picture your body suspended inside three sheets of glass that all pass through the middle of you, with a nail through each one. An axis is the nail: rotation happens around it. A plane is the sheet: it divides the body into front and back, left and right, or top and bottom. Every movement term you need can be worked out by asking two questions - which sheet does the movement travel in, and which nail does it turn around?

平面是想象中的玻璃片，轴是想象中的钉子。想象你的身体悬在三片都穿过你身体中线的玻璃中间，每片上都钉着一根钉子。轴就是那根钉子：旋转围绕它发生。平面就是那片玻璃：它把身体分成前后、左右或上下。你需要的每一个动作术语，都可以通过问两个问题推导出来--这个动作发生在哪个平面里？它绕哪根轴旋转？

The detail / 详细讲解

There are three planes and three axes, and they are perpendicular to each other.

有三个平面和三个轴，它们彼此垂直。

The sagittal plane divides the body into left and right, and it has a mediolateral axis running side to side.

矢状面把身体分成左右，其轴是横行的外侧到外侧（mediolateral）轴。

The frontal or coronal plane divides front from back, and it has an anteroposterior axis running front to back.

额状面（冠状面）把身体分成前后，其轴是前后向（anteroposterior）轴。

The transverse or horizontal plane divides top from bottom, and it has a vertical axis.

横断面把身体分成上下，其轴是垂直轴。

Every named movement in sport is movement in one of these three planes around one of these three axes.

运动中每一个被命名的动作，都是在这三个平面之一中、绕这三个轴之一发生的运动。

Flexion and extension happen in the sagittal plane around the mediolateral axis.

屈曲和伸展发生在矢状面内、绕外侧到外侧轴。

Flexion decreases the angle between two bones, which is what a knee bend, an elbow bend, a hip bring-up and a biceps curl all have in common.

屈曲减小两块骨之间的角度，膝屈、肘屈、髋部上抬和肱二头肌弯举的共同点正在于此。

Extension increases that angle and returns the joint to or past its neutral position.

伸展增大该角度，使关节回到或超过中立位。

The same pair also names the shoulder movements - flexion raises the arm forward, extension brings it back - and the trunk movements, where flexion is bending forward and extension is bending backward.

同一对术语也用于肩部动作--前屈把手臂向前抬，伸展把它带回--以及躯干动作，其中前屈是向前弯、伸展是向后弯。

Abduction and adduction also live in the frontal plane, around the anteroposterior axis.

外展和内收也发生在额状面内、绕前后向轴。

Abduction moves a limb away from the midline, adduction brings it back toward it.

外展把肢体移离身体中线，内收把它收回中线。

At the shoulder these are the arms out to the side and back down.

在肩部，这是手臂向体侧抬起和放下；

At the hip they are the leg out to the side and back in.

在髋部，这是腿向体侧张开和收回。

Lateral flexion of the trunk is the same idea applied to the spine, and it is one of the movements most likely to cause a problem when a one-sided sport develops a one-sided habit.

躯干的侧屈是同一概念应用于脊柱，而它是单项运动发展出单侧习惯时最容易导致问题的动作之一。

Rotation is the third family, and it is the one people forget.

旋转是第三类，也是人们最容易忘掉的一类。

Medial and lateral rotation happen in the transverse plane around the vertical axis, turning a limb inward or outward.

内旋和外旋发生在横断面内、绕垂直轴，使肢体内转或外转。

Pronation and supination are the forearm equivalents: turning the palm down or up.

前臂的旋前和旋后是它的对应动作：把掌心朝下或朝上。

Internal and external rotation of the shoulder, rotation of the trunk in a golf swing or a tennis serve, and the pronation and supination of the forearm when you take a racket or turn a doorknob all belong here.

肩关节的内旋与外旋、高尔夫挥杆或网球发球中的躯干旋转、以及拿球拍或拧门把手时的前臂旋前旋后，都属于这一类。

Anything that twists a segment about its own long axis is a rotation.

任何使某一节段绕自身长轴扭转的动作都是旋转。

Because planes and axes are perpendicular, an action can be pure or combined.

由于平面和轴彼此垂直，一个动作可以是单一平面的，也可以是组合的。

A knee bend during a squat is pure sagittal flexion.

深蹲时的屈膝是纯粹的矢状面屈曲。

A single-leg squat becomes a combination: flexion in the sagittal plane, and the trunk and pelvis have to rotate and tilt in the other two planes to keep the centre of mass over the foot.

单腿深蹲就变成组合：矢状面的屈曲

，加上躯干和骨盆必须在另外两个平面里旋转和倾斜，才能让重心保持在脚的上方。

Real sport is almost entirely made of combinations, and describing a movement properly means naming each plane and axis involved, not just the most obvious one.

真实运动几乎完全由组合构成，而准确描述一个动作意味着指出所涉及的每一个平面和轴，而不只是最明显的那一个。

The numbers / 算一算

Numbers to hold. Three planes, three axes, each axis perpendicular to its plane. A joint range of motion is measured in degrees, and the sagittal plane usually has the largest range at each joint - knee flexion about 0 to 135 degrees in a healthy adult, hip flexion similar, shoulder flexion around 180 degrees, and elbow flexion around 0 to 145. The transverse plane is generally the most restricted, which is why rotation injuries are comparatively uncommon and why a joint that has lost rotation often feels stiff in a way that a joint which has lost flexion does

not. The practical exam habit: for any movement, give the plane, the axis and the direction, and you cannot be marked wrong on two of the three.

需要记住的数字。三个平面、三个轴，每个轴都与其所在平面垂直。关节活动度以度为单位，而矢状面通常在每个关节拥有最大的活动范

Common mistakes / 易错点

- "Flexion and extension happen in the frontal plane." They are sagittal, around a mediolateral axis. The frontal plane carries abduction and adduction.
"屈曲和伸展发生在额状面。"它们在矢状面、绕外侧到外侧轴。额状面承载的是外展和内收。
- "Rotation is its own plane." Rotation is movement in the transverse plane around the vertical axis. It is easy to forget precisely because it has no visible bend to it.
"旋转有自己的平面。"旋转是横断面内绕垂直轴的运动。它容易被忘记，恰恰因为它没有可见的弯曲。
- "Abduction is moving a limb forward." That is flexion. Abduction is moving it away from the midline, which at the shoulder is out to the side.
"外展是把肢体向前移动。"那是屈曲。外展是移离中线，在肩部就是向体侧抬起。
- "Every movement is pure in one plane." Real movement is a combination. Naming only the most obvious plane is the most common way to lose a mark on a movement question.
"每个动作都纯粹属于一个平面。"真实动作是组合的。只指出最明显的那个平面，是动作类问题丢分最常见的方式。

Why it matters / 为什么重要

Planes and axes are the language that lets an athlete and a coach talk about a movement precisely enough to fix it. A gymnast whose ankle collapses in the frontal plane needs a different intervention from a sprinter losing extension in the sagittal plane, and the two problems can look superficially similar on video. It is also the vocabulary that makes a written answer safe: a question that asks you to "identify the plane and axis" is not testing whether you memorised a list, it is testing whether you can work something out, and naming all three parts is what separates a correct answer from a guess.

平面和轴是让运动员与教练能够足够精确地谈论一个动作、并把它修正的语言。踝关节在额状面"塌陷"的体操运动员，需要的处理与在矢

- Joint movement depends on the joint surface, axis and muscles crossing the joint.
关节运动取决于关节面、轴以及跨关节的肌肉。

Key terms / 核心术语

Flexion 屈曲

Decrease the angle at a joint.

关节角度减小。

Extension 伸展

Increase the angle at a joint.

关节角度增大。

Abduction 外展

Move away from the midline.

远离中线移动。

Adduction 内收

Move towards the midline.

向中线移动。

Formulas / 公式

During elbow flexion in a biceps curl, the movement occurs in the sagittal plane around a mediolateral axis.

弯举中肘屈曲发生在矢状面，并绕内外侧轴运动。

Worked examples / 实例

>

Compare the three planes and their perpendicular axes.

比较三个平面及其垂直轴。

Exam focus / 考核要点

Give a movement example for each plane.

为每个平面给出一个运动例子。

HL anthropometry and ergonomics · HL人体测量学与人体工效学

First, picture it / 先这样想

Equipment is designed for a body, and if the designer guessed wrong about the body, the equipment fights the person using it. Anthropometry is the discipline of not guessing: measuring the dimensions of real people and then designing to those numbers instead of to an average that nobody actually is. A bicycle frame built for the average of a population fits nobody in particular, and the person who suffers most is usually the one furthest from that average - which is often the person who most needs the equipment to work.

装备是为身体设计的，而如果设计者对身体猜错了，装备就会与使用它的人对抗。人体测量学就是一门“不猜”的学科：测量真实人体的尺寸，然后按这些数字而不是按某个“平均值”来设计--而那个平均值谁也不是。按人群平均值造的车架对任何具体的人都不合适，而受苦最多的通常正是偏离平均值最远的那个人--往往也是最需要装备发挥作用的那个人。

The detail / 详细讲解

The measures that matter are stature, sitting height, standing eye height, arm span, limb lengths, shoulder width, hip width, hand span and foot length, and each one exists because some piece of equipment depends on it.

重要的测量项目包括身高、坐高、站立眼高、臂展、四肢长度、肩宽、髋宽、手跨和足长，而每一项之所以存在，都是因为某件装备

Stature and sitting height set frame and saddle size.

身高和坐高决定车架与坐垫尺寸；

Arm span and limb lengths set handlebar and crank reach.

臂展和肢长决定把手与曲柄的触及范围；

Hand span sets grip size.

手跨决定握把尺寸；

Foot length sets shoe and cleat sizing.

足长决定鞋和鞋钉的尺码。

None of this is academic once you have tried to use a bicycle that is the wrong size, which is why equipment fitting is assessed at HL rather than treated as a detail.

一旦你用过尺寸不对的自行车，这些就不再是学术问题了--这正是装备适配被放在HL而不是当作细节的原因。

Ergonomics is the part that asks whether the equipment helps the task.

人体工程学是追问装备是否有助于完成任务的环节。

A good design lets the athlete adopt a posture that suits the movement; a poor one forces a compensating posture that costs something elsewhere.

好的设计让运动员采用适合该动作的姿势；差的设计则迫使他采用代偿性姿势，而代偿要在别处付出代价。

The clearest examples are the bench press, where the bar should sit where the shoulder can handle it, and the running shoe, where cushioning and support should match the surface and the athlete rather than the marketing.

最明显的例子是卧推--杠铃应放在肩部能承受的位置；

A shoe designed for a laboratory walk is not a shoe designed for intervals, even if it is comfortable for both.

以及跑鞋--缓震与支撑应匹配场地和运动员，而不是匹配宣传语。

为实验室步行设计的鞋并不是为间歇训练设计的鞋，即使两者穿着都舒服。

Individual variability is the part that makes anthropometry a discipline rather than a list.

个体差异是让人体测量学成为一门学科而不是一张清单的原因。

Limb length relative to torso length varies enormously between people, and that ratio changes how a movement feels and how much load a joint has to take.

四肢长度与躯干长度的比例在人与人之间差异极大，而这个比例会改变一个动作的感觉以及关节必须承担的负荷。

A sprinter with long femurs and a short torso has to achieve a different trunk position to get the same hip height, and that difference is biomechanical, not a matter of trying harder.

一个股骨长、躯干短的短跑运动员，必须采用不同的躯干位置才能获得同样的髋高，而这个差异是生物力学上的，而不是“更努力一点”。

This is also why a single "ideal" body proportion for a sport does not exist, and why talent identification that assumes one produces false negatives.

这也是为什么某项运动不存在唯一“理想”身体比例，以及为什么假设存在这种比例的选材工作会产生漏判。

Static and dynamic anthropometry are worth distinguishing.

静态与动态人体测量值得区分。

Static measures are taken in a standard position and describe body size.

静态测量是在标准姿势下进行的，描述身体尺寸。

Dynamic measures describe the body in motion - how far a joint travels, how high a shoulder sits at the top of a throw, how far a hand reaches in a specific direction.

动态测量描述运动中的身体--关节走多远、投掷最高点时肩有多高、手在某个方向能伸多远。

For equipment, static measures determine fit; for performance analysis, dynamic measures determine technique.

对装备而言，静态测量决定适配；

And the two interact, because equipment changes dynamic measures: a stiffer shaft changes the swing weight, a higher saddle changes hip angle at the top of the pedal stroke.

对表现分析而言，动态测量决定技术。

而两者会互相影响，因为装备会改变动态测量：更硬的杆身改变挥重，更高的坐垫改变踩踏顶点的髋角。

The numbers / 算一算

Numbers worth having. Human arm span is close to stature, usually within a few centimetres, which is the basis of span-based height estimation. Sitting height is roughly half of stature, and the ratio is why frame size follows height so reliably in cycling. Thigh length is about half of leg length, and the thigh-to-leg ratio varies more between individuals than leg length does - which is the statistical reason proportion matters more than height for technique. For shoes, the useful figure is simply foot length measured standing, with about 1 to 2 cm of clearance in front of the longest toe. For rowing or cycling, the governing measurement is not the torso but the inside of the leg, because that is what sets the seat and the crank or stretcher position.

需要具备的数字。人的臂展接近身高，通常只差几厘米，这也是用臂展估算身高的基础。坐高大约是身高的一半，而这个比例正是自行车。

Common mistakes / 易错点

- "Design to the average and most people are served." The average is nobody. Those furthest from it, who are often the ones who most need the equipment, are served worst.
"按平均值设计就能满足大多数人。"平均值谁也不是。偏离它最远的人--往往正是最需要装备的人--反而被服务得最差。
- "Talent identification has one ideal body type per sport." Proportion varies far more than height, and assuming a single ideal produces false negatives in people who would succeed.
"每个运动的选材都有一个理想体型。"比例的差异远大于身高，而假设存在唯一理想体型会让本可成功的人被漏掉。
- "A comfortable shoe works for any activity." Comfort is not the same as suitability, and a shoe built for one loading pattern can be wrong for another even when it feels fine.
"穿着舒服的鞋做任何运动都行。"舒服不等于合适，为一种受力模式设计的鞋即使感觉不错，用在另一种活动上也可能是错的。
- "Only static measurements matter for equipment." Static measures set fit, but equipment changes the dynamic measures too - shaft stiffness changes swing weight, saddle height changes hip angle.
"对装备来说只有静态测量重要。"静态测量决定适配，但装备也会改变动态测量--杆身硬度改变挥重，坐垫高度改变髋角。

Why it matters / 为什么重要

Anthropometry and ergonomics are where biomechanics becomes something an athlete can feel rather than something they have to accept. A sprinter who learns that their long femurs require a different trunk angle at the start understands why the textbook version does not work for them, and that understanding is usually the first step

to a technical fix. It is also the most immediately practical HL topic in the course: the equipment an athlete uses every day was specified using somebody else measurements, and knowing which measurements lets a coach, a teacher or an athlete argue for a change that costs nothing and prevents injury.

人体测量与人体工程学，是生物力学从"运动员必须接受的东西"变成"运动员能感觉到的东西"的地方。一位短跑运动员明白自己的长股骨

-

Cycling seat, handlebar and pedal positions can affect power and strain.

自行车的座椅、车把和踏板位置会影响功率与应力。

-

Appropriate footwear, helmet, racket or rowing-seat fit can improve control and reduce risk.

合适的鞋、头盔、球拍或划船座椅定位可改善控制并降低风险。

Key terms / 核心术语

Anthropometry 人体测量

Measurement of body dimensions used for design and equipment sizing.

用于设计和装备尺寸选择的身体尺寸测量。

Ergonomics 人体工程学

Design of tasks, equipment and environments for efficiency and reduced risk.

为提高效率、降低风险而设计任务、装备和环境。

Formulas / 公式

A taller athlete may need a longer frame or larger equipment, but the design must still allow control and safe movement.

较高的运动员可能需要更长的车架或更大的装备，但设计仍必须允许控制和安全运动。

Worked examples / 实例

>

How can anthropometry influence equipment design?

人体测量如何影响装备设计？

Exam focus / 考核要点

Give one ergonomic adjustment for a sport task.

为一个运动任务给出人体工程学调整。

Connective tissues and joints

结缔组织与关节

Guiding question / 引导问题

How do connective tissues and joint structures permit movement while maintaining stability?

结缔组织和关节结构如何在保持稳定的同时允许运动？

Connective tissues form the framework that supports, links and transmits forces. Joints are classified by the tissue connecting the bones, and synovial joints combine cartilage, synovial fluid, capsule, ligaments and tendons to balance mobility with stability.

结缔组织形成支撑、连接和传递力量的框架。关节按连接骨的组织分类，滑膜关节结合软骨、滑液、关节囊、韧带和肌腱，在活动度

Connective tissues and articulations · 结缔组织与关节

First, picture it / 先这样想

Connective tissue is the difference between a pile of parts and a body. Bone is the frame, ligament is the rope holding bone to bone, tendon is the rope holding muscle to bone, cartilage is the padding, and fascia is the wrapping that keeps the compartments separate. Every one of them has a job that is about connection rather than movement, which is why injuries to these tissues are almost always about being pulled, twisted or crushed rather than about a failure to move.

结缔组织是“一堆零件”和“一个身体”之间的差别。骨是框架，韧带是骨与骨之间的绳，肌腱是肌与骨之间的绳，软骨是缓冲垫，筋膜是让各腔室彼此分开的包裹。它们每一个的职责都与连接而不是运动有关，所以这些组织的损伤几乎总是因为被拉、被扭或被压，而不是因为“动不了”。

The detail / 详细讲解

Bone supports and protects, acts as a lever, and stores minerals.

骨提供支撑与保护、充当杠杆、并储存矿物质。

It is a living tissue that is continually being rebuilt, and the rebuilding is why it responds to load.

它是一种持续被重建的活组织，而这种重建正是它能对负荷作出反应的原因。

It also forms the levers that muscles pull on, which is why a fracture changes not just the structure but the mechanics of the whole region.

它同时构成肌肉施力的杠杆，所以骨折改变的不只是结构，而是整个区域的力学。

Its mineral role is the one most people forget: bone stores calcium and phosphate, and the fact that it is a reservoir rather than a fixed structure is what makes blood calcium regulation work.

它储存矿物质这一点最常被遗忘：骨储存钙和磷酸盐，而它是一个储备而不是固定结构，这正是血钙调节得以运作的原因。

Ligaments connect bone to bone, stabilise joints and limit excessive movement.

韧带连接骨与骨、稳定关节并限制过度活动。

They are strong, relatively inextensible, and they do not generate force - they restrict it.

它强、相对不可伸长，而且不产生力--它限制力。

That last point matters: a ligament cannot be strengthened usefully by the exercises you would use to strengthen a muscle, which is why ligament training is about control and proprioception rather than about load.

最后一点很重要：韧带无法用你增强肌肉的那些练习有效地增强，所以韧带训练的重点是控制与本体感觉而不是负荷。

When a ligament is damaged the joint it supports becomes less stable, and the body responds by increasing muscle co-contraction around it to substitute for the lost restraint.

韧带受损时，它所支撑的关节稳定性下降，身体的回应是增加周围肌肉的共同收缩，以替代失去的限制。

Cartilage reduces friction and absorbs shock.

软骨减少摩擦并吸收冲击。

It has no blood supply of its own, which is why it repairs slowly and why nutrition and loading history matter to it long before they matter anywhere else.

它本身没有血液供应，因此修复很慢，也因此营养和负荷历史对它的影响远早于对别处的影响。

Articular cartilage covers the ends of the bones inside a joint, and its job is to let two surfaces slide across each other for decades with very little wear.

关节软骨覆盖关节内骨的末端，它的工作是让两个表面在数十年里以极小的磨损相互滑动。

Tendons connect muscle to bone and transmit the force the muscle generates; they are less extensible than **ligaments but very strong in tension, and they are the tissue most often overloaded by sport.**

肌腱连接肌与骨，传递肌肉产生的力；它们比韧带

更不可伸长，但在受拉方向上非常强，也是运动中最常被过度负荷的组织。

Fascia wraps and separates structures, and it is the tissue most often described as mysterious.

筋膜包裹并分隔结构，也是最常被形容为神秘的组织。

Its practical function is to keep muscles and organs in their compartments, to transmit some force between adjacent structures, and to provide a highway for nerves and blood vessels.

它的实际功能是让肌肉和器官留在各自的腔室中、在相邻结构之间传递一部分力、并为神经和血管提供通道。

The practical consequence for injury is that fascia can transmit a force from one region to a distant one, so a problem in the foot can present as a symptom in the back, and treatment has to address the source rather than the site of the pain.

对损伤的实际推论是：筋膜能把力从一个区域传到远处，所以脚部的问题可能表现为背部的症状，而治疗必须针对源头而不是疼痛出现的地方。

Fascial restriction and mobility are often described in ways that overshoot the evidence, and it is worth being clear about what is established.

关于筋膜受限与活动度的说法常常超出证据，需要明确哪些是已确立的。

That fascia exists, is continuous and transmits some force is established.

"筋膜存在、是连续的、并传递一部分力"是已确立的。

That a specific myofascial restriction can be palpated and released to treat an arbitrary complaint is a much stronger claim, and the honest position is that the evidence for it is limited and mixed.

"特定的"肌筋膜受限"可以被触诊并被松开以治疗任意主诉"则是强得多的主张，而诚实的立场是它的证据有限且不一致。

For a middle-school reader the useful version is simpler: your body is connected by layers that carry load, and training one region can change another.

对一个初中读者来说，有用的版本更简单：你的身体由承载负荷的层连接起来，训练一个区域可以改变另一个区域。

The numbers / 算一算

Numbers to keep. A ligament is strong enough to resist forces of several times body weight, and is therefore not the tissue that fails first in most sprains - the first thing to fail is usually the muscle-tendon unit or the joint capsule. Tendons carry forces in running that can exceed several times body weight per stride, which is why the Achilles tendon is vulnerable in sudden acceleration. Bone adapts to load at roughly the rate of remodelling cycles, which is why an athlete bone is measurably denser than a non-athlete bone and why bone density responds over months. And the practical number for cartilage: it has effectively no blood supply, so its ability to repair is limited, and the reason joints that have been unloaded for a long time can be worse on return is that the cartilage atrophied while the rest of the body rested.

需要记住的数字。韧带强到能抵抗数倍体重的力，因此它并不是多数扭伤中最先失效的组织--最先失效的通常是肌腱单位或关节囊。跑步

Common mistakes / 易错点

- "Ligaments are what you strengthen with resistance training." Ligaments do not generate force, so they are not strengthened that way. Ligament training is about control, proprioception and graded exposure.

"韧带是用抗阻训练增强的。"韧带不产生力，所以不是这样增强的。韧带训练的重点是控制、本体感觉和渐进暴露。

- "Cartilage has no blood supply so it cannot heal." It is avascular, so its repair is slow and limited - but it is not entirely incapable, and loading within tolerance supports its maintenance.

"软骨没有血液供应所以不能愈合。"它确实无血管，修复缓慢且有限--但并非完全不能，适度负荷有助于维持它。

- "Fascial release can treat most pain." The evidence for palpating and releasing a specific restriction to treat an arbitrary complaint is limited and mixed, and the confident claims outrun the data.

"筋膜松解可以治疗大多数疼痛。"通过触诊并松开特定受限来治疗任意主诉的证据有限且混杂，而自信的说法超出了数据。

"Tendons are like ligaments but attached to muscle." Correct, and the difference is functionally important: tendons transmit generated force, ligaments restrict movement and neither generates force.

"肌腱就是连在肌肉上的韧带。"对，而且这个差别在功能上很重要：肌腱传递产生的力，韧带限制活动，两者都不产生力。

Why it matters / 为什么重要

Knowing which tissue is which turns an injury description from a label into a mechanism, and the mechanism is what determines both what to do and how long it will take. A hamstring strain in the muscle-tendon unit recovers on a different timeline from a ligament sprain, because muscle repairs fast and ligament does not. An overuse problem in a tendon responds to load management rather than to rest. And the subject matters in a wider sense, because the tissue that adapts least and recovers slowest is the one that decides whether a career is long or short. 知道哪种组织是哪一种，会把损伤描述从一个标签变成一个机制，而机制决定该做什么以及要多久。肌腱单位的腘绳肌拉伤，康复时间表

Fibrous joints are very stable and usually immovable, such as skull sutures. Cartilaginous joints allow slight movement and absorb shock, such as intervertebral discs. Synovial joints allow wide movement with low friction, such as knee and shoulder.

纤维关节非常稳定且通常不活动，如颅缝；软骨关节允许少量运动并吸收冲击，如椎间盘；滑膜关节以低摩擦提供广泛运动，如膝和

Key terms / 核心术语

Ligament 韧带

Bone-to-bone tissue that stabilises a joint.

连接骨与骨、稳定关节的组织。

Synovial joint 滑膜关节

A freely movable joint with cartilage, synovial fluid, capsule and ligaments.

具有软骨、滑液、关节囊和韧带的自由活动关节。

Formulas / 公式

A wrist uses smaller gliding movements while a shoulder uses multiaxial movement. Use joint class to explain the difference.

手腕进行小范围滑动，肩关节进行多轴运动。用关节类别解释差异。

Worked examples / 实例

>

Compare fibrous, cartilaginous and synovial joints.

比较纤维、软骨和滑膜关节。

Exam focus / 考核要点

Name the function of a ligament and a tendon.

说出韧带和肌腱的功能。

Synovial structure and joint classes · 滑膜结构与关节分类

First, picture it / 先这样想

A synovial joint is a hinge with a wet, self-oiling, self-repairing surface on each side, and everything else exists to hold those two surfaces together while letting them slide. Think of a door that has to swing for years without squeaking, sagging or wearing through: the hinges are the capsule and the ligaments, the oil is the synovial fluid, and the two smooth faces are cartilage. That is the whole design, and every part of it can fail in a characteristic way.

滑膜关节就像一副铰链：两个接触面上有能自我润滑、自我修复的湿润表面，而其他一切存在的意义就是在让它们滑动的同时把它们连在一起。想象一扇要开合多年、不尖叫、不下垂、不磨穿的门：铰链是关节囊和韧带，油是滑液，两个光滑的面是软骨。这就是整个设计，而其中每一部分都以各自特有的方式失效。

The detail / 详细讲解

Articular cartilage reduces friction and absorbs shock, and it is the part that wears out.

关节软骨减少摩擦并吸收冲击，也是会磨损掉的那一部分。

Synovial membrane produces the fluid; synovial fluid lubricates and nourishes the cartilage and removes waste, and it is also the main route by which nutrients reach a tissue that has no blood supply of its own.

滑膜产生滑液；

The capsule encloses the joint, ligaments strengthen it, and tendons outside the capsule pull the bones together and supply most of the joint stability.

滑液起润滑和滋养软骨并清除废物的作用，它同时也是营养到达一块自身没有血液供应的组织的主要途径。

关节囊包住关节，韧带加强它，关节囊外的肌腱把骨拉在一起并提供大部分关节稳定性。

Joints are classified by the tissue connecting the bones, and the classification is a way of predicting what the joint can and cannot do.

关节按连接两块骨的组织分类，而这个分类是一种预测关节能做什么、不能做什么的方法。

Fibrous joints are dense and immovable, which is right for the skull and for the sacroiliac region.

纤维关节致密且不动，这适合颅骨和骶髂区域。

Cartilaginous joints connect with cartilage and allow limited movement, as between vertebrae.

软骨关节以软骨连接，活动有限，例如椎骨之间。

Synovial joints are the ones with a cavity and synovial fluid, and they are the ones that move, which is why they are also the ones that get injured.

滑膜关节是带有关节腔和滑液的那一类，也正是能动的那一类--所以它们也是最容易受伤的那一类。

Synovial joints are then further classified by shape, and the shape is a shortcut to the movement each one permits.

滑膜关节再按形状细分，而形状是一条通往"这个关节允许什么动作"的捷径。

Hinge joints allow one main movement - flexion and extension - as at the knee and the elbow.

铰链关节允许一个主要动作--屈曲与伸展--如膝和肘。

Pivot joints allow rotation, as at the neck and the forearm.

枢轴关节允许旋转，如颈和前臂。

Gliding joints allow sliding surfaces, as between the wrist bones and the shoulder blade.

滑动关节允许面之间滑动，如腕骨之间和肩胛之间。

Condyloid or ellipsoid joints allow combined movements, as at the wrist.

髁状或椭圆球关节允许组合动作，如腕。

Ball-and-socket joints allow the widest range of all, which is why the shoulder is the most mobile and most frequently dislocated joint in the body.

球窝关节允许最大的活动范围，因此肩是全身活动度最大、也最常脱位的关节。

Stability is not a fixed property.

稳定性不是固定属性。

It comes from four things working together: the shape of the joint surfaces, the ligaments, the muscles acting across the joint, and the capsule.

它来自四样东西的协同工作：关节面的形状、韧带、跨关节作用的肌肉，以及关节囊。

That last arrangement is the one that matters practically: muscles are the only part of the system you can consciously change, so dynamic stability is trainable in a way that passive stability is not.

最后这一组合在实际中最重要：肌肉是这套系统中唯一可以有意识改变的部分，所以动态稳定性是可以训练的，而被动稳定性不能。

That is the mechanism behind most of the shoulder and knee rehabilitation work ever done, and it is why the exercise is always about control rather than about rest.

这就是有史以来大多数肩和膝康复工作的机制，也是为什么那里的锻炼始终关于控制而不是关于休息。

The numbers / 算一算

Numbers to keep. The knee is a hinge joint with two main movements, flexion and extension, and it carries a large share of body-weight load in many activities. The shoulder ball-and-socket has the greatest range of any joint in the body, approaching 180 degrees of flexion, and it is also the most commonly dislocated, because that mobility is bought with relatively less bony restraint. The sacroiliac joint is a fibrous joint with a small but real amount of movement. The intervertebral joints are cartilaginous and contribute a meaningful part of spinal flexion. And the practical number: synovial fluid is only a few millilitres, which is why a small amount going missing through a swollen joint still produces a measurable loss of lubrication.

需要记住的数字。膝是铰链关节，有两个主要动作--屈曲与伸展--并在许多活动中承担很大一部分体重负荷。肩的球窝关节拥有全身最大

Common mistakes / 易错点

- "A hinge joint only flexes and extends." It has one main movement, but it also has a small amount of rotation at the knee. "Main" is doing work in that sentence.
"铰链关节只屈曲和伸展。"它有一个主要动作，但膝关节还有少量旋转。"主要"这个词在句子里是有作用的。
- "The knee is the most stable joint because it is strong." Strength is not the same as stability, and the knee relies heavily on muscles for dynamic stability because its bony shape is not especially constraining.
"膝是最稳定的关节因为它强壮。"强壮不等于稳定，而膝的骨性形状并不特别受限，所以它高度依赖肌肉提供动态稳定。
- "Cartilage has no blood supply so nutrition does not matter." It is avascular, so synovial fluid is its main supply route, which is exactly why nutrition and loading history matter to it over years.
"软骨没有血液供应所以营养无关紧要。"它无血管，滑液是它的主要供给途径，这正是营养和负荷历史在多年尺度上对它重要的原因。
- "More mobility is always better." Mobility and stability trade off, and the joint with the most mobility is usually the one with the least bony protection and the highest dislocation risk.
"活动度越大越好。"活动度与稳定性互相权衡，活动度最大的关节通常也是骨性保护最少、脱位风险最高的那个。

Why it matters / 为什么重要

This section is the bridge between anatomy and injury, and it explains why the same load is harmless in one athlete and damaging in another. A shoulder that has been held in one position all week is less able to absorb an unexpected force than one that has been used across its range, and that difference is not about strength at all. It also explains why rehabilitation is exercise-based: the stability a joint needs during sport is largely muscular, and a muscle that has not been asked to do that job for weeks will not do it when it matters.

这一节是解剖学与损伤之间的桥梁，它解释了为什么同样的负荷对一位运动员无害、对另一位却有破坏性。一整周都被固定在一个位置的

- Ball-and-socket joints are multiaxial; hinge joints are uniaxial; pivot joints allow rotation; condyloid joints are biaxial; saddle joints allow opposition; plane joints glide.

球窝关节多轴；铰链关节单轴；枢轴关节旋转；髁状关节双轴；鞍状关节允许对掌；平面关节滑动。

Key terms / 核心术语

Mobility 活动度

The range of movement available at a joint.

关节可提供的运动范围。

Stability 稳定性

The ability to resist unwanted movement and transfer force safely.

抵抗不必要运动并安全传递力量的能力。

Formulas / 公式

Shoulder mobility supports throwing, while knee stability and ligament control are important for cutting and landing.
肩部活动度支持投掷，膝关节稳定性和韧带控制对变向和落地很重要。

Worked examples / 实例

>

How do synovial structures allow movement and stability together?

滑膜结构如何兼顾运动和稳定？

Exam focus / 考核要点

Name three synovial joint classes and a sport example for each.

说出三个滑膜关节类别及各自运动例子。

Mobility, stability and injury · 活动度、稳定性与损伤

First, picture it / 先这样想

Every joint is a compromise between two opposing goods: how far it can move, and how well it holds together. Turn a joint towards mobility and you get a shoulder that can throw above the head, and you also get a shoulder that comes out of the socket. Turn it towards stability and you get a hip that never dislocates, and you also get a hip that cannot squat to the floor. The design of a joint is the answer its species gave to the question of which trade-off was worth more for its survival.

每个关节都是两种对立好处之间的妥协：能移动多远，以及能保持多完整。把关节调向活动度，你会得到一个能过头投掷的肩，也会得到一个容易脱位的肩；调向稳定性，你会得到一个从不脱位的髋，也会得到一个蹲不到底的髋。关节的设计，就是一个物种在进化中回答“哪种权衡对生存更值得”这个问题时给出的答案。

The detail / 详细讲解

High mobility supports skill performance and carries a matching instability risk. It is what allows a gymnast to invert, a swimmer to reach full extension, a basketballer to absorb a landing in a position nobody planned. The cost is that the more a joint can move, the less bony surface is holding the surfaces together, so the soft tissues have to do more of the work - and soft tissues fail before bone does.

高活动度支持技巧性表现，也带来相应的稳定性风险。它让体操运动员能够倒立、游泳者能充分伸展、篮球运动员能在没人计划过的

High stability protects the joint and transfers force well, but reduces movement efficiency. A very stable joint moves less, requires more muscular effort for the same movement, and cannot access the range where the skill actually lives. This is a real cost rather than a theoretical one: a swimmer with restricted shoulder rotation spends energy on a stroke they cannot make long, and a cyclist with restricted hip motion cannot generate the power that the pedal stroke depends on.

高稳定性保护关节并很好地传递力，但降低运动效率。一个非常稳定的关节活动更少、同样动作需要更多肌肉努力，而且无法进入技

The relevant quantity is not mobility in the abstract but the mobility that is available at the moment it is needed. A joint with plenty of passive range that is not controlled under load is a liability, not an asset - which is the whole reason dynamic stability exists as a concept. Control through range is what allows a joint to use its mobility without paying the instability price, and it is trainable, which is the hopeful part.

真正相关的量不是抽象的活动度，而是需求发生时可用

的活动度。被动活动度很大、但在负荷下却不受控的关节是负债而不是资产--这正是“动态稳定”这个概念存在的全部理由。通过整个活

Injury follows when demand exceeds what the tissue has been prepared for, and "prepared for" is about both capacity and control. Capacity is how much force the tissue can take; control is how evenly that force is distributed.

The same tissue can be injured at a given load if it is fresh and stiff and unprepared, and survive a larger load if it has been progressively loaded and is used to the movement pattern. This is why conditioning is protective and why the last two weeks before a season matter more than the first two.

损伤发生在需求超过组织“准备程度”之时，而“准备程度”同时关乎容量与控制。容量是组织能承受多大的力；控制是这个力被分配得多

The numbers / 算一算

Numbers to keep. Shoulder range is the reference for mobility: roughly 180 degrees of flexion and 180 of abduction in a healthy adult, which is the widest of any joint. Knee flexion is about 135 degrees. Hip flexion is similar, and the combined hip and knee range is what a deep squat requires - somewhere near 180 degrees of total "hip open, knee bend" to get the torso upright at depth. The number that links mobility to injury is load tolerance, and it is the least measured of all: nobody routinely records how much force a given athlete joint has been prepared for, which is why the same training load injures one athlete and conditions another.

需要记住的数字。肩关节的活动范围是灵活性的参照：健康成人屈曲约180度、外展约180度，是全身最广的。膝屈约135度。髋屈与之相

Common mistakes / 易错点

- "Maximum flexibility is always the goal." Flexibility is only useful if it is available and controlled at the moment of demand. A joint that is loose but not controlled is a liability.

"柔韧性越大越好。"只有当柔韧在被需要的那一刻是可获得且受控的，它才有用。一个松弛但不受控的关节是负债。

- "Stronger muscles prevent all joint injury." Strength helps dynamic stability, but a very strong muscle guarding

an unstable joint can also stiffen the joint excessively. Capacity and control both matter, and they are not the same.

"更强的肌肉能预防所有关节损伤。"力量有助于动态稳定，但一块非常强健的肌肉在保护一个不稳定关节时，也可能让关节过度僵

"If it does not hurt it is fine." Pain is a late signal, and tissue can be overloaded well before it becomes painful, which is why monitoring and progressive loading exist.

"不疼就没问题。"疼痛是晚期信号，组织在变痛之前很久就可能已被过度负荷，这正是监测与渐进负荷存在的理由。

"Stretching protects against injury." Static flexibility alone has limited protective value; progressive exposure to load and good movement control are what actually protect tissue.

"拉伸能预防损伤。"单纯的静态柔韧性保护价值有限；真正保护组织的是对负荷的渐进暴露和良好的动作控制。

Why it matters / 为什么重要

The trade-off between mobility and stability is the most practically useful idea in biomechanics, because it explains a paradox that every athlete meets: the movement that makes them good is often the one that is most likely to hurt them. It changes what a coach watches, what a physiotherapist asks about, and what an athlete should be willing to give up. It also reframes injury as something that is usually predictable rather than random, because the combination of demand, capacity and control can be assessed - and each of those three is something a programme can change.

活动度与稳定性之间的权衡，是生物力学中最实用的概念，因为它解释了一位运动员都会遇到的悖论：让他变强的那个动作，往往也正是

Warm-up, strength, technique, recovery and appropriate equipment support joint health.

热身、力量、技术、恢复和合适装备支持关节健康。

Key terms / 核心术语

Worked examples / 实例

>

Cutting, landing and change of direction require ligament control and adequate force absorption.

变向、落地和改变方向需要韧带控制和足够的冲击力吸收。

Quick check / 快速检查

1. Give one example of a connective-tissue or joint injury.

举出一个结缔组织或关节损伤例子。

Exam focus / 考核要点

Explain how joint structure supports performance and resists injury.

解释关节结构如何支持表现并抵抗损伤。

Muscular function

肌肉功能

Guiding question / 引导问题

How do motor units, fibre types and contractions create force and movement?

运动单位、肌纤维类型和收缩如何产生力量与运动？

A motor unit is one motor neuron and all the fibres it stimulates. Recruitment, fibre type, contraction type and the sliding-filament mechanism explain how skeletal muscle produces and controls force.

运动单位是一个运动神经元及其刺激的所有肌纤维。募集、肌纤维类型、收缩类型和滑动肌丝机制解释了骨骼肌如何产生并控制力量。

Motor units and fibre types · 运动单位与肌纤维类型

First, picture it / 先这样想

A muscle is not one thing. It is a bundle of cables, and each cable is served by its own private switch. One nerve cell plus the muscle fibres it reaches is one motor unit, and the nerve cell fires or it does not - there is no half switch. This is why a single fibre always gives its full effort once the signal arrives, and why you get more force by turning on more switches, from the small ones first, rather than by asking any one of them for more.

肌肉不是一整块东西，而是一束缆线，每根缆线都有自己的专属开关。一个神经细胞加上它所支配的肌纤维构成一个运动单位，而神经细胞要么放电要么不放--没有“半开”的开关。这就是为什么单根纤维一旦收到信号就总是给出全力，而你要更大的力量，是靠从小的开关开始逐个打开更多，而不是要求其中某一个开得更大。

The detail / 详细讲解

The all-or-none principle states that a muscle fibre contracts fully once a sufficient threshold is reached, and does not contract at all below it.

全或无原则指出：肌纤维一旦达到足够的阈值就完全收缩，低于阈值则完全不收缩。

There is no graded response within a single fibre.

单根纤维内部不存在分级反应。

The grading you can produce - a light lift versus a heavy one - happens at the level of the motor unit, by recruiting more of them and by changing their firing rate.

你能做出的分级--轻举和重举之间的差别--发生在运动单位这个层面，靠招募更多运动单位以及改变它们的放电频率。

The size principle describes the order in which units are recruited: small first, large later.

大小原则描述运动单位被招募的顺序：先小后大。

Small motor units contain small muscle fibres, have low thresholds, and fire slowly, so the body uses them for fine, low-load, long-duration work - posture, and the small stabilising muscles.

小运动单位

含小肌纤维、阈值低、放电慢，所以身体把它们用于精细、低负荷、长时间的工作--姿势维持和小的稳定肌群。

Large motor units contain larger fibres, have higher thresholds and fire faster, so they are recruited last, for heavy, fast, powerful work.

大运动单位含较大的纤维、阈值高、放电快，因此最后被招募，用于重的、快的、有力的工作。

This ordering is why a training-inexperienced athlete performs well at low intensity and falls apart at high intensity.

这一顺序正是缺乏训练经验的运动员在低强度下表现良好、却在高强度下崩溃的原因。

Muscle fibre types explain the same phenomenon from the other direction.

肌纤维类型从另一个方向解释同一现象。

Type I slow oxidative fibres are highly vascular, rich in mitochondria, fatigue-resistant and suited to endurance.

I型慢氧化肌纤维血管丰富、线粒体多、耐疲劳，适合耐力。

Type IIa fast oxidative-glycolytic fibres are a hybrid: they produce more power and still have some endurance capacity.

IIa型快氧化糖酵解纤维是混合型：功率更高，同时仍有一定耐力能力。

Type IIx fast glycolytic fibres produce the highest power, fatigue very quickly, and are what a sprinter uses and a

marathoner barely touches.

IIx型快糖酵解纤维功率最高、极快疲劳，是短跑运动员使用而马拉松运动员几乎不用的类型。

The proportion is largely genetic, which is a large part of why sprinters and distance runners look different before they train.

这些类型的比例在很大程度上由遗传决定，这正是短跑运动员和长跑运动员在训练之前外貌就已经不同的一个主要原因。

Most human skeletal muscle is a mixture, and the interesting part is that the mixture is not fixed.

大多数人类骨骼肌是混合的，而有趣之处在于这个混合并非固定。

Fibre type shifts with training in a limited but real way: endurance work increases the oxidative capacity of type IIx fibres and can move some of them toward a IIa phenotype, and resistance work moves in the other direction.

纤维类型会随训练发生有限但真实的变化：耐力训练提高IIx纤维的氧化能力，并可使其一部分向IIa型表型移动；

The honest framing is that you are nudging a distribution rather than rewriting it, which is why a naturally slow-twitch athlete can become an excellent distance runner but rarely a world-class sprinter.

抗阻训练则朝相反方向移动。

诚实的框架是：你在推动一个分布，而不是重写它--这正是天生慢肌为主的运动员可以成为优秀的耐力选手、却很少成为世界级短跑运动员的原因。

The numbers / 算一算

Numbers to keep. A single motor unit ranges from roughly 10 to several thousand fibres depending on the muscle - small in the extraocular muscles where precision matters, large in the quadriceps where force matters. Type I fibres have high mitochondrial density and a rich capillary supply; type IIx have markedly less and fatigue far sooner. Fast-twitch units can fire in the range of 50 to 100 Hz during maximal effort compared with around 10 to 20 Hz for a slow unit at steady low load. And the practical number for training: recruitment order cannot be reversed, so a sprinter cannot train the size principle away - the big units are available, they simply come in last.

需要记住的数字。根据肌肉不同，一个运动单位大约包含10到数千根纤维--在重视精确的眼外肌中很小，在重视力量的股四头肌中很大。

Common mistakes / 易错点

-

"You can partially contract a single muscle fibre." Fibres are all-or-none. The grading comes from recruiting more units and changing firing frequency.

"你可以让一根肌纤维部分收缩。"纤维是全或无的。分级来自招募更多运动单位和改变放电频率。

-

"Fast-twitch fibres become slow-twitch with endurance training." There is a shift within a limited range, mostly as type IIx takes on a IIa phenotype and oxidative capacity rises. You shift a distribution, you do not rewrite it.

"耐力训练会把快肌纤维变成慢肌纤维。"变化发生在有限范围内，主要是IIx向IIa表型移动、氧化能力提升。你是在推动一个分布，而不是重写它。

-

"Bigger motor units are recruited first." The size principle is the opposite: small, low-threshold units first, large ones only at high intensity.

"大运动单位先被招募。"大小原则正相反：小的、低阈值的先，大的只在高强度时。

-

"Fibre type is fixed by genetics so training cannot change it." Training changes the oxidative capacity of the fibres and the distribution within limits. Genetics sets the range, not the outcome.

"纤维类型由基因决定，所以训练改不了。"训练会在一定限度内改变纤维的氧化能力和分布比例。基因设定的是范围，不是结局。

Why it matters / 为什么重要

Motor units and fibre types explain the practical asymmetry that every coach meets: the same session trains two athletes very differently, and the reason is not effort or motivation. They also explain why sprint and distance training look so different even though both are running. And they matter for injury, because the fast fibres that generate the most power are also the ones that generate the most strain on the tissue and the ones that fatigue soonest, which is exactly the combination that produces a hamstring strain in a sprinter who is not conditioned for top speed.

运动单位和肌纤维类型解释了每位教练都会遇到的现实不对称：同一节训练课对两位运动员的塑造差别很大，而原因不是努力程度也不是天赋。

-

Type I fibres are slow, oxidative and fatigue-resistant; Type IIa are fast and use both aerobic and glycolytic metabolism; Type IIx are very fast, glycolytic and powerful but fatigue quickly.

I型纤维慢、有氧、耐疲劳；IIa型快速，结合有氧和糖酵解；IIx型极快、依赖糖酵解、力量强但很快疲劳。

Key terms / 核心术语

Motor unit 运动单位

One motor neuron and the muscle fibres it innervates.

一个运动神经元及其支配的肌纤维。

Hypertrophy 肌肥大

Increase in fibre cross-sectional area after training.

训练后肌纤维横截面积增加。

Atrophy 肌萎缩

Decrease in fibre cross-sectional area after detraining or inactivity.

停训或不活动后肌纤维横截面积减少。

Formulas / 公式

Sprinters rely more on fast powerful fibres, while distance runners rely more on fatigue-resistant fibres.

短跑者更多依赖快速高功率纤维，长跑者更多依赖耐疲劳纤维。

Worked examples / 实例

>

Apply the size principle to a strength task.

把募集大小原则应用到力量任务。

Exam focus / 考核要点

Compare Type I, IIa and IIx fibres.

比较I型、IIa型和IIx型肌纤维。

Contractions and muscle interaction · 肌肉收缩与相互作用

First, picture it / 先这样想

A muscle can do three different jobs with the same two ends. Shortening against resistance is a concentric contraction, like a biceps lifting a weight toward the shoulder. Producing force with no change in length is an isometric contraction, like pushing a wall you cannot move. Lengthening while still controlling force is an eccentric contraction, like lowering that same weight back down - and eccentric is the odd one out, because it is the muscle resisting being lengthened rather than shortening.

同一块肌肉用同样的两个末端可以干三份不同的活。对抗阻力缩短是向心收缩，比如肱二头肌把重物向肩膀方向拉；产生力而长度不变是等长收缩，比如推一堵推不动的墙；在仍控制着力的情况下被拉长是离心收缩，比如把同一个重物放回去--而离心是那个异类，因为它是在抵抗被拉长，而不是在缩短。

The detail / 详细讲解

Concentric contraction shortens the muscle against resistance, and it is what most people picture when they think of muscle working.

向心收缩是肌肉对抗阻力缩短，也是多数人想到“肌肉在工作”时脑海中的画面。

The external load is smaller than what the muscle can produce, so the muscle shortens and moves the joint.

外负荷小于肌肉能产生的力，所以肌肉缩短并带动关节。

Everyday examples are the shortening phase of a biceps curl, the lifting phase of a squat, the pull phase of a swim stroke.

日常例子是肱二头肌弯举的缩短阶段、深蹲的上升阶段、划水动作的拉水阶段。

Isometric contraction produces force with no joint movement: the muscle develops tension while its length stays essentially the same.

等长收缩产生力而关节不动：肌肉建立张力而长度基本不变。

It is the contraction of a push-up at the top, of a wall sit, of the neck muscles holding posture, and of the abdominals holding the trunk still.

它是俯卧撑顶端的力量、靠墙静蹲、颈部维持姿势的收缩，以及腹肌固定躯干的收缩。

In sport it is everywhere, and it is the contraction type that produces the least movement while often still costing a great deal of effort.

在运动中它无处不在，也是那种几乎不产生动作、却常常仍要付出很大努力力的收缩类型。

Eccentric contraction lengthens active muscle while it controls the resistance, and this is the one that is least understood.

离心收缩是在控制阻力的情况下肌纤维被拉长，也是最不容易被理解的一种。

The muscle is doing work while getting longer, because it is resisting being pulled apart.

肌肉在做功的同时变长，因为它在抵抗自己被拉开。

It is the lowering phase of the curl, the descent in a drop jump, the braking phase at the end of a sprint, and the controlled landing after a jump.

它是弯举的下放阶段、跳深落下的下降阶段、冲刺末段的制动阶段，以及跳跃后的可控落地。

Eccentric contractions can generate substantially more force than concentric ones - up to roughly 1.5 to 2 times - because the muscle has to actively resist the stretch rather than simply overcome it.

离心收缩能产生的力量明显大于向心收缩--最多可达约1.5到2倍--因为肌肉必须主动抵抗拉伸，而不只是克服阻力。

Isokinetic contraction is the fifth description worth knowing: the muscle shortens at a constant velocity, which is why it is described as a type rather than a task.

等速收缩是第五个值得知道的描述：肌肉以恒定速度缩短，因此它是被当作"类型"而不是"任务"来描述的。

The practical significance is that it is a description of how fast the muscle is shortening, not of what it is doing, and it is how the phrase "constant velocity resistance" makes sense in a gym.

它的实际意义在于，它描述的是肌肉缩短有多快，而不是它在做什么，这也正是健身房里"恒速阻力"这个说法的由来。

Most real movements are combinations, and the combination is the point.

多数真实动作是组合的，而组合正是重点。

A squat is concentric on the way up and eccentric on the way down, with an isometric pause at the bottom if the athlete chooses one.

一次深蹲在上升段是向心、在下降段是离心，如果运动员选择停顿，底部还有一段等长。

When a muscle is asked to change from shortening to lengthening, there is a brief phase where tension is highest and the muscle is most vulnerable, and that transition is where many strains occur - which is why the bottom of a heavy lift and the end of a sprint are the moments worth watching.

当一块肌肉被要求从缩短转为拉长时，会有一个张力最高、肌肉最脆弱的短暂阶段，而很多拉伤就发生在这个过渡处--这也是为什么

The numbers / 算一算

Numbers to keep. Eccentric contractions can generate roughly 1.5 to 2 times the force of a concentric contraction of the same muscle at the same length, which is why they are both protective as stabilisers and hazardous when the load is unexpected. A muscle generating tension without changing length is still consuming energy, so isometric work has a real metabolic cost despite producing no movement. A controlled eccentric descent through about 2 to 3 seconds per repetition is the range generally used to minimise strain, and slower than that adds fatigue without obviously adding protection. And the practical figure for training: the force in a muscle rises steeply as it is lengthened toward its maximum, so a muscle is mechanically strongest in a lengthened position - which is why lengthened-position curls and pull-ups are prescribed for the biceps and lats.

需要记住的数字。同一块肌肉在同一长度下，离心收缩能产生的力约为向心的1.5到2倍--这正是它作为稳定肌既具保护性、又会在负荷意

Common mistakes / 易错点

- "Eccentric means the muscle shortens under load." Eccentric means the muscle is lengthening while controlling the load. Lengthening while active is the definition.

"离心是指肌肉在负荷下缩短。"离心是指肌肉在控制负荷的同时被拉长。主动状态下被拉长就是定义。

- "Isometric work is free because nothing moves." It produces no joint movement but it still consumes energy and still generates force, and it is metabolically real.

"等长训练不费劲因为什么都不动。"它不产生关节动作，但仍然消耗能量、仍然产生力，代谢上是实实在在的。

- "Eccentric training is always safe because the muscle controls it." Eccentric contractions produce the highest

forces of any type. That is protective when expected and hazardous when not.

"离心训练总是安全因为肌肉在控制它。"离心收缩产生所有类型中最高的力。预期之内时它是保护，意外时它是危险。

"Only the shortening phase of an exercise counts as the work." The lowering phase is where the highest force and the highest strain occur, and it is the phase most often skipped.

"只有动作的缩短阶段才算训练。"下放阶段是力量和应变最大的阶段，也是最常被跳过的阶段。

Why it matters / 为什么重要

Contraction types are how a coach translates an intention into a stimulus, and getting the type right is often the difference between rehabilitation and re-injury. A patient returning to a sport after an ACL repair needs quadriceps control in the lengthened position, which is an eccentric and isometric demand rather than a lifting one. An athlete learning to absorb a landing needs to be able to produce force without shortening, which is isometric. And the practical warning that follows from all of it: the moments that feel harmless are usually the ones generating the highest forces.

收缩类型让教练把意图翻译成刺激，而选对类型往往是康复与再次受伤之间的差别。ACL重建后回归运动的患者需要股四头肌在拉长位

The agonist is the prime mover; the antagonist opposes it. Reciprocal inhibition helps the agonist act without simultaneous strong antagonist contraction.

主动肌是主要产生动作的肌肉，拮抗肌与其作用相反。交互抑制帮助主动肌动作而不让拮抗肌同时强烈收缩。

Key terms / 核心术语

Concentric 同心收缩

Active muscle shortens.

主动肌肉缩短。

Eccentric 离心收缩

Active muscle lengthens under load.

主动肌肉在负荷下延长。

Formulas / 公式

In a biceps curl, the biceps is the agonist and triceps the antagonist during flexion; lowering the weight is eccentric for the biceps.

弯举屈肘时，肱二头肌是主动肌，肱三头肌是拮抗肌；放下重量时肱二头肌进行离心收缩。

Worked examples / 实例

>

Identify contraction type, motor-unit change and adaptation in a strength task.

在力量任务中识别收缩类型、运动单位变化和适应。

Exam focus / 考核要点

Describe and give an example of each contraction type.

描述每种收缩类型并举例。

Recruitment, adaptations and sliding filament · 募集、适应与肌丝滑行学说

First, picture it / 先这样想

The sliding-filament mechanism is a rowing boat with oars that pull past each other without ever touching. Two filaments - a thick one made of myosin and a thin one made of actin - slide past one another, the heads of the myosin grab the actin, pull it toward the centre of the sarcomere, and then let go and re-cock. Shorten the overlap and the muscle shortens. Straighten the filament and the muscle is floppy. There is no squeezing or stretching of the filaments themselves, which is the detail that took a century to establish.

滑行肌丝机制像一艘rowing小船，两支桨互相擦身而过却从不接触。粗肌丝由肌球蛋白构成，细肌丝由肌动蛋白构成，两者互相滑过；肌球蛋白的头抓住肌动蛋白，把它拉向肌节中央，然后松开并重新上膛。重叠减少，肌肉就缩短；肌丝被拉直，肌肉就松软。肌丝本身既不被挤压也不被拉伸--正是这个细节花了一个世纪才被确立。

The detail / 详细讲解

When calcium is released it binds to troponin, which moves tropomyosin out of the way, which exposes the binding sites on the actin filament.

当钙被释放时，它与肌钙蛋白结合，使原肌球蛋白移开，从而暴露出肌动蛋白细肌丝上的结合位点。

The myosin heads, already carrying ADP and phosphate from the previous cycle, bind to those sites, release the phosphate, and perform the power stroke - a rotation of the head that drags the actin toward the centre of the sarcomere.

携带上一轮留下的ADP和磷酸的肌球蛋白头部与这些位点结合，释放磷酸，然后完成动力冲程--头部的旋转把肌动蛋白拖向肌节中央

ADP is then released, the head detaches, ATP is bound to re-cock the head, and the cycle begins again.

随后ADP释放，头部脱离，结合ATP使头部重新上膛，循环再次开始。

The mechanical consequence is that a muscle only shortens because the overlap between the filaments increases; the filaments themselves do not change length, and the sarcomere is the unit that shortens.

力学上的推论是：肌肉之所以缩短，只是因为肌丝之间的重叠增加了；

That is the sliding-filament model, and it is the reason a muscle has to be pulled in at both ends to shorten at all: the sarcomeres are arranged in series, so shortening each one in turn shortens the whole fibre.

肌丝本身长度不变，缩短的单位是肌节。

这就是滑行肌丝模型，也是为什么一块肌肉必须两端都被拉住才能缩短：肌节是串联排列的，每个依次缩短，整根肌纤维就缩短了。

Calcium has to be removed for the muscle to relax, and it costs energy to do it.

要让肌肉放松，钙必须被移走，而这件事要消耗能量。

The calcium pump moves calcium back into the sarcoplasmic reticulum, and that pump requires ATP.

钙泵把钙送回肌浆网，而钙泵需要ATP。

So relaxation is an active, energy-consuming process, not a passive letting-go - which is why a muscle held in a contracted state consumes energy even though it is not doing any visible work.

所以放松是一个主动的、耗能的过程，而不是被动地“松手”--这就是为什么一块维持在收缩状态的肌肉，在没有任何可见动作的情况下

Hypertrophy increases cross-sectional area and can increase force, because more parallel filaments mean more cross-bridges and therefore more force.

肌肥大增加横截面积并能增加力，因为更多平行的肌丝意味着更多的横桥，从而产生更多力。

Atrophy reduces cross-sectional area and force, and it happens faster than hypertrophy can reverse it - the general pattern being that strength is lost more quickly than it is built, which is why maintaining is easier than achieving.

肌萎缩减少横截面积和力，而且发生得比肌肥大被逆转的速度更快--总体规律是力量流失比力量建立更快，这也正是“维持”比“获得”容

Both are responses to demand: the tissue changes to match what it is repeatedly asked to do.

两者都是对需求的反应：组织会改变自己去匹配它被反复要求做的事。

Adaptation has three names worth using precisely.

适应有三个需要精确使用的名称。

Hypertrophy is the increase in fibre size.

肌肥大是肌纤维尺寸的增加。

Hyperplasia, an increase in fibre number, is what you would expect and is not convincingly demonstrated in humans.

肌增生，即纤维数量的增加，是你直觉上会预期的，但在人类身上并未被令人信服地证明。

Strength can increase without hypertrophy, particularly early in training, through neural adaptation - better coordination, better timing, better recruitment - which is why the first two weeks of a new programme produce real strength gains with little visible change in muscle size.

力量可以在没有肌肥大的情况下提升，尤其在训练早期，途径是神经适应--更好的协调、更好的时机、更好的募集--这就是为什么一个

The numbers / 算一算

Numbers to keep. Myosin binds ATP once per cycle, and a single myosin head performs roughly one to several power strokes per second depending on the load, so a fibre with many heads generates a great deal of force. Muscle strength gains of 10 to 20% are realistic in the first four to eight weeks of a new programme, and much of that is neural rather than size. Strength declines during a few weeks of complete rest, and the decline is faster than the time taken to rebuild it. And the practical number for programming: the total cross-sectional area of the contractile proteins, not the number of fibres, is what determines force, which is why the training variable that matters most for size is volume over time rather than the heaviest load used.

需要记住的数字。肌球蛋白每个循环结合一次ATP，一个肌球蛋白头部每秒大约完成一到数次动力冲程（取决于负荷），所以拥有许多头

Common mistakes / 易错点

- "Muscle fibres stretch and shorten like elastic bands." The filaments slide past each other; they do not stretch or compress. The sarcomere is what shortens.
"肌纤维像橡皮筋一样伸缩。"肌丝是互相滑过，而不是伸缩。缩短的是肌节。
- "Hypertrophy is more fibres." No - hypertrophy is an increase in fibre size. An increase in fibre number is hyperplasia, which is not convincingly shown in humans.
"肌肥大是纤维数量增加。"不是--肌肥大是纤维尺寸增加。纤维数量增加是肌增生，在人类身上并未被令人信服地证明。
- "All early strength gains come from bigger muscles." A large share of the first weeks of gains is neural - coordination, timing and recruitment - with little change in size.
"早期所有力量增长都来自肌肉变大。"前几周增长中有很大一部分来自神经--协调、时机和募集--而尺寸变化很小。
- "Relaxation is passive." It requires ATP to pump calcium back into the sarcoplasmic reticulum, so a held muscle burns energy while doing nothing visible.
"放松是被动的。"它需要ATP把钙泵回肌浆网，所以一块维持收缩的肌肉在什么可见动作都没做的情况下也在烧能量。

Why it matters / 为什么重要

This is where muscle contraction and training adaptation become one subject, and it is the mechanism behind almost every change an athlete sees in the mirror. It is also where the most useful practical lesson lives: strength can be built faster than muscle, early gains are largely neural, and the stimulus has to keep progressing or the tissue stops responding. That single idea explains why programmes change, why the same programme stops working after eight weeks, and why the athlete who keeps progressing the stimulus keeps improving while the athlete who does the same sessions for a year stalls.

这里是肌肉收缩与训练适应合成同一门学科的地方，也是运动员在镜子里看到的几乎所有改变背后的机制。而最实用的一个教训也在这里

- A motor unit is recruited from small to large as force demand rises.
力量需求增加时，运动单位由小到大募集。

Key terms / 核心术语

All-or-none principle 全或无原则

A fibre contracts fully after threshold stimulation.
达到阈值刺激后肌纤维完全收缩。

Formulas / 公式

Strength training can increase force through recruitment and hypertrophy; slowing the lowering phase emphasises eccentric control.
力量训练可通过募集和肌肥大增加力量；放慢下降阶段可强调离心控制。

Worked examples / 实例

- >
Outline the sliding-filament theory.
概述滑动肌丝理论。

Exam focus / 考核要点

Why is ATP needed for both contraction and relaxation?
为什么收缩和松弛都需要ATP？

Levers in movement and sport

人体杠杆与运动

Guiding question / 引导问题

How do lever arrangements change force, speed and range of motion?

杠杆排列如何改变力量、速度和关节活动范围？

Every lever contains an effort, a fulcrum and a load. Their positions determine the class and the trade-off between force, speed and range of motion. Equipment can act as an external lever to extend the body's mechanical options.

每个杠杆都包含动力、支点和阻力。它们的位置决定杠杆类别，也决定力量、速度和活动范围之间的取舍。装备可作为外杠杆，扩展

Components and lever classes · 杠杆的构成与分类

First, picture it / 先这样想

Every movement your body makes is a lever, and a lever is just three things arranged in a line. The fixed point that does not move is the fulcrum. The force you supply is the effort. The thing being moved is the load. Where the fulcrum sits relative to the other two is what decides whether the lever gives you force or gives you speed, and that single arrangement is the whole reason a sprinter and a weightlifter are built so differently.

你身体做出的每一个动作都是一个杠杆，而杠杆不过是排成一条线的三样东西。不动的那一点是支点，你提供的力是动力，被移动的东西是阻力。支点相对另外两者的位置决定了杠杆给你力还是给你速度，而这一个安排就是短跑运动员和举重运动员体型差异巨大的全部原因。

The detail / 详细讲解

In the body, bones act as levers, joints as fulcrums, muscles as effort, and body parts or objects as load.

在身体中，骨充当杠杆，关节充当支点，肌肉充当动力，身体部位或物体充当阻力。

The effort is almost always produced by muscle pulling across a joint, which is why a joint angle changes the mechanical advantage of a movement: the moment arm between the muscle attachment and the joint centre changes with the angle, and so does the force the muscle has to produce.

动力几乎总是由肌肉跨关节拉出来的，这正是关节角度会改变动作机械优势的原因：肌肉附着点与关节中心之间的力臂随角度改变，

A first-class lever has the fulcrum between the effort and the load, so effort and load act on opposite sides of it.

一类杠杆的支点位于动力和阻力之间，因此动力和阻力作用在它的两侧。

The neck is the standard example, with the head as the load supported by muscles behind the joint.

颈部是标准例子，头部作为阻力由关节后方的肌肉支撑。

First-class levers trade speed for force, because the load arm is usually the longer one.

一类杠杆用速度换力，因为阻力臂通常更长。

A second-class lever has the load between the fulcrum and the effort, so the effort is always at one end and the load in the middle.

二类杠杆的阻力位于支点和动力之间，因此动力总在一端、阻力在中间。

The calf raising onto the toes is the example: the toes and ball of the foot are the fulcrum, the body weight on the toes is the load, and the calf muscles supply the effort from behind.

踮脚抬起小腿就是例子：脚趾和前脚掌是支点，压在脚趾上的体重是阻力，小腿肌肉从后面提供动力。

Because the effort arm is always longer than the load arm, a second-class lever always gives a mechanical advantage greater than 1, and it always gives force rather than speed.

由于动力臂总是长于阻力臂，二类杠杆的机械优势总是大于1，它总是给出力而不是速度。

A third-class lever has the effort between the fulcrum and the load, so the effort is in the middle.

三类杠杆的动力位于支点和阻力之间，所以动力在中间。

The biceps lifting a weight in the hand is the standard example: the elbow joint is the fulcrum, the biceps is the effort, and the weight in the hand is the load.

肱二头肌把手中重物举起是标准例子：肘关节是支点，肱二头肌是动力，手中的重量是阻力。

A third-class lever always has a load arm longer than the effort arm, so it always gives a mechanical advantage below 1, and it always trades force for speed and range - which is exactly what a limb needs, because limbs are for moving, not for lifting.

三类杠杆的阻力臂总是长于动力臂

，所以机械优势总是小于1，它总是用力换速度和幅度--而这正是肢体所需要的，因为肢体是用来移动的，不是用来举重的。

The useful insight is that the body mostly uses third-class levers, which are fast but weak, and then solves the weakness by changing the muscle attachments and by using multiple joints in sequence.

一个有用的洞见是：身体主要使用三类杠杆，它快但弱，然后通过改变肌肉附着点、并串联使用多个关节来解决这种弱势。

A single third-class lever would be hopeless for anything powerful.

单条三类杠杆对任何有力度的动作都毫无希望。

A chain of levers, each contributing a little, becomes a limb that can throw, jump or sprint - and that is the mechanical reason a body is built out of many small levers rather than one big one.

一串每一条都贡献一点点的杠杆，组成了能投掷、能跳跃、能冲刺的肢体--这正是一个身体由许多小杠杆而不是一个大杠杆构成的力学

The numbers / 算一算

Numbers to keep. Mechanical advantage is effort arm divided by load arm, so a value above 1 gives a force advantage, 1 is neutral, and below 1 gives a speed or range advantage. Third-class levers in the human body typically sit well below 1, often around 0.2 to 0.3, which is why muscles must generate several times the load they move. That number is not a design flaw but a design choice: a limb that optimised for mechanical advantage would be strong and immobile. The compensation is the muscle moment arm, and the way to improve it is to move the attachment further from the joint - which is why the deltoid insertion and the gluteal insertions are placed where they are, and why a strap or a changed hand position can change a lift by a surprising amount.

需要记住的数字。机械优势是动力臂除以阻力臂，所以大于1给出力量优势，等于1为中性，小于1给出速度或幅度优势。人体中的三类杠

Common mistakes / 易错点

-

"A second-class lever gives a speed advantage." The load sits between fulcrum and effort, so the effort arm is longer and the advantage is always above 1: force, not speed.

"二类杠杆给出速度优势。"阻力位于支点和动力之间，所以动力臂更长，优势总是大于1：给出的是力，不是速度。

-

"The body is built to maximise mechanical advantage." It is built mostly on third-class levers with advantages well below 1, because limbs are for moving. Strength comes from muscle size and from chaining levers, not from high advantage.

"身体是为了最大化机械优势而建造的。"它主要由机械优势远低于1的三类杠杆构成，因为肢体是用来移动的。力量来自肌肉体积和

-

"Mechanical advantage is the force you produce." It is a ratio of two arms, not a force. The force produced is the effort multiplied by the advantage.

"机械优势就是你产生的力。"它是两个臂的比值，不是力。产生的力是动力乘以优势。

-

"Moving the muscle attachment closer to the joint makes it stronger." Closer means a shorter moment arm and less force. Moving it further away increases force at the same muscle length.

"把肌肉附着点移近关节会让它更强。"移近意味着力臂更短、力更小。移远才会在肌肉长度相同的情况下增加力。

Why it matters / 为什么重要

Lever classification is the first real piece of analysis in biomechanics, and it is the piece that lets an athlete and a coach argue about technique instead of guessing. It explains why a biceps curl with the elbow close to the body is harder than one with the elbow away, why the arm straightens the load further from the shoulder and becomes weaker but faster, and why two athletes lifting the same weight can look completely different. It is also the basis of the strength-versus-speed trade-off that runs through every sport, and the reason that a sport requiring power and a sport requiring speed train the same muscles in opposite ways.

杠杆分类是生物力学中第一块真正的分析工具，而正是它让运动员和教练可以就技术进行讨论而不是靠猜。它解释了为什么肘部贴近身

Second-class levers always have a force advantage. Third-class levers always have a speed and range-of-motion advantage. First-class levers can vary depending on arm lengths.

二类杠杆总是具有力量优势；三类杠杆总是具有速度和活动范围优势；一类的效果取决于两臂长度。

Key terms / 核心术语

Effort arm 动力臂

Distance from fulcrum to effort.

支点到动力作用点的距离。

Load arm 阻力臂

Distance from fulcrum to load.

支点到阻力作用点的距离。

Formulas / 公式

The triceps push-down and knee extension are second-class examples; the biceps curl is third-class; the atlanto-occipital joint is first-class.

肱三头肌下压和膝伸展是二类例子；肱二头肌弯举是三类；寰枕关节是一类。

Worked examples / 实例

>

Classify the elbow and explain one equipment change to its lever system.

对肘关节分类，并解释一种器械调整。

Exam focus / 考核要点

Label effort, fulcrum and load on a lever.

在杠杆上标出动力、支点和阻力。

Mechanical advantage and equipment · 机械优势与装备

First, picture it / 先这样想

Equipment is a machine for borrowing force from somewhere. A longer handle is a way of borrowing it from distance, because force times distance is the same work whether you do it with a long gentle push or a short hard one. That is the whole idea behind mechanical advantage, and it is also the whole idea behind good equipment design: change the shape of the machine so that the athlete does not have to supply more force. Bad equipment makes the athlete work harder to achieve the same result, which is why it feels heavy even when the weight is identical.

装备是一台用来“借力”的机器。更长的手柄是一种从距离上借力的方式，因为力乘以距离就是同样的功，无论你用长而轻的推还是短而重的推。这既是机械优势的全部理念，也是好的装备设计的全部理念：改变机器的形状，让运动员不必提供更大的力。差的装备让运动员用更大的力气得到同样的结果--这就是为什么重量一样时它仍然感觉沉重。

The detail / 详细讲解

Mechanical advantage is effort arm divided by load arm.

机械优势是动力臂除以阻力臂。

Above 1 you gain force and lose speed and range.

大于1时你获得力量，损失速度和幅度；

At 1 it is neutral.

等于1时中性；

Below 1 you gain speed and range and must supply more force.

小于1时你获得速度和幅度，但必须提供更大的力。

The body runs on third-class levers, so it lives permanently below 1, and it pays for that in muscle size: the quadriceps and the biceps routinely generate several times the force of the load they move.

人体主要运行在三类杠杆上，所以永远处于小于1的状态，并为此用肌肉体积来支付：股四头肌和肱二头肌产生的力routinely是它们所

To improve advantage you either increase the effort arm or decrease the load arm.

要改善优势，要么增加动力臂，要么减小阻力臂。

In the body that means moving the muscle attachment further from the joint, or moving the body part being moved closer to the joint.

在身体中，这意味着把肌肉附着点移得离关节更远，或者把被移动的身体部位移得离关节更近。

Both are visible in technique: locking the elbow out changes the moment arm of the biceps, and standing up from a chair changes the moment arms of the hip and knee extensors.

两者在技术中都能看到：锁定肘关节会改变肱二头肌的力臂，而从椅子上站起来会改变髋和膝伸肌的力臂。

This is why a small postural change can produce a large change in what a lift feels like.

这就是为什么一个很小的姿势变化能带来举起重量时感觉上的很大变化。

Equipment applies the same arithmetic externally.

装备在外部应用同样的算术。rowing桨上更长的杠杆在入水瞬间提高机械优势；

A longer lever on a rowing shaft increases mechanical advantage at the catch, a lower gear on a bicycle reduces the force needed for a given power, a stiff shaft changes swing weight and therefore the force needed to accelerate it, and a pole vault changes its advantage through the whole movement as the pole bends and unbends.

自行车更轻的档位降低给定功率所需的力；更硬的桨身改变挥重从而改变加速它所需的力；

In each case the athlete feels something different and the underlying principle is the same: you are choosing where the work comes from.

撑杆跳在整个过程中随着杆的弯曲与回弹改变其优势。

在每一种情况下运动员感觉到的都不同，而底层原则相同：你在选择这份工作从哪里来。

Equipment that fits is a safety device as much as a performance one.

合适的装备既是性能装置，也是安全装置。

A shoe that does not fit the foot, or a bike set up for someone else proportions, creates small unpredictable movements at exactly the moment the athlete needs control, and small unexpected movements at high load are how most equipment-related injuries happen.

一双不合脚的鞋，或一台按别人身材调好的自行车，会在运动员最需要控制的时刻制造出微小而不可预测的动作，而在高负荷下微小

Correct setup is not a marginal gain; it is the removal of a source of variance.

正确的调校不是边际收益，而是消除一个变异的来源。

The numbers / 算一算

Numbers to keep. Effort arm divided by load arm gives the advantage, and for human third-class levers the value is commonly around 0.2 to 0.3, which is the reason muscles generate several times the external load. A meaningful equipment change can be worth a surprisingly large fraction of that: changing saddle height or crank length alters the effective leverage and therefore the force at the pedal for a given power, and those changes are usually worth more than any exotic equipment. The relationship to remember for power is that power is force times velocity, so equipment that increases one may reduce the other, which is the real reason a very light gear cannot produce a very high power at a low cadence.

需要记住的数字。动力臂除以阻力臂给出优势，而人体三类杠杆的这个值通常在0.2到0.3左右，这就是肌肉要产生数倍外负荷的原因。

Common mistakes / 易错点

- "Mechanical advantage is a force." It is a ratio of two arms. Force out is effort multiplied by the advantage, and the effort is what the athlete actually has to produce.

"机械优势是一个力。"它是两个臂的比值。输出的力是动力乘以优势，而动力才是运动员真正要产生的东西。

- "A longer lever always makes it easier." It increases force advantage and reduces speed and range. A longer lever is not automatically better; it is a different machine.

"杠杆越长总是越省力。"它增加力量优势，同时减少速度和幅度。更长的杠杆并非自动更好，而是另一台机器。

- "Only the weight on the bar matters." Lever geometry, body proportions, grip, joint angles and setup change the force required for a given weight, sometimes by a large amount.

"只有杠铃上的重量重要。"杠杆几何、身体比例、握法、关节角度和调校都会改变举起给定重量所需的力，有时幅度很大。

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"Expensive equipment is better for performance." Well-fitted standard equipment usually beats poorly fitted expensive equipment, because fit removes variance and variance is what causes loss of control and injury.

"昂贵的装备性能更好。"贴合良好的标准装备通常胜过不贴合的昂贵装备，因为贴合消除变异，而变异正是失去控制与受伤的原因。

Why it matters / 为什么重要

Mechanical advantage and equipment sit at the point where biomechanics becomes a shopping decision, which is why they matter more than students expect. The athlete who understands that a lever is a ratio of two distances can work out for themselves whether a piece of equipment is worth it, rather than taking a marketing claim on trust. It also explains the training of equipment over seasons: the correct bike, the correct pole, the correct shoe will not make an athlete better at the skill, but they will let the athlete practise that skill with less interference, and practice with less interference is what actually produces skill.

机械优势与装备处在生物力学变成购物决定的那个点上，这就是为什么它们比学生预期的更重要。理解杠杆是两个距离之比的运动员，可

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Longer bats, rackets, paddles and poles can increase reach, power, swing speed, propulsion or energy return, depending on the task.

更长的球棒、球拍、球板和撑杆可按任务需要增加触及范围、功率、挥拍速度、推进力或能量回收。

Key terms / 核心术语

Mechanical advantage 机械优势

Ratio of effort arm to load arm.

动力臂与阻力臂的比值。

Worked examples / 实例

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Moving the load farther from the fulcrum increases load arm and can reduce force advantage while increasing speed or range.

将阻力远离支点会增加阻力臂，可能降低力量优势并提高速度或范围。

Quick check / 快速检查

1. Which lever class gives the greatest force advantage and which gives the greatest speed advantage?

哪类杠杆力量优势最大？哪类杠杆速度优势最大？

Exam focus / 考核要点

Change $MA = EA / LA$

改变 $MA = EA / LA$

Newton's laws of motion

牛顿运动定律

Guiding question / 引导问题

How can forces explain linear and angular movement in sport?

力如何解释运动中的直线和角运动？

Newton's three laws link force to inertia, acceleration, action and reaction. Linear and angular variables, stability, impulse, momentum, collisions, friction, work and power connect those principles to sporting performance.

牛顿三定律将力与惯性、加速度、作用和反作用联系起来。直线和角运动变量、稳定性、冲量、动量、碰撞、摩擦、功和功率把这些

Three laws and motion variables · 三大定律与运动变量

First, picture it / 先这样想

Physics is the study of what a moving thing will definitely do, and there are only three rules to remember. One: things keep going unless something stops them. Two: for every push there is an equal push back. Three: if you make something spin faster, it takes more force to keep it spinning. That is the whole of Newton. Everything else in this topic is about using those three sentences to explain something a coach or an athlete can see.

物理学研究的是一个运动中的物体必然会做什么，而你只需要记住三条规则。一：物体将继续运动，除非有什么阻止它。二：每一次推都有一个大小相等、方向相反的推。三：如果你让某个东西转得更快，保持它旋转就需要更大的力。这就是牛顿的全部。本主题其余内容，就是用这三句话去解释教练或运动员能看见的东西。

The detail / 详细讲解

The first law is inertia: a body stays at rest, or in constant motion in a straight line, unless acted on by a net external force.

第一定律是惯性：物体保持静止，或沿直线做匀速运动，除非受到合外力的作用。

In sport this is why a sprinter does not stop instantly at the line, why a ball keeps going after a boot stops touching it, and why a skater glides.

在运动中，这解释了为什么短跑运动员冲过终点线不会立刻停下、为什么球在脚停止接触后仍会继续前进、为什么滑冰运动员能滑行。

It is also the reason a hamstring straining during late swing is doing so while the leg is moving fast - the tissue has to decelerate a large mass, and the force required grows with speed.

它也是腘绳肌在摆动后期拉伤的原因：组织必须让一个很大的质量减速，而所需的力随速度增长。

The second law is $F = ma$: the acceleration of a body is proportional to the net force and inversely proportional to its mass.

第二定律是 $F=ma$ ：物体的加速度与合外力成正比，与质量成反比。

Force equals mass times acceleration.

力等于质量乘以加速度。

It explains why the same push moves a light ball further and faster than a heavy one, why a heavier athlete needs more force to reach the same speed, and why force applied over a longer time produces more change in velocity.

它解释了为什么同样的力让轻球比重球移动得更远更快、为什么体重更大的运动员需要更大的力才能达到同样速度、以及为什么把力

That last part is the entire basis of every striking action in sport - a cricket swing or a badminton smash works by extending the time over which the same force acts.

最后那一点是运动中每一个击球动作的完整基础--板球挥棒或羽毛球扣杀之所以有效，正是因为延长了同一个力作用的时间。

The third law is action and reaction: for every action there is an equal and opposite reaction, and the two act on different bodies.

第三定律是作用与反作用：每一个作用都有一个大小相等、方向相反的反作用，而这两个力作用在不同的物体上。

That last clause is the part that confuses people.

最后这半句是最让人困惑的地方。

When a runner pushes backwards on the ground, the ground pushes forwards on the runner - not the other way

round.

当跑者向后蹬地时，地向前推跑者--而不是反过来。

The swimmer pushes the water backwards and the water pushes the swimmer forwards.

游泳者向后推水，水向前推游泳者。

In every case the two forces are on different objects, which is why they never cancel out and why a moving object can keep moving.

每一种情况下两个力都作用在不同物体上，这正是它们不会相互抵消的原因，也是运动物体能够持续运动的原因。

The motion variables are what the laws act on, and describing them accurately is what earns the mark.

运动变量是这些定律作用的对象，而准确描述它们正是得分的要点。

Linear: displacement, velocity, acceleration.

直线变量：位移、速度、加速度。

Angular: rotation, angular velocity, angular acceleration.

角变量：旋转、角速度、角加速度。

The distinction is not academic - a javelin throw is mostly linear along the shaft with a small angular component, while a hammer throw is almost entirely angular.

这个区分不是学术性的--标枪投掷主要沿枪杆是直线运动、只带少量角运动，而链球则几乎完全是角运动。

Circular motion adds a fourth idea, centripetal acceleration, which points toward the centre of the circle and is what keeps a turning body on its path.

圆周运动还增加了第四个概念：向心加速度，它指向圆心，正是它让转动的物体保持在圆周轨迹上。

And knowing the variables is what lets you find the error in a movement.

而知道这些变量，正是找出动作中错误的方法。

A long jumper whose jump is short has a problem with velocity at take-off.

跳远距离短，问题在起跳瞬间的速度。

A discus thrower whose release is wild has a problem with release angle, which is a question about direction.

铁饼出手失控，问题在出手角度，也就是方向问题。

A gymnast landing badly usually has a problem with linear and angular deceleration, in that order.

体操运动员落地不稳，通常问题在于线性和角减速度的先后顺序。

Naming the variable turns a vague observation into a diagnosis, and a diagnosis is what a technical session can act on.

指出变量，就把一个含糊的观察变成了诊断，而诊断才是技术训练课能据以行动的东西。

The numbers / 算一算

Numbers to keep. Gravity is 9.8 m/s^2 near the surface and is the constant in every projectile calculation. A sprinter reaches roughly 10 m/s , which is 36 km/h , and a top speed of 12 m/s is about 43 km/h . A 100 m sprint lasts around 10 seconds from gun to line, so average speed is 10 m/s even though the top speed is higher. A golf ball struck at 70 m/s with a launch angle of about 20° travels a long way, which is why golfers change lofted and compressed clubs rather than simply swinging faster. The useful relationship is that impulse equals force multiplied by time, so a bat or a racket that stays in contact for longer sends the ball faster at the same force.

需要记住的数字。地面的重力加速度约为 9.8 米每二次方秒 ，是所有抛体计算中的常量。短跑运动员约达到 10 米每秒 ，即每小时 36 公里 ，

Common mistakes / 易错点

- "Inertia is the same as momentum." Inertia is a property of mass - its resistance to a change in motion. Momentum is mass multiplied by velocity, and it is conserved in a collision.
"惯性和动量是一回事。"惯性是质量的属性--抵抗运动变化的能力。动量是质量乘以速度，并且在碰撞中守恒。
- "Action and reaction forces cancel out." They never do, because they act on different bodies. They only appear to cancel when you look at a system as a whole.
"作用力和反作用力会相互抵消。"它们永远不会，因为它们作用在不同物体上。只有当你把整个系统作为一个整体来看时，它们才
- "A heavier object falls faster." It does not, in the absence of air resistance. Galileo settled this, and it is the

foundation of every projectile calculation.

"更重的物体下落更快。"在没有空气阻力的情况下不会，伽利略已经解决了这个问题，而这是所有抛体计算的基础。

"Describing a movement does not need the right terms." Naming the plane, the direction and the variable is the difference between an observation and a diagnosis, and it is where the marks are.

"描述一个动作不需要用准确的术语。"指出平面、方向和变量，就是观察与诊断的区别，而分数就在这里。

Why it matters / 为什么重要

Newton is the part of biomechanics that lets an athlete argue with a coach using arithmetic rather than opinion. It is also the part that makes every other topic in Theme B possible: impulse needs the third law, drag needs the second, leverage needs the first. The practical reason to learn it properly rather than approximately is that these three sentences are the vocabulary of every technical session for the rest of an athlete career, and a mis-named variable sends the correction to the wrong part of the movement.

牛顿力学是生物力学中让运动员能与教练用算术而不是用意见争论的部分。它也是主题B其他所有内容成为可能的基础：冲量需要第三定律。

Linear motion is described by speed, velocity, acceleration and momentum $p = mv$. Angular motion is described by angular velocity, angular acceleration, moment of inertia and torque.

直线运动用速度、速率、加速度和动量 $p = mv$ 描述；角运动用角速度、角加速度、转动惯量和扭矩描述。

Key terms / 核心术语

Inertia 惯性

Resistance to a change in state of motion.

改变运动状态的抵抗。

Momentum 动量

Quantity of motion, $p = mv$.

运动量， $p = mv$ 。

Worked examples / 实例

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A swimmer pushes backwards on the block, and the block pushes the swimmer forwards with an equal and opposite force.

游泳者向后推蹬板，蹬板以大小相等、方向相反的力推动游泳者向前。

Quick check / 快速检查

1. Compare linear velocity and angular velocity.

比较直线速度和角速度。

Exam focus / 考核要点

State Newton's three laws with a sport example.

说出牛顿三定律并给出运动例子。

Stability, impulse and collisions · 稳定性、冲量与碰撞

First, picture it / 先这样想

Two ideas sit behind this topic, and both are about giving way. Stability is about giving way the least. Impulse is about giving way the most, briefly. A wrestler wants to be immovable; a boxer's fist wants to move through you and then stop. In between is everything a body does when something hits it, and the whole thing is a trade-off between resisting and absorbing.

这个主题背后有两个概念，都与"让步"有关。稳定性讲的是让步最少；冲量讲的是让步最多、但时间极短。摔跤运动员希望自己纹丝不动；拳击手的拳头希望穿过你然后停下。介于两者之间的，是身体被击中时做的一切，而这整件事就是在"抵抗"与"吸收"之间的权衡。

The detail / 详细讲解

Stability depends on the line of gravity relative to the base of support, and the rule is simple: the closer the line of

gravity stays inside the base, the more stable the body.

稳定性取决于重力线相对于支撑面的位置，规则很简单：重力线越落在支撑面之内，身体越稳定。

Standing on one leg, the base of support is one foot, so stability is poor.

单腿站立时支撑面只有一只脚，所以稳定性很差；

Spreading the feet widens the base.

两脚分开可以加宽支撑面；

Lowering the centre of gravity improves it further, which is why a sprinter starts from a crouch.

降低重心可以进一步提高稳定性，这正是短跑运动员采用起蹲姿势的原因。

Moving the base under the centre of mass is what a skater does, and it is why a person who is falling can catch themselves by moving their feet in the direction of the fall.

把支撑面移到重心下方是滑冰运动员所做的事，也正是为什么一个正在摔倒的人可以通过把脚往摔倒的方向迈出去来救回自己。

Impulse is force multiplied by the time it acts, and it equals the change in momentum.

冲量是力乘以它作用的时间，等于动量的变化。

Force multiplied by a short time and force multiplied by a long time are the same force with very different consequences.

乘以短时间的力和乘以长时间的力是同一个力，后果却大不相同。

A cricket ball arriving at 40 m/s has an enormous momentum; a tennis ball arriving at the same speed has far less, because the mass is different.

以每秒40米飞来的板球有巨大的动量；

Then a racket that extends the contact time increases the impulse, which is why a larger, softer racket face sends the ball faster for the same swing speed.

以同样速度飞来的网球则小得多，因为质量不同。

而球拍延长了接触时间就增加了冲量--这就是更大的、更软的拍面在相同挥拍速度下能把球打得更快的原理。

Collisions are where both ideas meet.

碰撞是这两个概念的交汇点。

A collision can be elastic, where momentum and kinetic energy are both conserved, or inelastic, where momentum is conserved but energy is converted into heat, sound and deformation.

碰撞可以是弹性的，动量和动能都守恒；

Almost every sporting collision is inelastic, which is why a ball that hits a wall and bounces back is slower than it arrived: it cannot be, because energy left the ball as sound and heat in the wall.

也可以是非弹性的，动量守恒但能量转化为热、声音和形变。

几乎所有运动中的碰撞都是非弹性的，这正是球撞墙反弹后速度比原来慢的原因：它不可能更快，因为能量以声音和热的形式留在了墙上。

Injury happens at the interface, and it is a race.

损伤发生在两者的交界处，而且那是一场赛跑。

The stiffer the tissue, the more it resists deformation and the more load it passes on to the next structure.

组织越僵硬，它抵抗形变的程度越大，传给下一个结构的负荷就越多。

A stiff ankle does not absorb the landing, so the knee takes it.

一个僵硬的踝关节不吸收落地，于是膝关节来承受。

Flexible, well-conditioned tissue absorbs and protects, which is why progressive loading and strength work on the tissues around a joint are injury prevention rather than performance work.

可活动的、状态良好的组织会吸收并保护，所以对关节周围组织做渐进负荷和力量训练是伤病预防，而不只是表现性训练。

And the practical variables are the ones you can change.

而真正有用的变量都是可以改变的。

A larger contact area lowers the peak force, which is why landing with two feet rather than one reduces injury risk even when the total energy is identical. A shorter contact time raises the peak force.

更大的接触面积会降低峰值力，所以两脚落地而不是单脚落地，即使总能量相同也能降低受伤风险。

A stiffer surface raises the peak force.

更短的接触时间会提高峰值力，更硬的地面也会提高峰值力，而姿势会改变整条链的几何关系。

Position changes the geometry of the whole chain. Every one of those is a decision a coach or an athlete can make in the moment before impact.

这些全都是教练或运动员在撞击前那一刻可以做出的决定。

The numbers / 算一算

Numbers to keep. A cricket ball travels at roughly 35 to 40 m/s, a tennis serve at around 40 to 45 m/s, and a soccer ball kicked hard at about 30 m/s. Impulse from a short-duration collision is large even when the force looks modest, because the product matters. For balance, the practical figure is the base of support: a single foot gives roughly the area of one foot, and the margin of stability is how far the centre of gravity can be displaced before the line of gravity leaves the base - a few centimetres on one foot, several times that with a wide stance. A useful coaching number is that landing with two feet instead of one can roughly halve the peak force on the lower limb for the same landing velocity.

需要记住的数字。板球的速度约为每秒35到40米，网球发球约每秒40到45米，用力踢出的足球约每秒30米。即使力看起来不大，短时间

Common mistakes / 易错点

- "Stability is the same as strength." Stability is geometry - where the line of gravity falls relative to the base. Strength affects how much you can resist once you start to move.
"稳定性就是力量。"稳定性是几何关系--重力线落在支撑面的什么位置。力量影响的是一旦开始移动之后你还能抵抗多少。
- "Impulse is a large force." Impulse is force multiplied by time, so a moderate force sustained briefly and a large force applied instantly can produce the same change in momentum.
"冲量就是很大的力。"冲量是力乘以时间，所以短暂维持的中等力和瞬间施加的大力可以产生同样的动量变化。
- "A ball must bounce back at the same speed." Only in an ideal elastic collision does kinetic energy return. Every real sporting collision is inelastic, so the rebound is always slower.
"球必须以同样速度弹回。"只有在理想的弹性碰撞中动能才会返回。真实的运动碰撞都是非弹性的，所以反弹速度总是更慢。
- "Stiff tissue is stronger tissue." Stiffness is not strength. Stiff tissue resists deformation and passes load on to the next structure, which is how a stiff ankle becomes a knee problem.
"僵硬的组织就是更强的组织。"僵硬不等于力量。僵硬组织抵抗形变，把负荷传给下一个结构--僵硬的踝关节变成膝关节问题就是

Why it matters / 为什么重要

Stability, impulse and collisions are the physics of every impact in sport, and they are the difference between an injury that happens and an injury that could have been designed out. Every technique in every contact sport is an attempt to control these variables: widen the base, soften the landing, use two feet, extend the contact time, meet the load with a larger surface. None of it requires exceptional strength. It requires knowing which variable to change, and that knowledge is worth more to an athlete than almost any other technical detail, because it is the difference between a career with injuries and a career without.

稳定、冲量和碰撞是运动中每一次撞击的力学，而它们决定了某个伤病是发生了，还是本来可以被设计掉。每一种接触运动的每一项技术

- A penalty kick uses a force over a contact time to change ball momentum. A tennis forehand applies force away from the axis to create torque and spin.
罚球通过一段时间的力改变球的动量；网球正手在远离轴的位置施力，产生扭矩和旋转。

Key terms / 核心术语

Torque 扭矩

Turning effect of a force about an axis, $T = r \times F$.

力绕轴产生的转动作用， $T = r \times F$ 。

Worked examples / 实例

>

Apply impulse to a longer or shorter contact: the same force for longer time changes momentum more, while power is work per unit time.

将冲量应用于更长或更短接触：相同力作用更久会改变更多动量，而功率是单位时间内做的功。

Quick check / 快速检查

1. What quantities are conserved in a collision?

碰撞中什么量守恒?

Exam focus / 考核要点

Explain how impulse, work and power differ in a jump.

解释跳跃中冲量、功和功率的区别。

Friction, work and power · 摩擦、功与功率

First, picture it / 先这样想

Three everyday words that are actually precise physics. Friction is the tax you pay for touching something. Work is what you spend when you move a force through a distance. Power is how fast you spend it. A cyclist climbing a hill spends a lot of work and very little power. A cyclist winning a sprint spends almost no work and enormous power. Same word, completely different demands.

三个日常词语，其实是精确的物理概念。摩擦是你为了接触某样东西而缴纳的税；功是你用一个力移动一段距离所付出的东西；功率是你付出它的速度。一位爬坡的自行车手花了很多功，却几乎不花功率；一位冲刺获胜的自行车手几乎不花功，却花掉巨大的功率。同样的词，完全不同的需求。

The detail / 详细讲解

Friction exists because surfaces are not perfectly smooth, and it can help or hurt depending on which surface moves.

摩擦之所以存在，是因为表面并非绝对光滑；

The coefficient of friction is the ratio of the friction force to the normal force, and the number you choose depends entirely on what is in contact.

它究竟帮忙还是拖后腿，取决于哪一侧在滑动。

A stud on a boot is built for high friction with a ball.

摩擦系数是摩擦力与正压力之比，而取什么数值完全取决于接触的是什么。

A skin suit is built for low friction with the water.

足球鞋底的鞋钉是为了与球之间的高摩擦而设计的；

And the classic example: friction is what lets you grip the ground when you push off, and it is also what a slide costs you in ski jumping, luge and curling.

竞速泳衣是为了与水之间的低摩擦而设计的。

而经典的例子是：摩擦让你蹬地时能抓住地面，它也是跳台滑雪、luge和冰壶运动中“滑”的代价。

Work is force multiplied by the distance it acts through, and a force that acts through no distance does no work at all.

功是力乘以它作用通过的距离，而作用在零距离上的力不做功。

This is worth dwelling on because it overturns an intuition.

这一点值得停下来想，因为它推翻了一个直觉：沉重的等长收缩产生了力但没有关节移动，所以严格按定义它不是力学意义上的功，

A heavy isometric hold produces force but no joint movement, so by the strict definition it is not work in the mechanical sense, even though the muscles are working hard and metabolically costly. Work in the mechanical sense is what changes the position of something, and it is why a slow heavy lift and a fast light lift can cost very different amounts.

力学意义上的功，是使某个东西位置发生改变的东西，这也正是缓慢的大重量举起和快速的轻重量举起代价可以截然不同的原因。

Power is work divided by time, or force multiplied by velocity, and it is the variable that decides explosive events.

功率是功除以时间，也就是力乘以速度，它是决定爆发性事件的关键变量。

It is the reason long jumps are long: the jumper needs high horizontal velocity at take-off, not more force.

这正是跳远跳得远的原因：跳远运动员需要的是起跳时的高水平速度，而不是更大的力；

It is the reason a heavier athlete can be beaten by a lighter one in a jump.

这也正是体重较大的运动员在跳跃项目上会被体重较小的击败的原因。

And it is the reason power training has to be trained with high intent, because power is specific - a slow effort at the same force produces no power gain.

而这同样是力量速度训练必须带着强烈意图去练的原因，因为功率是专项性的--同样的力做得慢，就不会产生任何功率增长。

The three interact in a way that matters for programme design.

这三者以一种对训练安排很重要的方式相互作用。

Friction determines how much force a surface will give you.

摩擦决定地面能给你多大的力；

Work determines how much total effort a task costs.

功决定一个任务总共消耗多少努力；

Power determines how fast you can spend it.

功率决定你能多快地把它们花掉。

A sprinter needs high power, low work, and specific friction at the start.

短跑运动员需要高功率、低功，以及起跑时特定的摩擦条件；

A marathon runner needs low power, very high work, and much less reliance on friction.

马拉松跑者需要低功率、极高的功，以及远没那么依赖摩擦。

An athlete who trains all three in the wrong proportions for their event will improve at something that is not their event.

一个按错误的配比训练这三者的运动员，会提升某项不是他比赛项目的能力。

And elasticity is the store that makes power possible, because power is a rate and a rate needs something to draw from.

而弹性是让功率成为可能的蓄能库，因为功率是一个速率，而速率需要有东西可供支取。

The tendon behaves like a spring: it stores energy during the loading phase and returns it during the shortening phase, and a meaningful fraction of jump and sprint power is returned this way rather than produced entirely by the muscle.

肌腱表现得像一根弹簧：在加载阶段储存能量，在缩短阶段释放，而跳跃与冲刺功率中有相当一部分是这样返回的，而不是完全由肌肉产生。

That is a direct practical link between tendon stiffness and explosive performance, and it is why a rigid, untrained tendon is both less powerful and more injury-prone.

这就在肌腱刚度与爆发表现之间建立了一个直接的实际联系，也解释了为什么一根僵硬、未经训练的肌腱既力量更小、又更容易受伤。

The numbers / 算一算

Numbers to keep. Ground friction on a sprint start is what converts a backwards push into a forwards acceleration, and it scales with the normal force and the surface. Mechanical work in a 100 m sprint is the horizontal force applied through the distance covered, and it accumulates across every stride even though the athlete feels only the explosive pushes. Power in a sprint can be many hundreds of watts per kilogram during the push-off phase and is essentially zero in the glide phase, which is why average power and peak power give very different pictures. And the practical number for programming: tendon stiffness adapts over weeks, muscle strength over weeks, and neural drive over days, which is why the first response to a new programme is neurological and the visible change comes later.

需要记住的数字。短跑起跑时的地面摩擦把向后的蹬地转化为向前的加速，而它随正压力和地面条件变化。100米中的力学功是水平力乘

Common mistakes / 易错点

- "Friction is a loss that is always bad." Friction is a loss at one interface and a gain at another. It is a loss for a ski and a gain for a sprint start.
"摩擦是损失，总是坏事。"在一个界面上摩擦是损失，在另一个界面上它是收获。对滑雪板是损失，对短跑起跑是收获。
- "Holding a heavy weight still burns a lot of energy, so it must be work." Physically it is not work in the mechanical sense, because no distance is covered. It is metabolically costly, which is a different statement.
"静止举着很重的东西消耗很多能量，所以一定是功。"在力学意义上它不是功，因为没有移动任何距离。它在代谢上代价很高，这
- "Power training is just strength training done faster." Power is specific. A slow effort at the same force produces little power and little power gain.
"力量训练练得更快就是力量速度训练。"功率是专项性的。同样力做得慢，功率很小，功率增长也很小。

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"More friction is always better for the athlete." The right amount depends entirely on the event. Grip is useful and sliding is the entire point in some sports.

"对运动员来说摩擦越大越好。"合适的量完全取决于项目。抓地在一种项目里有用，而在另一些项目里，滑动本身就是全部意义。

Why it matters / 为什么重要

Friction, work and power are the three numbers an athlete can change most directly, which is why they sit underneath almost every technical decision. The coach who knows them can explain why a start failed, why a jump was short, why one athlete can produce more than another at the same weight, and why a programme that only builds strength will not make anybody faster. It is also where equipment design lives, because every surface, every wheel, every boot and every suit is an attempt to set these three variables. Understanding them turns equipment choice and technique choice into the same decision.

摩擦、功和功率，是运动员最能直接改变的三个量，所以它们几乎处在每一个技术决策的底下。懂它们的教练能解释为什么起跑失败、为

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A sprinter can use a smooth surface and footwear to reduce unwanted friction while applying force in the direction of travel.

短跑者可以使用平整地面和合适的鞋减少不必要摩擦，同时沿运动方向施力。

Key terms / 核心术语

Coefficient of restitution 恢复系数

Ratio of relative separation speed to relative approach speed after collision.

碰撞后分离相对速度与接近相对速度之比。

Formulas / 公式

A ball can travel farther with higher release speed or a larger weight-to-drag ratio; increasing contact time can change impulse and power.

球以更高出手速度或更大重量/阻力比可飞得更远；增加接触时间会改变冲量和功率。

Worked examples / 实例

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How can technique and equipment optimise power?

技术和装备如何优化功率？

Exam focus / 考核要点

Compare friction, work and power in a sport.

在运动中比较摩擦、功和功率。

Fluid mechanics

流体力学

Guiding question / 引导问题

How do air and water forces alter projectile and body movement?

空气和水的力如何改变抛体与身体运动？

Fluid mechanics examines how gravity, drag, lift, buoyancy and thrust act in air or water. Initial velocity, release angle and height, environmental density, equipment and technique all influence performance and fairness.

流体力学研究空气或水中重力、阻力、升力、浮力和推力的作用。初速度、出手角和高度、环境密度、装备与技术都影响表现和公平。

Projectile motion and environmental forces · 抛体运动与环境作用力

First, picture it / 先这样想

Once something leaves your hand, it is on its own. This is the part of throwing that surprises people: you do not direct a projectile after release, you only choose where to let it go. Its path is then a curve, and the curve has two independent parts - how fast it left, which decides how far, and how high it left, which decides how long it stays up. Everything else is the air pushing back.

一旦东西离开你的手，它就自己管自己了。这是投掷项目最让人惊讶的部分：出手之后你无法再引导抛体，你只能选择让它从哪里离开。此后它的路径是一条曲线，而这条曲线由两个彼此独立的部分决定：出手速度决定远近，出手角度决定在空中待多久。其余一切，都是空气在往回推。

The detail / 详细讲解

A projectile follows two independent motions at once.

抛体同时进行两个彼此独立的运动。

Vertically it is pulled toward the ground by gravity, so it accelerates downward at about 9.8 m/s^2 and its height follows a predictable curve.

竖直方向上它被重力拉向地面，因此以约 9.8 m/s^2 的加速度向下加速，高度遵循一条可预测的曲线；

Horizontally, with no horizontal force and ignoring air resistance, it continues at constant speed.

水平方向上，在没有水平力且忽略空气阻力的情况下，它以恒定速度继续。

The independence is the key idea: gravity does not slow the horizontal motion, so a projectile thrown faster horizontally travels further without any change in its vertical behaviour.

独立性是关键：重力不会减慢水平运动，所以水平方向抛得更快的抛体会飞得更远，而它的竖直行为完全不变。

From those two ideas come the two numbers that matter.

由这两个概念得出两个重要数字。

Launch angle sets the optimum: on level ground without air resistance, 45 degrees maximises range, because the extra height gained at a steeper angle is exactly cancelled by the extra time spent getting there.

出手角度决定最优值：在水平地面、忽略空气阻力时，45度使射程最大，因为更陡角度带来的额外高度恰好被到达所需额外时间抵消。

But the optimum changes with everything real.

但最优值会随一切真实条件改变。

With air resistance, a javelin or a discus prefers a flatter angle of roughly 30 to 40 degrees, because a steep launch wastes energy fighting the air.

存在空气阻力时，标枪或铁盘更喜欢约30到40度这样较平的角度，因为太陡的出手会把能量浪费在与空气的对抗上。

A discus is released at around 30 to 35 degrees and a javelin at around 35 to 40.

铁盘的出手角约30到35度，标枪约35到40度。

Launch speed sets distance, and the relationship is roughly proportional: double the speed, roughly double the distance in the absence of drag.

出手速度决定距离，且大致成正比：在没有阻力时速度翻倍，距离大约也翻倍。

That is why throwing events are dominated by athletes who can produce high velocity, and why a technical improvement in the delivery usually shows up as a small increase in speed that produces a large increase in

range.

这就是投掷项目由能产生高速度的运动员主导的原因，也是为什么投掷技术上的改进通常表现为速度的小幅提升、却带来射程的大幅提升。

Then the environment starts to matter, and the two forces are the aerodynamic ones.

然后环境开始起作用，这两种力就是气动力。

Air resistance grows sharply with speed and with the area presented to the air, which is why a javelin is thrown point-first and why an athlete in a low-drag suit gains a few percent.

空气阻力随速度和迎风面积急剧增长，所以标枪是尖头先出、所以穿低阻力比赛服的运动员能获得几个百分点的优势。

Lift and side force come from the shape and angle of the object, and they curve a path sideways or upwards in ways the athlete cannot control directly.

升力和侧力来自物体的形状与角度，它们把路径向侧方或上方弯曲，运动员无法直接控制。

The numbers / 算一算

Numbers to keep. Gravity is 9.8 m/s squared. Without air resistance a 45 degree launch maximises range, and the formula is range equal to speed squared times sine of twice the angle, divided by gravity - which is worth understanding rather than memorising, because it shows immediately that angle and speed enter differently. Shot put release is around 12 to 13 m/s, discus around 25 m/s, javelin around 25 to 30 m/s, and hammer lower still at around 20 to 22 m/s because the ball is heavy. A javelin that is thrown 5 degrees flatter or steeper can lose several metres of range, and a release that is 5 percent slower loses considerably more, because speed enters the range formula squared.

需要记住的数字。重力为9.8米每二次方秒。没有空气阻力时45度出手使射程最大，公式是射程等于速度的平方乘以二倍角的正弦再除以重力。

Common mistakes / 易错点

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"45 degrees is always the best release angle." It is the optimum only on level ground with no air resistance.

Real throwing angles are 30 to 40 degrees because drag penalises a steep launch.

"45度总是最佳出手角。"它只是在水平地面、无空气阻力时的最优值。真实投掷角是30到40度，因为阻力会惩罚太陡的出手。

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"You can steer the projectile in flight." After release the only forces are gravity and air. Everything after release is decided before release.

"你可以在飞行中操控抛体。"出手后只有重力和空气两种力。出手之后发生的一切，在出手之前就已经决定了。

-

"A heavier implement flies further." It flies less far at the same speed, because drag depends on shape and area while the driving force is limited by what the athlete can produce. Weight in an implement is a trade, not a benefit.

"更重的器械飞得更远。"在相同速度下它飞得更近，因为阻力取决于形状和面积，而驱动力受运动员能产生的量限制。器械加重是

-

"Launch angle matters more than launch speed." In the range formula speed is squared and angle is not, so speed is the more powerful variable - which is why velocity is the priority in every throwing programme.

"出手角比出手速度更重要。"在射程公式中速度是平方项而角度不是，所以速度是更有力的变量--这就是为什么速度是每一个投掷项目

Why it matters / 为什么重要

Projectile motion is the most immediately satisfying thing in biomechanics, because the cause and the effect are separated in time and completely predictable. A coach can say "you are releasing at 32 degrees" and the athlete can go away and do something about it, which is not true of almost any other technical correction. It is also the clearest link between an athlete and their equipment, since release angle and release speed are both set by the implement. And it is the foundation of the fairness question that follows, because once you accept that the path depends on measurable variables, a rule about equipment becomes a rule about measurement.

抛体运动是生物力学中最直接令人满意的部分，因为原因和结果在时间上被分开，而且完全可预测。教练可以说"你的出手角是32度"，运

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Temperature, humidity, air pressure, wind, water salinity and altitude change fluid density and therefore force.

Headwinds increase drag; tailwinds reduce relative drag; salt water increases buoyancy; altitude reduces air density while also reducing oxygen availability.

温度、湿度、气压、风、水盐度和海拔改变流体密度及作用力。逆风增加阻力，顺风减小相对阻力，盐水增加浮力；海拔升高降低空

Key terms / 核心术语

Drag 阻力

Fluid resistance opposing relative motion.

阻碍相对运动的流体阻力。

Worked examples / 实例

>

A javelin released high and fast can travel farther, but a tailwind and low drag also affect official distance.

标枪出手更高、更快可飞得更远，但顺风 and 低阻力也会影响正式距离。

Quick check / 快速检查

1. Name three environmental factors that influence fluid forces.

说出三个影响流体作用力的环境因素。

Exam focus / 考核要点

What determines the flight of a projectile?

什么决定抛体的飞行？

Buoyancy, lift and drag · 浮力、升力与阻力

First, picture it / 先这样想

Water and air are fluids, and in a fluid the difference between floating and sinking is a single comparison. Ask whether the thing displaces enough fluid to outweigh the fluid it pushes aside. A boat wins that comparison; a stone loses it. The same question in the air explains why a plane flies and why a parachutist falls slowly. Drag is the tax the fluid charges for the shape you are presenting to it, and the clever shapes are the ones that cheat the tax.

水和空气都是流体，而在流体中，浮起与下沉的区别只是一次比较：问一样东西排开的流体是否足够抵消它压开的那些流体。船赢得这个比较，石头输掉。同样的问题放到空气里，就解释了飞机为什么能飞、跳伞者为什么下得慢。而阻力是流体对你呈现给它的形状所征收的税，聪明的形状就是那些能避开这笔税的形状。

The detail / 详细讲解

Buoyancy is the upward force a fluid exerts on an object that displaces it, and it is governed by one principle: the buoyant force equals the weight of the fluid displaced.

浮力是流体对排开它的物体施加的向上力，它由一条原理支配：浮力等于所排开流体的重量。

Because the object displaces a volume of fluid, and the fluid has a mass per unit volume, an object floats when the fluid it displaces outweighs the object.

因为物体排开一定体积的流体，而流体有单位体积的质量，所以当排开的流体比物体本身更重时，物体就浮起来。

That is the whole of Archimedes principle, and it is why a body fat percentage can be estimated by floating, and why a swimmer with a lower density floats more easily.

这就是阿基米德原理的全部，它解释了为什么可以用浮起来估计体脂率，也解释了密度较低的游泳者为什么更容易漂浮。

Lift in air works the same way in principle and differently in practice.

空气中的升力原理相同而实际不同。

Air moving past a surface at an angle creates a pressure difference, and that difference has a component perpendicular to the surface, which is lift.

空气以一定角度掠过表面会产生压力差，这个压力差中有一个垂直于表面的分量，那就是升力。

The wing shape, the angle of attack and the speed together determine it, and the four forces on a flying object are **lift up, weight down, thrust forward and drag back.**

翼型、攻角和速度共同决定它，而飞行物体上的四个力是：升力向上、重力向下、推力向前、阻力向后。

For a jumper or a skier the "wing" is the body, and the same physics applies to both, which is why a ski jumper in a tuck and a badminton shuttle after the hit are described in the same language.

对跳台滑雪者或撑杆跳运动员来说，那“翼”就是身体，而同样的物理适用于两者--这正是为什么收起身体的跳台滑雪者和被击打后的羽

Drag is the resistance a fluid offers to an object moving through it, and it depends on three things: how fast you are going, how much area you present, and how streamlined the shape is.

阻力是流体对在运动中运动的物体提供的抵抗，它取决于三件事：你的速度、你呈现的面积、以及形状有多流线。

The practical consequence is a hard ceiling on speed, because drag rises with roughly the square of velocity - double the speed and the aerodynamic resistance becomes roughly four times as large.

实际推论是一个关于速度的硬上限，因为阻力大致按速度的平方增长--速度翻倍，空气阻力就变成大约四倍。

That relationship is why elite sprinters and swimmers work so hard on reducing drag, and why once you are near the limit, tiny reductions in frontal area are worth more than large gains in power.

这个关系解释了为什么顶尖短跑和游泳运动员如此努力地减少阻力，也解释了为什么一旦接近极限，前面积的微小减少比力量的巨大增加更值得追求。

The two forces are not independent.

这两个力并非彼此独立。

A ski jumper in a V-style position increases lift by presenting a large surface to the air, but that same position also increases drag.

V字姿势的跳台滑雪者通过呈现较大迎风面来增加升力，但同样的姿势也增加了阻力。

The competition between the two is decided by the ratio of lift to drag, and the whole technical history of ski jumping, cycling and swimming has been a series of attempts to move that ratio.

两者的竞争由升阻比决定，而滑雪、骑行和游泳的整个技术史，就是一连串改善这个比值的尝试。

A cyclist in a tuck loses almost all frontal area and accepts a small loss of lift; a swimmer with a imperfect streamline pays drag for every degree of deviation.

收起身体的自行车手几乎失去全部迎风面积，同时接受了少量升力的损失；而动作划水稍有偏差的游泳者，会为每一个偏差付出阻力。

The numbers / 算一算

Numbers to keep. Air density is about 1.2 kg per cubic metre at sea level and falls with altitude and temperature, which is one reason a jumper goes further in Denver. Water is roughly 1000 times denser than air, so the buoyant force on a swimmer is enormous compared with the same body in air, and that is why swimming is a low-gravity-supported activity. Drag in air rises with the square of speed, while buoyant forces depend on displaced volume and are essentially independent of speed. And the practical number for ski jumping: the speed at the bottom of the hill can exceed 90 km/h before the athlete leaves the track, and the aerodynamic forces in flight are dominated by the position held in the air rather than by any further acceleration.

需要记住的数字。海平面空气密度约为每立方米1.2公斤，并随海拔和温度下降，这也是丹佛的跳台滑雪者能飞得更远的原因之一。水的密度是空气的1000倍。

Common mistakes / 易错点

- "A heavier object sinks faster." A dense object sinks faster because its density is higher, not because it is heavier. A steel ship floats because it is a hollow shape that displaces a great volume of water.
"更重的物体下沉更快。"密度大的物体下沉更快，是因为它的密度更高，而不是因为它更重。一艘钢船能浮，是因为它是中空形状，能排开大量的水。
- "Lift is a separate force that planes have and people do not." Lift and buoyancy are the same principle. The "wing" of a jumper or a swimmer is their own body, which is why their technique is described in the same language.
"升力是飞机才有而人没有的另一种力。"升力和浮力是同一个原理。跳台滑雪者或游泳者的"翼"就是他们自己的身体，所以他们的技术描述在同一个语言里。
- "Swimming fast is about power." It is at least as much about drag. A swimmer who improves technique can gain more than one who adds power, because drag rises with the square of speed.
"游得快靠的是力量。"至少有一半靠的是阻力。一位改善了技术的游泳者，收益可能超过一位增加力量的，因为阻力按速度的平方增长。
- "Dive with a straight body to go faster." A streamlined position reduces drag, but the body position that reduces drag most is the one appropriate to the moment. Speed through water is never free.
"伸直身体入水能更快。"流线型姿势减少阻力，但减少阻力最多的体位是适合当下那个时刻的体位。在水中获得速度从来不是没有代价的。

Why it matters / 为什么重要

Buoyancy, lift and drag are the reason three otherwise strange activities exist. A ski jumper has no aerodynamic engine at all - the entire flight is a shape holding an attitude against moving air. A swimmer is a body in a buoyant medium where almost everything they do is a negotiation with drag. And an aircraft is a wing that is continuously

trading lift against drag. Understanding those three forces turns those sports from puzzling into adjustable, because each one has a small number of variables that a coach and an athlete can actually change.

浮力、升力和阻力，解释了三项本来看起来很古怪的运动为什么会存在。跳台滑雪者没有任何空气动力引擎--整段飞行就是一个形状在对

-

Streamlined body position reduces form drag; a smooth stroke can reduce resistance and improve propulsion.

流线型身体位置可减少形状阻力；平滑划水可减少阻力并提高推进力。

Key terms / 核心术语

Buoyancy 浮力

Upward force from a fluid.

流体产生的向上作用力。

Lift 升力

Force that can redirect movement.

可改变运动方向的作用力。

Formulas / 公式

A swimmer lowers the head and presents a smoother body line; evaluate the likely effect on drag.

游泳者降低头部并使身体线条更流线；评价其对阻力的可能影响。

Worked examples / 实例

>

Explain one technique that reduces drag and one ethical or regulatory issue.

解释一种减阻技术和一个伦理或规则问题。

Exam focus / 考核要点

Define buoyancy, lift and drag with an example.

定义浮力、升力和阻力并举例。

Angle of attack, Magnus effect and fairness · 攻角、马格努斯效应与公平性

First, picture it / 先这样想

Angle of attack is the angle at which a moving object meets the air, and it is the single most abused idea in sport. Turn it too shallow and you generate drag without lift. Turn it too steep and you generate lift until the airflow separates and the lift collapses. In between is a narrow working range, and most of the technical skill in sailing, diving and swimming lives inside it. And there is a second, stranger effect: a spinning ball does not just push air, it drags some of it around with it.

攻角是一个运动物体迎向空气的角度，也是运动项目里最常被滥用的概念。角度太平，你产生阻力却没有升力；角度太陡，你产生升力直到气流分离、升力崩塌。两者之间是一个狭窄的工作区间，而帆船、跳水与游泳中的大部分技术能力都活在这个区间里。还有第二个更奇特的效果：一个旋转的球不只是推开空气，它还把一部分空气带着一起转。

The detail / 详细讲解

Angle of attack is the angle between the chord line of a wing and the direction of the oncoming airflow.

攻角是翼的弦线与迎面气流方向之间的夹角。

As it increases, lift increases roughly linearly, which is the principle a foil or a sail works on.

随着攻角增大，升力大致线性增加，这正是帆或翼板工作的原理。

But past a critical angle the flow can no longer stay attached to the surface, it separates, and the lift falls off sharply while drag rises steeply.

但超过某个临界角后，气流无法继续附着在表面上，它会分离，升力急剧下降而阻力陡升。

That collapse is called stall, and it is why a plane that has been diving needs to level off before it can climb again, and why a swimmer or a sailor who over-rotates a limb or a sail loses everything at once.

这个崩塌叫“失速”，这就是为什么俯冲中的飞机必须先改平才能重新爬升，也是为什么游泳者或帆手过度旋转肢体或帆会一次性失去所

That relationship is why the same angle means different things in different sports.

这种关系解释了为什么同一个角度在不同项目中含义不同。

A swimmer maximising lift wants a high angle; a swimmer minimising drag wants a low one.

追求最大升力的游泳者要高科技，追求最小阻力的要低技术角。

A sail trimming for speed wants an efficient angle, while a sail driving through an upwind race wants maximum lift.

为速度调帆的选手要有效攻角，而在顶风比赛中顶风驱动的帆手要最大升力。

And a ski jumper adjusts body angle in flight to keep the airflow attached, which is why the V-style exists: it presents a large surface at an angle that is still attached.

跳台滑雪者在空中调整身体角度以保持气流附着，这正是V字姿势存在的原因：它以仍然处于附着区的角度呈现一个大面积。

The Magnus effect is a spin producing a sideways force.

马格努斯效应是旋转产生的一个侧向力。

A ball spinning one way drags air around with it on that side, which makes the airflow faster on one side and slower on the other, and the pressure difference pushes the ball sideways.

球向一个方向旋转，就把那一侧的空气带着一起转，使该侧气流更快、另一侧更慢，压力差就把球推向侧面。

This is why topspin dips a ball and backspin floats it, why a swerve in football or a banana kick from a defender works, and why a spinning table tennis ball changes direction after the bounce.

这就是上旋球会下坠、下旋球会飘、足球中的任意球或防守者的香蕉球有效、以及旋转的乒乓球在弹跳后会改变方向的原因。

The sign of the effect follows the direction of rotation, which is why the same spin can be a weapon or an error depending on who applies it.

效应的符号跟随旋转方向，所以同一个旋转在谁施加、施于何处不同，可以是武器也可以是失误。

And that leads to the fairness question, which is a genuinely interesting problem in sport science.

这就引出了公平性问题，而它是运动科学中一个真正有意思的问题。

The trajectory of a spinning ball is measurably different from a non-spinning one over a very short distance.

旋转球的轨迹与不旋转球相比，在极短距离上就有可测量的差异。

Topspin tennis serves travel differently from flat serves, and in baseball a pitcher's spin axis has measurable effects the batter cannot counter with the same technique used against a non-spinning pitch.

上旋网球发球的飞行方式不同于平击发球；

If some players have access to more or more effective spin, the difference is not a small advantage; it is a change in the physics of the contest.

在棒球中，投手的旋转轴有可测量的效果，而打者无法用对付不旋转球同样的技术来应对它。

如果某些选手能获得更多或更有效的旋转，这个差别就不是小的优势，而是比赛物理规律的改变。

The numbers / 算一算

Numbers to keep. The critical angle of attack for a typical wing is somewhere around 12 to 20 degrees, depending on the section, and stall begins past it. Tennis serve topspin commonly produces thousands of revolutions per minute, and the resulting Magnus force is a significant fraction of the ball weight at high spin rates. In baseball, spin axis tilt of a few degrees produces a measurable change in movement, which is why a four-seam fastball and a two-seam fastball from the same release point can have visibly different trajectories. The practical number for fairness is the spin rate in revolutions per minute, because it is measurable, repeatable and clearly outside the normal course of a skill that was always played without it.

需要记住的数字。典型翼面的临界攻角大约在12到20度之间，取决于翼型，失速在超过它之后开始。网球发球的上旋通常达到每分钟数千转。

Common mistakes / 易错点

- "More angle of attack is always more lift." Up to a critical angle. Past it the flow separates and lift collapses while drag rises - that is stall.
"攻角越大升力越大。"只到临界角为止。超过它气流分离、升力崩塌而阻力上升--那就是失速。
- "Spin is a cosmetic effect on a ball." The Magnus force is measurable and is a significant fraction of ball weight at high spin rates, and it changes the trajectory over metres rather than centimetres.
"旋转对球只是装饰效果。"马格努斯力是可测量的，在高转速下是球重的一个相当可观的比例，而且它在数米而非数厘米的尺度上。
- "Effective spin is a skill anyone can develop." Spin rate is highly individual, it is partly a function of the implements and balls used, and a few degrees of spin-axis tilt can produce an advantage that technique cannot

neutralise.

"有效旋转是任何人都能发展出来的技术。"转速个体差异极大，部分由所用器械和球决定，而几度的旋转轴倾角就能产生技术无法

"A sailing trim that works upwind is the right trim in every condition." Maximum lift and minimum drag are different targets, and the right angle changes with wind, sea state and the leg of the race.

"顶风有效的调帆在所有条件下都是对的。"最大升力和最小阻力是不同的目标，而正确的角度随风况、海况和比赛段落变化。

Why it matters / 为什么重要

Angle of attack and the Magnus effect are where biomechanics becomes a question of fairness, which is a rare and important thing for a student to meet. The physics of a spinning ball is not a matter of opinion, and once it is measured, the question of whether every competitor should have access to it becomes concrete rather than philosophical. The same is true of equipment more broadly: a technology that measurably changes the physics of the contest belongs in a different category from a technology that only makes existing physics easier to exploit. Knowing which category something falls into is a genuinely professional skill.

攻角与马格努斯效应，是生物力学变成一个公平性问题的所在--这是学生很少遇到、却很重要的一件事。旋转球的物理不是意见问题，而

Equipment rules may limit technological advantage, and wind readings affect official records in athletics.

装备规则可能限制技术优势，风速读数会影响田径正式纪录。

Key terms / 核心术语

Angle of attack 攻角

Angle between an object and oncoming fluid flow.

物体与来流之间的夹角。

Magnus effect 马格努斯效应

Pressure difference caused by spin.

旋转造成的压力差。

Formulas / 公式

A spinning football can curve, but a high angle can also increase drag; equipment and environmental effects must be interpreted fairly.

旋转足球可以弯曲飞行，但过大攻角也会增加阻力；装备和环境影晌应公平解读。

Worked examples / 实例

>

How do angle of attack and spin influence lift?

攻角和旋转如何影响升力？

Exam focus / 考核要点

Describe three types of drag and their implications.

描述三种阻力及其影响。

Movement analysis and applications

动作分析与应用

Guiding question / 引导问题

How can breaking a movement into phases improve performance, safety and accessibility?

将运动分解为阶段如何提高表现、安全和无障碍参与？

Phase analysis divides a discrete skill into preparatory, force-production, critical-instant and follow-through or recovery phases. Observation, video, technology and athlete feedback then form an evidence loop for diagnosis, intervention and refinement.

阶段分析将离散技能分为准备、发力、关键瞬间以及随动或恢复阶段。观察、视频、技术和运动员反馈形成证据循环，用于诊断、干

Phases and the diagnosis loop · 动作阶段与诊断循环

First, picture it / 先这样想

Breaking a skill into phases is like putting frames around a film so you can stop the projector. In a jump shot, or a serve, or a swing, the same movement always happens in the same order - and each phase has its own failure mode. The useful consequence is that "the shot is bad" becomes "the landing is late", and a vague complaint becomes something a coach can act on this week.

把一个技术拆成阶段，就像给一部电影加上画格，好让你能把放映机停下来。在跳投、发球或挥拍中，同一个动作总是按同样的顺序发生--而每个阶段都有自己典型的失误方式。有用的推论是："这个投篮不行"变成了"落地晚了"，一个含糊的抱怨变成了教练本周就能处理的东西。

The detail / 详细讲解

Every discrete skill can be divided into phases, and the usual pattern is preparation, then force production, then a **critical instant**, then **follow-through**.

每一个离散的技术动作都可以拆成阶段，通常的模式是准备、发力、关键时刻、以及随动。

The critical instant is the shortest - often under a tenth of a second - and it is disproportionately important, because it is where the outcome is effectively decided and where the most force is generated.

关键时刻是最短的--常常不到十分之一秒--却承担了不成比例的重要作用，因为结果实际上就在那里被决定，最大力也正是在那里产生

Everything before it is preparation and everything after it is a consequence.

它之前的一切是准备，它之后的一切是结果。

The practical value is diagnostic, because each phase has a characteristic error.

它的实际价值在于诊断，因为每个阶段都有典型的错误。

A preparation error produces a shot that is off-line from the start.

准备错误产生一开始就偏的投篮；

A force-production error produces a shot that is straight but short.

发力错误产生方向对但距离不够的投篮；

A critical-instant error produces a shot that starts on line and then deviates.

关键时刻错误产生先对线然后偏掉的投篮；

A follow-through error produces a shot that reaches the target and then gets pushed off it.

随动错误产生到达目标又被推开的投篮。

Those four failure patterns look identical from the sideline and require completely different corrections, which is exactly why "shoot better" is not a coaching instruction.

这四种失误模式在场边看起来一模一样，却需要完全不同的纠正--这正是"投得更好"不构成一条教练指令的原因。

Video changes what is possible here, because most of these errors happen faster than the coaching eye can resolve.

录像改变了这里能做的事，因为大多数这类错误发生得比教练的眼睛所能分辨的更快。

What used to be an impression becomes a measurement: the release point relative to the body, the angle of the ankle at contact, the time between the last plant and the jump.

原本是印象的东西变成了测量：出手点相对身体的位置、触球时脚踝的角度、以及最后一次支撑与起跳之间的时间。

And a measured error can be tracked, so a correction can be shown to have worked or to have not.

而被测量的错误可以被追踪，所以纠正是否有效是可以被证明的。

The loop is what turns a phase analysis into progress: record, identify which phase failed, change one thing about that phase, record again.

循环才是把阶段分析变成进步的东西：录像、确定是哪个阶段出了问题、只改那个阶段的一件事、再录像。

Changing one thing at a time matters, because when two things change together and the shot improves, you have learned nothing you can use.

每次只改一件事很重要，因为当两件事同时改变而投篮变好时，你学不到下周能用的东西。

And the loop is iterative rather than linear, because fixing one phase frequently reveals that a different phase, which was fine before, is now the limiting one.

而这个循环是迭代的而不是线性的，因为修好一个阶段经常会暴露出另一个阶段--它之前本来没问题--现在成了限制因素。

The numbers / 算一算

Numbers to keep. The critical instant in a jump shot or a serve is typically under 0.1 seconds, which is why it cannot be coached by talking to the athlete during the attempt - the feedback has to come from a recording. Planting-to-jump intervals in a jump shot are on the order of a second, and shortening that interval without losing the plant is one of the more reliable ways to increase take-off. The practical measurement habit is to fix one camera position, one attempt and one phase at a time, because comparing recordings from different angles is worse than not recording at all. And the number that matters most is not in the technique: it is the conversion or completion rate, because a technically perfect phase that does not change the outcome is not what was being trained.

需要记住的数字。跳投或发球中的关键时刻通常不到0.1秒，这正是为什么它无法靠在动作过程中对运动员说话来指导--反馈必须来自录像。

Common mistakes / 易错点

- "The shot is bad, so work on the shot." That is not an instruction. Splitting the movement into phases converts it into a specific, correctable problem.
"投篮不行，所以练投篮。"这不是一条指令。把动作拆成阶段，才能把它变成一个具体、可纠正的问题。
- "Every phase is equally important." The critical instant is a small fraction of the movement and a large fraction of the outcome, and some phases matter far more than others for a given task.
"每个阶段同样重要。"关键时刻只占动作很小一部分，却占结果很大一部分，而且对特定任务来说有些阶段远比另一些重要。
- "Change two things and see what happens." If both change and the shot improves, you cannot attribute the improvement, and next time you will not know what to repeat.
"同时改两样然后看效果。"如果两样都改而结果变好，你就无法归因，下次也不知道该重复什么。
- "Video is a substitute for coaching." Video supplies the measurement. Coaching supplies the choice of which phase to work on and how to progress it.
"录像 是 教练的替代品。"录像提供测量。教练提供"该练哪个阶段、以及如何递进"的选择。

Why it matters / 为什么重要

Phase analysis is the most transferable technique in this course, because it applies to any skill that has an order to it, and because it converts an argument into a plan. Two coaches watching the same athlete can both see that the shot is short and still be right about different things, and phase analysis is what lets them find out which is the actual problem. It is also the honest answer to a question every athlete asks, which is "what exactly should I change", and it is the foundation of the diagnosis loop that follows.

阶段分析是这门课程里迁移性最强的技术，因为它适用于任何有先后顺序的技术，也因为 它把一场争论变成了一个计划。两位教练看着同

- Observe, analyse phases, diagnose causes, intervene and review. Small improvements in each phase can create a large overall change.

观察、分析阶段、诊断原因、干预并复查。每个阶段的小改进可能带来整体大变化。

Key terms / 核心术语

Critical instant 关键瞬间

The event that determines the skill outcome.

决定技能结果的关键事件。

Formulas / 公式

If contact is late, the intervention may be to change preparation or timing, then retest rather than adding an unrelated cue.

若击球偏晚，干预可以是改变准备或时机，然后重新测试，而不是添加无关提示。

Worked examples / 实例

>

Explain how phase analysis improves performance and safety.

解释阶段分析如何提高表现和安全。

Exam focus / 考核要点

Name the four phases of a discrete skill.

说出离散技能的四个阶段。

Evidence sources and compensation · 证据来源与代偿

First, picture it / 先这样想

Diagnosis has two halves, and the second half is the one people skip. Finding out that a knee hurts is half the job; working out whether it is weak, stiff, or strong-but-overworked changes what you do about it entirely. The two halves need different skills - one needs a good history, the other needs a good pair of hands and the willingness to be wrong. The tissue has been adapting quietly for years; compensation is what it did to keep working.

诊断有两个半程，而人们常跳过第二个半程。查出膝盖会痛只是一半工作；弄清楚它是弱、还是僵、还是强壮但被过度使用，会完全改变你的处理方式。两半需要不同的技能--一半需要好的病史采集，另一半需要一双好手，以及愿意承认自己会错的意愿。组织已经安静地适应了很多年，而代偿就是它为了继续工作所做出的改变。

The detail / 详细讲解

The assessment side is a differential: a list of possible causes kept live rather than narrowed to one early.

评估这一侧是鉴别诊断：把一组可能的原因保持开放，而不是过早收窄到一个。

Range of movement, strength, control, and the quality of the movement pattern all contribute information, and the useful habit is to test hypotheses rather than confirm one.

关节活动度、力量、控制能力以及动作模式的质量都提供信息，有用的习惯是检验假设而不是证实某一个。

The treatment side is entirely different: it asks what the tissue is doing, not what is broken.

治疗这一侧则完全不同：它问的是组织在做什么，而不是什么坏了。

Tissue adapts to the demands placed on it, and it adapts by changing structure, which is called compensation.

组织会对施加在它身上的需求作出适应，而它通过改变结构来适应，这就叫代偿。

So a painful knee might be a knee that has been doing someone else work.

所以一个疼的膝盖，可能是正在替别人干活的膝盖。

If the hip is weak, the gluteal muscles stop controlling femoral rotation, the femur rotates slightly inward in every step, and the medial knee compartment takes load it was not designed for.

如果髋部无力，臀肌就不再控制股骨旋转，每一步股骨都会略微内旋，内侧膝间室就承担了它并非为之设计的负荷。

Nothing is torn.

什么都没有撕裂。

The tissue is healthy and it is simply being used in a way it will eventually not tolerate.

组织是健康的，只是正在以它最终无法长期承受的方式被使用。

That is compensation, and it explains the majority of running injuries that have no incident.

这就是代偿，它解释了大多数没有明确外伤事件的跑步损伤。

The evidence sources differ in kind, not just in quality.

不同证据来源的差别不只是质量，而是种类。

A reliable athlete history is often the single most useful thing available, because it tells you the mechanism, the onset, the behaviour and what has already been tried.

可靠的运动员病史常常是现有最有用的单一信息，因为它告诉你机制、起病方式、行为特点以及已经尝试过什么。

Physical testing gives you strength and range numbers.

体格测试给出力量和活动度的数字。

Movement observation gives you the pattern.

动作观察给出模式。

Imaging gives structure.

影像给出结构。

The error to avoid is weighting one source too heavily: a normal scan does not exclude a mechanical cause, and a small imaging finding that is common in asymptomatic people does not explain a pain.

要避免的错误是过度依赖某一种来源：正常的影像并不能排除力学原因，而无症状人群中同样存在的小影像发现也并不能解释疼痛。

Management follows the diagnosis, and the sequence matters: reduce the load, restore range, rebuild strength, then retrain the pattern and the capacity.

处理方案跟随诊断，而顺序很重要：降低负荷、恢复活动度、重建力量，然后重新训练动作模式和容量。

Skipping a step is the commonest reason recovery stalls, and it is nearly always done because the athlete feels fine.

跳过某一步是恢复停滞的最常见原因，而几乎总是因为运动员自己感觉已经好了。

The tissue feels fine long before it is ready, because the other tissues have started compensating again.

在组织真正准备好之前，它会感觉良好--因为其他组织已经重新开始代偿了。

The numbers / 算一算

Numbers to keep. A reliable mechanism and a clear onset date are worth more than most tests, because they tell you whether to expect a tissue problem, a capacity problem or a control problem, and each implies a different timeline. Strength deficits of 20 to 30% are commonly meaningful enough to change a movement pattern, which is why testing a specific deficit and then retesting it is more informative than a general impression of weakness. And the practical rule for return to activity: progress the load, not the time. Symptoms settling in a week means the tissue is tolerating that load, not that it is ready for the next step.

需要记住的数字。一个可靠的机制和清楚的起病时间，比大多数检查更有价值，因为它们告诉你该预期组织问题、容量问题还是控制问题。

Common mistakes / 易错点

- "The scan was clear, so it is not a mechanical problem." A normal scan does not exclude a mechanical cause, and a large share of persistent pain is mechanical rather than structural.
"检查没问题，所以不是力学问题。"正常的检查不能排除力学原因，而相当大比例的持续性疼痛是力学性的而不是结构性的。
- "Pain equals damage." Pain is a poor guide to tissue state. Pain can persist after tissue is healed, and it can be absent while a tissue is being progressively overloaded.
"疼痛等于损伤。"疼痛是组织状态的糟糕指示。组织愈合后疼痛可以持续，而组织被渐进过度负荷时疼痛也可能不存在。
- "Fix the pain, then return to sport." Pain relief is not readiness. Returning on pain relief alone is the most common route to re-injury.
"治好疼就可以回运动了。"疼痛缓解不等于准备好。仅凭疼痛缓解就回归，是再伤最常见的路径。
- "Tightness needs stretching." Some tightness is protective, and forcing range before the tissue is ready can move the problem rather than remove it.
"紧就需要拉伸。"有些紧张是保护性的，在组织准备好之前强行增加活动度会把问题挪走而不是消除它。

Why it matters / 为什么重要

Assessment and diagnosis are the most portable skills in sports science, because they transfer to any population, any sport and any body, and because they are the part of the work that compounds. Knowing how to take a history, form hypotheses, test them and change one variable at a time is a way of thinking, not a piece of knowledge about a knee. It is also the part students most often skip in favour of memorising named conditions,

which is the wrong trade: the conditions are looked up, and the thinking is what the athlete in front of you actually needs.

评估与诊断是运动科学中最可迁移的技能，因为它适用于任何人群、任何项目、任何身体，而且因为它是那份会复利的工作。知道如何开

-

Common compensation signs include asymmetry, reduced range, poor timing or sequencing, excessive joint load, pain and discomfort. Possible impacts include lower performance, higher energy cost and overuse injury.

常见代偿包括不对称、活动范围减少、时机或顺序差、关节负荷过大、疼痛和不适；可能影响表现、增加能量成本和过度使用损伤。

Key terms / 核心术语

Notational analysis 记谱分析

Systematic recording of observed movement features.

系统记录观察到的动作特征。

Formulas / 公式

Use athlete feedback with video rather than diagnosing pain or pathology from video alone.

将运动员反馈与视频结合，不应仅凭视频诊断疼痛或病理。

Worked examples / 实例

>

List two evidence sources used in movement analysis.

列出动作分析中的两种证据来源。

Exam focus / 考核要点

How does the diagnosis loop work?

诊断循环如何运作？

Rehabilitation and accessibility · 康复与无障碍参与

First, picture it / 先这样想

Rehabilitation has one job that has nothing to do with the injury: put the athlete back to the thing they love, fully. That sounds obvious and it is routinely not what happens, because the easier goal is to make the pain go away, and a painless athlete is not a returning athlete. Accessibility is the same idea applied to everyone: a programme built for the average body will fail the people furthest from it.

康复只有一个目标，而且与损伤本身无关：把运动员完整地送回他热爱的那件事。这听起来显而易见，而它例行地不是实际发生的事，因为更容易的目标是让疼痛消失，而一个不疼的运动员并不是一个回归的运动员。无障碍是同一个想法应用到所有人身上：一个为“平均身体”设计的方案，会让偏离它最远的人失败。

The detail / 详细讲解

Rehabilitation progresses through stages, and the stages are dictated by biology rather than by calendar.

康复分阶段推进，而这些阶段由生物学而不是日历决定。

Control before strength, strength before power, power before sport-specific load, and sport-specific load before competition.

先控制再力量，先力量再功率，先功率再专项负荷，先专项负荷再比赛。

The commonest failure is skipping control, because an athlete who is strong but not in control will return to a mechanism they cannot yet handle, and the mechanism is what injured them in the first place.

最常见的失败是跳过控制阶段，因为一个强壮但缺乏控制的运动员会回到他尚不能应对的机制上--而那个机制正是最初让他受伤的东西。

Rehabilitation also has to address the reason the injury happened, not just the tissue that failed.

康复还必须处理“为什么会发生”这件事，而不只是处理“失败的组织”。

If a shoulder dislocated because the rotator cuff was not controlling external rotation, then rehabing the ligaments alone returns an athlete to the same instability.

如果肩关节脱位是因为肩袖没有控制外旋，那么只康复韧带会把运动员送回同样的不稳定之中。

The broader objective is capacity, not the absence of symptoms, and capacity is what allows an athlete to absorb the next unexpected load rather than failing on it.

更广义的目标是容量，而不只是症状的消失；容量才是让运动员能吸收下一次意外负荷、而不是在负荷上失败的东西。

Accessibility is the same principle applied to who gets to participate.

无障碍是同一个原则应用到"谁有资格参与"上。

Population guidelines are for populations, and within any population the individual variation is larger than the guideline difference.

人群指导是针对人群的，而任何人群内部的个体差异都大于不同人群之间的指导差异。

An older adult with reduced bone density, an adolescent still growing, a wheelchair athlete, a pregnant athlete and an athlete with a chronic condition are not edge cases to be handled by a separate plan - they are the reason the plan has to be adjustable.

一位骨密度下降的老年人、一位仍在生长的青少年、一位轮椅运动员、一位孕期运动员和一位有慢性病情的运动员，都不是"需要另一"

The adjustments that matter are usually small and almost never glamorous: an alternative route to the same physiological stimulus, a different surface, extra time, a modified progression, equipment that fits, or a rule interpreted sensibly.

真正重要的调整通常很小，而且几乎从不光鲜：一条达到同样生理刺激的替代路径、不同的地面、更多时间、修改过的进阶、合适的

The barrier is rarely the physical challenge.

障碍很少是身体上的挑战。

It is more often the assumption that the standard programme is the only legitimate one, and that adapting it is a concession rather than a design decision.

更常见的障碍是那个假设--标准方案是唯一正当的方案，而调整它是让步而不是设计决定。

The numbers / 算一算

Numbers to keep. Bone and tendon adapt over months and the protocol should reflect that, not a four-week plan borrowed from a muscle injury. Strength symmetry of roughly 90% between limbs is a commonly used practical benchmark before returning to higher-risk sport, though it is a guide rather than a law and it must be read alongside control and sport-specific exposure. In the population side, the guideline is 150 minutes of moderate activity a week for adults with added strengthening twice a week, and the honest observation is that a large share of people fall well short of it, which means the biggest public-health gains come from moving the least-active rather than from optimising the already-active.

需要记住的数字。骨和肌腱以数月适应，方案应当反映这一点，而不是照搬一份针对肌肉损伤的四周计划。回归高风险项目之前，两侧力

Common mistakes / 易错点

-

"Rehab ends when the pain ends." Pain is a symptom, not a test of readiness. Returning at pain resolution is the commonest route to re-injury.

"不疼了康复就结束。"疼痛是症状，不是准备度的测试。在疼痛消失时回归，是再伤最常见的路径。

-

"Return to training is return to sport." Training load and competition load are different, and an athlete cleared for one is not automatically cleared for the other.

"回到训练就是回到比赛。"训练负荷和比赛负荷不同，被允许回到一个并不自动意味着被允许回到另一个。

-

"Adaptations are for the permanently disabled." Adaptable programming is for everyone: life stage, pregnancy, chronic conditions, injury, and simply being a beginner all call for a plan that adjusts.

"适应性方案是给永久性残障的。"可调方案是给所有人的：生命阶段、怀孕、慢性病、损伤，以及仅仅是初学者，都需要一个能调

-

"The guideline is the target for everyone." Guidelines are population averages. The individual spread within a population is wider than the difference between populations, so the individual is where the plan has to flex.

"指导值就是每个人的目标。"指导值是人群平均。人群内部的个体分布比不同人群之间的差异更宽，所以方案必须在个体层面有弹

Why it matters / 为什么重要

Rehabilitation and accessibility are the two places where sports science makes a claim about a person rather than about a measurement, and both are graded on judgement. A coach can quote a protocol perfectly and still return an athlete too early; a teacher can hand out the same programme to everyone and fail the people who needed it most. What separates good practice here is not knowledge of protocols - it is the habit of asking what the individual can actually do, what they are actually trying to get back to, and what would let them keep it. That habit is worth

more in a career than any single technique, and it is the one most likely to be examined as a case question.

康复与无障碍，是运动科学从"测量"转向"人"的两处，而两者都以判断力评分。教练可以把方案背得一字不差，仍然让运动员回来得太早

-

Adapt the constraints to enable success, not the athlete to fit an unsuitable task.

调整限制条件以促进成功，而不是要求运动员适应不合适任务。

Key terms / 核心术语

Accessibility 无障碍参与

Removing barriers and adapting tasks or equipment so participation is possible.

去除障碍并调整任务或装备，使参与成为可能。

Formulas / 公式

A player with limited grip strength can use a larger or lighter racket, then progress the task gradually.

握力有限的球员可使用更大或更轻的球拍，再逐步增加任务难度。

Worked examples / 实例

>

Give one technique, equipment and task adaptation.

各给出一个技术、装备和任务调整。

Exam focus / 考核要点

How can analysis increase accessibility?

动作分析如何提高无障碍参与？

Causes of injury

损伤原因

Guiding question / 引导问题

Why do internal and external factors make some athletes more susceptible to injury?

为什么内部和外部因素会使一些运动员更易受伤？

Injury risk is the interaction of internal factors, external factors and exposure. A risk factor increases likelihood but does not guarantee injury. Acute trauma follows a sudden or excessive force, while cumulative trauma follows repeated force without adequate recovery.

损伤风险由内部因素、外部因素和暴露共同作用。风险因素增加可能性但不保证受伤。急性创伤由突然或过大的力导致，累积性损伤由重复力导致且无充分恢复。

Internal and external risk factors · 内部与外部风险因素

First, picture it / 先这样想

Risk factors are the things that quietly raise the odds. Some live inside the athlete - how the body is built, how it has healed, how it is loaded. Some live outside - the surface, the rules, the equipment, the schedule. The useful habit is counting them, because a single risk factor is usually survivable and several at once is where injuries actually happen.

风险因素是那些悄悄提高几率的因素。有些在运动员体内--身体如何构建、曾经如何愈合、如何被负荷。有些在体外--地面、规则、装备、赛程。有效的习惯是把它们数清楚，因为单一风险因素通常可以承受，而几个同时出现时，伤病才真正发生。

The detail / 详细讲解

Internal factors are the ones you cannot change and the ones you can train.

内部因素分为你无法改变的和可以训练的。

Non-modifiable: age, sex, previous injury, body structure, inherited tissue properties, and the fact that adolescent tissue is still adapting.

不可改变：年龄、性别、既往损伤、身体结构、遗传的组织特性，以及青少年组织仍在适应这一事实。

Modifiable: strength, flexibility where it is genuinely limited, movement control, load history, sleep, nutrition, and fatigue.

可改变：力量、在真正受限处的柔韧性、动作控制、负荷史、睡眠、营养和疲劳。

The distinction matters because effort spent on a non-modifiable factor is effort wasted, and effort spent on fatigue is often the highest-yield of all.

这个区分很重要，因为花在不可改变因素上的努力是浪费，而花在疲劳上的努力往往回报最高。

External factors are the ones the environment imposes.

外部因素是环境施加的。

Surface and friction, weather, altitude, equipment, protective gear, the rules of the game, the number of collisions in a fixture, and the density of the competition calendar.

地面与摩擦、天气、海拔、装备、护具、比赛规则、一场比赛中的碰撞次数，以及赛程密度。

Some are chosen, some are negotiated, and some simply are what they are.

有些是可以选择的，有些是可以协商的，有些就只是它们本来的样子。

The coach's real leverage is usually in the ones that are negotiable, and the discipline is noticing which is which before proposing a change.

教练真正的抓手通常在那些可以协商的因素上，而所需的纪律是在提出改变之前先分清哪一类是哪一类。

Risk factors interact multiplicatively rather than additively, and that is why a list of small factors is dangerous.

风险因素之间是乘法关系而不是加法关系，这正是“一串小因素”很危险的原因。

An athlete with a previous injury, playing on a hard surface, in a congested fixture list, with inadequate sleep, is not carrying one elevated risk - they are carrying a compound one.

一位有既往损伤、在硬地面、在拥挤赛程中、睡眠不足的运动员身上负担的不是一个升高了风险，而是一个复合风险。

The same is true in reverse, which is the hopeful part: removing a single factor can undo an accumulation, and the cheapest factor to remove is very often the one nobody is looking at.

反过来也同样成立，这正是有希望的一面：移除一个因素就可以化解累积，而最便宜的移除对象往往正是没人注意到的那个。

The practical tool is pre-participation screening, and its honest limitation is that a screen is a snapshot rather than a prediction.

实用工具是赛前筛查，而它诚实的局限在于：筛查是一次快照而不是一次预测。

A negative screen does not guarantee a safe season, and its value lies in finding the small number of cases where a precaution genuinely changes the outcome.

阴性结果不保证一个赛季安全，它的价值在于找出少数几个“加一道预防措施确实会改变结果”的个案。

Used well it is cheap and it prevents; used as a ritual or as a source of false reassurance it is worse than not doing it.

用得好它便宜而且能预防；把它当仪式或当成虚假的安心来源，它比不做还糟。

The numbers / 算一算

Numbers to keep. The most powerful single predictor of future injury is previous injury at the same site, which is why a history of ankle sprain or ACL reconstruction deserves more attention than any current test. Load is the other one: a large spike in weekly training volume relative to the athlete's own recent history is associated with increased injury risk, and the important word is relative - the same absolute week is safe for one athlete and dangerous for another. Rule of thumb figures used in practice are a body-mass increase per week in children as a rough growth marker, and a guideline of roughly 10 percent progression used by many programmes, while the honest statement is that no single progression figure is universal and the athlete's own history is the better guide.

需要记住的数字。对未来损伤最强的单一预测因素是同一位置的既往损伤，这正是踝关节扭伤或前交叉韧带重建史比任何当前检查都更

Common mistakes / 易错点

- "Screening catches the athletes who will be injured." It identifies a small number of people for whom a precaution may help. Most injuries occur in athletes who passed every screen.
"筛查能抓出将来会受伤的运动员。"它只识别出少数几个加一道预防措施可能有帮助的人。大多数伤病发生在通过了所有筛查的运动员中。
- "If the screen is clear, the athlete is safe to compete." A screen is a snapshot of one day, not a prediction for a season.
"筛查没问题，运动员就可以参赛。"筛查是某一天的一张快照，不是对一个赛季的预测。
- "Risk factors add up, so a few small ones do not matter." They interact, and the interaction is where the risk actually lives.
"风险因素会相加，所以几个小的没关系。"它们相互作用，而风险恰恰活在这种相互作用里。
- "Extrinsic factors are the ones you cannot change." Many of them are negotiable - the surface, the schedule, the equipment, the rules. Treating them as fixed removes the most useful lever a coach has.
"外源性因素是你改变不了的。"其中很多是可以协商的--地面、赛程、装备、规则。把它们当成固定不变的，等于拿掉了教练最有用的杠杆。

Why it matters / 为什么重要

This is the section that makes injury prevention a discipline rather than a slogan. Risk factors are only useful if they are counted, ranked and, where possible, changed - and the counting is what stops a coach from spending the season on the one factor that was never going to matter while the actual cause goes unaddressed. It is also the most directly applicable part of the whole course, because every athlete in every programme has these factors and almost none of them has ever been written down. The habit of writing them down is the transferable skill here, and it is one that a student can practise on a team they are already part of.

这一节使伤病预防成为一门纪律而不是一句口号。风险因素只有在被清点、排序并在可能的情况下被改变时才有价值--而正是清点这件事

- Previous injury can reduce strength, balance or proprioception and alter mechanics, increasing reinjury risk.
既往损伤可降低力量、平衡或本体感觉并改变动作，增加再伤风险。

A risk factor is a probability factor, not a diagnosis.

风险因素是概率因素，不是诊断。

Key terms / 核心术语

Exposure 暴露

The amount and type of activity or load experienced.

经历的活动或负荷的类型与数量。

Previous injury 既往损伤

An earlier injury that can affect current risk and function.

影响当前风险和功能的既往损伤。

Formulas / 公式

Manage controllable factors such as load, technique, recovery, equipment and surface; do not claim that every factor can be changed.

管理负荷、技术、恢复、装备和场地等可控因素；不要声称所有因素都能改变。

Worked examples / 实例

>

Classify a scenario and identify controllable risk factors.

对情境分类并识别可控风险因素。

Exam focus / 考核要点

Give two internal and two external risk factors.

给出两个内部和两个外部风险因素。

Acute and cumulative trauma · 急性与累积性创伤

First, picture it / 先这样想

Two different ways to break something. Acute trauma is one big event - a fall, a collision, a sudden twist - and it arrives with a cause you can name. Cumulative trauma is what happens when nothing is big enough to notice, over and over, until the structure can no longer carry the load. The second is far more common in sport and far harder to see, because there is no day you can point at.

有两种不同的“弄坏”方式。急性损伤是一次大事件--摔倒、碰撞、突然扭转--它带着一个你能说出来的原因。累积性损伤是当没有任何一次大到引人注目、却一次又一次发生，直到结构再也无法承担负荷时产生的。在运动中第二种远为常见，也远更难看见，因为没有哪一天是你能指出来的。

The detail / 详细讲解

Acute trauma is characterised by a single identifiable mechanism and a rapid onset.

急性损伤的特征是存在一个可识别的机制并且起病迅速。

The tissue is loaded beyond its capacity in one event, and the classification of the injury follows the tissue involved - a muscle strain graded by how much is torn, a ligament sprain graded by how much of the ligament is affected, a fracture by the bone involved.

组织在一次事件中被加载超过其承受限度，损伤分级取决于受累组织--肌肉拉伤按撕裂程度分级，韧带扭伤按受累范围分级，骨折按受累范围分级。

Acute injuries are relatively easy to diagnose and relatively predictable to manage, and the assessment question is usually about severity rather than about cause.

急性损伤相对容易诊断，处理也相对可预测，评估的问题通常是严重程度而不是原因。

Cumulative trauma is characterised by an absence of a single event.

累积性损伤的特征是没有单一事件。

The tissue is repeatedly loaded above its capacity to remodel, so it never gets the chance to return to full strength before the next demand arrives.

组织被反复加载到超出其重塑能力之上，因此来不及在下一次需求到来前恢复到完全强度。

Over months or years the structure adapts - sometimes usefully, in tendons and bone - and sometimes badly, in the way a tendon becomes less well organised, or a nerve becomes sensitive, or bone density falls.

数月或数年间结构发生适应--在肌腱和骨中有时是有利的，在肌腱变得组织排列不良、神经变得敏感、骨密度下降时则是有害的。

The result is a tissue that fails during an activity that would not have injured it a year earlier.

结果是一种组织，它会在一项一年前不会造成损伤的活动上失效。

The clinical pattern of cumulative trauma is also different.

累积性损伤的临床模式也不同。

Acute injuries have an obvious moment and often an obvious story.

急性损伤有一个明显的时刻，通常还有一个明显的故事。

Cumulative injuries present as gradual onset, changing symptoms that wax and wane, and an athlete who cannot name what happened because nothing did.

累积性损伤表现为逐渐起病、症状时轻时重地波动，以及一位说不出发生了什么的运动员--因为并没有发生什么。

That presentation is easy to dismiss and easy to mislabel, which is why a history of gradual onset with no incident is itself an important piece of information.

这种表现容易被忽视、也容易被贴错标签，所以"逐渐起病、无明确事件"这个病史本身就是一条重要信息。

And the two interact, which is the least comfortable truth in this topic.

而两者会相互作用，这是这个主题里最不舒服的真相。

A tissue that has been cumulatively altered is more likely to fail acutely, and an acute failure leaves the tissue and the athlete less able to tolerate the cumulative load that follows.

被累积性改变过的组织更容易发生急性失效，而一次急性失效会让组织与运动员都更难承受随后到来的累积负荷。

Return too early, and a single incident becomes the start of a chronic problem.

回来得太早，一次急性事件就变成慢性问题的开端。

That sequence - cumulative change, acute failure, incomplete recovery, resumed loading, cumulative change again - is the mechanism behind a large share of injuries that are described as bad luck.

这个顺序--累积改变、急性失效、恢复不彻底、重新加载、再次累积改变--正是大量被描述为"运气不好"的伤病背后的机制。

The numbers / 算一算

Numbers to keep. The distinction shows up in the time scale: acute effects are measurable in hours to weeks, cumulative effects in months to years. Muscle strength in a healthy adult is typically maintained with relatively little activity, so a strength loss of more than roughly 10 to 15 percent is meaningful and worth measuring rather than estimating. And the practical figure that runs through this whole section is load tolerance: the amount a tissue will accept before it fails, and the goal of any programme is to keep demand inside that tolerance rather than to find where it is.

需要记住的数字。这个区分体现在时间尺度上：急性效应以数小时到数周计量，累积效应以数月到数年计量。健康成年人的肌肉力量只靠

Common mistakes / 易错点

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"Cumulative means a single big event repeated." Cumulative means many small or moderate events that individually stay below the threshold for adaptation to keep up.

"累积是指一个大的事件反复发生。"累积是指许多小的或中等的事件，每一个单独来看都低于适应的阈值。

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"If there was no injury event, there is no injury." The absence of an event is one of the main clinical features of a cumulative injury, not evidence against it.

"如果当时没有受伤事件，就没有受伤。"没有事件正是累积性损伤的主要临床表现之一，而不是反对这一诊断的证据。

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"Acute injuries are unpredictable." The mechanism is usually known and the classification scheme is well defined, which makes the assessment largely a question of severity.

"急性损伤是不可预测的。"机制通常是已知的、分级体系也很明确，所以评估在很大程度上是一个关于严重程度的问题。

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"An acute injury is healed when the pain stops." Recovery of a re-injured or chronically altered tissue requires more than symptom resolution, and that gap is where many chronic problems begin.

"急性损伤不疼了就是好了。"重新受损或被慢性改变过的组织，其恢复需要的不只是症状消退，而许多慢性问题正是从这个差距开

Why it matters / 为什么重要

Acute versus cumulative is the distinction that decides what a coach should do. An acute injury is largely a medical and diagnostic problem, and the coach's job is to be accurate about severity and disciplined about the timeline. A cumulative injury is a systems problem, and the coach's job is to find the load that is producing it, which usually means looking at what happens every week rather than what happened once. Students who understand the difference stop asking "how do I tape this ankle" for problems that taping cannot address, and start asking what made the ankle the tissue that failed.

急性与累积的区分，决定了教练该做什么。急性损伤在很大程度上是医学和诊断问题，教练的职责是准确判断严重程度并严守时间表。易

Overuse examples include shin splints, tendinopathy and stress fractures. Biomechanical maladaptation can produce persistent pain, reduced performance, increased risk and longer recovery.

过度使用损伤例子包括胫骨疼痛、肌腱病和应力性骨折。生物力学适应不良可造成持续疼痛、表现下降、风险增加和恢复延长。

Key terms / 核心术语

Acute trauma 急性创伤

Sudden injury from unexpected or excessive force.

由突然或过大力导致的损伤。

Cumulative trauma 累积性损伤

Gradual injury from repeated load and inadequate recovery.

由反复负荷和恢复不足造成的渐进损伤。

Formulas / 公式

Hamstring pain worsening over several weeks as mileage rises is cumulative, not necessarily a single acute tear.

随着里程增加，数周内逐渐加重的腘绳肌疼痛属于累积性问题，不一定是单次急性撕裂。

Worked examples / 实例

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Difference between acute and overuse injuries.

区分急性和过度使用性损伤。

Exam focus / 考核要点

Why is poor technique an injury risk factor?

为什么技术不佳是损伤风险因素？

Tissues and concussion · 组织与脑震荡

First, picture it / 先这样想

The body is built from a small number of tissue types, each with its own rules, and injuries are mostly a story about which rule was broken. Concussion is the odd one out, because the tissue that gets hurt is not the tissue that produces symptoms. The brain tissue can be injured mildly while the function that complains - memory, balance, mood - is housed elsewhere and reacts to something the injury itself did not touch.

身体由少数几种组织类型构成，每一种都有自己的规则，而损伤大多是在讲“哪条规则被打破了”的故事。脑震荡是其中的异类，因为受损的组织并不是产生症状的组织。脑组织可以被轻微损伤，而发出抱怨的那个功能--记忆、平衡、情绪--却housed在别处，并且是对脑震荡本身并未触及的某种东西作出反应。

The detail / 详细讲解

Connective tissues are defined by their fibre type and their collagen.

结缔组织由纤维类型和胶原构成。

Tendon and ligament are dense regular collagen, which resists unidirectional pull.

肌腱和韧带是致密规则胶原，抵抗单向牵拉。

Bone is a composite that gains stiffness from mineral and gains toughness from collagen, and because it is the only one of these that remodels reliably in response to load, it is the one that responds best to progressive strength work.

骨是复合材料，矿物质提供刚度、胶原提供韧性，而因为它是这些组织中唯一能可靠地响应负荷而重塑的，所以它是对渐进力量训练

Cartilage has no blood supply of its own, which is why it is slow to repair and why feeding it is largely about maintaining the joint rather than healing it.

软骨本身没有血供，因此修复缓慢，而“供养”它主要靠维持关节而不是修复它。

Muscle and nerve behave differently again.

肌肉与神经的行为又不同。

Muscle has a good blood supply, heals well, and is remarkably adaptable - it will accept almost any pattern you ask of it.

肌肉血供良好、愈合良好，而且适应性极强--你要求它什么样的模式它几乎都能接受。

Nerve is the opposite: it regenerates slowly, and its recovery is governed less by how it was injured than by whether the cell body survived and whether the pathway was re-established.

神经正相反：它再生缓慢，而它的恢复与其说取决于如何受伤，不如说取决于胞体是否存活、神经通路是否重新建立。

That difference is why a muscle strain is a weeks problem and a nerve injury is a months problem, regardless of how similar they can look.

这个差异正是为什么肌肉拉伤是以周计的问题而神经损伤是以月计的问题，无论两者看起来多么相似。

Concussion is a disturbance of brain function caused by an impact, and the injury is diffuse rather than focal.

脑震荡是由撞击引起的大脑功能紊乱，损伤是弥漫性的而非局灶性的。

The mechanism is an acceleration and deceleration of the head, which produces strain on the tissue rather than a clean tear.

机制是头部的加速与减速，从而在组织上产生应变而不是干净的撕裂。

Most concussion injuries are mild, and imaging is usually normal - which is precisely the difficulty, because there is nothing on the scan to measure and the diagnosis is entirely one of history and observed behaviour.

大多数脑震荡是轻度的，影像学通常正常--而这恰恰是困难所在，因为扫描上没有任何可测量的东西，诊断完全依赖于病史和观察到的行为。

The symptoms span several systems, and the breadth is itself informative.

症状跨越多个系统，而这种广度本身就是信息。

Headache and dizziness, nausea, confusion, difficulty concentrating, sensitivity to light or noise, sleep disturbance, and a change in mood or behaviour.

头痛和眩晕、恶心、意识模糊、难以集中、对光或声音敏感、睡眠障碍，以及情绪或行为的变化。

The emotional and cognitive symptoms are frequently the last to resolve and the first to be dismissed, which is why a young athlete who is irritable and quiet in a week after a head knock is easy to send back too early.

情绪和认知方面的症状常常是最后消退的、也最容易最先被忽视的--所以一个头部撞击后一周里变得易怒而沉默的年轻运动员，很容易

The numbers / 算一算

Numbers to keep. A concussion is diagnosed on the presence of symptoms after an impact, not on any measurement, which is why the numbers that matter are behavioural: the number of days to symptom resolution, and the number of days from symptom resolution to full contact. Those two intervals, and the requirement that both are symptom-free before progressing, are what the staged return framework encodes. On tissue time scales the useful comparison is that muscle heals over weeks, tendon and ligament over weeks to months, and nerve over months, while bone and cartilage are slower still and are governed more by mechanical environment than by rest.

需要记住的数字。脑震荡的诊断依据是撞击后出现症状，而不是任何测量值，所以真正重要的数字是行为性的：症状消退所需的天数，以及

Common mistakes / 易错点

- "A normal scan means the brain is fine." Concussion is a disturbance of function with no structural damage to see, which is why imaging is normal in the great majority of cases and cannot be used to clear an athlete.
"检查正常说明脑子没问题。"脑震荡是功能紊乱而没有可见的结构损伤，所以绝大多数病例影像学都正常，不能用它来放行运动员。
- "Symptoms are physical, so they cannot be emotional." Mood and behaviour change is a recognised post-concussion symptom and often persists longest, which is why it gets missed.
"症状是身体性的，所以不可能是情绪上的。"情绪和行为变化是被承认的脑震荡后症状，而且常常持续最久，所以它常被漏掉。
- "Once the headache is gone, the athlete can return." Headache is one symptom among several, and cognitive and emotional symptoms frequently lag behind it.

"头不疼了就可以回归。"头痛只是众多症状之一，而认知和情绪症状经常落在它后面。

"Tissues all heal the same way." The time scales differ by an order of magnitude between muscle, tendon, ligament and nerve, and that difference determines the timeline.

"各种组织的愈合方式都一样。"肌肉、肌腱、韧带和神经的愈合时间尺度相差一个数量级，而这个差异决定了时间表。

Why it matters / 为什么重要

Tissue types explain why two injuries that look identical on the sideline have completely different timelines, and that single idea prevents a large number of return-too-early decisions. Concussion sits in the same section because it makes the general point sharply: the injured tissue and the impaired function are different things, and a normal scan says almost nothing about the second. Together these two ideas give a student the framework for handling any soft-tissue problem they will meet - what the tissue does, how long it takes, and what would be evidence that it is ready rather than merely quiet.

组织类型解释了为什么场边看起来一模一样的两个损伤，时间表却完全不同--仅仅是这一个概念，就能避免大量"回来太早"的决定。脑震

Concussion symptoms may include headache, dizziness, confusion, nausea and sensitivity to light. It must be managed correctly.

脑震荡症状可能包括头痛、头晕、困惑、恶心和对光敏感，必须正确管理。

Key terms / 核心术语

Functional injury 功能性损伤

A change in function without visible structural damage, such as concussion.

没有可见结构损伤的功能改变，如脑震荡。

Formulas / 公式

Classify the tissue first, then connect the likely injury mechanism to the structure affected.

先判断组织，再把可能的损伤机制与受影响结构联系起来。

Worked examples / 实例

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How do tissue function and injury mechanism connect?

组织功能与损伤机制如何联系？

Exam focus / 考核要点

Name one injury example for each of two tissues.

为两种组织各举一个损伤例子。

Interventions related to injury

损伤干预

Guiding question / 引导问题

How can injury risk be reduced and recovery managed safely?

如何降低损伤风险并安全管理恢复?

Interventions should reduce abnormal force, improve force absorption, use correctly fitted protective equipment, prepare the body and use correct technique. Healing, rehabilitation and return to activity progress gradually, while concussion return is staged and symptom-led.

干预应减少异常力、改善冲击力吸收、使用正确贴合的防护装备、做好身体准备并使用正确技术。愈合、康复和重返活动逐步进行，

Prevention and protection · 预防与防护

First, picture it / 先这样想

Two different jobs that get confused for each other. Prevention stops the injury happening. Protection reduces what happens when it does. They need different evidence, they fail in different ways, and the most common error in sport is putting effort into protection because it is visible and easy, while the prevention work that would have stopped the injury is invisible and slow.

两份不同的工作常常被混为一谈。预防是让伤病的发生不发生；保护是减少发生之后的后果。两者需要的证据不同，失败的方式不同，而运动中最常见的错误是把精力投在保护上--因为它可见、容易做--而真正能阻止那次伤病的预防工作是不可见且缓慢的。

The detail / 详细讲解

Warm-ups are the most available preventive measure in any sport, and the mechanism is specific rather than general.

热身是任何运动中最随手可得的预防措施，而它的机制是具体的而非泛泛的。

A warm-up raises tissue temperature, which makes collagen more extensible, and it rehearses the movement pattern at higher speed so the nervous system has practised what it is about to be asked to do.

热身提高组织温度，使胶原更容易延展；

Both effects are temporary, which is the practical point: a warm-up that ends twenty minutes before the event has largely worn off.

它还在更高速度下排练动作模式，让神经系统预先演练了即将被要求做的事。

两种效果都是暂时的，而这正是实际要点：一场在比赛前二十分钟结束的热身，其效果大多已经消退。

Load management is the most effective preventive measure and the least visible.

负荷管理是最有效的预防措施，也是最不可见的。

It is the progressive, individual history of what the athlete has already absorbed, and it is where the compound risk from earlier in this section actually lives.

它是对“这位运动员已经吸收了多少”这份个体历史的渐进把控，也正是前面提到的复合风险真正栖身之处。

The tools are monitoring, rest, and the discipline of not adding a second hard session on top of a hard week.

工具是监测、休息，以及不在一个已经吃力的星期之上再加第二次高强度训练的纪律。

It is invisible because nothing bad happens when it is done correctly, which is exactly the problem with invisible interventions.

它不可见，是因为做对了不会发生任何坏事--而这正是不可见干预的问题所在。

Protective equipment works on the energy-absorbing principle, and its design has converged on two features: spreading the impact over a larger area and stiffening so that peak force is reduced.

防护装备依靠能量吸收原理工作，而它的设计已经收敛到两个特征：把撞击分散到更大面积上，以及增加刚度以降低峰值力。

The cost is that stiffness transfers load to the next structure along the chain, so a helmet protects the brain at the expense of the neck, and a stiffer shin guard transfers load upward.

代价是刚度会把负荷传给链条上的下一个结构，所以头盔保护了大脑却让颈部承担更多，更硬的胫甲把负荷向上传递。

The honest reading is that protective gear reduces one specific risk while changing the distribution of the rest, and

the evidence for a net reduction in injury is good for some equipment and weak for others.

诚实的解读是：防护装备降低了某一个特定风险，同时改变了其余风险的分布；而它在净减少损伤方面的证据，对某些装备很好，对另一些则很弱。

Rules are prevention at the level of the game rather than the body.

规则是在游戏层面而不是身体层面的预防。

They work by removing a hazard from the environment entirely rather than making an athlete better at tolerating it, and they are therefore unusually effective per unit of effort - head-contact rules in youth football, the removal of cumulative matches from congested fixture lists, and pitch size limits in some age categories.

它们通过把危险从环境中彻底移除来起作用，而不是让运动员更能忍受它--因此它们每单位努力的效果异常之高：青少年足球中的头部

They are also the layer most often contested, because the people who object to them are usually the ones who benefit most from ignoring them.

它们也是最常被质疑的一层，因为反对它们的人通常正是最受益于无视它们的人。

The numbers / 算一算

Numbers to keep. Warm-up effects on tissue temperature and on movement quality decline measurably over roughly the following 20 to 30 minutes, which is why a long, warm-up-destroying ceremony at half-time is a real cost rather than a superstition. Rule-based prevention is best judged on exposure: the number of head contacts per thousand touches of the ball, or the number of matches in a congested period, are the numbers a rule is designed to move, and they can be counted. And the practical figure for equipment is that its benefit scales with the energy it can absorb before the force reaches the protected structure, so coverage and material matter more than the shape of the shell.

需要记住的数字。热身对组织温度和动作质量的影响，在其后大约20到30分钟内会可测量地衰减--这正是中场那种漫长、会把热身效果破

Common mistakes / 易错点

- "A warm-up an hour before the event counts." The effect is largely gone well before the event begins, and the movement rehearsal is what matters most in the final minutes.
"比赛前一小时的热身算数。"效果在比赛开始前很久就基本消失，而最后几分钟里的动作预演才是最重要的部分。
- "Protective gear reduces risk with no trade-off." Stiffening shifts load to neighbouring structures, so it reduces one risk and redistributes others.
"防护装备降低风险没有代价。"增加刚度会把负荷转移给邻近结构，所以它降低了一个风险、重新分配了其他风险。
- "Rules are a matter of preference." A rule that removes a hazard entirely is more effective per unit of effort than almost anything available at the level of the individual athlete.
"规则只是偏好问题。"一条彻底移除危险环境的规则，每单位努力的效果都高于在运动员个体层面几乎所有可用的手段。
- "If the programme is invisible, it is not working." Load management produces no visible benefit when it is done correctly, which makes it the easiest effective intervention to cut.
"如果方案看不见效果，说明它没起作用。"负荷管理做对了不会产生可见的收益，这使它成为最容易砍掉的有效干预。

Why it matters / 为什么重要

This section is where a student has to sort interventions by the mechanism they act through, and that sorting is most of the skill. A warm-up, a load-management decision, a piece of equipment and a rule change all "prevent injury" in a loose sense, but they work on different time scales, need different evidence, and can be traded against each other. A coach who can name the mechanism for each measure they introduce is making a claim that can be tested; one who cannot is repeating received wisdom. It is also the section where the honest answer is often "the most effective thing here is a decision about the fixture list", which is inconvenient and usually correct.

这一节要求学生按干预所作用的机制来分类，而这些分类工作本身就是技能的大部分。热身、一个负荷管理决定、一件装备和一次规则变

- Protective equipment must be sport- and position-specific, correctly sized, comfortable, secure, compliant and reassessed as the athlete grows or changes role.
防护装备应符合项目和位置、尺寸正确、舒适、牢固、符合标准，并随着运动员生长或角色变化重新评估。

Key terms / 核心术语

Prehabilitation 预康复

Preparing vulnerable or weak areas before injury or high demand.

在受伤或高负荷前准备薄弱或易受伤区域。

Force absorption 冲击力吸收

The ability to spread or reduce an external load.

分散或降低外部负荷的能力。

Formulas / 公式

A basketball player uses strength and landing technique to reduce ACL and ankle risk; a runner increases weekly distance gradually and varies surfaces.

篮球运动员用力量和落地技术降低膝前交叉韧带和踝部风险；跑者逐步增加周里程并变换地面。

Worked examples / 实例

>

How can prehabilitation, warm-up and technique reduce risk?

预康复、热身和技术如何降低风险？

Exam focus / 考核要点

What should be checked when selecting protective equipment?

选择防护装备时应检查什么？

PRICE, healing and rehabilitation · PRICE原则、愈合与康复

First, picture it / 先这样想

PRICE is the first twenty-four hours, and it is a good piece of memorised shorthand that a coach can apply immediately. It is also the smallest part of the story. The tissue has a biological timetable that runs for weeks and does not care what was done in the first day, and the job after the first day is to load that tissue in a way that makes it come back stronger than it left rather than merely healed.

PRICE 是最初的二十四小时，它是一段很好的可以立刻上手的缩写口诀。但它也是整个故事中最小的一部分。组织有一个持续数周的生物学时间表，而它并不在意第一天做了什么；第一天之后的工作，是用一种能让组织变得比原来更强、而不只是“愈合了”的方式去给它加载。

The detail / 详细讲解

The letters are Protection, Rest, Ice, Compression and Elevation, and the logic of each is worth understanding rather than just applying.

这几个字母代表保护、休息、冰敷、加压和抬高，每一个背后的逻辑都值得理解而不只是执行。

Protection prevents further damage, which is obvious.

保护防止进一步损伤，这很显然；

Rest removes the load that is preventing repair.

休息移除了妨碍修复的负荷；

Ice reduces pain and therefore lets an athlete move the part gently instead of guarding it completely.

冰敷减轻疼痛，从而让运动员能轻柔地活动该部位而不是完全护住它；

Compression limits swelling, and elevation works with gravity to do the same.

加压限制肿胀，抬高则配合重力达到同样效果。

The modern caution is that rest in the true sense, complete immobility, is now known to be worse than early controlled movement for most soft-tissue injuries.

现在的注意点是：严格意义上的“休息”--完全不动--已被证明对大多数软组织损伤比对早期受控活动更糟。

The tissue timetable is what replaces the acronym as the organising idea.

真正取代这个缩写、成为组织性想法的，是组织的时间表。

Inflammation is a necessary early phase that clears damaged tissue, and suppressing it aggressively removes something the repair needs.

炎症是一个必要的前期阶段，负责清理受损组织，而过度抑制它会移除修复所需的東西。

Repair then lays down new tissue along the lines of stress applied, which is the principle that makes progressive loading the core of rehabilitation rather than an optional extra.

随后修复沿着所施加应力的方向laid down新组织--正是这个原则使渐进加载成为康复的核心而不是可选附加物。

Remodelling is the slow final phase, in which the tissue reorganises to its mature form, and it is the phase that is most often cut short by a return to sport that felt fine.

重塑是最后的慢阶段，组织在其中重新组织为成熟形态，而这一阶段最常因为"感觉已经好了"就回归比赛而被提前砍掉。

Progressive loading is the spine of the whole process, and the principle is specific: the tissue should be loaded, then allowed to adapt, then loaded slightly more.

渐进加载是整个过程的脊柱，而原则是具体的：给组织加载、让它适应、再稍微多加载一点。

Too little loading produces tissue that heals but does not regain its capacity, which is why an athlete can have a fully healed tendon that cannot tolerate sport.

加载太少会产生"愈合了但没恢复容量"的组织，这正是为什么一位运动员可能拥有一根完全愈合却无法承受运动项目的肌腱；

Too much, too fast produces the tissue failing again.

太多太快则会让组织再次失效。

The judgement is not about a fixed protocol but about the gap between what the tissue can currently take and what is being asked of it.

这个判断不是关于某份固定方案，而是关于"组织当前能承受多少"与"正在要求它做多少"之间的差距。

And there is a fourth thing after the tissue, which is the capacity of the system rather than the tissue.

而组织之后还有第四件事，那就是系统能力而不是组织本身。

An ankle that has healed may still be owned by a hip that cannot control it, and a shoulder that has healed may still sit in a scapula that does not move.

一个愈合的踝关节，可能仍然被一个无法控制它的髋关节所支配；

Rehabilitation that only treats the site of injury leaves that in place, which is why the recurrence rate stays high in programmes that stop when the tissue is healed rather than when the movement is normal.

一个愈合的肩关节，可能仍然处在一个不活动的肩胛骨中。

只处理损伤部位组织的康复会把这个留在原地--这就是为什么那些在"组织愈合"而不是"动作恢复正常"时停止的方案，复发率一直偏高。

The numbers / 算一算

Numbers to keep. The first phase of the tissue timetable runs over roughly the first few days, repair over weeks, and remodelling over months, and those three intervals are a more reliable guide than any acronym. The practical loading number is that a tissue should be challenged and then given time to adapt before the next increase, and the honest statement is that the correct interval is decided by the response, not by a calendar. Return-to-sport figures are the ones to be careful with: a criterion-based return based on strength symmetry, control and sport-specific exposure is more defensible than a fixed number of weeks, and the most widely used practical benchmark of roughly 90 percent strength symmetry between limbs is a guide rather than a law.

需要记住的数字。组织时间表的第一阶段大约在最初几天，修复以周计，重塑以月计--这三个间隔比任何缩写都是更可靠的指引。加载方

Common mistakes / 易错点

- "RICE is the first step, so rest is always the answer." Total immobility is now understood to be worse than early controlled movement for most soft-tissue injuries.
"RICE 是第一步，所以休息永远是答案。"完全不动现在被理解为对大多数软组织损伤比早期受控活动更糟。
- "Ice is the treatment." Ice manages pain and swelling; it is not a treatment for the tissue, and pain is a signal that also needs to be respected rather than permanently removed.
"冰敷才是治疗。"冰敷处理的是疼痛和肿胀；它不是对组织的治疗，而疼痛也是一个需要被尊重、而不是被永久消除的信号。
- "The scan is clear, so the tissue is back to full capacity." Healing tissue and restored capacity are different, and imaging cannot tell you which one you have.
"检查清楚了，说明组织已经恢复全部容量。"愈合中的组织和恢复的容量是两回事，而影像学无法告诉你处在哪一个阶段。

"Rehab ends when the injury site looks normal." The system around the site is usually what failed, and stopping at the site leaves the cause in place.

"损伤部位看起来正常了，康复就结束。"周围系统通常才是失败的一方，在部位停下就把原因留在了原地。

Why it matters / 为什么重要

PRICE is worth knowing precisely because it is what an athlete or a parent will say in the first hour, and knowing the acronym makes the first hour better. But the reason this section matters is the reframe underneath it: recovery is a loading problem, not a waiting problem, and the two produce completely different athletes at the end of the same number of weeks. A student who has understood that the tissue remodels along the lines of the stress it receives has understood the mechanism behind almost every rehabilitation decision, and can reason about a case they have never seen before rather than looking up a protocol for it.

PRICE

值得精确掌握，因为它是运动员或家长在第一个小时会说的话，而认识这个缩写能让第一个小时变得更好。但这一节真正重要的地方在于

Rehabilitation progresses from restoring movement to restoring power and sport-specific performance. Heat, ice, ultrasound and electrical stimulation may assist pain or recovery but are not substitutes for progressive loading. 康复从恢复活动度逐步发展到恢复力量和专项表现。热疗、冰疗、超声和电刺激可帮助疼痛或恢复，但不能替代渐进负荷。

Key terms / 核心术语

Protection 保护

Prevent further injury while symptoms settle.

在症状稳定时防止进一步损伤。

Remodelling 重塑

The later stage in which tissue is reorganised and adapted.

组织重新组织并适应的后期阶段。

Formulas / 公式

A clinician may use surgery for severe tears or fractures, but the course is individualised and followed by post-operative protocol.

严重撕裂或骨折可能需要手术，但过程应个体化并遵循术后方案。

Worked examples / 实例

>

Place a treatment in prevention, acute management, rehabilitation or return to sport.

将处理归入预防、急性管理、康复或重返运动。

Exam focus / 考核要点

What is the sequence from healing to return to performance?

从愈合到重返表现的顺序是什么？

Staged concussion return · 脑震荡分阶段回归

First, picture it / 先这样想

A staged return is a staircase, and the rule is that you cannot skip a step. Each stage is cleared on symptoms, not on time, and the clock only starts once the previous stage has been passed symptom-free. The staircase exists because the risk in concussion is not "being unfit"- it is being symptom-free but still vulnerable, and the only way to test that is to add load and watch what happens.

分级回归是一座楼梯，而规则是你不能跳过任何一级。每一级都按症状而不是按时间来通过，而且时钟只有在这一级无症状通过之后才开始走。这座楼梯存在的原因在于，脑震荡的风险不是"身体不够格"--而是"没有症状但仍然脆弱"，而检验这点的唯一办法就是加上负荷，然后观察会发生什么。

The detail / 详细讲解

The framework is a sequence of increasing exposure.

这个框架是一段逐步增加的暴露序列。

Relative rest until symptoms settle, then light aerobic activity, then sport-specific exercise away from others, then non-contact training drills, then full contact practice, and only then return to normal competition.

相对休息直到症状消退，然后轻度有氧活动，然后在无人参与的情况下做专项练习，然后非接触性训练，然后完整接触的合练，最后

Each stage is passed when the athlete has been symptom-free at that stage, and the progression is deliberately conservative because the failure mode is a repeat injury during an apparently successful return.

每一级都在该级无症状的情况下通过，而这个递进被刻意设计得保守，因为其失败模式正是在一次看起来成功的回归过程中发生再次

Symptom-free is the operative phrase and it needs a definition, because "I feel fine" is a poor instrument.

"无症状"是那个操作性短语，而它需要被定义，因为"我感觉挺好的"是一个糟糕的工具。

Symptom-free means no headache, no dizziness, no nausea, no confusion, no difficulty concentrating, and no change in mood or behaviour - assessed by someone other than the athlete where possible, because an adolescent has a strong interest in reporting no symptoms.

无症状意味着没有头痛、没有眩晕、没有恶心、没有意识模糊、没有难以集中，以及情绪或行为没有变化--在可能的情况下由运动员以

Symptoms that return at a stage mean the load is too high at that stage, and the correct response is to drop back, not to push through.

在某一级出现症状复发，意味着那一级的负荷太高，而正确的反应是退回去，而不是硬撑过去。

The reason the stages are graded rather than binary is that concussion risk is about metabolic and vascular vulnerability, and those are dose-dependent.

之所以分级而不是二分，是因为脑震荡的风险与代谢和血管层面的脆弱性有关，而这两者都是剂量依赖的。

The brain after a knock is more susceptible to a second impact for some time, and the more demanding the activity, the more it can go wrong.

大脑在受撞击之后的一段时间内对第二次撞击更为敏感，而活动越苛刻，可能出错的余地就越大。

A graded return also produces better information: if symptoms return at stage four, you have learned something specific about this athlete, and that knowledge is worth more than the two weeks the conservative route appears to have cost.

分级回归还能产生更好的信息：如果症状在第四级复发，你就了解了关于这位运动员的一件具体事情，而这份知识比保守路线表面上

There are two things that sit outside the staircase and are both easy to forget.

有两件事在楼梯之外，而且都很容易被忘记。

The first is that a child or adolescent takes longer at every stage, and symptom resolution is more reliably judged by the school day - a bad Monday morning is more informative than a good afternoon at training.

第一，孩子和青少年在每一级都需要更长时间，而症状是否消退通过上学日来判断更可靠--一个糟糕的周一早晨比一次好的训练日下午

The second is that the risk of a repeat injury stays elevated for some time after full return, which is a question about the first few weeks back rather than about the staircase itself.

第二，在完全回归之后的一段时间内，再次损伤的风险仍然偏高--这是一个关于回归后最初几周的问题，而不是关于这座楼梯本身的问

The numbers / 算一算

Numbers to keep. The variable that matters is symptom-free days at each stage, not the total length of the return, and the practical numbers people quote are typically something like 24 to 48 hours of relative rest until symptoms settle, then a minimum of 24 hours at each subsequent stage, with each stage symptom-free before progressing. Those figures are conventions rather than evidence and should be treated as minimums, not targets - the honest framing is that the criteria are symptoms, and the numbers only describe how those criteria are usually administered. Children and adolescents should sit at each stage longer than adults, and the reason is physiological, not administrative.

需要记住的数字。真正重要的变量是每一级上无症状的天数，而不是回归的总时长，而人们引用的实用数字通常是：先做24到48小时的

Common mistakes / 易错点

- "Symptom-free means the athlete says they feel fine." An athlete is a poor reporter of their own symptoms, and the emotional and cognitive ones are the easiest to miss.

"无症状意味着运动员自己说感觉挺好。"运动员是自身症状的糟糕报告者，而情绪和认知方面的症状最容易漏掉。

- "They have been symptom-free for a week, so they can play." Symptom-free is a stage requirement, not a discharge from the entire protocol, and the first weeks back carry their own elevated risk.

- "他们已经无症状一周了，可以上场了。"无症状是每一级的要求，而不是整个流程的终止条件，而回归后的最初几周还有它自己的

- "The stages are the same for everyone." Children and adolescents progress more slowly at every stage, and progress is governed by response rather than by a schedule.

"这些级别对所有人都一样。"儿童和青少年在每一级都推进得更慢，而推进由反应而不是由时间表决定。

- "If symptoms return, push through the discomfort." Symptoms returning at a stage mean the load is too high at that stage. The response is to drop back a level.

"如果症状复发，就忍过去。"某一级出现症状复发意味着那一级负荷太高。反应是退回去一级。

Why it matters / 为什么重要

Concussion management is the clearest case in the course of a rule that is easy to state and hard to follow: symptom-free, progressing, and never skipping. It rewards the habits the rest of the topic teaches - objective criteria over time, one change at a time, and the willingness to go backwards. It is also the area where the cost of getting it wrong is highest and least reversible, which is why it is examined carefully and why the person making the return decision should be explicit about who assessed the symptoms. A student who can explain why a graded return exists, rather than reciting the stages, is ready to make these decisions.

脑震荡管理是这门课程中"规则容易说清、却难以遵守"的最清楚例子：无症状、递进、绝不跳过。它奖励的是这个主题其余部分所教的习

- A symptom return means step back to the previous tolerated stage rather than pushing through.

症状复发意味着退回上一个可耐受阶段，而不是硬撑。

Key terms / 核心术语

Return to sport 重返运动

The final stage after symptom-free progression through daily, cognitive and sport demands.

经过日常、认知和运动需求且无症状后的最后阶段。

Formulas / 公式

Drowsiness, repeated vomiting, worsening headache, confusion, seizure, neck pain, weakness, tingling or slurred speech are red flags.

嗜睡、反复呕吐、头痛加重、困惑、癫痫、颈痛、无力、刺痛或言语含糊是红旗症状。

Worked examples / 实例

>

Sequence a safe return from concussion.

排列脑震荡后的安全回归顺序。

Exam focus / 考核要点

What should happen if symptoms return?

症状复发时应怎么办？

Personality

人格

Guiding question / 引导问题

How do stable traits and social experiences shape sporting behaviour?

稳定特质和社会经验如何塑造运动行为?

Personality consists of relatively stable patterns of thought, feeling and behaviour. It emerges through interaction between traits and environment, influences tendencies rather than guarantees outcomes, and can change across the lifespan.

人格是相对稳定的思维、情感和行为模式。它通过特质与环境的互动形成，只影响行为倾向而不保证结果，并且可以随人生阶段变化。

Trait-environment interaction · 特质与环境交互作用

First, picture it / 先这样想

Traits are not verdicts. A trait is a tendency that shows up more often under some conditions than others, and the conditions are part of the explanation. Two athletes can both be described as competitive, and one of them will get enormous energy out of a championship while the other gets out of bed at the thought of it. The trait was never the whole story - the situation was doing half the work.

特质不是判决。特质是一种倾向，在某些条件下比另一些条件下更容易显现，而这些条件本身就是解释的一部分。两位运动员都可以被描述为“好胜”，其中一位会从锦标赛里获得巨大的能量，另一位一想到它就不想起床。特质从来不是全部--情境在其中起了一半的作用。

The detail / 详细讲解

The core idea is that personality and performance are not independent.

核心概念是人格与表现并非彼此独立。

Behaviour is a product of person and situation meeting, which is why the same athlete behaves differently at a training session, at a championship, at school and at home.

行为是人与情境相遇的产物，所以同一位运动员在训练课上、在锦标赛上、在学校和在家里的行为并不一样。

A coach who attributes a performance to a trait alone has explained half of it, and - more importantly - has concluded that nothing can be done, because traits are treated as fixed.

一位把一次表现完全归因于特质的教练只解释了其中一半，更重要的是，他由此得出了“无计可施”的结论，因为特质被视为固定的。

That conclusion is where trait psychology has done most damage in sport, and the alternative is interactionism.

这个结论正是特质心理学在运动中造成最大伤害的地方，而替代方案是交互论。

It says that a trait expresses itself differently depending on the situation - a high need for achievement may look like extra practice in a supportive environment and like avoidance of competition in a hostile one.

它认为特质会依情境

而以不同方式表达--在高成就需要的人身上，在支持性的环境中可能表现为额外训练，而在充满敌意的环境中可能表现为回避竞争。

So the useful question is never "is this athlete competitive" but "competitive in what, exactly, and under whose conditions".

所以有用的问题从来不是“这位运动员好胜吗”，而是“究竟在哪件事上好胜，在谁的条件下好胜”。

The other half of the interaction is that the environment shapes the trait over time.

交互的另一半是环境会随时间塑造特质。

Environments that are autonomy-supportive, that give feedback about controllable things, and that treat mistakes as information produce more self-directed, persistent learners.

提供自主支持、对可控因素给出反馈、把错误当作信息的环境，会培养出更自主、更持久的学习者；

Environments that are controlling, that praise ability, and that treat mistakes as verdicts produce the opposite.

而控制型的、赞美能力的、把错误当判决的环境，则培养出相反的结果。

This is the hopeful half of interactionism: the environment is a lever, and it is one a coach controls.

这是交互论中充满希望的一半：环境是一个杠杆，而且是教练能够控制的杠杆。

And this is where the evidence is strongest, because it comes from intervention studies rather than from personality descriptions.

而这里是证据最强的部分，因为它来自干预研究而不是人格描述。

Coaches trained in a specific style produce measurably different athlete outcomes from coaches trained in another style, holding the coaching content constant.

接受某种特定风格训练的教练，与接受另一种风格训练的相比，会产生可测量的不同运动员结果，而教练内容保持不变。

That is a much stronger claim than "some athletes are more motivated", and it is the version of the idea that belongs in a sport psychology course.

这比"有些运动员更有动力"强得多的主张，也才是这个属于运动心理学课程的概念应该采用的版本。

The numbers / 算一算

Numbers to keep. Individual differences are normally large - the spread between athletes on any given psychological measure is often wide enough that a group average describes very few individuals in it. That is the practical justification for individualising: if the average is not a good description of any one athlete, a programme built only for the average is wrong for most of the people doing it. On the environment side, the consistent finding across studies is that feedback and motivational climate reliably shift persistence, effort and self-talk - the size of those effects varies, but they are among the more dependable findings in the field.

需要记住的数字。个体差异通常很大--在任何给定的心理测量上，运动员之间的分布常常宽到某个群体平均值几乎不描述其中任何个体。

Common mistakes / 易错点

- "They are a natural, so they will be a natural." Trait descriptions predict tendencies, not outcomes, and a prediction that cannot be acted on is not a prediction worth making.
"他天生就是这块料，所以将来也会是。"特质描述预测的是倾向而不是结果，而一个无法据以行动的预测不是值得做出的预测。
- "Motivation is a personality trait." It is far more reliably a function of the environment the athlete is in, which makes it a target rather than a fixed input.
"动机是一种人格特质。"它更可靠地是运动员所处环境的一个函数，因此它是一个靶点，而不是一个固定的输入。
- "Environment does not matter much once the athlete is selected." Selection is itself a result of the environment the athlete developed in, and coaching style changes outcomes with content held constant.
"运动员被选出来之后环境就不那么重要了。"选拔本身就是运动员所处发展环境的结果，而教练风格在内容不变的情况下依然会改
- "Individualise everything." The traits a person brings are real and worth knowing. The point is not to ignore them but to read them in relation to a situation rather than in isolation.
"把一切都个体化。"一个人带来的特质是真实且值得了解的。重点不是忽略它们，而是结合情境来读它们，而不是孤立地读。

Why it matters / 为什么重要

This is the first section of the psychology theme and it establishes how everything after it should be read. It is the difference between explaining an athlete as a fixed thing and working with a person in a situation, and that difference shows up in every decision made afterwards: whether to change the task, the feedback, the environment or the athlete. The most valuable sentence a student can take from it is that trait descriptions are hypotheses about tendencies, while situation design is the part with evidence and leverage behind it.

这是心理学主题的第一节，它确立了此后所有内容应当如何被阅读。它是"把运动员解释成一个固定的东西"与"在一个情境中与一个人一起

- Personality can influence perception and appraisal, goals, motivation, coping, emotion regulation, decision making, risk taking and interpersonal behaviour.

人格可影响知觉和评价、目标、动机、应对、情绪调节、决策、风险选择和人际行为。

Key terms / 核心术语

Trait 特质

A relatively stable tendency within the person.

个人内部相对稳定的倾向。

Context 情境

The social and physical situation in which behaviour occurs.

行为发生的社会和物理情境。

Formulas / 公式

Two athletes with similar trait scores can behave differently because the task, pressure, environment and development differ.

两名运动员特质得分相近，却可能因任务、压力、环境和发展阶段不同而表现不同。

Worked examples / 实例

>

Explain trait-environment interaction.

解释特质与环境的互动。

Exam focus / 考核要点

Why can personality not simply predict performance?

为什么人格不能简单预测表现？

Big Five and measurement · 大五人格与测量

First, picture it / 先这样想

When someone says "psychology measures personality", they usually mean they asked a person to describe themselves, and then the answer was turned into a number. That is a defensible thing to do and it is also where most of the criticism comes from. The question is not whether the numbers mean anything - it is what a number about a person is actually a measurement of.

当有人说“心理学测量人格”时，他们通常是指让一个人描述自己，然后把回答变成一个数字。这是一件站得住脚的事，也正是大多数批评的来源。问题不在于这些数字是否有意义--而在于关于一个人的数字实际上测量的是什么。

The detail / 详细讲解

The Big Five - openness, conscientiousness, extraversion, agreeableness and neuroticism - came out of a statistical technique rather than a theory.

大五人格--开放性、尽责性、外向性、宜人性和神经质--是从一种统计技术而非一种理论中产生的。

Thousands of words describing personality were collected from many sources, reduced, and the clusters that survived were stable across studies and across languages.

描述人格的成千上万个词从许多来源被收集、缩减，而最终存活的维度在各研究之间、各语言之间都是稳定的。

That is why it is described as a taxonomy: it describes the dimensions on which people reliably differ, without claiming to explain why.

这就是为什么它被称为一种分类学：它描述的是人们可靠地存在差异的那些维度，而不声称解释为什么。

The usual measure is a self-report questionnaire in which the person rates agreement with statements.

通常的测量是一份自评问卷，让被试对陈述表示同意程度。

That design produces three familiar problems.

这一设计带来三个众所周知的问题。

Social desirability means people describe how they would like to be seen, especially in a coaching context where they want to be taken seriously.

社会赞许性意味着人们描述的是希望被怎么看，尤其在需要被认真对待的教练情境中；

Mood means the answer partly reflects the day rather than the person.

情绪意味着答案部分反映的是当天状态而不是这个人；

And self-knowledge means some people are simply better at guessing what the questionnaire wants.

而自我认识能力意味着有些人就是更擅长猜出问卷想要什么答案。

The alternative approaches exist because self-report is not the only route to the same construct.

存在其他方法，正是因为自评不是通往同一构念的唯一路径。

A structured interview gives a trained observer more room to follow up, at the cost of time and of the observer's

own bias.

结构化访谈给了受过训练的观察者更多追问的空间，代价是时间以及观察者自身的偏差。

Behavioural observation samples what the athlete does rather than what they say, and it is strong on the traits that show up in conduct - conscientiousness especially - and weak on the ones that do not.

行为观察取样的是运动员做什么而非说什么，它对在行为中显现的特质--尤其是尽责性--很强，而对不显现的特质较弱。

Peer and coach ratings add another perspective, and they capture something self-report structurally cannot: how the person comes across to others in this role.

同伴和教练评分提供了另一个视角，并且捕捉到了自评在结构上无法捕捉的东西：这个人身在这个角色中，给别人带来的印象如何。

The numbers / 算一算

Numbers to keep. Standard test-retest reliability for well-established trait questionnaires over a short interval is typically in the region of 0.7 to 0.9, which is good enough to trust a pattern across a season but not good enough to call a single result a fact about a person. And the figure most worth remembering is the interval: trait measures are stable over weeks to months and much less stable over days, which is exactly why measuring an athlete on a bad day and concluding something about their personality is a mistake the design invites.

需要记住的数字。成熟人格问卷在短间隔下的重测信度通常在0.7到0.9之间，这足以让人相信一个赛季中的模式，但不足以把一次结果称为事实。

Common mistakes / 易错点

- "The Big Five is a theory of personality." It is a taxonomy derived from word data. It describes dimensions reliably; it does not explain their causes.
"大五人格是人格理论。"它是一种从词汇数据中导出的分类学。它可靠地描述维度；它不解释其成因。
- "The score tells you what the athlete is like." A self-report score is a sample of how the athlete described themselves on that day, filtered through desirability and mood.
"分数告诉你这位运动员是什么样的人。"一份自评分数是运动员在那一天如何描述自己的一个样本，并经过了赞许性和情绪的过滤。
- "Only self-report counts, because only the person knows." A valid finding is one that holds across methods. Convergent evidence from a different route is what upgrades a result from impression to fact.
"只有自评算数，因为只有本人知道。"一个有效的发现是在不同方法下都成立的那个。来自另一条路径的汇聚性证据，正是把一个印象升级为事实。
- "No measurement, so no conclusion." The opposite error is as common: treating an unreliable number as precise. Measure, and then treat the result as an estimate with a range.
"没有测量，所以没有结论。"相反的错误同样常见：把一个不可靠的数字当作精确值。要测量，然后把结果当作带范围的估计。

Why it matters / 为什么重要

Measurement is the least glamorous and most consequential topic in any applied field, because it decides what can be claimed. A coach who believes the questionnaire measured the person rather than the answer will draw firmer and worse conclusions than one who knows the limits, and the difference shows up every time a plan is built on a single number. The transferable skill is knowing when a measure is strong for the question being asked and when it is not - which is exactly the judgement the rest of this theme requires, applied to athletes instead of questionnaires.

在任何应用领域里，测量都是最不光鲜却最有后果的话题，因为它决定了什么结论可以被主张。一位相信问卷测到的是人而不是答案的教练会做出更坚定也更糟糕的结论，而知道局限性的教练则不会，这种差异会在每次制定计划时显现出来。可迁移的技能是知道何时测量对提出的问题足够有力，何时不够 - 这正是这个主题其余部分所需要的判断，应用于运动员而不是问卷。

- Examples of use include Big Five Inventory, NEO PI-R, Sport Personality Scale and Competitive State Anxiety Inventory-2. Ethical use includes informed consent, confidentiality, appropriate interpretation and feedback for development.
应用例子包括大五人格问卷、NEO PI-R、运动人格量表和竞技状态焦虑量表-2。伦理使用包括知情同意、保密、适当解释和支持发展的反馈。

Key terms / 核心术语

Reliability 信度

Consistency of results.

结果的一致性。

Validity 效度

Evidence that a measure assesses what it intends to assess.

测量工具确实评估目标概念。

Formulas / 公式

Use the measure to identify a development opportunity, not to label or rank an athlete.

使用测量识别发展机会，而不是给运动员贴标签或排名。

Worked examples / 实例

>

What qualities should a personality measure have?

人格测量应具备哪些质量？

Exam focus / 考核要点

Name the five Big Five traits.

说出大五人格的五个特质。

Social learning and development · 社会学习与发展

First, picture it / 先这样想

Most of what a coach teaches is not invented by the coach, and much of what an athlete does was learned before they ever walked into a session. Behaviour is learned by watching, by trying, and by finding out what happens afterwards. That is good news and bad news: it means coaching works, and it also means the things a coach models are being learned whether the coach intends it or not.

教练所教的大部分内容不是教练发明的，而运动员所做的大部分事情在他走进训练课之前就已经学过了。行为是通过观察、尝试，以及事后发现结果如何而被学会的。这是好消息也是坏消息：它意味着教练是有效的，同时也意味着教练所示范的东西--无论教练是否有意--都在被学习。

The detail / 详细讲解

Social learning is observation plus outcome.

社会学习是观察加上结果。

An athlete watches a model, forms an expectation about what will happen if they copy it, and then tests that expectation.

运动员观察一个榜样，形成一个关于“如果我照做会发生什么”的预期，然后检验这个预期。

Attention decides what gets copied, retention decides what survives to the attempt, reproduction decides whether it can be performed, and motivation decides whether it is performed at all when a cheaper option exists.

注意力决定什么会被复制，记忆决定什么能保留到尝试时，动作再生决定它是否能被做出来，而动机决定在存在更省力选项时它是否

That chain explains why demonstrating a skill works better than describing it, and also why a demonstration the athlete cannot yet reproduce produces frustration rather than learning.

这条链解释了为什么示范一个技术比描述它更有效，也解释了为什么一个运动员尚无法再现的示范产生的是挫败而不是学习。

Bandura's work is the reference point, and the two findings that matter most in sport are vicarious learning and self-efficacy.

班杜拉的研究是参照点，而其中对运动最重要的两个发现是替代学习和自我效能。

Vicarious learning is straightforward: seeing a peer succeed raises the belief that success is possible for someone like me.

替代学习很直接：看到同伴成功会提高“像我这样的人也可能成功”的信念。

Self-efficacy is the belief that I can execute a specific behaviour in a specific situation, and it is the more useful of the two, because it is situation-specific rather than general - which is why one athlete can be confident on the pole vault bar and not on the approach run.

自我效能是“我能在某个具体情境中做出某个具体行为”的信念，它是两者中更有用的那个，因为它针对具体情境而不是笼统--这正是为

Development is the developmental-psychology side, and its contribution to sport is a timing message.

发展是发展心理学这一侧，它对运动的贡献是一个关于时机的信息。

The adolescent brain is not a worse adult brain, it is a differently organised one, and the practical consequences

are about when to teach and when to let an athlete specialise.

青少年的脑子不是"差一点的成人脑子", 而是一个组织方式不同的脑子, 其实际后果是关于何时教、以及何时让运动员进行专项化。

The window for certain motor and perceptual learning is favourable in childhood, some windows close, and the adolescent peak in some domains is accompanied by a real decline in others - which is the argument against early specialisation and for sampling many skills during this period.

某些动作和知觉学习在儿童期有有利窗口, 某些窗口会关闭, 而某些领域在青春期的峰值伴随着另一些领域的真实下降--这就是反对过早专业化。And the coaching implication is uncomfortable: the athlete is being taught constantly, including by everything the coach does off the clock.

而对教练而言的推论有些令人不适: 运动员一直在被教导, 包括被教练在场下所做的一切所教导。

What the coach rewards, what the coach ignores, how the coach reacts to a mistake in front of the group, and what the culture punishes are all part of what the athlete learns.

教练奖励什么、忽视什么、教练如何在众人面前对一个错误作出反应、以及这种文化惩罚什么, 这些全都是运动员所学内容的一部分。

Awareness of that is not censorship - it is recognising that a coach teaches by example whether or not there is a lesson plan.

对这一点的觉察不是审查--而是认识到无论有没有教案, 教练都以身作则地教导着。

The numbers / 算一算

Numbers to keep. Self-efficacy is measurable per task, which is what makes it far more useful than a general confidence rating, and the practical habit is to measure the specific behaviours that the next competition will require. On the developmental side, the two figures worth knowing are the conventional windows for general motor learning in childhood and for peak rate of learning in adolescence - the practical reading being that the period in which an athlete is most plastic is not the period in which they are most physically developed, and the two are easily confused.

需要记住的数字。自我效能可以按任务测量, 这正是它比笼统的自信评分有用得多的原因, 而实用的习惯是测量下一次比赛真正需要的动作。

Common mistakes / 易错点

- "Some athletes are more motivated; the rest are not." Motivation is affected by the confidence the athlete has in their own ability to perform, and that confidence is built by mastery experiences, modelling and the environment the coach creates.
"有些运动员更有动力, 其余的不是。"动机会受运动员对自己能力信心的影响, 而这种信心由成功经验、榜样以及教练创造的环境所塑造。
- "Self-efficacy is confidence." It is specific to a behaviour in a situation. An athlete can have high self-efficacy for a task they have mastered and low self-efficacy for a nearly identical one they have not practised.
"自我效能就是自信。"它针对具体情境中的具体行为。运动员对一个已经掌握的任务可以有很高的自我效能, 而对一个几乎相同但尚未掌握的任务则自我效能较低。
- "Motivation is either intrinsic or extrinsic." Performance can be directed toward mastery, toward winning, toward approval, or toward avoiding shame, and the mix changes with the situation and the stage of a career.
"动机要么内在要么外在。"表现可以指向掌握、指向获胜、指向获得认可, 或指向避免羞耻, 而这个组合会随情境和职业生涯阶段而变化。
- "Coaching is the planned session." The planned session is a small fraction of what is learned, and the unplanned part is the part the athlete copies most closely.
"教练工作就是计划的训练课。"计划的训练课只是所学内容的一小部分, 而没计划的那部分才是运动员模仿得最像的部分。

Why it matters / 为什么重要

This section supplies the mechanism for why coaching works at all, which is an odd thing to say in a course that is largely about what to coach. Observation, attention, retention and motivation give a technical explanation for the gap between a demonstration and a performance, and self-efficacy gives a handle on why the same athlete behaves differently on the same skill on different days. It is also where the developmental material earns its place: knowing when a window is open is a practical intervention that has more effect than almost anything an adult can add later.

这一节提供了"教练为什么有效"的机制, 而在一门主要关于该教什么的课程里, 这听起来有点奇怪。观察、注意、记忆和动机构成了对"动作"的编码。

- Personality can change through maturation, experiences, success, failure, feedback, intentional practice, self-reflection, life events and supportive social environments.

人格可通过成熟、成功失败经验、反馈、有意练习、自我反思、生活事件和支持性社会环境改变。

Key terms / 核心术语

Self-fulfilling development 自我实现式发展

A supported tendency becomes stronger through practice and experience.

通过练习和经验，支持的倾向变得更稳定。

Formulas / 公式

A coach who models respectful effort and gives constructive feedback can support positive habits and persistence.

教练示范尊重努力并提供建设性反馈，可以支持积极习惯和坚持。

Worked examples / 实例

>

How can personality develop over time?

人格如何随时间发展？

Exam focus / 考核要点

Why is social learning relevant to sport?

为什么社会学习与运动相关？

Mental toughness

心理韧性

Guiding question / 引导问题

How do individuals respond differently to pressure and challenge?

个体如何以不同方式回应压力和挑战？

Mental toughness supports consistent performance under pressure, adversity and challenge. The workbook's five core attributes are challenge appraisal, commitment, confidence, perceived control and resilience. It is developable and interacts with health, sleep and attribution.

心理韧性支持压力、逆境和挑战下的稳定表现。工作册的五个核心属性是挑战评估、投入、自信、知觉控制和恢复力。它可以发展，

The five core attributes · 五个核心属性

First, picture it / 先这样想

Motivation is not a single thing sitting inside an athlete waiting to be measured. It is a moving target made of several processes, and the useful skill is learning to tell which one you are looking at. When someone says "he is unmotivated", the sentence is almost always incomplete - because you can name at least four different things they might have meant, and each of them calls for a different response from a coach.

动机不是运动员身体内部等着被测量的某个单一东西。它是由若干过程构成的移动靶标，而有用的技能是学会分辨你正在看的是哪一个。当有人说“他没有动力”时，这句话几乎总是不完整的--因为你至少能说出四种他可能指的意思，而每一种都要求教练做出不同的回应。

The detail / 详细讲解

The set usually taught splits into four processes.

通常所教的一组划分是四个过程。

Activation is the energy to begin: sleepiness, anxiety about an event, low arousal after travel all reduce it, and it is the easiest of the four to misread as motivation.

激活是开始行动所需的能量：睡意、对某项赛事的焦虑、旅行之后的低唤醒都会降低它，而它是这四个中最容易被误读为动机的一个。

Direction is choosing an outcome, and the two basic directions are mastery and winning, which pull in different directions under pressure.

方向是对结果的选择，两个基本方向是掌握与获胜，而在压力下它们会朝不同方向拉扯。

Intensity is how much of the available energy goes into the activity, and persistence is how long it stays there.

强度是可用能量中投入该活动的比例，而坚持是这份能量停留的时间。

The fifth attribute sits alongside them and is often the most useful in practice, which is whether the reason is internal.

第五个属性与它们并列，在实践中常常最有用，那就是理由是否是内部的。

Not internal as in "does it come from inside" - everything the athlete does is experienced from inside - but internal as in "does the athlete endorse it".

内部的含义不是“它是否来自内心”--运动员做的一切在体验上都来自内心--而是“运动员是否认可它”。

An athlete can generate high intensity for a goal they do not value, and the fuel runs out quickly.

一位运动员可以为一个自己并不看重目标产生很高的强度，而燃料很快就耗尽。

The internal reason is a property of the athlete's endorsement, and it is the one that predicts persistence most reliably.

内部理由是运动员认可与否的属性，也是最可靠预测坚持的那个。

The value of the framework is diagnostic rather than decorative.

这个框架的价值在于诊断而不是装饰。

当一位运动员退出时，问“四个过程中的哪一个发生了变化”，就把一个观察变成了一组假设：激活指向睡眠、营养、疾病和负荷；

When an athlete withdraws, asking which of the four processes has changed turns one observation into a set of hypotheses: activation points to sleep, nutrition, illness and load; direction points to whether the goal itself still matters to them; intensity points to whether the activity is valued; persistence points to whether the environment

keeps removing the reason to continue.

方向指向目标本身对他是否还重要； 强度指向这项活动是否被看重；

These have almost nothing in common in terms of what you do about them.

坚持指向环境是否一直在移除他继续下去的理由。 这些在"该怎么做"上几乎没有共同之处。

And the framework is deliberately a description rather than a prescription.

而这个框架刻意是描述性的而不是规定性的。

It tells you that a thing changed; it does not tell you what the change means or how to produce it.

它告诉你某个东西变了；

Knowing that persistence has dropped tells you where to look.

它不告诉你这个变化意味着什么、也不告诉你如何产生它。

It does not tell you that the answer is praise, or more rest, or a different goal, or a better relationship with the coach

- and choosing between those is where the judgement actually lives.

知道坚持下降了，就告诉你该去哪里看。

它并不告诉你答案是表扬、更多休息、换一个目标，还是与教练建立更好的关系--而在这些之间做选择，才是判断力真正所在的地方。

The numbers / 算一算

Numbers to keep. The measurement habit that fits this framework is to track the four processes separately rather than as one "motivation" score, because they move for different reasons and a combined number hides which one moved. On duration, the two time frames that matter are the seconds around a start and the weeks across a season, and the same framework describes both - an athlete can be well activated and well directed and simply have no persistence left. The useful practical figure is that a change in one process is a much stronger clue than a change in a combined score, and it costs nothing to record them separately.

需要记住的数字。与这个框架相配的测量习惯是把四个过程分别追踪，而不是合并成一个"动力"分数，因为它们出于不同原因变化，而一

Common mistakes / 易错点

-
"Unmotivated" is a diagnosis. It names none of the processes, so it points at nothing and licenses whatever the coach already believed.
"没有动力"是一个诊断。它没有指明任何过程，所以它没有指向任何东西，只是让教练以为自己原本的看法得到了印证。

-
"Activation, direction, intensity and persistence are one thing." They are four processes with different causes and different responses, and treating them as one guarantees the wrong intervention.
"激活、方向、强度和坚持是一回事。"它们是四个原因不同、应对不同的过程，把它们当作一个，就保证了会用错干预。

-
"Intrinsic means coming from inside." All experience is felt from inside. The meaningful contrast is whether the athlete endorses the reason, not where it is located.
"内在意味着来自内心。"所有体验都从内部被感受到。有意义的对比是运动员是否认可那个理由，而不是理由的位置在哪里。

-
"Motivation is only about how much effort appears." Effort is the visible output. Activation, direction and persistence are upstream of it and are where the causes sit.
"动机只是关于看起来使出了多少努力。"努力是可见的产出。激活、方向和坚持在它上游，原因在那里。

Why it matters / 为什么重要

This framework is the most immediately usable thing in the whole psychology theme, because almost every coaching conversation eventually turns into a question about why an athlete is or is not giving what the session requires. Having four named options makes that question answerable, and having a fifth that asks whether the athlete endorses the reason stops a coach from mistaking compliance for commitment. It also builds a habit that carries into every other topic: name the process, form the hypotheses, and resist the temptation to reach for the same response regardless of which process actually changed.

这个框架是整个心理学主题中最可直接使用的东西，因为几乎每一次教练谈话最终都会变成一个问题：这位运动员为什么给出、或者没有

-
A pressure moment may require all five attributes, but the relevant emphasis depends on the task and person.
压力情境可能需要全部五个属性，但重点取决于任务和个人。

Key terms / 核心术语

Challenge appraisal 挑战评估

Seeing pressure as growth rather than threat.

把压力看作成长而不是威胁。

Resilience 恢复力

Bouncing back and learning from adversity.

从逆境中恢复并学习。

Formulas / 公式

A basketball player views a close final quarter as a chance to prove themselves, focuses on controllables and resets after an error.

篮球运动员把关键的最后一节看作证明自己的机会，关注可控因素，并在失误后重新调整。

Worked examples / 实例

>

Name and apply the five core attributes.

说出并应用五个核心属性。

Exam focus / 考核要点

How do the five Cs support pressure performance?

五个C如何支持压力下表现？

Malleability and self-fulfilling belief · 可塑性与自我实现预言

First, picture it / 先这样想

If a coach treats an athlete as a poor performer, and the coach is in a position of authority, something odd happens: the athlete starts to behave in ways that are consistent with the expectation. Not because the coach was right, and not because the athlete is weak - but because people tend to act consistently with what they believe is expected of them, and because the coach has the power to arrange the evidence.

如果一位教练把运动员当成水平差的选手来对待，而这位教练又处在权威位置上，就会发生一件奇怪的事：运动员开始表现出与这个预期一致的行为。不是因为教练是对的，也不是因为运动员软弱--而是因为人们倾向于表现得与自己认为别人对自己的期待一致，也因为教练有能力安排好那些证据。

The detail / 详细讲解

Self-fulfilling belief works in both directions and it is not a personality weakness - it is ordinary social psychology operating at full strength.

自我实现信念在两个方向上都起作用，而且它不是人格弱点--它是普通社会心理学在满负荷运转。

A positive expectation produces behaviour that confirms it: the coach expects effort, gives more responsibility, the athlete responds, and the coach now has evidence.

正向预期产生确认它的行为：教练期待努力，就给更多责任，运动员作出回应，教练于是有了证据。

A negative expectation runs the same loop in reverse, and it is more damaging because the coach both creates the belief and controls the opportunities that could disprove it.

负向预期以同样的回路反向运行，而且破坏性更大，因为教练既创造了那个信念，又控制了本可以推翻它的机会。

The mechanism is prediction.

机制是预期。

When an athlete expects to fail, perception changes too - the same ambiguous feedback is read as criticism, other people's success is read as luck, and available information is selectively attended to in ways that confirm the expectation.

当一位运动员预期会失败时，知觉也会改变--同样含混的反馈被读成批评，别人的成功被读成运气，而可获得的信息被选择性地以确认

Low expectation also removes the challenges that would generate a different experience, so the athlete gets less opportunity to be competent, which makes the low expectation look reasonable.

低预期还会移除那些本会产生不同体验的挑战，于是这位运动员获得"胜任"的机会更少，而这又让低预期显得合理。

The loop closes and becomes self-sustaining.

这个回路闭合了，变成了自我维持的。

Malleability is the good half of the same idea, and it is the half that makes a coaching career worth doing.

可塑性是同一个想法中好的一半，也正是它让教练这份职业值得做下去。

The characteristics people describe as fixed in early adolescence - how much confidence they have, how they react to failure, whether they find competition attractive or threatening - are more changeable than they look, and they change most in response to what the environment does.

人们认为在青春期早期就已固定下来的那些特征--他们有多少自信、他们如何对失败作出反应、他们是觉得比赛有吸引力还是有威胁

A coach who is told "this one is not a fighter" has been given a description of a past environment, not a fact about a person.

一位被告知"这个孩子不是战斗型"的教练，得到的其实是对过去环境的描述，而不是关于这个人的事实。

The practical consequences run in both directions.

实际推论在两个方向上都成立。

For an athlete labelled weak, the intervention is to engineer competence experiences: achievable tasks, honest feedback, a visible record of improvement, and responsibility that is real rather than symbolic.

对一位被贴上"弱"标签的运动员，干预是设计胜任体验：可完成的任务、诚实的反馈、可见的进步记录，以及真实而非象征性的责任。

For an athlete who is over-confident and under-prepared, the symmetric move is to raise difficulty, protect them from public comparison, and give them the experience of being genuinely stretched.

而对一位过度自信、准备不足的运动员，对称的举措是提高难度、保护他们不被当众比较、并给他们真正感到吃力的体验。

Both are the same skill: arranging experiences that make the belief and the performance match each other in the right direction.

两者是同一个技能：安排体验，让信念与表现在正确的方向上彼此匹配。

The numbers / 算一算

Numbers to keep. The measurable part of self-fulfilling belief is that the performance gap attributed purely to expectation is not trivial - expectancy effects in sport and education are typically reported as meaningful, on the order of a fraction of a standard deviation rather than a rounding error. The harder number is developmental: the characteristics treated as fixed in early adolescence are the ones that remain most changeable, which is the argument for treating a coach's judgement about a young athlete as provisional rather than as a forecast.

需要记住的数字。自我实现信念中可测量的部分是：完全归因于预期的那部分表现差距并不微小--在运动和教育中报告的预期效应通常是

Common mistakes / 易错点

- "They are not a fighter, so that is just who they are." That is a description of how they have been treated so far. Malleability means it is a starting hypothesis, not a conclusion.
"他不是战斗型的，所以他就是这样的人。"这只是对他至今所受对待的描述。可塑性意味着它是一个起始假设，而不是一个结论。
- "Self-fulfilling belief is about confidence." It is about behaviour, opportunity and the evidence available. Confidence is one part of the loop, not the whole of it.
"自我实现信念是关于自信的。"它是关于行为、机会和可获得证据的。自信是这个回路的一部分，而不是全部。
- "Boost confidence and performance will follow." Confidence without competence and without appropriate challenge is fragile, and it fails at exactly the moment it is needed.
"提升自信，表现自然跟上。"没有能力、也没有适当挑战支撑的自信是脆弱的，而且恰在最需要它的时刻失效。
- "Only the low-expectation case matters." Over-confident, under-prepared athletes are the mirror image and need the opposite treatment: harder tasks, less comparison, more honest feedback.
"只有低预期那一种情况重要。"过度自信而准备不足的运动员是镜像情况，需要相反的处理：更难的任务、更少的比较、更诚实的

Why it matters / 为什么重要

Malleability and self-fulfilling belief are the two halves of the same ethical problem, and they are the part of the theme a student is most likely to be examined on in a case question. The negative half says that a coach's low expectation can cost an athlete years of confidence, and the positive half says the same mechanism can be run in the athlete's favour deliberately. Neither half requires advanced theory to apply: it requires noticing which way the

loop is running in front of you, and then changing what you put in it.

可塑性与自我实现信念是同一个伦理问题的两半，也是学生最有可能在案例题上被考查的部分。负面的一半说：教练的低预期可能让一位

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The MTQ48 is presented as a 48-item self-report tool measuring the five attributes. A score identifies strengths and areas for development rather than proving a guaranteed outcome.

MTQ48是测量五个属性的48项自评工具。得分可识别优势和发展领域，而不是证明必然结果。

Key terms / 核心术语

Malleable 可塑的

Capable of developing through experience and practice.

可通过经验和练习发展。

Formulas / 公式

An athlete who believes a difficult routine is learnable may practise more carefully, succeed and become more confident.

运动员相信困难动作可以学会时，可能更认真练习、成功并增强自信。

Worked examples / 实例

>

Explain a self-fulfilling prophecy in sport.

解释运动中的自我实现预言。

Exam focus / 考核要点

What is the MTQ48 used for?

MTQ48用于什么？

Learned helplessness, attribution and health · 习得性无助、归因与健康

First, picture it / 先这样想

Two things hide behind a sentence like "I just am not good at this". One is what happened, and the other is what the athlete concluded from it. The first is history and the second is a belief - and the belief is the part that decides whether the next attempt gets made, and whether the athlete takes advice, and whether they come back after a bad day.

在“我就是不擅长这个”这句话后面藏着两样东西：一样是发生过的事，另一样是运动员由此得出的结论。前者是历史，后者是信念--而信念才是决定下一次尝试会不会发生、会不会接受建议、以及糟糕的一天之后还会不会回来的那一部分。

The detail / 详细讲解

Learned helplessness is the specific and most damaging outcome.

习得性无助是其中最具体、也最具破坏性的结果。

It forms when effort repeatedly produces no change: the athlete tries, fails, tries again, fails again, and the reasonable conclusion is that effort does not work here.

它的形成条件是努力反复地不产生变化：运动员尝试、失败、再尝试、再失败，于是得出一个看似合理的结论--在这里努力没有用。

That conclusion is accurate - effort genuinely did not work - and it generalises far beyond the situation that produced it.

这个结论是准确的--努力确实没有用--而它会远远超出产生它的那个情境去做泛化。

A badminton player who has been beaten repeatedly by a stronger opponent does not conclude "I lost those matches"; they conclude "I am not a badminton player".

一位反复被更强对手击败的羽毛球运动员得出的结论不会是“我输了那些比赛”，而是“我不是打羽毛球的”。

Attribution is the frame that decides what happens next, because the explanation an athlete holds determines what they try.

归因是决定接下来会发生什么的框架，因为运动员持有的解释决定了他会尝试什么。

Attributing a loss to something stable and internal - "I have no talent" - discourages further effort, because effort

could not possibly change it.

把一次失败归因于稳定且内部的"我没有天赋", 会打消进一步努力, 因为努力不可能改变它;

Attributing it to something unstable or controllable - "my positioning was wrong", "I was not ready" - points toward the next attempt.

把它归因于不稳定的或可控的"我的站位不对""我还没准备好", 则指向下一次尝试。

The third category, blaming circumstances entirely, feels good briefly and protects the athlete from examining anything.

第三类--完全归咎于外界的情形--短期感觉不错, 并且保护运动员不去审视任何东西。

This is a skill worth teaching deliberately, because the attributions athletes default to are learned from us.

这是一项值得有意教养的技能, 因为运动员的默认归因是从我们这里学来的。

A coach who treats every mistake as evidence of low ability teaches stability and internality, and the athlete learns them.

一位把每个错误都当作能力不足证据的教练教会了稳定性和内归因, 于是运动员就学会了这些。

A coach who asks what the athlete controlled, and what information they had at the time, teaches controllability and information.

一位追问"运动员控制了哪些""当时他掌握了什么信息"的教练教会了可控性和信息。

The same mistake, coached differently, produces a different athlete three months later.

同一个错误, 不同的教练方式, 会在三个月后产生不同的运动员。

And it connects to health because helplessness and depression share a mechanism, while mastery and self-efficacy protect against both.

而它与健康相连, 是因为无助与抑郁共享一个机制, 而掌握感和自我效能则对两者都有保护作用。

The practical consequence for a coach is that the most effective mental-health support in a sporting setting is not a referral but the daily structure of the environment: challenges that are real, feedback that is specific and about controllable things, and a record of progress the athlete can see.

对教练而言的实际推论是: 在运动环境中, 最有效的心理健康支持不是一次转介, 而是环境的日常结构: 真实的挑战、针对可控之事的

That structure is protective because it repeatedly supplies the evidence that effort produces change.

这种结构之所以有保护作用, 是因为它反复提供了"努力会产生变化"的证据。

The numbers / 算一算

Numbers to keep. The proportions are the part worth having in mind: in most athlete samples a substantial minority of competitors report a helplessness pattern after repeated non-response, and the practical figure for a coach is not the percentage but the number of athletes in any group who have stopped trying. The paired measurement is equally useful - ask the athlete to rate both effort and outcome after each session, because the pattern where effort is high and outcome is flat is the signature of the problem and the earliest point at which it can be caught.

需要记住的数字。比例是值得放在脑子里的部分: 在多数运动员样本中, 反复经历无效之后报告出无助模式的是相当大的一部分人, 而

Common mistakes / 易错点

- "They need to work harder." If effort has repeatedly produced no change, more effort of the same kind confirms the belief. The variable that needs changing is the task, the feedback or the difficulty, not the effort.
"他们需要更努力。"如果努力反复没有产生变化, 更多同类努力只会确认那个信念。需要改变的是任务、反馈或难度, 而不是努力。
- "Never make excuses." The attributions an athlete offers are information about their belief system, and dismissing them removes the only way to find out what the belief actually is.
"不要找借口。"运动员给出的归因是关于其信念系统的信息, 否定它们会移除了解这个信念究竟是什么的唯一途径。
- "Motivation is a separate topic from mental health." Helplessness and depression share a mechanism, and mastery and self-efficacy protect against both.
"动机是与心理健康分开的另一个话题。"无助与抑郁共享一个机制, 而掌握感与自我效能对两者都有保护。
- "Praise the effort, whatever happened." Effort praise is only accurate when the effort did produce something. Praise that is disconnected from reality teaches the athlete that your feedback is noise.

"不管结果如何都表扬努力。"只有当努力确实产生了东西时，表扬努力才是准确的。与现实脱节的表扬会教会运动员：你的反馈是

Why it matters / 为什么重要

This is the section where the psychology theme becomes a welfare topic rather than a performance topic, and it is the part most likely to matter in a real career. The athlete who has stopped trying is not being difficult; they have made a correct inference from the evidence they were given, and the coach's job is to change the evidence rather than to argue with the inference. It is also the place where the least glamorous intervention - small achievable tasks, specific feedback about controllable things, a visible record of progress - does the most good.

这一节是心理学主题从表现问题变成福祉问题的地方，也是真实职业生涯中最可能产生意义的部分。那位已经停止尝试的运动员并不是在

-

Higher mental toughness is linked in the workbook with better stress management, fewer depressive or burnout symptoms, better sleep, positive mood and recovery.

工作册将较高心理韧性与更好的压力管理、更少的抑郁或倦怠症状、改善睡眠、积极情绪和恢复联系起来。

Key terms / 核心术语

Locus of control 控制点

Whether an explanation focuses on internal or external causes.

解释聚焦内部原因还是外部原因。

Stability 稳定性

Whether a cause is seen as changeable over time.

原因是否被认为会随时间改变。

Formulas / 公式

A player who misses a penalty may think "my routine needs adjusting" and practise, rather than "bad luck means I cannot change anything."

罚球未进的球员可以想"我的流程需要调整"并训练，而不是认为"坏运气意味着无法改变"。

Worked examples / 实例

>

How do internal, unstable attributions support motivation?

内因、不稳定归因如何支持动机？

Exam focus / 考核要点

Name two health or wellbeing associations of mental toughness.

说出心理韧性的两项健康或福祉联系。

Motor learning processes

运动学习过程

Guiding question / 引导问题

How are skills acquired, practised and perfected?

技能如何获得、练习和完善？

Motor learning is a relatively permanent change in capability; performance is what can be produced now and can fluctuate. Practice, feedback, schemas, stages, constraints, transfer and the environment shape learning.

运动学习是能力的相对永久变化；表现是当下能完成的内容，会波动。练习、反馈、图式、阶段、限制条件、迁移和环境共同塑造学习。

Learning, performance and schemas · 学习、表现与图式

First, picture it / 先这样想

Two performances by the same athlete, sometimes on the same day, are not the same act. One is running something that has become automatic; the other is working out a problem in real time. Coaches constantly slide between them without noticing, and almost every frustrating coaching session is a moment when the coach treated a second performance like a first one.

同一位运动员的两次表现--有时就在同一天--并不是同一件事。一件是把某样东西跑到自动化；另一件是实时解决一个问题。教练在两者之间不断滑动而没有察觉，而几乎每一次令人沮丧的教练课，都是教练把“第二次表现”当成了“第一次表现”的那一刻。

The detail / 详细讲解

Learning is the relatively permanent change that comes from practice and experience, and performance is what shows up on the day.

学习是由练习和经验带来的相对持久的变化，而表现是当天呈现出来的东西。

A skill is learned when the athlete can produce it under conditions that are not ideal - when they are tired, when the crowd is loud, when the surface is different - because performance is the thing that gets tested and learning is the thing that made it possible.

当运动员能在不理想的条件下做出某项技术时--在他们疲劳时、在观众很吵时、在地面不同时--这项技术才算学会了，因为表现是那个被检验的东西，而学习是让表现成为可能的东西。

That gap is where schemas come in.

这个落差就是图式发挥作用的地方。

A schema is a mental structure that organises what the athlete already knows, and in sport the main ones are perceptual: a schema for a shot is a pattern of body positions that the athlete has stored, and the athlete does not consciously think it up - they recognise the situation, and the pattern arrives.

图式是组织运动员已有知识的心理结构，而在运动中主要的是知觉图式：投篮的图式是运动员储存下来的一组身体位置模式，运动员

A well-developed schema is fast and almost automatic; a poorly developed one means every shot is a fresh problem to solve.

发展良好的图式快速且几乎自动化；发展不良的图式意味着每一次投篮都是一个需要从头解决的新问题。

The consequence is that practice and performance respond to different things.

推论是：练习和表现会对不同的东西作出反应。

Practice responds to feedback, to the quality of the attempt, and to how the task is set.

练习对反馈、对尝试的质量、以及任务如何设置作出反应；

Performance responds to how the athlete is that day and to how much pressure the situation adds.

表现则对运动员当天的状态、以及情境增加了多少压力作出反应。

A coach who is puzzled that an athlete performs badly in a competition despite performing well in training is usually looking at two different processes and has assumed they are the same one.

一位对“运动员训练中表现好、比赛中表现差”感到困惑的教练，通常是在看两个不同的过程，却假定它们是同一个。

The other half of the gap is what happens under pressure, and the useful idea is that pressure does not damage the skill - it changes the conditions under which the athlete accesses it.

这个落差的另一半是压力下会发生什么，而有用的概念是：压力并不破坏技术--它改变了运动员提取技术时的条件。

Under pressure, attention narrows, the number of cues the athlete can use falls, and a skill that had been stored as a fast pattern has to be reconstructed deliberately.

在压力下，注意变窄，运动员能使用的线索数量减少，而一项原本以快速模式储存的技术必须被刻意重新建构出来。

That is why a well-learned skill is more robust under pressure, and why more practice is not always the answer: practice that helps an athlete find the skill under pressure is different from practice that only makes it smooth in training.

这正是为什么一个学得扎实的技术在压力下更稳健，也为什么"多练"并不总是答案：帮助运动员在压力下找到技术的练习，与只是让技术更流畅不同。

The numbers / 算一算

Numbers to keep. The practically important number is the size of the competition effect - the drop in performance from training conditions to competition conditions, which is consistently reported in sport and which is usually a few percentage points, not a collapse. That is the figure a coach should hold in mind before deciding that an athlete "cannot perform under pressure", and it is the reason the correct first response to a bad competition is usually a review of preparation rather than a search for a psychological explanation.

需要记住的数字。实际重要的数字是"比赛效应"的大小--从训练条件到比赛条件的表现下降幅度，它在运动研究中被反复报告，而且通常很小。

Common mistakes / 易错点

- "They do it in training, so the skill is learned." It is learned in the sense that it is available. Whether it is available under pressure is a separate question, and it is the one competition answers.
"他们训练中能做到，所以技术就是学会了。"它是在"可用"这个意义上被学会了。在压力下是否可用是另一个问题，而比赛检验的才是答案。
- "More repetitions will fix it." Repetitions in calm conditions mostly produce smoothness. What pressure exposes is the ability to access a skill when attention and cues are reduced.
"多做几组就能解决。"在平静条件下多做重复主要产生的是顺滑。压力所暴露的是在注意和线索都减少时提取技术的能力。
- "A schema is a plan." A schema is a stored pattern that the athlete retrieves, not a plan they construct. That is why it feels instant and why it can be wrong without the athlete knowing.
"图式是一份计划。"图式是运动员提取出来的储存模式，而不是他们构建的计划。这正是它感觉是瞬间发生的、也正是它可能在比赛中出错的原因。
- "Pressure damages the technique." Pressure narrows attention and removes cues. The technique is usually intact; the conditions for reaching it are not.
"压力破坏技术。"压力使注意变窄并移走线索。技术通常完好无损；到达它的条件不是。

Why it matters / 为什么重要

This section is the hinge between the practice environment and the competition environment, and it exists because so much coaching is done in the first and evaluated in the second. A student who can state that learning and performance are different processes, with different determinants, has a framework for most of the recurring problems in sport: why a well-drilled athlete freezes, why a confident athlete misses an easy chance, and why adding training does not always change the result. It is also the theoretical justification for pressure rehearsal, which is one of the few interventions with consistent support.

这一节是练习环境与比赛环境之间的铰链，它之所以存在，是因为几乎所有教练工作都在前者中完成、却在后者中被评估。一个能说出"图式"的学生，已经掌握了理解大多数运动表现问题的框架。

- Schema features include generality, flexibility, efficiency and updating with experience.
图式具有概括性、灵活性、效率，并随经验更新。

Key terms / 核心术语

Motor learning 运动学习

Relatively permanent change in capability.

能力的相对永久变化。

Performance 表现

What can be produced at a particular moment.

某一时刻能够完成的内容。

Formulas / 公式

A volleyball player uses a schema to choose a pass based on the position of the set and the conditions.

排球运动员利用图式，根据来球位置和情境选择传球。

Worked examples / 实例

>

Differentiate motor learning and performance.

区分运动学习和表现。

Exam focus / 考核要点

Outline the information-processing model.

概述信息加工模型。

Linear and non-linear pedagogy · 线性与非线性教学法

First, picture it / 先这样想

Linear pedagogy means one correct answer per situation, delivered by a coach who already knows it.

Non-linear means a range of acceptable solutions, with the athlete exploring to find one. Most sport is taught linearly, and most elite sport turns out to be non-linear, and the gap between those two facts is where a coaching career either works or does not.

线性教学法意味着每个情境只有一个正确答案，由一位已经知道答案的教练传授。非线性意味着存在一系列可接受的解法，由运动员去探索找到。多数运动是用线性方式教的，而多数顶尖运动最终是非线性的，而这两个事实之间的落差，正是一位教练的职业生涯要么奏效、要么不奏效的地方。

The detail / 详细讲解

Linear pedagogy assumes the coach holds the solution.

线性教学法假定教练手里握着答案。

The athlete is told what to do, demonstrates it, repeats it until it is reliable, and is then moved to the next thing.

运动员被告知该做什么，示范它，重复它直到它变得可靠，然后被移到下一件事。

It is efficient: it works, it scales to a group of twenty, it produces competent athletes quickly, and it is how almost every technical skill in every sport is currently taught.

它是高效的：它有效、能扩展到二十人的一个组、能快速培养出有能力的运动员，而这也是目前几乎所有项目中每一项技术都被教授的方式。

Its limit is that it produces athletes who are competent at the thing they were told, and who are then asked to do something no one has described.

它的局限是：它培养出的运动员在被告知的那件事上有能力，然后却被要求去做一件没有人描述过的事。

Non-linear pedagogy starts from a constraint rather than a solution.

非线性教学法从一个约束而不是一个答案出发。

The coach sets a boundary - keep the ball above the net, get the shot over the line, land on one foot - and lets the athlete find their own way of satisfying it.

教练设定一条边界--把球保持在网以上、把球打到线内、用一只脚落地--然后让运动员自己找到满足它的方法。

Within that boundary there are many valid solutions, and the athlete is not looking for the coach's version, because the coach's version is not in the room.

在那条边界之内有许多有效解法，而运动员并不是在找教练的版本，因为教练的版本并不在场。

That changes what practice is for: it is no longer executing, it is problem-solving under a rule.

这改变了练习的目的：它不再是执行，而是在一个规则下解决问题。

The reason elite sport is non-linear is variance.

顶尖运动之所以是非线性的，原因是变化。

In a championship the opponent does something the athlete has not seen, the conditions differ, and the athlete has to adapt in seconds.

在锦标赛上对手会做出运动员没见过的事，条件会不同，而运动员必须在几秒内适应。

A movement repertoire built non-linearly has a much better chance of containing something useful, because

exploration is what puts unusual solutions into the repertoire in the first place.

以非线性方式建立的动作库有更好的机会包含某个有用的东西，因为探索本身就是把不寻常的解法放进动作库的原因。

Linear training produces one solution used repeatedly; non-linear training produces many, only some of which are ever needed - and the athlete cannot know which.

线性训练产生一个被反复使用的解法；

非线性训练产生许多解法，其中只有一部分会被用到--而运动员无法知道是哪一部分。

The two are not alternatives to be chosen between, and the practical reading is that they serve different purposes.

两者不是在二者选一，实用解读是它们服务于不同目的。

Early in learning, the athlete needs a reliable solution and needs it to work; that is what linear teaching is for, and it would be a mistake to withhold it in the name of discovery.

在学习早期，运动员需要一个可靠的解法、并且需要它真的管用；

Once a skill is reliable, the linear approach stops adding anything, and the non-linear approach is what keeps it usable under novelty.

这正是线性教学的目的，而以“发现”之名不去给它是错误的。

Most programmes that fail do so by staying linear too long rather than by becoming non-linear too early.

一旦某项技术变得可靠，线性方式就不再增加任何东西，而非线性方式才是让它在陌生情境中仍然可用的东西。

大多数失败的计划是因为在线性上停留太久，而不是因为过早变得非线性。

The numbers / 算一算

Numbers to keep. The number that matters here is the size of the movement repertoire, and the honest observation is that a linear approach produces a repertoire of roughly one while a non-linear one produces a repertoire of many, most of which are never used. The other useful number is the number of constraints a session can hold: too few and the athlete solves one problem, too many and the session becomes unfocused, so the practical figure is a small number of clear constraints with the athlete generating the solutions between them.

需要记住的数字。这里重要的是动作库的大小，而诚实的观察是：线性方法产生的动作库大约只有一个解法，而非线性方法产生的是许多。

Common mistakes / 易错点

- "Non-linear is a modern progressive idea." It is how any athlete has always had to solve a real game, whether or not anyone taught it that way.
"非线性是一种现代的先进理念。"它就是任何运动员一直以来都必须解决真实比赛的方式，无论有没有人那样教过。
- "Exploration wastes time." Exploration costs time early and saves it later, when the athlete meets a situation the linear method never prepared them for. The error is exploring before the athlete has a reliable solution at all.
"探索浪费时间。"探索在早期消耗时间，而在运动员遇到线性方法从未准备过的情境时节省时间。真正的错误是在运动员还没有任何可靠解法之前探索。
- "Let the athlete work it out and you will not have coached them." You have coached them. The coach decides what the athlete is allowed to be wrong about, and what they are not - and that is the most important coaching decision there is.
"让运动员自己琢磨，那你就没有在指导他。"你指导了。教练决定的是"允许运动员在什么上犯错、不允许在什么上犯错"--而这是最重要的决策。
- "Non-linear and linear are two methods and the programme should pick one." They serve different stages. Most programmes fail by staying linear too long, not by exploring too early.
"线性和非线性是两种方法，计划应该选一种。"它们服务于不同阶段。多数计划是在线性上停留太久而失败的，而不是因为探索得太早。

Why it matters / 为什么重要

This section is the closest thing in the course to a genuine professional dispute, and being able to argue it well is worth more than remembering either side. The linear view has the historical record on its side - it produced competent, reliable technicians at a scale no other approach has matched. The non-linear view has the ecological record on its side - because real competition contains situations that were never described, and only a varied repertoire survives them. A student who can state both records accurately, and say what each implies for a particular athlete at a particular stage, is showing the judgement the section exists to develop.

这一节是整门课程中最接近真正专业分歧的内容，而能够把它论证清楚，比记住任何一方都更有价值。线性一方掌握着历史记录--它以其

Ecological dynamics sees perception and action as coupled and movement as emerging from the performer, task and environment.

生态动力学把知觉和动作视为耦联，运动从表演者、任务和环境的互动中产生。

Key terms / 核心术语

Schema teaching 图式教学

Structured practice that develops and refines generalised movement memories.

发展并完善概括性动作记忆的结构化练习。

Constraints-led approach 限制引导法

Practice that manipulates constraints to guide exploration.

通过调整限制条件引导探索的练习。

Formulas / 公式

A coach can combine a clear technical model with a task that changes ball size, space or time so the learner explores adaptations.

教练可以把清晰技术模型与改变球大小、空间或时间的任务结合，让学习者探索调整。

Worked examples / 实例

>

Compare schema teaching and a constraints-led session.

比较图式教学和限制引导课程。

Exam focus / 考核要点

When should practice move from a simple movement to tactics or strategy?

练习何时应从基本动作转向战术或策略？

Stages, PRP and transfer · 阶段、PRP与迁移

First, picture it / 先这样想

A skill passes through stages, and the mistake is to train it as if it were always at the same one. Early, the athlete needs to get the shape roughly right and needs the movement to be safe. Middle, they need it to work every time, and reliability is the whole job. Late, they need it to work while everything else about them is wrong - tired, rushed, behind, watched. Most coaches are excellent at the middle stage and then quietly assume the late stage will follow.

一项技术会经历若干阶段，而错误是把它当作一直处在同一个阶段来训练。早期，运动员需要大致正确的动作形状、并且需要这个动作是安全的；中期，他们需要它每次都管用，可靠性就是全部工作；后期，他们需要它在自身其他一切都不同时仍然管用--疲劳、匆忙、落后、被注视。大多数教练在中期阶段非常出色，然后悄悄地假定后期阶段会随之而来。

The detail / 详细讲解

The cognitive stage is about forming a correct idea of the movement, and its signature is inconsistency plus a lot of conscious effort - the athlete is thinking about the movement while doing it.

认知阶段的目的是形成一个正确的动作概念，它的特征是不稳定加上大量的有意识努力--运动员在做的时候还在想着这个动作。

The associative stage is about making it reliable, and its signature is fewer errors and less thinking.

联结阶段的目的是让它变得可靠，特征是错误减少、思考减少。

The autonomous stage is about performing it without attending to it, which frees attention for the environment.

自主阶段的目的是在不注意它的情况下完成它，从而把注意释放给环境。

The stages are not optional labels; each one has a different job, and practising stage three while still in stage one is not advanced training, it is noise.

这些阶段不是可选的标签；

每一个都有不同的工作，而在仍处于第一阶段时就去练第三阶段，不是高级训练，而是噪声。

PRP - periodised resistance training or periodised training generally - exists because of a limit that is easy to miss.

PRP--周期化抗阻训练、或一般意义上的周期化训练--之所以存在，是因为有一个容易被忽略的限制。

Specificity is the idea that an adaptation is specific to what was trained, and the practical consequence is that general preparation cannot be replaced by specific preparation.

专门性指的是适应与所训练内容相对应，而实际推论是一般准备不能被专项准备所替代。

A genuinely specific session, done when the athlete is fatigued, is worth considerably less than the same session done fresh.

运动员在疲劳状态下做的一次真正专项的训练课，其价值明显低于状态新鲜时做的同一堂课。

That is why the order of the week matters, and why an "extra" technical session added on top of a hard day usually costs more than it gives.

这就是为什么一周的顺序很重要，也为什么在吃力的日子之上额外加一堂“顺带”的技术课，通常代价大于收益。

Transfer is the third piece, and it is the one that decides whether training shows up on the day.

迁移是第三块内容，也是决定训练是否会体现在比赛日的那一块。

Transfer has three faces: whether a change carries from one task to a similar task, from gym strength to sport performance, and from training to competition.

迁移有三个面向：一种改变能否从一项任务迁移到相似任务、从健身房力量迁移到运动表现、以及从训练迁移到比赛。

All three are imperfect, and each has a known pattern of what transfers well and what does not - strength transfers to strength-dominant skills better than to skill-dominant ones, and vertical jumps transfer well while squat strength transfers less to sprinting than people expect.

三者都不完美，而且每一个都有已知的“什么迁移得好、什么迁移不好”的规律--力量向力量主导的技术迁移得比向技术主导的技术更好，垂直起跳迁移得很好，而深蹲力量向冲刺的迁移则比人们预期的要差。

Putting the three together is the actual job, and the sensible structure is general preparation early, specific preparation later, and a taper that reduces volume while keeping intensity.

把三者合在一起才是真正的工作，而合理的结构是：早期一般准备，后期专项准备，以及一个降低总量但保持强度的减量期。

The taper sounds counter-intuitive and it is the most consistently supported part of the whole picture: fitness is largely built early and lost quickly, so a well-managed peak arrives from removing fatigue rather than from adding work, and athletes who arrive fresh frequently out-perform athletes who arrive tired having "done everything".

减量听起来违反直觉，而它是整幅图景中证据最一致的部分：体能主要在早期建立、却很快流失，所以一个管理良好的峰值来自去除疲劳。

The numbers / 算一算

Numbers to keep. The taper is the number to know: a reduction in volume of roughly 30 to 60 percent over the one to two weeks before competition, with intensity maintained or slightly increased, is the range most consistently associated with better performance. On stages, the honest position is that the boundaries are fuzzy and the practical guide is behavioural rather than numerical - the athlete is still thinking about the movement, they are thinking but reliably, or they are not thinking about it. And the ordering rule that follows: general work when fresh, specific work when fresh enough, and never specific work on top of fatigue.

需要记住的数字。减量是必须知道的数字：在比赛前一到两周内把总量降低约30%到60%、同时保持或略微提高强度，是与更好表现相关的。

Common mistakes / 易错点

- "Autonomous means relaxed." Autonomous means it no longer needs conscious attention. Relaxation and automaticity are different, and an athlete can be very relaxed and still consciously controlling every phase.
"自主就是放松。"自主意味着它不再需要有意识的注意。放松和自动化是两回事，运动员可以非常放松，同时仍然有意识地控制每
- "Specific work is always better than general work." Specific work is better than general work when fresh. On top of fatigue, the same session is worth less, and adding it usually costs more than it gives.
"专项训练总是优于一般训练。"状态新鲜时，专项训练优于一般训练。在疲劳之上，同一堂课的价值更低，而把它加上去通常代价
- "Strength transfers to everything." It transfers well to strength-dominant skills and less well to skill-dominant and speed-dominant ones, and the transfer is never automatic.
"力量会迁移到一切。"它对力量主导的技术迁移得好，对技术主导和速度主导的迁移得较差，而且这种迁移从来不是自动的。
- "Peak fitness means training hardest into the competition." Fitness is built early and lost quickly, so arriving fresh usually beats arriving tired having done everything.

"达到最佳体能就是赛前练得最狠。"体能早期建立、很快流失，所以以新鲜状态抵达通常胜过疲惫地做完所有训练。

Why it matters / 为什么重要

These three ideas are the skeleton of every programme in sport, and they matter more than the exercises do. Knowing that a stage has a job prevents a coach from practising a skill wrongly; knowing that specificity and freshness are a package prevents a well-intentioned extra session from undoing a week; and knowing how transfer fails prevents a strength block from being credited with a sprint improvement it did not produce. Together they are what lets a student read a training plan critically rather than admiring it, which is the difference between following a programme and being able to design one.

这三个概念构成了运动中每一份训练计划的骨架，而它们比具体练什么更重要。知道一个阶段有它的工作，能防止教练错误地练某项技术。

-

Six transfer types are skill-to-skill, practice-to-performance, bilateral, stage-to-stage, abilities-to-skills and principles-to-skills. Positive transfer helps; negative transfer can hinder.

六种迁移是技能到技能、练习到表现、双侧、阶段到阶段、能力到技能和原则到技能。正迁移促进学习，负迁移会阻碍。

Key terms / 核心术语

Worked examples / 实例

>

Practising a chest pass can help a bounce pass; balance learned earlier can help advanced gymnastics.

练习胸前传球可帮助反弹传球；早期学习的平衡能力可帮助高级体操。

Quick check / 快速检查

1. Name the three stages of learning.

说出学习的三个阶段。

Exam focus / 考核要点

Why should learners avoid overlearning?

为什么学习者应避免过度学习？

Attentional control

注意控制

Guiding question / 引导问题

How does attentional focus influence skilled performance?

注意焦点如何影响熟练表现？

Attention can be internal or external and broad or narrow. The focus should match the task: external focus often supports automatic, fluid performance, while internal focus is useful for early learning and technique correction.

注意可分为内部或外部、广义或狭窄。焦点应与任务匹配：外部焦点通常支持自动、流畅表现，内部焦点适合早期学习和技术纠正。

Internal, external, broad and narrow · 内部、外部、宽域与窄域注意

First, picture it / 先这样想

Attention is a limited resource, and every technical cue a coach gives is a request for some of it. Some of those requests are useless - asking an athlete to focus on something the athlete cannot actually perceive or control is a request that cannot be met. The skill is not in giving more cues. It is in knowing which cues are worth the resource and which are a waste of it.

注意是一种有限的资源，而教练给出的每一条技术提示都是在向它索取一部分。其中一些索取是无效的--让运动员去关注某个他实际上无法感知或无法控制的点，是一个无法被满足的请求。技巧不在于给出更多提示，而在于知道哪些提示值得这份资源、哪些是浪费。

The detail / 详细讲解

An internal cue points attention to something inside the athlete - a body position, a feeling, a rhythm, a thought.

内部提示把注意指向运动员体内的东西--一个身体位置、一种感觉、一个节奏、一个想法。

An external cue points it to something outside - a target, a landing spot, a line, an opponent's position, the sound of the catch.

外部提示把它指向体外的东西--一个目标、一个落点、一条线、对手的位置、接球的声音。

Both work, and they fail differently.

两者都有效，而且失败方式不同。

Internal cues are portable to any environment but risk the athlete monitoring a sensation that does not correspond to the outcome.

内部提示可移植到任何环境中，但可能让运动员去监测一种与结果并不对应的感觉；

External cues are easy to check and to correct but are sometimes unavailable at the moment they are needed.

外部提示容易检查、容易纠正，但有时在需要的时刻并不存在。

Broad and narrow describe how much of the performer the cue addresses.

宽与窄描述提示指向表演者的多大范围。

A broad attentional focus is on the whole movement or the whole picture; a narrow one is on a single detail.

宽的注意焦点落在整个动作或整体图景上；

A broad focus tends to support planning and rhythm and is what you want in early learning and in open, fast situations.

窄的落在单一细节上。

A narrow focus supports precision - which is exactly what an elite performer needs, and exactly what an early learner cannot use.

宽焦点倾向于支持计划性和节奏感，在早期学习以及开放、快速的情境中正是你想要的；

窄焦点支持精确性--而这正是精英表演者需要的，也正是初学者无法使用的。

Narrow attention has a real cost, which is that it removes information.

窄的注意有真实的代价，即它移走了信息。

A performer locked onto one detail performs that detail well and stops seeing everything else - the opponent, the space, the pattern.

一位锁定在某个细节上的表演者会把那个细节做得很好，并停止看到其他一切--对手、空间、局势。

This is why an athlete who is excellent at a technical correction can suddenly look worse: the correction was right, the cost was paying attention to it, and the cost showed up exactly where the athlete stopped looking.

这就是为什么一位在某个技术纠正上做得很好的运动员可能突然看起来更差：那个纠正是对的，代价是去注意它，而代价恰恰出现在纠正的地方。

Which is why distractors are a legitimate technique rather than an exception.

因此干扰项是一项正当的技术而不是例外。

A good distractor is a signal that captures attention for a moment and then releases it, and it works because attention is a limited resource that will be spent on something.

一个好的干扰项是一个抓住注意片刻、随后又释放它的信号，而它之所以有效，是因为注意是一种会被花在某处的有限资源。

In shooting and archery the distractor might be the target; in a racquet sport it might be the strings on the opponent's racket, which pull the eye at the moment before contact.

在射击和射箭中，干扰项可能是靶心；

The point is not to deceive the opponent's eyes.

在球拍项目中，可能是对手球拍上的线，它在触球前一刻把视线拉走。

It is to move your own athlete's attention to where it needs to be at the moment it needs to be there.

重点不是欺骗对手的眼睛，而是把你自己的运动员的注意移到它在需要的那一刻所在的地方。

The numbers / 算一算

Numbers to keep. The useful number is the count of cues per attempt: more than a very small number competes with itself, because the athlete cannot hold several narrow attentional focuses at once and the attempt to do so is itself a distractor. One cue per attempt, changed every few attempts, is a more effective design than three cues held constant. And the timing point is quantitative too - the window in which a distractor can be applied is short and sits immediately before the key instant, which is why the cue has to be timed to the movement rather than to the session.

需要记住的数字。有用的数字是每次尝试的提示数量：超过很小的数量，提示之间会互相竞争，因为运动员无法同时维持多个狭窄的注意力焦点。

Common mistakes / 易错点

- "More cues means more help." Several cues compete for the same limited resource, and the athlete ends up attending to none of them properly.
"提示越多帮助越大。"多条提示会争夺同一份有限资源，运动员最后哪一条都没有充分关注。
- "External cues are objective, so always prefer them." They are easy to check, and they are sometimes unavailable at the moment they are needed - an athlete cannot look at a target that is behind them.
"外部提示更客观，所以总是优先用它们。"它们容易检查，但有时在需要的时刻并不存在--运动员没法去看一个在他身后的目标。
- "External focus makes you focus on the target." An external focus directs attention toward the goal. A distractor is a distractor because it deliberately takes attention away from where it was going.
"外部焦点会让你专注于目标。"外部焦点把注意导向目标。干扰项之所以是干扰项，正因为它故意把注意从原本要去的地方拉走。
- "A distraction technique is about the opponent." In sport psychology a distractor is a tool the athlete uses on their own attention.
"干扰战术是针对对手的。"在运动心理学中，干扰项是运动员用在自己注意上的工具。

Why it matters / 为什么重要

Attention is the scarcest thing an athlete has on the day, and this section is about spending it deliberately. The coaching skill it describes is genuinely hard: knowing which cues a particular athlete can actually use, at what stage of learning, and at what moment in the movement. A coach who gives three cues at once has usually given none, and a coach who gives one well-timed cue has done something quite difficult. It is also a section with immediate practical transfer, because the same reasoning applies to attention under fatigue, under pressure and in tactical decision-making.

注意是运动员在比赛日最稀缺的东西，而这一节讲的是如何有意识地花掉它。它所描述的教练技能是真的难：知道某位特定运动员实际上的注意力分配。

External focus is generally more effective for skilled performance, but internal focus can help initial learning.
外部焦点通常更有利于熟练表现，但内部焦点有助于初期学习。

Key terms / 核心术语

External focus 外部注意

Attention on the movement effect or environmental target.

关注动作效果或环境目标。

Attentional narrowing 注意狭窄

Reduced processing of peripheral information under high arousal.

高唤醒时减少外围信息加工。

Formulas / 公式

For a penalty, "see the target and trust the technique" uses an external focus; a long list of body instructions can overload attention.

罚球时"看到目标并相信技术"是外部焦点；过多身体指令会增加注意负荷。

Worked examples / 实例

>

Define internal and external attention.

定义内部和外部注意。

Exam focus / 考核要点

Give a penalty example of internal and external focus.

分别给出罚球中的内部和外部注意例子。

Distractors and control strategies · 干扰项与控制策略

First, picture it / 先这样想

A control strategy is a decision you make beforehand about what to do with your attention, and making it beforehand is the entire point. Deciding mid-event is not strategy, it is panic. The most useful thing a coach can do for an athlete who freezes is to decide, in advance and out loud, where the athlete's attention will be - because a decision made in advance can be executed, and a decision made under pressure cannot.

控制策略是你事先做出的、关于要把注意放在哪里的决定，而"事先"正是全部要点。在事件中途做决定不是策略，那是慌乱。对于一位会僵住的运动员，教练能做的最有用的事情，就是事先并且明说地把这位运动员的注意安排在哪里--因为事先做出的决定可以被执行，而压力下做出的决定不能。

The detail / 详细讲解

The basic control strategies divide by what the attention is aimed at.

基本的控制策略按注意指向什么来划分。

Association puts attention on the movement itself - the feeling in the hand, the position of the elbow, the rhythm of the prep.

联想式把注意放在动作本身--手上的感觉、肘部的位置、准备动作的节奏。

Dissociation puts attention on the outcome or the environment - the target, the score, the opponent, the next point.

分离式把注意放在结果或环境上--靶心、比分、对手、下一个分。

And dissociation can be external, pointing outside the body entirely, or internal, pointing to a sensation such as breathing.

分离式还可以是外部的，完全指向身体之外；

All of them work, which is why the choice is not "which is best" but "which suits this athlete in this situation".

或者是内部的，指向呼吸这样的感觉。

三者都有效，所以选择不是"哪个最好"，而是"哪个适合这位运动员在这个情境下"。

The practical value of a pre-commitment strategy is that it moves the decision out of the moment of pressure.

事先承诺策略的实际价值，在于它把决定移出了压力时刻。

An athlete who has decided in advance to watch the target does not need to decide anything at the decisive instant, and removing a decision from that instant is worth more than any technique the athlete could execute there.

一位事先决定要看靶心的运动员，在决定性瞬间不需要做任何决定，而从一个瞬间里移走一个决定，比那位运动员在那里能执行任何动作都更有价值。

This is the mental equivalent of the biomechanical argument for a pre-determined take-off position: a movement that is already determined needs no decision at the moment it happens.

这在心理上等价于生物力学中关于预先确定起跳位置的论据：一个已经确定的动作，在它发生的那一刻不需要任何决定。

Arousal regulation is the other side, and it is a matter of turning the volume up or down before the event rather than during it.

唤醒调节是另一面，它是在事件之前而不是之中把音量调高或调低的问题。

Breathing is the most portable tool because it is the only physiological process the athlete can deliberately change and the only one that is also under voluntary control at will.

呼吸是最可移植的工具，因为它既是运动员能够刻意改变的唯一生理过程，也是唯一真正处于随意控制之下的过程。

Slow, long exhalation shifts the balance toward parasympathetic activity, which is why a breathing routine is a more reliable tool than an athlete hoping to relax by deciding to relax.

缓慢、绵长的呼气把平衡移向副交感活动，这就是为什么一个呼吸例程比一位运动员"决定放松一下来放松"要可靠得多。

And then there is the category that includes the familiar pre-performance routines, the visualisation, and the **self-talk, which all work through the same door: they occupy the mind during the window in which worry would otherwise occupy it.**

然后是那一类，包括熟悉的赛前例程、心理意象和自我对话

，而它们都通过同一扇门起作用：它们在"担忧原本会占据的那段窗口"里占住了脑子。

The best of them are not mysterious. They are simple, repeatable, and short, and the reason they work is that attention has to be spent on something, and a routine gives it somewhere better to go.

其中最好的并不神秘--它们简单、可重复、简短，而它们之所以有效，是因为注意总得被花在某处，而一个例程给了它一个更好的去处。

The numbers / 算一算

Numbers to keep. The window is the number: the interval between arriving at the venue and the event beginning is when a pre-commitment strategy is set, and the interval between the event and the first contact with the task is when it must still be available. The second practical figure is the length of a breathing cycle used for regulation - the longer and slower the exhale, the larger the shift, which is why counting the exhale rather than the inhale is the usual instruction.

需要记住的数字。窗口就是关键数字：到达场地到比赛开始之间的间隔，是设定事先承诺策略的时候；比赛开始到与任务第一次接触之间的间隔。

Common mistakes / 易错点

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"Just relax." Relaxation is not a strategy because it names no behaviour. A breathing count, a gaze point and a phrase are strategies; "relax" is a wish.

"放松一下就行。"放松不是一个策略，因为它没有指定任何行为。数呼吸、一个注视点、一句话才是策略；"放松"只是一个愿望。

-

"Internal focus is always better." Internal cues support monitoring and rhythm; external cues support targeting and judgement. The right choice depends on the task and the stage.

"内部焦点总是更好。"内部提示支持监测和节奏；外部提示支持瞄准和判断。正确的选择取决于任务和阶段。

-

"Turn the thought around during the event." A thought that has to be reversed takes time to reverse, which is exactly the resource the situation does not have.

"在过程中把想法转过来。"一个需要被反转的想法需要时间才能反转，而那恰恰是当下没有的资源。

-

"Visualisation is preparation." Visualisation is attention management. It works by occupying the mind during a specific window, and it works less well as a general rehearsal of technique.

"意象就是准备。"意象是注意管理。它靠在某个特定窗口占住脑子而起作用，而作为技术的总体预演，它的作用要小得多。

Why it matters / 为什么重要

This section is the most immediately usable in the whole theme, because almost every athlete can use a control

strategy and most have never been taught one. The reason these techniques work is worth understanding precisely because it is not mysterious: attention will be spent, and a strategy decides where. That is the same principle as the attentional section before it, applied deliberately, and it gives a coach something concrete to build rather than an attitude to encourage. It is also the section that translates directly into a competition routine, which is one of the few mental skills an athlete can practise, measure and refine.

这一节是整个主题中最可直接使用的一节，因为几乎每位运动员都能用上控制策略，而大多数从未被教过。理解这些技术之所以有效的力

- Pre-performance routines, breathing, focus cues, self-talk and process goals can clear irrelevant information and refocus. High arousal can create tunnel vision, fewer cues and poorer decisions.
赛前流程、呼吸、焦点提示、自我对话和过程目标可以清除无关信息并重新集中。高唤醒会造成隧道视野、线索减少和决策变差。

Key terms / 核心术语

Controlled distraction 受控分散

A deliberate cue or routine that redirects attention.

有意的提示或流程，用于重新引导注意。

Self-talk 自我对话

Internal statements used to guide attention and behaviour.

用于引导注意和行为的内部语言。

Formulas / 公式

A player bounces the ball twice, breathes and looks at the target before the attempt.

球员在罚球前运球两次、呼吸并看向目标。

Worked examples / 实例

>

Classify a distraction and select one attentional-control strategy.

判断干扰类别并选择一种注意控制策略。

Exam focus / 考核要点

How does high arousal affect attention?

高唤醒如何影响注意？

Task matching and self-talk · 任务匹配与自我对话

First, picture it / 先这样想

A task can be made to ask for less, and asking for less is not cheating. If the goal is to hit the ball, the task "get the bat to the ball" is a different size of question from "hit the ball perfectly", and a human being can nearly always attempt the first one. Same athlete, same ability, same day - different task, different outcome, because the difficulty is a property of the question rather than a property of the athlete.

一项任务可以被设定成要求更少的东西，而要求更少并不是作弊。如果目标是击中球，"让球棒碰到球"和"完美地击中球"是两个大小不同的问法，而一个人几乎总能尝试前者。同样的运动员、同样的能力、同一天--不同的任务、不同的结果，因为难度是问题的属性，而不是运动员的属性。

The detail / 详细讲解

Task matching is the systematic version of that.

任务匹配是这一想法的系统化版本。

One common approach reduces the number of units in the task - a three-touch sequence rather than two, a longer runway, a larger target, an easier surface, lower stakes, fewer obstacles.

一种常见做法是减少任务中的单元数量--三次触球而不是两次、更长的助跑道、更大的目标、更容易的地面、更低的赌注、更少的障碍。

Another approach reduces the information the athlete has to process.

另一种做法是减少运动员必须处理的信息量。

A third changes what counts as success so that a partial success is still a success.

第三种做法是改变"什么算成功"，让部分成功仍然算成功。

None of them make the athlete more talented; they make the problem solvable today, which is what creates the practice in which talent then grows.

它们都不会让运动员更有天赋；它们让问题在今天变得可解，而这正是创造出让天赋得以成长的那种练习。

Self-talk is the internal version of the same idea.

自我对话是同一个想法的内部版本。

Instructions like "bend your knees" or "eyes on the ball" are task-relevant self-talk; motivational self-talk like "push" or "come on" and negative self-talk like "don't mess this up" are the other two categories.

"屈膝"或"眼睛看球"这样的指令是与任务相关的自我对话；

Instructional self-talk is the one that is reliably associated with better performance, and it is the one most athletes under-use, because they reach by default for motivational self-talk, which raises effort reliably and improves technique rarely.

"顶住""加油"这样的动机性自我对话和"别搞砸了"这样的负性自我对话是另外两类。

指令性自我对话是与更好表现稳定相关的那个，也是大多数运动员使用不足的那个--因为他们会默认去抓动机性自我对话，而后者可靠

Timing matters more than people expect, and the distinction is between process and outcome.

时机比人们预期的更重要，而区分在于过程与结果。

Process self-talk refers to something the athlete can do - "smooth", "through the ball", "tall" - and it directs attention to controllable factors.

过程性自我对话指向运动员能做的事--"顺""穿球""高一点"--它把注意引向可控因素。

Outcome self-talk refers to results - "win", "don't lose", "beat him" - and it points attention at something the athlete does not control.

结果性自我对话指向结果--"赢""别输""击败他"--它把注意指向运动员无法控制的东西。

A good deal of the anxiety and distraction that athletes report at competition is outcome self-talk that has been rehearsed until it feels like strategy.

运动员在比赛中报告的大量焦虑与分心，其实就被反复排练到感觉像策略的结果性自我对话。

The combination is where the useful work happens, and the honest summary is that the strongest version uses all three kinds.

三者的结合才是有用的工作发生的地方，而诚实的总结是：最强的版本同时使用三种。

Instructional talk for the technique, motivational talk for the effort, and a plan for the outcome talk that displaces it - and the plan is usually to notice it and redirect, which sounds trivial and works because noticing is an attention event and attention is what the process talk was competing for in the first place.

指令性对话用于技术，动机性对话用于努力，而对结果性对话则有一个把它挤开的计划--这个计划通常就是"注意到它并把注意转回来"

The numbers / 算一算

Numbers to keep. The useful figures here are the design ones. Task matching works by adding a constraint or removing one, and one meaningful change at a time is the rule that makes the gain attributable. Self-talk research is clearest on the effect size of instructional versus motivational talk, and the honest reading is that the difference is real but smaller than the athlete's experience of it - which is another reason to select on task rather than on taste. The measurement habit is to count the proportions of the three types over a session, because most athletes use motivational talk far more than instructional talk and the imbalance is usually invisible to them.

需要记住的数字。这里有用的是设计方面的数字。任务匹配通过增加或减少一个约束来起作用，而一次只做一次有意义的改变，是让收益

Common mistakes / 易错点

- "Positive thinking is always better than negative." Motivational and negative self-talk both raise arousal and narrow attention. Instructional talk is the category reliably associated with better technique.
"积极的自我对话总是好过消极的。"动机性和负性自我对话都会提高唤醒并使注意变窄。指令性对话才是与技术表现稳定相关的那个
- "Task matching is making it easy." It makes today's problem solvable, which is a different thing from making it easy. The difficulty is deferred, not removed, and the next task is a fraction harder.
"任务匹配就是把它变简单。"它让今天的问题变得可解，这和把它变简单是两回事。难度是被推迟了，而不是被消除了，而下一个
- "Outcome self-talk motivates." It points attention at something outside the athlete's control, usually at the most

consequential moment.

"结果性自我对话能激励人。"它把注意指向运动员无法控制的东西，而且通常是在最关键的时刻。

-

"Change several things to see what works." Change one constraint, or one type of self-talk, and measure - otherwise the improvement cannot be attributed.

"同时改几样看看哪样有用。"只改一个约束或一类自我对话，然后去测量--否则无法归因。

Why it matters / 为什么重要

Task matching and self-talk are the two techniques in this theme that a coach can prescribe concretely and a student can practise immediately, which makes them a good place to convert theory into behaviour. The underlying idea is the same in both and is the most useful sentence in the section: difficulty belongs to the task, and attention belongs to whatever the athlete is rehearsing saying to themselves. Change either one and a stuck athlete moves, which is a considerably more encouraging finding than the alternative - that stuck athletes need to want it more.

任务匹配与自我对话，是这个主题中教练可以具体开方、学生可以立刻练习的两项技术，这使它们成为把理论转化为行为的好地方。两者

-

"Don't miss" can increase anxiety; "calm and focused" or "see the target" is more controllable and task-relevant.

"不要失误"可能增加焦虑；"保持冷静专注"或"看到目标"更可控、更贴近任务。

Key terms / 核心术语

Goal setting 目标设定

Setting process or performance goals to focus controllable actions.

设定过程或表现目标以关注可控行动。

Formulas / 公式

Practice with crowd noise or pressure cues so attentional control is trained, not assumed.

在人群噪声或压力提示下练习，使注意控制得到训练而不是靠假设。

Worked examples / 实例

>

Choose the attentional width that fits a task.

为任务选择合适的注意广度。

Exam focus / 考核要点

Give two external and two internal distractors.

给出两个外部和两个内部干扰。

Achievement motivation

成就动机

Guiding question / 引导问题

Why do some athletes approach achievement challenges while others avoid them?

为什么一些运动员主动迎接成就挑战，而另一些回避？

Achievement motivation results from personality, situational factors, emotional factors and behaviour. Task orientation is self-referenced and values effort, learning and persistence; ego orientation is other-referenced and values winning and ability demonstration.

成就动机由人格、情境、情绪和行为共同影响。任务定向以自我为参照，重视努力、学习和坚持；自我定向以他人为参照，重视获胜。

Need achievement and orientations · 成就需要与定向

First, picture it / 先这样想

Achievement motivation is a need to be good at something, and like every need it has two faces: a drive to improve, and a drive to look good. Athletes who look like they want to improve sometimes want the second one badly. Telling those two apart is the single most practical judgement in this whole section, because the two look identical from the outside and need opposite coaching.

成就动机是需要在某件事上做得好，而像每一种需要一样，它有两副面孔：改进的驱动力，和显得优秀的驱动力。看起来想要进步的运动员，有时其实非常想要后者。把这两者分辨出来是整节中最实际的判断，因为从外面看它们完全一样，却需要相反的教练方式。

The detail / 详细讲解

The need for achievement is a need for excellence, and it shows up as a preference for tasks of moderate difficulty - hard enough to be a real test, easy enough that success is possible.

成就需要是对卓越的需要，它表现为对中等难度任务的偏好--难到构成真正的测试，又容易到成功是可能的。

That preference is the diagnostic feature.

这种偏好就是诊断特征。

An athlete who chooses tasks that are comfortably easy is not being cautious; they are choosing tasks that will confirm what they already believe about themselves.

一位选择自己轻松就能做好的任务的运动员不是在谨慎；

And an athlete who chooses tasks that are almost certainly beyond them is doing the same thing in the other direction.

他们是在选择会确认自己既有信念的任务。

而选择几乎肯定超出自己能力的任务的运动员，做的是同一件事、方向相反。

From that preference the two orientations follow.

由这种偏好派生出两种取向。

Achievement orientation is a need to master, to grow, to be able to do what the athlete could not do last month.

成就取向是需要掌握、需要成长、需要能够做到上个月做不到的事；

Performance orientation is a need to be superior, to be better than others, to be seen as competent.

绩效取向是需要优越、需要胜过他人、需要被视为有能力。

Both athletes will train hard, both will follow instructions, and both will look committed - which is why the two are so hard to tell apart from behaviour alone and why the test is which tasks they choose and how they explain failure.

两种取向的运动员都会刻苦训练、都会遵循指令、都会显得投入--这正是为什么单从行为上极难分辨，也为什么真正的检验在于他们如何解释失败。

Where the orientations diverge is in how they respond to a bad outcome.

两种取向的分歧在于它们如何回应一个坏结果。

A mastery-oriented athlete treats a loss as information about what to work on, and their effort afterwards is usually higher.

掌握取向的运动员把一次失败当作“该练什么”的信息，之后的努力通常更高；

A performance-oriented athlete treats a loss as information about themselves, and the effort afterwards is usually

lower - because more effort would make the verdict more public.

绩效取向的运动员把失败当作关于自己的信息，之后的努力通常更低--因为更多努力会让这个判决更加公开。

The pattern is the same in a bad grade, a missed selection and a dropped game, and it is usually misread as a loss of motivation rather than as a change of direction.

同样的模式出现在一次差成绩、一次落选和一场失利之后，而且它通常被误读为动力下降，而不是方向的改变。

The two also coexist, and that is the more accurate picture.

两者也同时存在，而那才是更准确的图景。

Most athletes want both, and what varies is which one is dominant in a given situation.

大多数运动员两者都想要，变化的是在某个具体情境中哪一个占主导。

The practical consequence is that the question "is this athlete achievement or performance oriented" is the wrong question, because it forces a binary on a person who has both.

实际推论是，"这位运动员是成就取向还是绩效取向"是个错误的问题，因为它把一个两者都有的人塞进二分法。

The right question is which one this athlete is being pulled toward here, today, in this task - and the answer is usually visible in the task they choose next.

正确的问题是：这位运动员在这里、在今天、在这项任务中被拉向哪一个--而答案通常就写在他接下来选择的任务里。

The numbers / 算一算

Numbers to keep. The cleanest number in this area is the task-difficulty choice, because it is a decision rather than a self-report and it is available on every set. The second is the effort following a failure, which is easy to record and hard to argue with. And the third is the proportion of training time spent on tasks the athlete could already do perfectly, which is a surprisingly good index of whether a session is producing mastery or merely producing repetition.

需要记住的数字。这个领域里最干净的数字是任务难度的选择，因为那是一个决定而不是自评，而且每一组动作都能拿到。第二个是失败

Common mistakes / 易错点

- "High achievers just want to win." Winning is what performance orientation wants. Achievement orientation wants to be able to do something they could not do before, and the two athletes can look identical while wanting opposite things.
"高成就者只是想赢。"赢是绩效取向想要的。成就是向想要能够做到自己之前做不到的事，而这两个运动员可以看起来一模一样、
- "Motivation dropped after the loss." The effort drop after a loss is usually a change of direction rather than of motivation, and it is reversible by changing what the next session asks for.
"输掉之后动力下降了。"失败之后的努力下降通常是方向的改变而不是动力的改变，而且可以通过改变下一堂训练课的要求来逆转
- "Choose the right orientation for your athlete." You cannot convert one into the other on demand. You can build a climate in which one of them is easier to act on, and that is the available lever.
"为你的运动员选一个正确的取向。"你无法按要求把一种转化成另一种。你能做的是营造一种让其中一种更容易被践行的氛围，而
- "Both orientations are fine, so ignore the difference." They predict opposite responses to failure and opposite reactions to feedback, which is not a detail.
"两种取向都很好，所以不用管区别。"它们对失败的预测反应相反、对反馈的反应也相反，这不是细节。

Why it matters / 为什么重要

This is the first of the motivation-theory sections, and it earns its place because the distinction is observable, teachable and immediately actionable. A coach who can read which direction an athlete is being pulled can choose a task that helps - a harder, more learnable task for a mastery-oriented athlete, and a clear, non-comparative standard for a performance-oriented one. It also reframes a frustrating athlete: a performance-oriented athlete who appears to have lost heart after a loss is usually running a coherent strategy that was designed for a different world, and it is easier to work with than to correct.

这是动机理论部分的第一节，它之所以值得占一个位置，是因为这个区分是可观察的、可教的，而且可以立刻行动。一位能读出运动员正

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Task orientation focuses on self-referenced improvement, effort, learning, responsibility and enjoyment. Ego orientation focuses on outperforming others, winning, avoiding failure and protecting self-worth through outcomes.

任务定向关注自我参照的进步、努力、学习、责任和享受；自我定向关注胜过他人、获胜、避免失败以及通过结果保护自尊。

Key terms / 核心术语

Task orientation 任务定向

Self-referenced focus on learning, effort and improvement.

以自我为参照，关注学习、努力和进步。

Ego orientation 自我定向

Other-referenced focus on superiority and winning.

以他人作为参照，关注优越和获胜。

Formulas / 公式

A task-oriented athlete asks for feedback, reflects and stays committed after a difficult session.

任务定向运动员寻求反馈、反思，并在困难训练后保持投入。

Worked examples / 实例

>

Compare task and ego orientation.

比较任务定向和自我定向。

Exam focus / 考核要点

List three features of each orientation.

分别列出两种定向的三个特征。

High ego with low ability and coach influence · 高自我定向、低能力与教练影响

First, picture it / 先这样想

An athlete can be absolutely certain they are good at something and be wrong. Certainty and competence are independent, which is why confidence is not a diagnosis. The dangerous combination is high certainty with low competence, because it is stable - the athlete does not seek information that would correct it - and the coach is then left trying to change a belief that will not move on evidence.

一位运动员可以完全确定自己在某件事上很厉害，同时是错的。确定感与能力彼此独立，这就是为什么自信不是一个诊断。危险的组合是高确定感加低能力，因为它是稳定的--这位运动员不会去寻找能纠正它的信息--而教练于是只能去改变一个不会因为证据而移动的信念。

The detail / 详细讲解

High ego with low ability is a specific and common pattern in sport, and it has a shape.

高自我观加低能力在运动中是一个具体而常见的模式，它有形状。

The athlete takes credit for good outcomes regardless of the cause, takes blame for poor ones regardless of the cause, seeks information selectively, and reacts to criticism by disputing it rather than testing it.

这位运动员把好结果归功于自己，无论原因是什么；把坏结果归咎于自己，同样无论原因；

None of this is dishonesty.

选择性地寻找信息；并以反驳而不是检验来回应批评。

It is a self-assessment that is accurate from the inside and inaccurate from the outside, maintained by an athlete who is usually performing well enough at the start to be rewarded for the belief.

这里面没有不诚实。

这是一份从内部看准确、从外部看不准确的自我评估，而维持它的是一个在早期通常表现足够好、因而因这个信念而得到奖励的运动员。

Why it persists is the useful part for a coach.

它为何能持续下来，是对教练有用的部分。

Early performance and early results both support it, so the belief is repeatedly confirmed.

早期的表现和早期成绩都支持它，所以这个信念被反复确认。

The athlete is also protected from the evidence that would correct it: they seek feedback that flatters, they interpret setbacks as bad luck, and they are reinforced by teammates and spectators who read the confidence as competence.

这位运动员同时也受到那些本应纠正它的信息的保护：他们寻找让人愉快的反馈、把挫折解释为运气不好、并且被把这份自信读成能力。

The result is a self-assessment that is robust against exactly the information that should undermine it.

结果是一份恰恰对最应该削弱它的信息免疫的自我评估。

A coach's influence here is limited in a specific way, and understanding the limit is the useful skill.

教练在这里的影响力在某个具体方向上是有限的，而理解这个局限才是有用的技能。

Confrontation rarely works, because disputing the belief invites the athlete to defend it and produces a public contest the athlete will not lose socially.

直接对抗很少有效，因为反驳那个信念会让运动员去捍卫它，并产生一场社交上他不会输掉的公开争执。

更有效的是间接方式：提高挑战的标准，让这个信念被任务而不是被争论检验；

What has more effect is indirect: raising the standard of the challenge so that the belief is tested by the task rather than by argument, structuring situations where the athlete cannot attribute the outcome to luck, and giving feedback that is about the specific behaviour rather than about the person.

安排运动员无法把结果归因于运气的情境；并给出针对具体行为而不是针对个人的反馈。

And the honest ethical position is that the athlete is entitled to believe what they believe until the performance contradicts it.

而诚实的伦理立场是：运动员有权相信他所相信的，直到表现与之矛盾。

The coach's job is to arrange the performance to contradict it, and to do so without humiliating the athlete in public, because humiliation produces compliance rather than learning and it produces it only until the athlete is somewhere the coach cannot see.

教练的工作是安排一次与之矛盾的表现，并且要在不公开羞辱这位运动员的情况下做到--因为羞辱产生的是顺从而不是学习，而且只持续到教练看不到的地方。

The most effective long-run approach is the slow one: make the standard visible, make the feedback honest, and let the athlete draw the conclusion.

长期来看最有效的办法是那个慢的办法：让标准变得可见、让反馈保持诚实，然后让运动员自己得出结论。

The numbers / 算一算

Numbers to keep. The most useful measurement is objective and simple: put the athlete and a coach or a partner on the same task and compare, because the belief is about a gap that only a comparison reveals. Second, the feedback record - what was said after each session, and whether the athlete accepted it, changed anything, and argued - is a surprisingly good early indicator. And the honest limitation is that no quick intervention reliably reduces an overinflated self-assessment, which is why the practical expectation should be set in months rather than in sessions.

需要记住的数字。最有用的测量是客观而简单的：让运动员与教练或搭档做同一项任务并比较，因为这个信念关乎的是一个只有比较才能发现的东西。

Common mistakes / 易错点

- "Break the delusion." Confrontation produces a public argument the athlete cannot lose socially, and it produces compliance rather than learning.
"打破这个错觉。"对抗会产生一场社交上运动员输不了的公开争论，而且产生的是顺从而不是学习。
- "They know they are good, they have just lost the plot." The belief is maintained by selection: the athlete looks for information that flatters and reads setbacks as luck.
"他们知道自己有水平，只是最近没状态。"这个信念是靠选择维持的：这位运动员寻找让人愉快的信息，并把挫折读成运气。
- "Feedback will fix it." Feedback about the person is disputed. Feedback about the specific behaviour, delivered consistently, is testable - and that is why it works.
"反馈能解决。"关于这个人的反馈会被反驳。关于具体行为的反馈则可被检验--这也正是它有效的原因。
- "Confidence is always good." Confidence unsupported by competence is a liability, because it removes the athlete's incentive to seek the information that would improve them.

"自信总是好事。"没有能力支撑的自信是一种负担，因为它移走了运动员去寻找那些能让他们进步的信息的动机。

Why it matters / 为什么重要

This section is uncomfortable because it is about a mistake coaches make from benevolence rather than from malice - assuming that confidence is a good thing to build. It matters because the pattern is common, stable, and hard to reverse, and because the coach's most natural response is the one that works least. The transferable judgement is about sequencing: get the athlete into situations where the standard is visible and the evidence is honest, and let the belief meet reality on its own schedule rather than on the coach's.

这一节令人不舒服，因为它讲的是教练出于善意而非恶意所犯的错--假定自信是一件值得建立的好事。它之所以重要，是因为这个模式常

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Reward structure, norms, social climate and task design send messages about what is valued.

奖励结构、规范、社会氛围和任务设计会传递"什么被重视"的信息。

Key terms / 核心术语

Perceived ability 知觉能力

The athlete's belief about their capability.

运动员对自身能力的信念。

Approach behaviour 趋近行为

Taking on challenge and practising effort.

接受挑战并投入努力。

Formulas / 公式

Change "You are the best" to "Your preparation and decision improved this set."

把"你是最棒的"改为"你的准备和决策让这一局有所改善"。

Worked examples / 实例

>

How can coaches encourage approach behaviour?

教练如何鼓励趋近行为？

Exam focus / 考核要点

What outcomes may occur with high ego orientation and low perceived ability?

高自我定向和低知觉能力可能导致哪些结果？

Coaching the environment · 对环境的教练指导

First, picture it / 先这样想

The environment is doing something to this athlete whether you are designing it or not. Every session either tells the athlete that mistakes are dangerous, or that mistakes are information. Every piece of feedback either points at the person or at the action. A team culture is not a set of values on a wall - it is a set of messages, sent constantly, about what will be tolerated and what will be rewarded.

无论你是否在设计，环境都在对这位运动员做着什么。每一堂训练课要么在告诉运动员"错误是危险的"，要么在告诉它"错误是信息"。每一条反馈要么指向这个人，要么指向这个动作。团队文化不是墙上的一套价值观--它是一套关于什么可以被容忍、什么会被奖励的信息，被持续不断地发出。

The detail / 详细讲解

Climate is the term for the aggregate message a coaching environment sends, and the useful way to think about it is as a set of standing answers to a few questions.

"氛围"这个词指的是教练环境所发出的总体信息，而有用的思考方式是把它看作对几个问题的固定回答：努力是否被看重，还是只有结果

Is effort valued or only result?

一个错误是被纠正，还是被当作数据对待？

Is a mistake met with correction or with data?

努力是否在全队都得到同等回报，还是只有成绩得到？

Is effort rewarded equally across the team, or only the results?

安静一些的运动员的意见被听见吗？

Are the quiet athletes heard from? A team can answer these questions without ever discussing them, and its answers are visible in who speaks, who volunteers, and who stops trying.

一个团队可以在从不讨论这些问题的同时回答它们，而它的答案就写在谁会发言、谁会自愿参加、以及谁停止尝试里。

A mastery climate rewards effort, strategy and learning, and treats a mistake as information about what to work on next.

掌握氛围奖励努力、策略与学习，并把错误当作“接下来该练什么”的信息；

An ego climate rewards outcome, treats ability as fixed, and treats a mistake as evidence about the person.

自我氛围奖励结果，把能力视为固定，并把错误当作关于这个人的证据。

Neither is better in the abstract - a performance climate is appropriate for a task where the outcome is genuinely the point, and it is destructive in a task where the point is to get better.

抽象地说没有哪一种更好--当结果本身就是目的的任务中，表现氛围是合适的；

The error is applying one of them where the other belongs, and the evidence is fairly consistent that the effect of a climate on performance is larger than most coaches expect.

而在“变得更好”才是目的的任务中，它是破坏性的。

错误在于把其中一种用在了该用另一种的地方，而关于氛围对表现的影响大于多数教练预期这一点的证据相当一致。

The coach's behaviour is the most powerful lever in it, which is the slightly discouraging part, because behaviour is much easier to copy than a philosophy.

教练的行为是其中最有力量的抓手，这多少令人沮丧，因为行为比一种理念容易模仿得多。

Praise that is specific and about the process ("that was a good choice, you set up early") teaches something an athlete can use.

具体且针对过程的表扬（“这是个好选择，你提早建立了”）教会了运动员可以使用的某样东西；

Praise that is about the person ("you are a natural") teaches that performance is an identity, which is wonderful on the day it is true and unusable on the day it is not.

关于这个人的表扬（“你是天生的”）教会的是“表现是一种身份”，在它为真的那天很美妙，而在它不为真的那天无法使用。

There is a second-order effect that coaches often miss, which is that a climate changes who is willing to try.

有一个教练常常错过的二阶效应：氛围会改变谁愿意尝试。

In an ego climate, a beginner who makes a mistake is publicly told something about their ability, and the rational response is to stop trying in public and to protect the self-image.

在自我氛围中，一个犯错的初学者会被当众告知一些关于自己能力的事，而理性的反应是停止在公开场合尝试、并保护自我形象；

In a mastery climate the same mistake is data and the beginner is asked what to try next.

在掌握氛围中，同一个错误是数据，初学者被追问接下来该试什么。

The difference in how many people are willing to attempt a difficult task is large, and it is the main reason the two climates produce such different teams from the same talent pool.

在“有多少人愿意尝试一项困难任务”上的差别很大，这也是同样一批天赋在两种氛围下产生如此不同团队的主要原因。

The numbers / 算一算

Numbers to keep. The figure worth knowing is the proportion of the team's messages that are about the process rather than the outcome, because that proportion is a proxy for the climate and it is countable. A second practical measure is what happens to a beginner in the first weeks - whether they are still attempting things that are beyond them at the end of the month is a much better indicator of the climate than any questionnaire. And the third is the frequency of feedback that names a specific action, which is a habit rather than a philosophy and can therefore be changed by a coach without a workshop.

需要记住的数字。值得知道的数字是团队信息中关于过程而非结果的比例，因为这个比例是氛围的代理指标，而且可以数出来。第二个实

Common mistakes / 易错点

- "The climate is what the club values." It is what the club actually rewards and tolerates, and those are frequently different things.

"氛围就是俱乐部所倡导的价值观。"它是俱乐部实际奖励和容忍的东西，而这两者常常是不同的。

- "A mastery climate is always better for development." It is better for learning. Where the outcome is the point, a

performance climate is appropriate, and the error is using the wrong one for the task.

"掌握氛围对发展总是更好。"它对学习更好。在结果本身是目的的地方，表现氛围是合适的，而错误在于对任务用错了那一个。

"Praise effort." Praising effort as an outcome of trying is useful. Praising a trait, or praising effort after every attempt regardless of what happened, teaches nothing and is read as noise.

"表扬努力。"表扬努力这个尝试的结果是有用的。把努力当作一种特质来表扬，或者不管结果如何每次尝试都表扬，则什么也没教。

"One difficult session does not change a culture." It does not, in the sense that it does not stick. It does change behaviour immediately, and behaviour is what the athlete experiences that day.

"一节训练课不会改变一种文化。"从它不会固化的意义上说，是的。它会立刻改变行为，而行为正是运动员当天体验到的东西。

Why it matters / 为什么重要

This section is where a coach stops being the person who teaches a skill and becomes the person who designs the conditions in which a team learns, and that is a larger and more durable kind of influence than a session plan. It is also the section that makes the whole motivation theme actionable, because a climate is a thing a coach can deliberately set rather than a thing they can only hope for. The practical starting point is unglamorous: watch what your feedback actually rewards, and what your team actually tolerates, because that is the climate regardless of what the handbook says.

这一节是一位教练从"教会某项技术的人"变成"设计团队学习条件的人"的地方，而这种影响比一份训练课计划更大也更持久。它也是让整

Motivation is malleable because it changes with experience, feedback, goals and social climate.

动机具有可塑性，会随经验、反馈、目标和社交氛围改变。

Key terms / 核心术语

Social climate 社会氛围

The interpersonal and normative environment of the group.

团体的互动和规范环境。

Formulas / 公式

A team that rewards learning and effort is more likely to retain athletes and develop potential than a group focused only on winning.

奖励学习和努力的团队比只关注获胜的团队更可能保留运动员并发展潜能。

Worked examples / 实例

>

Give two ways coaches can manipulate situational factors.

给出教练调节情境因素的两个方法。

Exam focus / 考核要点

What is the difference between task and ego orientation?

任务定向和自我定向有什么区别？

Self-determination

自我决定

Guiding question / 引导问题

How does satisfaction of psychological needs influence the quality of motivation?

心理需要的满足如何影响动机质量？

Self-determination theory focuses on the quality of motivation. Autonomy, competence and relatedness support autonomous motivation, wellbeing and performance; need frustration is associated with controlled motivation or amotivation and poorer outcomes.

自我决定理论关注动机质量。自主、胜任和联结支持自主动机、福祉和表现；需要受挫与受控动机或无动机及较差结果相关。

Three basic needs and continuum · 三种基本需要与连续体

First, picture it / 先这样想

Three needs are usually treated as three separate boxes, and the more accurate picture is a continuum. All three can be satisfied or frustrated in any situation, in any proportion, and a session can be autonomy-supportive at the start and controlling by the end without anyone noticing the switch. The reason this matters is that frustration is not the absence of satisfaction - it is a separate thing, and it has its own consequences.

三种需要通常被当作三个分开的方框，而更准确的图景是一条连续体。在任何情境中，三种需要都可以被满足或被受挫，比例任意，而一堂训练课可以在开头支持自主、在结尾变成控制型，却没有人注意到这个切换。这一点之所以重要，是因为受挫并不等于满足的缺失--它是另一件事，并且有它自己的后果。

The detail / 详细讲解

The three are autonomy, competence and relatedness.

这三种需要是自主、胜任和归属。

Autonomy is experiencing action as self-endorsed rather than as imposed - it is about whether the reason for doing something feels like yours.

自主是把行动体验为自我认可而非被强加--它关乎做某件事的理由是否感觉像是你自己的。

Competence is feeling effective, which means the task is a genuine test that can be met, not a set of impossible hurdles and not a set of trivialities.

胜任是感到自己有效，这意味着任务是一个能够被达到的真正考验，而不是一组不可能的障碍，也不是一组琐碎的事。

Relatedness is feeling connected to and cared for by the people involved.

归属是感到与相关的人有连接并被关心。

All three are needs rather than wants, which means their frustration produces a specific and predictable set of responses.

三者都是需要而不是愿望，这意味着它们的受挫会产生一组特定的、可预测的反应。

Autonomy support and control are opposites on one dimension, and the confusion between them causes most coaching errors.

自主支持与控制在同一个维度上是相反的，而两者之间的混淆造成了大多数教练错误。

Autonomy support means providing rationale, acknowledging the athlete's perspective, and offering meaningful choice - it does not mean the athlete decides everything.

自主支持意味着提供理由、承认运动员的视角、并提供有实质意义的选择--它并不意味着由运动员决定一切。

Control means pressuring them toward an outcome regardless of their view, using guilt, conditional approval or threats.

控制意味着无论他们怎么想都把他们推向某个结果，使用内疚、有条件的认可或威胁。

An athlete can be given a decision that is not theirs while being treated with respect, and an athlete can be given complete choice while being made to feel guilty about it.

一位运动员可以被交给一个不属于他的决定而同时被尊重对待；

一位运动员也可以被给予完全的选择权，同时却因此感到内疚。

The frustration responses are as interesting as the satisfaction responses, because they are what actually gets observed.

受挫时的反应与满足时的反应一样有意思，因为被真正观察到的是前者。

Frustrated autonomy produces a compliance that looks identical to motivated behaviour, and it can be very hard to tell the two apart from the outside.

自主受挫产生一种与有动机的行为看起来一模一样的顺从，而从外面很难把两者区分开。

Frustrated competence produces helplessness and withdrawal - the athlete stops attempting tasks because attempting has been associated with evidence of inadequacy.

胜任受挫产生无助与退缩--这位运动员停止尝试任务，因为尝试已经被与"证据显示自己不足"联系起来。

Frustrated relatedness produces disengagement from the people rather than from the task, and it often looks like apathy about the sport.

归属受挫产生的是对人的疏离而不是对任务的疏离，而且它常常表现为对这项运动本身显得冷漠。

The numbers / 算一算

Numbers to keep. The three needs are not equally satisfied in most environments, and the useful measurement is simply to ask the three separately, because a single "how supported do you feel here" question produces a single average that hides which need is missing. The second practical figure is the proportion of decisions in a session that the athlete actually influenced - autonomy support is measurable as a count of genuine choices offered, not as an intention. And the third is the rate at which competence is challenged: task difficulty that is repeatedly too low produces the same withdrawal as difficulty that is repeatedly too high, and the symptom looks identical.

需要记住的数字。在大多数环境中这三种需要的满足程度并不相等，而有用的测量就是分别问这三个问题，因为一个"你在这里感到多被

Common mistakes / 易错点

- "Autonomy support means letting them do what they want." It means explaining the why, hearing their view, and offering real choice. It is not abdication, and a clear instruction can be autonomy-supportive.
"自主支持就是让他们做想做的。"它意味着解释原因、听取他们的看法、并提供真实的选择。它不是放任，而且一个清晰的指令也
- "Compliance means they are motivated." Compliance is what controlled behaviour looks like from outside. The two are indistinguishable in the moment and opposite in origin.
"顺从说明他们有动力。"顺从就是控制型行为从外面看的样子。两者在当下无法区分，成因却相反。
- "Frustration just means not enough satisfaction." Frustration is a distinct state with its own responses, and it is not fixed by adding a bit more of the same satisfaction.
"受挫只是意味着满足不够。"受挫是一个有自己反应的独立状态，而它不会通过再多给一点同样的满足来解决。
- "All three needs matter equally to everyone." Their relative strength varies by person and by situation, so the practical question is which one is currently the bottleneck rather than whether all three are being addressed.
"三种需要对人人都同样重要。"它们的相对强度因人因情境而异，所以实际问题是眼下哪一个瓶颈，而不是三种是否都被涉及。

Why it matters / 为什么重要

This is the framework that the rest of the motivation theory hangs from, and its value is that it is descriptive rather than prescriptive - it names what is happening in a room instead of telling you what to do. That matters because the same visible behaviour can come from two opposite places: a tired athlete and a frustrated athlete look the same, a compliant athlete and a motivated athlete look the same, and distinguishing them is a prerequisite for responding appropriately. A coach with this framework can diagnose a session the way a mechanic diagnoses an engine, and it transfers to any environment in which people are asked to perform.

这是其余动机理论所依托的框架，而它的价值在于它是描述性的而不是规定性的--它为房间里正在发生的事情命名，而不是告诉你该做什么

The continuum moves from amotivation through controlled and external/introjected regulation to identified, integrated and intrinsic motivation.

连续体从无动机，经过受控、外部和内摄调节，走向认同、综合和内在动机。

Key terms / 核心术语

Autonomy 自主

Experiencing choice and volition.

体验选择和意愿。

Competence 胜任

Feeling effective and able to improve.

感到有效并能改进。

Relatedness 联结

Feeling connected, valued and supported.

感到连接、被重视和支持。

Formulas / 公式

A coach offers choice, explains the reason for a task, gives optimal challenge, and builds inclusive team support.

教练提供选择，解释任务原因，给予适当挑战，并建立包容的团队支持。

Worked examples / 实例

>

How do need satisfaction and need frustration affect motivation quality?

需要满足和受挫如何影响动机质量？

Exam focus / 考核要点

Name the three basic psychological needs.

说出三种基本心理需要。

Controlled, autonomous and four extrinsic regulations · 受控动机、自主动机与四种外在动机调节

First, picture it / 先这样想

Motivation comes in a spectrum rather than two boxes, and the useful thing about the spectrum is that it is a continuum a person can move along. The same activity can be driven by enjoyment at one moment and by avoiding a bad feeling at the next, and the practical value of the classification is that each rung suggests a different response - while the "intrinsic versus extrinsic" question, as usually asked, is answered by a yes or a no and therefore helps nobody.

动机存在于一条光谱而不是两个方框，而这条光谱的有用之处在于它是人可以沿着移动连续体。同一项活动在这一刻可以由乐趣驱动，在下一刻由回避一种不舒服的感受驱动；而这个分类的实际价值在于每一级都提示一个不同的应对--而通常被提出的“内在还是外在”这个问题是用是或否来回答的，因此对谁都没有帮助。

The detail / 详细讲解

The continuum runs from intrinsic regulation at one end to external regulation at the other, with four identified extrinsic forms in between.

这条连续体一端是内在调节，另一端是外在调节，中间有四种被识别出的外在形式。

Internal regulation means the activity is done because it is inherently interesting or satisfying.

内在调节意味着做这项活动是因为它本身就有趣或令人满足；

External regulation means it is done because the outcome leads to a separable consequence - a reward or a punishment.

外在调节意味着做它是因为结果会带来一个可分离的后果--奖励或惩罚。

The problem is not the extrinsic end itself; rewards and consequences are real, and sometimes they are the honest reason for doing something.

问题不在于外在这一端本身；

The problem is the fixedness.

奖励和后果是真实存在的，而且有时它们就是做某件事的诚实理由。问题在于固定性。

The four extrinsic forms are external, introjected, identified and integrated.

这四种外在形式是外在的、内摄的、认同的和整合的。

External regulation is doing it for the reward or to avoid the punishment.

外在调节是为了奖励而做，或为了避免惩罚。

Introjected is doing it because you would feel guilty or ashamed not to - the reason is internal but the control is still from outside.

内摄调节是因为不做会感到内疚或羞耻--理由是内部的，但控制仍然来自外部。

Identified is doing it because you have decided the value is real, so it functions more like intrinsic.

认同调节是因为你已经认定这个价值是真的，所以它功能上更接近内在。

Integrated is doing it because it fits the rest of who you are, so it is part of a coherent self.

整合调节是因为它与你是一个什么样的人相契合，所以它是一个连贯自我的一部分。

What the framework buys is leverage, because each form implies a different intervention.

这个框架提供的是抓手，因为每一种形式都意味着不同的干预。

Extrinsic rewards tend to undermine the activity itself when the activity was already interesting, so adding them to a session the athletes enjoy usually reduces it.

当一项活动本身已经有趣时，外在奖励往往会破坏该活动本身，所以给它加到你本来就很喜欢的运动员的一堂课上通常会减少它。

That is a robust finding and it surprises people.

这是一个稳健的发现，而且让人意外。

But the same extrinsic reward, given for something genuinely tedious, can be the honest way to get it done - and the sensible position is not "extrinsic motivation is bad" but "know which rung the athlete is on and what would move them up rather than down".

但同样的外在奖励，用在真正乏味的事情上，可以是把它做完的诚实办法--而合理的立场不是"外在动机是坏事"，而是"知道运动员在哪

And the most important practical point is that the two ends are not opposed, so a coach can use both.

而最重要的实际点是两端并不对立，所以教练可以同时用两者。

The athlete who finds the training itself tedious can be given a separable reward, and the athlete who finds it enjoyable can be left alone or given information about their own progress as the reward.

觉得训练本身乏味的运动员可以被给予一个可分离的奖励，而觉得它有趣的运动员可以被放任不管，或者把关于他自己进步的反馈作

The error is applying one strategy uniformly across a team whose members are standing on different rungs of the same ladder.

错误是对一个成员分别站在同一架梯子不同级上的团队统一套用一种策略。

The numbers / 算一算

Numbers to keep. The continuum is best understood as something a person moves along over months rather than a fixed trait, which is why a single classification at the start of a season is worth much less than a second one at the end. The practical measurement is a short scale covering several rungs, administered twice, because the change is the finding. And the reward figure worth knowing is the shift in intrinsic motivation reported after tangible rewards are withdrawn - the interest does not always come back to where it was, which is the practical reason for being careful with rewards for activities that were already enjoyed.

需要记住的数字。最好把这条连续体理解为一个人在数月里沿着移动的东西，而不是一个固定特质，所以赛季开始时的一次分类，远不如

Common mistakes / 易错点

- "Intrinsic is good, extrinsic is bad." The continuum is a description, not a moral ranking. A reward for something genuinely tedious is often the correct response.
"内在是好的，外在是坏的。"连续体是一种描述，不是一张道德排名。对真正乏味的事情给予奖励，往往是正确的回应。
- "Rewards always increase motivation." Tangible rewards tend to reduce the intrinsic motivation of an activity that was already interesting.
"奖励总会提高动机。"有形奖励往往会减少一项本来就已经有趣的活动上的内在动机。
- "Motivating means adding more." The question is which rung the athlete is on. Adding more of the same thing is often what moves them down.
"激励就是加更多。"问题是运动员在哪一级。增加同样的东西，往往正是把他往下推的动作。
- "Classify them once and file it." Regulation shifts with the task, the season and the reasons the athlete is there, so a single classification at the start of a programme is out of date by the end of it.

"分类一次然后归档。"调节会随任务、赛季以及运动员做这件事的理由而改变，所以在一份计划开始时的一次分类，到计划结束时

Why it matters / 为什么重要

This section converts a question most people ask badly into one they can act on. The reason it is worth a full section is that the common framing - intrinsic versus extrinsic - is a false dichotomy that makes rewards look either essential or forbidden, and both readings lead to the same coaching errors. Knowing that the continuum has named rungs, and that a person moves along it, gives a coach two things: a reason not to add rewards where interest already exists, and a legitimate reason to use them where it does not. That is a more useful position than either half of the dichotomy.

这一节把一个多数人问得很糟的问题，变成了一个可以据此行动的问题。它值得占一整节的原因是：常见的框架--内在还是外在--是一个

A controlling reward can reduce intrinsic motivation through the overjustification effect; informational feedback and choice can support it.

控制性奖励可能通过过度理由效应降低内在动机；信息性反馈和选择可以支持内在动机。

Key terms / 核心术语

Overjustification effect 过度理由效应

A reduction in intrinsic motivation after controlling rewards are introduced.

引入控制性奖励后内在动机下降。

Formulas / 公式

"Train because I must avoid being dropped" differs from "train because it reflects who I am and what I value."

"训练是因为我必须避免被淘汰"不同于"训练因为它反映了我是谁和我重视什么"。

Worked examples / 实例

>

Explain how a controlling reward can reduce intrinsic motivation.

解释控制性奖励如何降低内在动机。

Exam focus / 考核要点

Order the four extrinsic regulations by self-determination.

按自我确定程度排列四种外在调节。

Six mini-theories · 六个小理论

First, picture it / 先这样想

Self-determination theory is a single claim with several faces, and the six mini-theories are the specific mechanisms that make it true. Knowing the theory gives you the destination; knowing the mini-theories gives you the specific levers. It is the difference between "support their autonomy" as a principle and knowing exactly which part of a session will not be autonomy-supporting and why.

自我决定理论是一个带有若干副面孔的单一主张，而六个小理论就是让它为真的那些具体机制。知道这个理论给你的是目的地；知道这些小理论给你的是具体的抓手。它是"支持他们的自主"这条原则与"确切知道一堂训练课的哪一部分不是支持自主的、以及为什么"之间的差别。

The detail / 详细讲解

The six map onto the pathway from extrinsic to intrinsic.

这六个对应着从外在到内在的路径。

Cognitive evaluation theory concerns whether an event is perceived as controlling or informational - a consequence that informs is very different from a consequence that pressures.

认知评价理论关乎一个事件被感知为控制性的还是信息性的--一个提供信息的后果，与一个施加压力的后果，是很不一样的。

Organismic integration theory concerns internalisation, how a value goes from "someone told me" to "this matters to me".

有机体整合理论关乎内化，即一个价值如何从"有人告诉我"变成"这对我重要"。

The behavioural outcomes that support them are self-determination of behaviour, perceptions of competence, and

autonomy support - three things an athlete actually experiences rather than reasons about.

支持它们的行为学结果是行为自我决定、胜任知觉和自主支持--三者是运动员真正体验到、而不是去推理的东西。

The two remaining mini-theories describe the contexts.

剩下两个小理论描述情境。

Basic psychological need theory identifies the three needs that the environment must satisfy.

基本心理需要理论指出环境中必须满足的三种需要。

Goal content theory addresses the fact that not all goals are equally good for the self, and separates goals that build mastery and health from goals that undermine them - which is why "just set a goal" is inadequate advice, since the content of the goal matters as much as having it.

目标内容

理论处理这样一个事实：并非所有目标对自我都同样好，并把那些建立掌握与健康的目标与那些损害它们的目标区分开--这就是为什么

The work motivation piece is the part most relevant to a coach's daily problem, and it draws a line between intrinsic motivation, which is doing something because it is inherently satisfying, and identified regulation, which is doing it because you have concluded that the value is real.

其中与教练的日常问题最相关的是"工作动机"那一块，它在内在动机（因为某件事本身令人满足而去做）和认同调节（因为你已经认同

The distinction matters because an athlete can have very little of the first and plenty of the second, and the second is far more durable - a professional who identifies with the value of their work will turn up on a day when they do not feel like it, and an athlete who has not made that identification will not.

这个区分之所以重要，是因为一位运动员可能前者很少、后者很多，而后者要持久得多--一位认同自身工作价值的专业人士，会在自己
而一位还没完成这种认同的运动员不会。

And the practical synthesis is that the mini-theories together describe a path an environment can enable.

而实际的综合在于：这些小理论合在一起描述了一条环境可以促成打开的路径。

An athlete can be moved from doing something for a reward, through doing it because it is approved of, through doing it because it is useful, to doing it because it is who they are - and none of those steps happens automatically.

一位运动员可以从"为了奖励而做"，经过"因为被认可而做"，经过"因为它有用而做"，走到"因为这就是我是谁"--而这些步骤没有一步会

Each requires something specific from the environment, which is the reason the framework is useful rather than merely interesting.

每一步都需要环境中某种具体的东西，这正是这个框架有用而不只是有趣的原因。

The numbers / 算一算

Numbers to keep. The single most useful figure in the whole framework is the proportion of a session in which the athlete's reasons are informational rather than controlling, because it is countable and it moves for identifiable reasons. Second, the standard against which the activity is judged: a task experienced as genuinely useful to the athlete is identified regulation, and one experienced as an imposition is controlled no matter how well it is delivered. Third, the honest limitation - the mini-theories are small and overlapping, and the practical guidance is more consistent than the empirical separation between them suggests, so a coach should use the framework as a checklist of questions rather than as six separate prescriptions.

需要记住的数字。整个框架中最有用的单一数字，是一堂训练课中运动员的理由属于信息性而非控制性的时间比例，因为它可数，而且会

Common mistakes / 易错点

-

"SDT is one theory, so it will give one answer." The six mini-theories are specific mechanisms with different questions, and a coach who only knows the headline claim will apply it indiscriminately.

"自我决定理论是一个理论，所以它会给出一个答案。"六个小理论是不同的机制、对应不同的问题，只知道那句概括性主张的教练

-

"Just set some goals." Goal content matters as much as having a goal. Goals that build mastery and goals that erode it are both goals.

"设一些目标就行了。"目标的内容与是否设目标同样重要。建立掌握的目标和侵蚀掌握的目标都是目标。

-

"Intrinsic motivation is what we are aiming for." Many athletes are sustained by identified regulation, which is more durable and does not require the activity to be fun on the day.

"内在动机才是我们的目标。"许多运动员是靠认同调节维持的，而它更持久，也不要求这项活动当天就是有趣的。

-
"The mini-theories have been experimentally separated." They have been studied separately far more often than they have been convincingly distinguished, and the practical advice is more solid than the distinctions.

"这些小理论已经被实验区分开了。"它们被分开研究的次数远多于被令人信服地分辨的次数，而实用建议比那些区分更扎实。

Why it matters / 为什么重要

This is the section that gives the motivation theme its vocabulary, and vocabulary matters more than it looks because these are the terms a student will use in an exam and a coach will use in a staff room. But the deeper reason it is taught is that it converts a general commitment to good practice into specific answers. "Support their autonomy" is easy to agree with and hard to execute; knowing that the same activity becomes controlling when a consequence is added to it, and informational when it is explained, is executable. It is the difference between having a value and knowing what it requires of you on a Tuesday.

这一节给了动机主题它的词汇表，而词汇表比看上去更重要，因为这些正是学生在考试中会用的词、也是教练在办公室里会用的词。但

-
Theories interconnect: needs influence motivation, causal orientations support internalisation, goals and relationships shape contexts, and contexts influence intrinsic motivation.

这些理论相互连接：需要影响动机，因果取向支持内化，目标与关系塑造情境，情境影响内在动机。

Key terms / 核心术语

Goal content 目标内容

Whether goals emphasise learning and mastery or comparison and ego.

目标强调学习掌握还是比较和自我。

Internalisation 内化

The process of becoming more self-determined.

变得更自我确定的过程。

Formulas / 公式

An athlete attributes success to controllable effort, values training, and experiences supportive relationships, supporting quality motivation.

运动员把成功归因于可控努力，重视训练，并经历支持性关系，从而支持动机质量。

Worked examples / 实例

>

Briefly explain the six mini-theories.

简要解释六个小理论。

Exam focus / 考核要点

How do the mini-theories connect to a sport context?

六个小理论如何联系运动情境？

Motivational climate

动机氛围

Guiding question / 引导问题

How does the environment created by a coach shape motivation?

教练创造的环境如何塑造动机？

Motivational climate is the psychological environment created through session design, instruction, relationships and feedback. A mastery climate emphasises effort, cooperation, improvement and enjoyment; an ego climate emphasises comparison, winning and ability.

动机氛围是通过课程设计、指导、关系和反馈形成的心理环境。掌握氛围强调努力、合作、进步和享受；自我氛围强调比较、获胜和

Mastery and ego climates · 掌握氛围与自我氛围

First, picture it / 先这样想

A climate is the set of questions a team answers without discussing them. Two answers dominate sport: effort and learning count, or ability and winning count. Both teams will work hard on Saturday. What differs is what Saturday was for, and that difference follows the athletes into every subsequent competition, in both directions.

氛围是一个团队在不讨论它们的情况下所回答的一组问题。体育中有两种答案占主导：努力与学习算数，或者能力与获胜算数。两个团队周六都会很拼。不同的是周六是为了什么，而这一差别会跟着运动员进入之后的每一场比赛，双向皆然。

The detail / 详细讲解

A mastery climate - also called a task climate - treats success as defined by the task: you succeeded if you improved, executed the plan, or solved the problem.

掌握氛围（也称任务氛围）把成功定义为由任务来判定：进步了、执行了计划、或解决了问题，就算成功。

An ego climate - also called a performance climate - treats success as defined relative to others: you succeeded if you were better.

自我氛围（也称表现氛围）把成功定义为相对于他人的判定：你更强了才算成功。

The criteria are not equivalent because one is a judgement about process and the other is a judgement about a comparison, and comparisons are inherently zero-sum.

这两个标准并不等价，因为一个是关于过程的判断，另一个是关于比较的判断，而比较天然为零和的。

Somebody is above you in an ego climate whether you did well or not.

在自我氛围中，总有人排在你前面，不管你做得好不好。

The behavioural consequences run consistently.

行为后果是有一致性的。

Under a mastery climate, effort, persistence and the seeking of difficult tasks all increase, because a difficult task is where improvement is available.

在掌握氛围下，努力、坚持和寻求困难任务都会增加，因为困难任务正是“改进可得”的地方。

Under an ego climate, the athlete selects tasks that maximise the chance of looking good, avoids tasks where they might be plainly outclassed, and - critically - attributes outcomes to ability rather than to what they did, because ability is a durable identity and an outcome is not.

在自我氛围

下，运动员会选择最能“看起来好”的任务、回避那些可能明显被比下去的任务，并且--关键地--把结果归因于能力而不是自己做了什么。

There is a counterintuitive downside to a pure mastery climate that a coach should know: if winning never matters, the athlete can become a master of a sport they no longer want to compete in.

纯粹的掌握氛围

有一个反直觉的缺点，教练应当知道：如果获胜从来都不重要，运动员可能变成一个他已不再想参加比赛的项目的高手。

And there is a real upside to a well-mixed environment, which is that a technically excellent, resilient athlete still needs a standard to aim at.

而一个搭配得当的环境确实有好处，因为一位技术出色、抗压的运动员仍然需要一个瞄准的标准。

The evidence generally favours mastery climates as the default for learning and for youth, and it does not say that a performance climate is harmful in every context - the practical claim is narrower than the popular one.

证据总体上支持把掌握氛围

作为学习阶段和青少年阶段的默认选择，它并没有说表现氛围在所有情境中都有害--实际主张比流行的那种说法要窄。

The transition problem is worth naming, because most athletes meet it.

过渡问题值得点名，因为大多数运动员都会遇到它。

A youth programme is usually mastery by default - the results do not matter yet - and then the athlete arrives at a club where outcomes are the only thing anyone reports.

青少年计划通常默认是掌握导向的--成绩还不重要--然后这位运动员来到一个俱乐部，在那里结果是所有人唯一会汇报的东西。

A sudden move to a performance climate removes the reason the athlete was good, and the effect is often a drop in performance rather than a rise.

突然转向表现氛围，会移除这位运动员当初变强的理由，而其后果往往是表现下滑而不是上升。

The practical handling is to keep a mastery standard available alongside the competitive one: a technical criterion, a personal target, something the athlete can succeed at that is not "be the best in the room".

实际处理方式是：在一个竞争性标准旁边保留一个掌握标准--一个技术上的判据、一个个人目标、某件他能成功做到、而它不是"成为"

The numbers / 算一算

Numbers to keep. The two measures that are actually informative are the proportion of training time spent on tasks the athlete has already mastered and the proportion of the coach's feedback that names a specific action rather than an outcome. Both are countable, both change when the climate changes, and neither requires anyone to be happy. The third figure is the realistic expectation about the transition: a well-run programme is a majority mastery environment with a performance overlay on competition days, not the reverse, and a plan that inverts that ratio usually loses talent before it produces results.

需要记住的数字。真正有信息量的两个指标是：花在运动员已掌握任务上的训练时间比例，以及教练反馈中点名具体动作而非结果的比例。

Common mistakes / 易错点

- "Just make it a mastery climate and the problem is solved." Without any competitive standard, an athlete can become a master of a sport they no longer want to compete in.
"只要营造掌握氛围问题就解决了。"在没有任何竞争性标准的情况下，一位运动员可能变成一个他已不再想参加比赛的项目的高手。
- "Ego climates build tough competitors." They build athletes who attribute outcomes to ability, which makes the first sustained failure far more damaging than it needs to be.
"自我氛围培养出更坚韧的竞争者。"它培养出的是把结果归因于能力的运动员，这让第一次持续失败造成的伤害远大于必要的程度。
- "Changing the climate is a matter of telling athletes to be process-focused." A climate is what the environment rewards and tolerates. A coach who says one thing and rewards another has an ego climate and a confusing message.
"改变氛围就是告诉运动员要专注过程。"氛围是环境所奖励和容忍的东西。一位一边说一套、一边奖励另一套的教练，拥有的是一套。
- "The junior programme should be performance-based because that is what senior sport is." A sudden inversion of the ratio removes the reason the athlete was good, and the result is usually a drop in performance.
"青少年计划应该以表现导向，因为成人体育就是这样。"比例的突然倒置会移除这位运动员当初变强的理由，而结果通常是表现下

Why it matters / 为什么重要

The mastery and ego distinction is the most useful single idea for a coach who has to build a team rather than repair an individual, because a climate can be set deliberately while a personality cannot. It is also the most misapplied, since the slogan version - "just create a mastery climate" - skips the part that requires judgement, which is the balance. And the transition problem makes it practical rather than theoretical: an athlete trained entirely on mastery arriving at a results-driven club is one of the most common and most avoidable causes of a talented adolescent dropping out.

掌握与自我的区分，是一位需要"建设一支队伍"而非"修理一个个体"的教练最有用的单一概念，因为氛围可以有意识地设定，而人格不行。

- Climate shapes what athletes think is valued and how they define success.

氛围塑造运动员认为什么有价值以及如何定义成功。

Key terms / 核心术语

Mastery climate 掌握氛围

A climate focused on learning, effort, cooperation and personal improvement.

关注学习、努力、合作和个人进步的氛围。

Ego climate 自我氛围

A climate focused on winning, comparison and superiority.

关注获胜、比较和优越的氛围。

Formulas / 公式

Rewriting "you are the best" to "your effort and strategy improved" reinforces mastery.

把"你是最棒的"改为"你的努力和策略有所进步"可强化掌握氛围。

Worked examples / 实例

>

Define motivational climate and list four mastery features.

定义动机氛围并列四个掌握特征。

Exam focus / 考核要点

How can climate affect retention?

氛围如何影响留存?

TARGET framework · TARGET框架

First, picture it / 先这样想

TARGET stands for task, authority, recognition, grouping, evaluation and timing, and it is a checklist for a session rather than a philosophy. Each letter answers a question a coach has to answer anyway: what is the task, who decides, who gets noticed, who trains with whom, how is it judged, and when. It is useful precisely because it is boring - it converts vague intentions about a session into six decisions that can be made before the athletes arrive.

TARGET 代表任务、决策权、认可、分组、评价和时间，而它是一份训练课清单而不是一种理念。每个字母都回答一个教练本来就得回答的问题：任务是什么、谁来决定、谁被看见、谁和谁一起练、如何评价、以及什么时候。它之所以有用恰恰因为它很枯燥--它把对一堂训练课的模糊意图，转化成六项可以在运动员到来之前做出的决定。

The detail / 详细讲解

Task is the biggest and the most consequential.

任务是其中最大、也最有后果的一个。

A session designed around personal improvement produces very different behaviour from one designed around beating the next team in the table, and the difference is not motivational fluff - it determines which tasks the coach sets, how hard they are, and whether failure is informative.

围绕个人进步设计的训练课，会产生与围绕"击败积分榜上下一队"设计的训练课完全不同的行为，而这个差别不是动机上的空话--它决

Authority is the second: who decides what the session contains.

第二是决策权：谁决定这堂课的内容。

Giving athletes a genuine decision about something real is cheap and changes the perceived reason for the session.

把某件真实的事情交给运动员决定，成本很低，却会改变他们对这堂课之理由的感受。

Recognition is easy to get wrong because praise feels like a good thing to give.

认可很容易做错，因为表扬感觉是一件好事。

Recognition that is purely public and result-based makes status the currency of the session and undermines everything that is not visible.

纯粹公开且基于结果的认可，会让地位成为这堂课的货币，并破坏一切不可见的东西。

Recognition that names what a person did well and is distributed to whoever did it well, including people who will never win anything, is a different intervention.

指出某个人做得好、并且把它分配给真正做得好的人--包括那些永远不会赢得任何东西的人--则是另一种干预。

Grouping determines who compares themselves to whom, and mixed-ability grouping consistently produces better learning than grouping by level.

分组决定谁拿自己和谁比较，而混水平分组一致地产生比按水平分组更好的学习。

Evaluation is the one with the longest memory, because athletes remember how they were judged.

评价是记忆最久的那一项，因为运动员会记得自己是怎么被评价的。

Formative evaluation during the session, about specific actions, is a different experience from summative evaluation at the end, about a score or a place.

训练课中关于具体动作的形成性评价，与结束时关于分数或名次的总结性评价，是两种不同的体验。

And timing is the one most often forgotten: the first minutes of a session set the arousal and the attention for everything after them, and a session that opens with a long explanation and an easy warm-up has already spent the part of the athlete that was most ready.

而时间是最常被忘记的一项：一堂课的头几分钟设定了之后一切的唤醒水平和注意力；

而一堂以长时间讲解和轻松热身开场的课，已经花掉了运动员最准备好的那部分。

The numbers / 算一算

Numbers to keep. The useful figures are structural and countable: how many of the six elements were consciously decided before the session, how long the opening segment runs, how many athletes received a piece of recognition that was not tied to winning, and how many distinct ability levels are present in each training group. The first of those is the honest measure of whether the framework is in use - most sessions are run on three or four of the six by habit and never revisited, and writing the six down is what turns a good session into a repeatable one.

需要记住的数字。有用的数字是结构性的、可数的：六个要素中有几个在训练课之前被有意识地决定、开头段有多长、有多少位运动员得

Common mistakes / 易错点

- "TARGET is a motivational model." It is a session-design checklist. The value is in the six decisions, not in the acronym.
"TARGET 是一个动机模型。"它是一份训练课设计清单。价值在那六项决定里，而不在这个缩写本身。
- "Praise is always good." Recognition tied only to results and given publicly makes status the currency of the session, and it erodes everything invisible.
"表扬总是好的。"只与结果挂钩、并且公开给出的认可，会让地位成为这堂课的货币，并侵蚀一切不可见的东西。
- "Group by ability so everyone can be challenged." Grouping by level removes the comparison that makes mixed groups learn better.
"按能力分组，这样每个人都能被挑战。"按水平分组移走了那个让混水平组学得更好的比较。
- "The warm-up is where the class begins." The opening minutes set arousal and attention for everything after them, and a long explanation spends the part of the athlete that was most ready.
"热身就是这堂课开始的地方。"开头的几分钟为之后一切设定唤醒和注意力，而一段长讲解会花掉运动员最准备好的那部分。

Why it matters / 为什么重要

TARGET is worth teaching because it is the rare motivational concept that a coach can apply to the next session, on Monday, without changing anything about themselves. Most of the framework is a matter of noticing decisions that are being made by default and deciding them instead, and the six elements interact - a recognition practice that rewards only results will undo a grouping structure that was carefully mixed. It is also a good vehicle for teaching the general principle that a session is a designed environment, which is the same principle the previous section arrives at from a different direction.

TARGET

值得教，是因为它是少有的那种"下一个训练课、周一、不需要改变自己任何东西就能应用"的动机概念。这个框架的大部分内容其实就是

Coaches can manipulate these dimensions through what they emphasise, who decides, what is praised, how groups are formed, how performance is evaluated and how time is allocated.

教练可通过强调什么、谁作决定、认可什么、如何分组、如何评价和如何分配时间来调节这些维度。

Key terms / 核心术语

TARGET TARGET框架

Task, Authority, Recognition, Grouping, Evaluation and Time.

任务、权威、认可、分组、评价和时间。

Formulas / 公式

A coach sets a meaningful challenge, asks athletes for input, praises process, rotates positions, gives individual feedback and ensures fair practice time.

教练设定有意义挑战，征求运动员意见，表扬过程，轮换位置，给予个人反馈并确保公平练习时间。

Worked examples / 实例

>

What does each letter in TARGET stand for?

TARGET每个字母代表什么？

Exam focus / 考核要点

Give one mastery and one ego example for a TARGET dimension.

为TARGET的一个维度各给一个掌握和自我例子。

Links to motivation and SDT · 与动机和自我决定理论的联系

First, picture it / 先这样想

Two frameworks that sound different are often the same idea described at two altitudes. Achievement motivation is describing what the athlete is pulled toward; self-determination theory is describing what the environment must do to make that pull sustainable. Reading them together, the question "what does this athlete want" gets replaced by "what has the environment been providing, and what happens to the want when it is not provided".

两个听起来不同的框架，往往是同一个想法在两个海拔高度上的描述。成就动机在描述运动员被拉向什么；自我决定理论在描述环境必须做什么才能让那种拉扯可持续。把它们合起来读，"这位运动员想要什么"这个问题就被替换成了"环境一直在提供什么、以及当它不提供时那种想要会怎样"。

The detail / 详细讲解

The clean link is between the achievement orientations and the three needs.

最清晰的连接在成就取向与三种需要之间。

A mastery orientation is what an athlete approaches the need for competence with: it is a preference for a task that is a real test, and it is the behavioural form of wanting to be effective.

掌握取向是运动员用来接近胜任需要的方式：它是对一项构成真实考验的任务的偏好，它是"想要有效"的行动形式。

A performance orientation is the approach to the same need when effectiveness is defined comparatively.

绩效取向是当"有效"被定义为比较性地时，接近同一个需要的方式。

So the two achievement orientations are not a separate theory - they are two ways of serving the same need, and a coach who understands this can read an athlete's orientation as information about which need is currently under-supplied.

所以两种成就取向并不是一个独立理论--它们是服务于同一个需要的两种方式；

而理解这一点的教练可以把运动员的取向当作"当前哪一个需要供给不足"的信息来读。

The second link is between the needs and the environment, and this is where SDT does the work that achievement motivation cannot.

第二个连接在需要与环境之间，而这正是自我决定理论完成了成就动机做不到的工作的地方。

Achievement motivation describes a direction; self-determination theory describes the conditions under which a direction holds.

成就动机描述一个方向；

An athlete with a mastery orientation whose competence is chronically under-fed will not simply keep trying - they will drift toward the comparative standard, because that is the only source of the feeling left.

自我决定理论描述一个方向得以维持的条件。

The change looks like a change of orientation, and it is a change of supply.

一位掌握取向、但胜任感长期供给不足的运动员不会一直坚持--他会漂移到那个比较性的标准上，因为那是唯一剩下的那种感觉的来源。这个变化看起来像是取向的改变，而它实际上是供给的改变。

The third link is directional.

第三个连接是有方向的。

The continuum from controlled to integrated regulation describes the quality of an athlete's engagement, and it moves in the same conditions the needs theory describes: it strengthens where the environment supplies competence, autonomy and relatedness, and it weakens where it does not.

从控制调节到整合调节的连续体，描述的是运动员投入的质量，而它恰好在需要理论所描述的同样条件下移动：环境供给胜任、自主和

So the same intervention - a task that is a genuine test, chosen by the athlete, alongside people they are connected to - is simultaneously a competence intervention, an autonomy intervention, a relatedness intervention and a regulation intervention.

所以同一个干预--一项由运动员自己选择的、构成真实考验的任务，和他们有连接的人一起做--同时是一次胜任干预、一次自主干预、

That is the practical pay-off of reading the frameworks together.

这就是把两个框架合起来读的实用回报。

And the honest limitation is that the frameworks were developed for different purposes and the evidence does not always line up neatly.

而诚实的局限是：这两个框架是为不同目的发展出来的，证据并不总是整齐吻合。

Achievement motivation is a content theory - it predicts which direction an athlete is pulled.

成就是内容理论--它预测运动员被拉向哪个方向。

Self-determination theory is a process theory - it predicts when that pull holds.

自我决定理论是过程理论--它预测那种拉扯何时能维持。

A coach who treats them as one framework gains a way to ask better questions and loses nothing, but a student writing about it should be able to say which claim comes from which theory, because the exam will ask.

把它们当作一个框架的教练会因此获得提出更好问题的能力而没有损失任何东西，但一个在写作中要用到它们的学生，应当能够说出哪

The numbers / 算一算

Numbers to keep. The three needs are the most useful measurement because they map directly onto things an environment does, and the practical figure is the athlete's rating of the session on each of the three separately, taken twice - the first and the last week of a block, so the change is visible. The second figure is the proportion of the session in which the athlete's reasons are informational rather than controlling. The third, for an honest limit, is that the achievement-orientation and the need-supply measures do not always covary, and a programme that improves one without noticing the other will sometimes appear to work for the wrong reason.

需要记住的数字。三种需要是最有用的测量对象，因为它们直接对应环境所做的事，而实用数字是运动员在一个训练周期第一周和最后一

Common mistakes / 易错点

- "These are two competing theories." They are not competing. One is a content theory about direction, the other a process theory about what sustains it, and reading them together is where the practical value is.

"这是两个相互竞争的理论。"它们并不竞争。一个是关于方向的内容理论，另一个是关于什么维持方向的过程理论，而把它们合起来

- "An athlete switched from mastery to performance orientation." Often the orientation did not switch; the supply of competence did, and the athlete moved to the only source of the feeling still available.

"这位运动员从掌握取向变成了绩效取向。"往往是取向没有变，胜任的供给变了，于是这位运动员转投到唯一还剩下的那种感觉的

- "Self-determination theory explains the content of motivation." It explains when motivation holds. The content of motivation is what achievement motivation theory is about.

"自我决定理论解释动机的内容。"它解释的是动机何时能维持。动机的内容是成就动机理论所讨论的。

-

"Knowing both means you can blend them." The value is being able to name which claim comes from which theory, and being able to ask a better question of a session than you could before.

"两个都知道就可以把它们融合起来。"价值在于能够说出哪个主张来自哪个理论，以及能够对一堂训练课问出比之前更好的问题。

Why it matters / 为什么重要

This section is the payoff for the whole motivation theme, and it is the section that turns a set of parallel theories into a way of thinking. A coach who has read the frameworks separately accumulates a list of concepts. A coach who has read them together can look at a session and ask a single question that spans all of them: is the athlete being given a task that is a real test, did they have a say in it, and are they doing it alongside people they are connected to. That question is answerable, it is short, and it is a better use of a coach's attention than almost any individual technique in this theme.

这一节是整个动机主题的回报，也是把一组平行理论变成一种思维方式的章节。分别读过这些框架的教练，会积累起一份概念清单；合起来

-

In team sports mastery climate builds culture and cohesion, in individual sports it builds confidence, and in youth sport it protects long-term enjoyment and participation.

团队运动中掌握氛围建立文化和凝聚力；个人运动中建立自信；青少年运动中保护长期享受和参与。

Key terms / 核心术语

Long-term participation 长期参与

Continued involvement supported by enjoyment, learning and belonging.

由享受、学习和归属支持的持续参与。

Formulas / 公式

A mastery climate supports both achievement goals and quality motivation rather than replacing one with the other.

掌握氛围同时支持成就目标和动机质量，而不是用其中之一替代另一个。

Worked examples / 实例

>

How does a mastery climate satisfy SDT needs?

掌握氛围如何满足自我决定理论的需要？

Exam focus / 考核要点

Why can a mastery climate improve long-term participation?

掌握氛围为何能提高长期参与？

Arousal and anxiety

唤醒与焦虑

Guiding question / 引导问题

How do activation and anxiety influence performance?

激活和焦虑如何影响表现？

Arousal is physical and psychological activation. Drive, inverted-U and catastrophe theories describe different relationships with performance. Anxiety has cognitive and somatic dimensions, and the individual zone of optimal functioning differs by person, task and context.

唤醒是身体和心理的激活。驱力、倒U和灾难理论描述不同的表现关系。焦虑包含认知和躯体维度，最佳功能区因人、任务和情境而

Arousal and theories · 唤醒与相关理论

First, picture it / 先这样想

Arousal is not a thing you want to be low or high. It is a dial, and the useful question is what this task wants the dial set to. A sprinter wants it high, a surgeon-like putting golfer wants it low, and treating both as "relax before you perform" gives one of them the wrong instruction. The whole history of arousal research is the history of learning that the relationship is curved, not linear.

唤醒不是一件你想要它"低"或"高"的东西。它是一个旋钮，而有用的问题是这项任务希望这个旋钮被设在哪里。短跑运动员希望它高，像外科医生一样稳的推杆球手希望它低，而把两者都当作"上场前放松"处理，就给了其中一个错误的指令。唤醒研究的整个历史，就是逐渐学会这个关系是弯曲的而不是线性的历史。

The detail / 详细讲解

The inverted-U hypothesis is the starting point: performance improves with arousal up to an optimum and then deteriorates.

倒U形假设是起点：表现随唤醒提升到某个最优、之后开始下降。

It is an improvement on the simple linear idea, and it is still the one most people remember, which is unfortunate because it is nearly useless on its own.

它比简单的线性想法更进一步，而且它仍然是大多数人记得的那个，这很遗憾，因为它单独看几乎没用。

Every task has its own optimum, the optimum moves with difficulty and with the individual, and the curve says nothing at all about the direction of causation.

每项任务有自己的最优点，这个最优会随难度和随个体移动，而且这条曲线对因果方向完全没有说明。

Drive theory came first and argued that arousal is a performance of the nervous system, so more arousal means more capacity to respond.

驱力理论最早出现，主张唤醒是神经系统的一种表现，所以更多唤醒意味着更多反应能力。

That is reasonable for a simple reflex and increasingly wrong as the task gets complex, because complex performance depends on the opposite of a large pool of response - it depends on selecting one response and suppressing the rest.

这对一个简单反射来说是合理的，而任务越复杂就越不正确，因为复杂表现依赖的是"大量反应池"的反面--它依赖于选择一个反应并抑

This is why a high-arousal athlete is often excellent at sprinting and poor at tactical decision-making, and it is the same mechanism as the narrowing of attention under pressure.

这就是为什么一位高唤醒的运动员常常在短跑上很出色、在战术决策上很差，而这与压力下注意变窄是同一个机制。

Catastrophe theory added the part that explains the thing everyone remembers.

灾难理论补充了那个能解释大家都记得的现象的部分。

Performance deteriorates gradually as arousal rises past the optimum - that is the inverted-U - and then, for a sufficiently complex task, it collapses abruptly rather than deteriorating smoothly.

表现随着唤醒越过最优点而逐渐下降--那就是倒U形--然后，对足够复杂的任务来说，它不是平滑下降而是突然崩塌。

The collapse happens at a lower level of arousal for a difficult task than for an easy one, and it happens at a lower level for an unskilled performer than for a skilled one.

对于困难任务，崩塌发生在比简单任务更低的唤醒水平；

That is why a novice who is slightly too activated does noticeably worse, while an expert in the same state can be merely slightly worse.

对于不熟练者，崩塌发生在比熟练者更低的水平。

这就是为什么一位稍微过度激活的初学者会明显表现更差，而处在同样状态的专家可能只是稍微差一点。

The practical consequence is that the intervention depends on the task and the person, and a single team-wide pre-performance routine is usually wrong for somebody.

实际推论是：干预取决于任务和人，而一个全队统一的赛前例程通常对某个人来说是错的。

Turning arousal down is achievable and cheap - long breathing, a repeatable routine, a familiar warm-up, reducing pre-event thinking time - and it is what most anxious athletes need.

降低唤醒

是可以做到而且成本低的--长呼吸、可重复的例程、熟悉的热身、减少赛前思考时间--而这正是多数焦虑型运动员需要的。

Turning it up is also achievable and it is what some athletes need before a flat, unfocused effort.

提高唤醒同样可以做到，而且有些运动员在出现平淡、不集中的表现之前需要它。

The mistake is prescribing by the coach's own preference rather than by the task's demand.

错误在于按教练自己的偏好而不是按任务的需求来开处方。

The numbers / 算一算

Numbers to keep. Two figures matter. The first is the habitual pre-competition arousal level of the athlete, which is stable across events and is a better predictor of how a routine should be designed than any measure taken on the day. The second is the cognitive anxiety score, which is a better predictor of performance decrement than the somatic one - an athlete who reports "I am not worried, I just cannot stop thinking about it" is more impaired than one who reports a racing heart. The measurement of both is short and the second one is the more useful of the two.

需要记住的数字。有两个数字重要。第一个是运动员习惯性的赛前唤醒水平，它在各场比赛之间是稳定的，因此比任何当天测得的指标都

Common mistakes / 易错点

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"Arousal is the same as anxiety." They are separable. An athlete can be highly aroused and calm, or anxious with a resting heart rate, and the two respond to different things.

"唤醒就是焦虑。"两者是可分的。一位运动员可以高度唤醒而平静，也可以焦虑而心率处于静息状态，而这两者对不同的事物作出

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"The inverted-U tells you the optimum." It does not. Every task has a different optimum, the optimum moves with difficulty, and the curve says nothing about which causes which.

"倒U形告诉你最优点。"它没有。每项任务有不同的最优点，这个最优随难度移动，而且这条曲线对谁导致谁什么也没说。

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"Relax before you perform." Correct for many tasks and for many athletes, and wrong for the sprinter who needs activation and wrong for the athlete who goes flat without it.

"上场前放松。"对许多任务和许多运动员是正确的，而对需要激活的短跑运动员是错的，对没有它就会变得平淡的运动员也是错的

-

"Arousal problems are all performance anxiety." Some are preparation problems, some are fatigue, some are illness, and treating them all as nerves removes the opportunity to fix the actual cause.

"唤醒问题全都是比赛焦虑。"有些是准备问题，有些是疲劳，有些是疾病，而把它们一律当作紧张，就失去了修复真正原因的机会

Why it matters / 为什么重要

Arousal is the section where a coach is most likely to give advice that is wrong for a particular athlete, because the advice feels obvious and the obvious version is only right about half the time. The useful skill is twofold: knowing that the relationship is curved rather than linear, and knowing that a collapse is qualitatively different from a decline. The second point is the one with the most practical consequence, because it explains why the last two per cent of arousal are not a small problem, and why a routine that works for most of a squad can be actively harmful for the one athlete who should be doing something else.

唤醒是教练最有可能给出对某位特定运动员来说是错误建议的一节，因为那个建议听起来显而易见，而显而易见的版本只有大约一半时间

The optimal zone changes with task difficulty, experience, time, fatigue and situation. Power tasks may benefit from higher arousal; precision tasks often need lower to moderate arousal.

最佳区域会随任务难度、经验、时间、疲劳和情境变化。力量型任务可能受益于较高唤醒，精准任务通常需要较低到中等唤醒。

Key terms / 核心术语

Arousal 唤醒

Physical and psychological activation.

身体和心理激活。

IZOF 个体最佳功能区

The arousal range in which an individual performs best.

个人表现最佳的唤醒范围。

Formulas / 公式

A weightlifter may use high arousal, while a golfer may regulate down to support precision.

力量举者可能使用较高唤醒，高尔夫球手则可能调节到较低水平以支持精准。

Worked examples / 实例

>

Compare drive, inverted-U and catastrophe predictions.

比较驱力、倒U和灾难理论预测。

Exam focus / 考核要点

What is IZOF and why is it individual?

最佳功能区是什么？为什么因人而异？

Anxiety dimensions and catastrophe · 焦虑维度与灾难理论

First, picture it / 先这样想

"Nervous" is two different things wearing the same coat. One is a body state - a racing heart, sweating, shallow breath - and one is a thought process: a running list of what could go wrong, each item convincing. They look identical from outside and they need opposite treatment, because the body one is often helped by movement and the thought one is only helped by a different way of thinking about the uncertainty.

"紧张"是两件穿着同一件外套的事。一件是身体状态--心慌、出汗、呼吸浅；另一件是一个思维过程：一份关于可能出错之处的滚动清单，每一条都很有说服力。从外面看它们一模一样，而它们需要相反的处理，因为身体那件常常靠活动来改善，而思维那件只能靠用另一种方式去看待那份不确定性来改善。

The detail / 详细讲解

Cognitive anxiety and somatic anxiety separate on the way in, on the way through, and on the way out.

认知焦虑与躯体焦虑在输入时、在过程中、在输出时都是分开的。

An athlete can have high somatic and low cognitive - physically wound up and thinking clearly - or the reverse, preoccupied with failure while the body is quiet.

一位运动员可以躯体高而认知低--身体紧绷而思路清晰--或者反过来，满脑子失败而身体平静。

The prediction differences are real, and the most consistent finding is that cognitive anxiety is the better predictor of performance decrement.

预测上的差别是真实的，而最一致的发现是认知焦虑是表现下降更好的预测因子。

The practical consequence is that asking an athlete how they feel gets you somatic information whether you wanted it or not, and you have to ask a different question to get the other kind.

实际推论是：问一位运动员"你感觉怎么样"，无论你想不想要，拿到的都是躯体信息；

而要拿到另一种，你必须问一个不同的问题。

Catastrophe theory's contribution to this section is the part that makes it memorable and the part that is most often taught badly.

灾难理论在这一节中的贡献既是让人记得住的部分，也是最常被教错的部分。

The finding is that for complex tasks performance deteriorates as arousal rises, and past a critical point it collapses

- and the critical point is lower for harder tasks, for less skilled performers, and for tasks performed while being watched.

这个发现是：对复杂任务，表现随着唤醒上升而下降，越过一个临界点之后崩塌--而这个临界点对更困难的任务、对不够熟练的表演者

That is why a routine that works in training and fails in a competition is usually not about skill.

这就是为什么一套在训练中管用、在比赛中失效的例程，通常与技能无关。

It is about where the athlete is relative to that collapse threshold, and the competition moved the line.

它关乎这位运动员相对于那个崩塌阈值处在什么位置，而比赛把这条线移动了。

The second finding is about direction, and it is counter-intuitive: performance under pressure often deteriorates because the athlete is trying too hard.

第二个发现在方向上是反直觉的：压力下的表现常常因为运动员"太用力"而变差。

The attempt to control the uncontrollable - to get the perfect start, to make the perfect contact - consumes the attentional resource that the task itself requires, and what an observer sees as trying hardest is the athlete spending capacity on effort rather than on the task.

试图控制那些无法控制的东西--要一个完美的起动、要一次完美的触球--消耗掉了任务本身所需要的注意资源，而旁观者看到的"最拼"，

This is why a technique that works when relaxed can fail when desperate, and why "calm down" is rarely the instruction that helps.

这就是为什么一个放松时有效的技术在拼命时会失效，也为什么"冷静点"很少是有帮助的那条指令。

And the honest framing is that some anxiety is useful.

而诚实的表述是：一定程度的焦虑是有用的。

Performance is better with some arousal than with none, an athlete who reports zero concern before a final is often under-prepared rather than calm, and a small amount of cognitive anxiety sharpens attention for a genuinely important task.

表现好过完全没有唤醒，一位在决赛前报告"完全不担心"的运动员往往是准备不足而不是平静，而少量的认知焦虑会为一项真正重要的任务锐化注意。

The clinical target is a level the athlete can function at, not the absence of anxiety - and that reframe is what makes a plan sustainable, because a plan aimed at eliminating a normal response will eventually require the athlete to suppress it, which is both unpleasant and unreliable.

临床上的目标是一个运动员能够正常发挥的水平，而不是焦虑的消失--正是这个重构让一份计划变得可持续，因为一份以"消除一个正

The numbers / 算一算

Numbers to keep. The most useful routine in this area is a pre-event routine with a fixed length, because the variable that matters is the time spent in pre-event thinking, and shortening it is one of the few interventions with consistent support. The second figure is the arousal level at which an athlete's performance actually collapses, which is measurable in training by deliberately over-activating them in progressively harder tasks - and knowing where that line sits tells the coach how much headroom the athlete has on the day. The third is the proportion of a routine that is physical rather than cognitive, and the practical pattern is that the physical part can be shortened while the cognitive part should be brief and complete rather than long and open-ended.

需要记住的数字。这个领域里最实用的例程是一个长度固定的赛前例程，因为真正重要的变量是花在赛前思考上的时间，而缩短它是少数

Common mistakes / 易错点

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"They are nervous, so calm them down." Anxiety has two components and they respond differently. Somatic anxiety often helps from movement and familiarity; cognitive anxiety needs a different way of thinking about uncertainty.

"他们紧张，让他们冷静点。"焦虑有两个成分，反应也不同。躯体焦虑常常靠活动和熟悉感来改善；认知焦虑则需要用另一种方式

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"Zero nerves is the goal." Some arousal improves performance, and an athlete who reports no concern before a final is often under-prepared rather than calm.

"目标是完全不紧张。"一定程度的唤醒会改善表现，而一位在决赛前报告"毫不担心"的运动员往往是准备不足而不是平静。

-

"Under pressure they try too hard, so coach them to try harder." The problem is usually the opposite: effort spent on controlling the uncontrollable consumes the attention the task needs.

"压力下他们太拼了，那就让他们更拼。"问题通常恰恰相反：花在控制不可控之事上的努力，消耗掉了任务所需要的注意。

"A routine that works in training is not the issue." The collapse threshold is lower in front of an audience, on a harder task and for a less skilled performer, which is exactly why the same routine fails at a competition.

"在训练中管用的例程不是问题所在。"在被观众注视时、在更困难的任务上、对不够熟练的表演者，崩塌阈值都更低--这正是同一

Why it matters / 为什么重要

This section is where anxiety stops being a character flaw and becomes a measurable, adjustable, task-dependent phenomenon - which is both more accurate and more useful. The two things a student should take from it are the separation of cognitive from somatic anxiety, because it changes what the athlete should be asked, and the catastrophe finding, because it explains why the last increment of arousal is not a small thing. Together they replace the vague instruction to relax with a set of specific options, and they explain to an athlete whose technique has collapsed in front of a crowd that the problem is a threshold that moved rather than a technique that failed.

这一节让焦虑不再是性格缺陷，而成为一个可测量、可调整、依赖任务的现象--这既更准确也更有用。学生应当从中带走的两件事是：认

Somatic anxiety is not the same as cognitive anxiety: a high heart rate may be useful for a sprint but worry can harm precision.

躯体焦虑不同于认知焦虑：心率升高可能有助于冲刺，但担忧可能损害精准。

Key terms / 核心术语

Cognitive anxiety 认知焦虑

Worry, negative thoughts and self-doubt.

担忧、负面想法和自我怀疑。

Catastrophe 灾难性崩溃

A sudden performance collapse when cognitive anxiety is high.

认知焦虑高时的表现突然崩溃。

Formulas / 公式

A 100 m sprinter can use high somatic arousal but should reduce cognitive worry to avoid the catastrophic drop.

100米冲刺者可以利用较高躯体唤醒，但应减少认知担忧以避免灾难性下降。

Worked examples / 实例

>

How do cognitive and somatic anxiety differ?

认知焦虑和躯体焦虑有何不同？

Exam focus / 考核要点

What happens in catastrophe theory?

灾难理论认为会发生什么？

Measurement and regulation · 测量与调节

First, picture it / 先这样想

You cannot adjust a dial you have not looked at. Anxiety in sport is among the most self-reported things there is, and self-report is cheap, immediate and shaped by the moment - which means an athlete asked on the morning of a final gives you their morning, not their pattern. The practical work is to make the dial readable over weeks, and then to teach the athlete to read it themselves.

你无法调整一个你从没看过的旋钮。运动中的焦虑是最依赖自我报告的东西之一，而自我报告便宜、即时、并且被当下塑造--这意味着一位在决赛当天早上被问到的运动员，给出的是他那天早上的状态，而不是他的模式。实际工作是把旋钮变得在数周尺度上可读，然后教会运动员自己读它。

The detail / 详细讲解

The measurement options are short scales, interviews, observation and physiological measures, and they answer different questions.

测量选项包括短量表、访谈、观察和生理测量，而它们回答的是不同的问题。

Self-report scales are the workhorse because they are quick and because the athlete can fill them in daily without a coach present.

自评量表是主力，因为它们快，而且运动员可以在没有教练在场的情况下每天填写。

Observation by a coach is valuable precisely because the athlete cannot game it.

教练的观察之所以有价值，恰恰因为运动员无法对它作假。

Physiological measures are objective and available, but in sport they are often confounded - a raised heart rate might be anxiety, or it might be the third repetition of a sprint.

生理测量是客观的、可获得的，但在运动中它们常常被混淆--心率升高可能是焦虑，也可能是冲刺的第三次重复。

The critical habit is to measure across time rather than at the moment.

关键习惯是在时间跨度上测量而不是在当下测量。

A single administration on a competition day tells you about that day; a series tells you the athlete's baseline, which is the only thing against which the day can be interpreted.

在比赛当天测一次告诉你关于那一天的事；

And the baseline varies by task and by individual, which is why comparing two athletes' scores is nearly useless while comparing one athlete's score against their own past is informative.

一系列测量告诉你这位运动员的基线，而基线是唯一可以用来解读那一天的参照。

基线还会随任务和随个体变化，所以比较两位运动员的分数几乎没有用处，而把一位运动员的分数与他自己过去的分数相比才有信息。

Regulation has more tools than most coaches use, and they fall into three groups.

调节的工具比多数教练用到的要多，而它们可归为三组。

Reduce the input: pre-event thinking time, warm-up familiarity, the amount of uncertainty in the warm-up itself.

减少输入：赛前思考时间、热身的熟悉度、热身本身的不确定性。

Raise the floor: breathing with a long exhale, muscle tension and release, a physical routine that gives attention somewhere to be.

抬高底线：带长呼气的呼吸、肌肉的紧张与放松、以及给注意一个去处的身体例程。

And change the interpretation: reframing the meaning of the situation, which is the only one of the three that touches cognitive anxiety, and the one most often left out on the grounds that a young athlete cannot be talked round.

而改变解读：重新界定这个情境的含义--这三组中唯一触及认知焦虑的那一组，也是最常被略去的那一组，理由是“年轻运动员讲不通”。

And the sequencing matters as much as the choice.

而顺序与选择本身同样重要。

Input and floor work can be put in place in a week and are what most anxious athletes need first, because they reduce the amount of distress to be interpreted.

输入与底线的做法一周内就能建立，也是多数焦虑型运动员首先需要的，因为它们减少了需要被解读的痛苦量。

The cognitive work takes longer, is less reliably effective, and is worth starting before a competition rather than during a build-up to one.

认知层面的工作花时间更长、效果没那么可靠，而且值得在比赛之前就开始，而不是在备战的最后阶段才开始。

Which is also the honest reason to build a routine early: a routine assembled the week of a competition is a routine the athlete has never once rehearsed under the pressure it was made for.

这也是要把例程提前建好的诚实理由：在比赛那一周临时凑出来的例程，是一个运动员从未在它本来要应对的压力下演练过的例程。

The numbers / 算一算

Numbers to keep. The most useful numbers here are frequency and length rather than scores. Daily measurement is only worth doing if it stays under about two minutes, because a measurement that takes longer will not be done, and the whole value of the baseline comes from consistency rather than from depth. The second figure is the length of the pre-event routine, which in the most consistent work on this is measured in minutes rather than described qualitatively. The third is the proportion of sessions in which a routine was actually completed as intended, which is the figure that distinguishes a plan that exists from a plan that runs.

需要记住的数字。这里最有用的数字是频率和长度，而不是分数。每日测量只有在保持在约两分钟以内时才值得做，因为更长的测量不会

Common mistakes / 易错点

- "Ask him how nervous he is." That gives you somatic information and a snapshot. A pattern across weeks is what tells you how to design a routine.

"问问他有多紧张。"那给你的是躯体信息和一张快照。数周尺度上的模式才告诉你该怎么设计一套例程。

- "Heart rate is anxiety." A raised heart rate in sport is often exertion, and treating it as anxiety leads to intervening with the wrong tool.

"心率就是焦虑。"运动中的心率升高常常是运动负荷所致，而把它当作焦虑会导致用错工具。

- "A competition day score is the anxiety measure." It is a reading of one day. The measure is the athlete's own baseline, which is the only reference that makes the day interpretable.

"比赛当天的分数就是焦虑的测量。"那是对一天的读数。测量是这位运动员自己的基线，而基线是唯一能让那天变得可解读的参照。

- "Breathe deeply to control nerves." Long exhale is the effective half. Deep slow breathing that emphasises the in-breath can increase arousal, and a routine built on it will not do what it is meant to.

"深呼吸来控制紧张。"有效的是长呼气那一半。强调吸气的深慢呼吸会增加唤醒，而建立在它之上的例程不会达到它应有的效果。

Why it matters / 为什么重要

This section is the most operational in the psychology theme, and it is the one a coach can complete properly in a fortnight. The core idea is unglamorous: build a baseline, read the dial, and choose from a menu of tools rather than reaching for reassurance. The reason it is worth the time is that most anxiety problems in sport are not severe enough to need clinical attention but are chronic enough to cost an athlete a career, and they are addressed almost entirely by things a coach does daily. The most valuable single habit is also the cheapest: a short daily rating, kept for weeks, which turns a vague impression of "he is anxious" into a pattern that can be designed for.

这一节是心理学主题中最具操作性的一节，也是教练两周之内就能把它正确做完的一节。核心想法并不光鲜：建立基线、读旋钮、然后设计。

- Routines, breathing, self-talk and imagery can help regulate arousal.

流程、呼吸、自我对话和意象可以帮助调节唤醒。

Key terms / 核心术语

Galvanic skin response 皮电反应

Skin sweat-gland activity used as a physiological indicator.

皮肤汗腺活动，可作为生理指标。

Formulas / 公式

Before a penalty, use breathing and a short routine, then ask the athlete about worry and confidence as well as heart rate.

罚球前使用呼吸和简短流程，同时询问担忧、自信和心率。

Worked examples / 实例

>

Name three objective anxiety measures.

说出三种客观焦虑指标。

Exam focus / 考核要点

Why use both subjective and objective measures?

为什么同时使用主观和客观指标？

Stress and coping

压力与应对

Guiding question / 引导问题

How should athletes match coping strategies to different stressors?

运动员如何将应对策略与不同压力源匹配？

A stressor creates demand that taxes psychological or physiological resources. Strain can be positive eustress or negative distress. Coping can address the stressor, manage the response or involve avoidance, and effectiveness depends on controllability, person and situation.

压力源产生需求，消耗心理或生理资源。压力反应可以是积极的亢奋，也可以是消极困扰。应对可以处理压力源、管理反应或进行回避。

Stressors and strain · 压力源与压力反应

First, picture it / 先这样想

A stressor is an event and strain is the cost of handling it, and the useful thing about the distinction is that two athletes can face the same event and pay different prices. The demand is a property of the situation. The cost is a property of what the athlete brought with them - their coping, their support, their sleep, their history with this exact kind of event. Confusing the two produces the very common advice to reduce the stressor when the athlete needed resources.

压力源是事件，应激是处理它的代价，而这一区分的有用之处在于：两位运动员可以面对同一个事件、付出不同的代价。需求是情境的属性，代价是运动员带进来的东西的属性--他们的应对方式、他们的支持、他们的睡眠、他们与这一类事件的过往经历。混淆这两者，就会产出那种非常常见的建议：去减少压力源，而这位运动员真正需要的是资源。

The detail / 详细讲解

The distinction is between demand and resource, and the working model is that strain appears when demand exceeds the resources available to meet it.

这个区分就是需求与资源之间的区分，而可用的模型是：当需求超过可用于应对它的资源时，应激就出现了。

That is why the same injury, the same exam or the same selection decision strains some athletes badly and barely touches others.

这就是为什么同一次受伤、同一次考试或同一个选拔决定，会让一些运动员承受很大代价、而几乎不影响另一些人。

It is also why adding resources can work without changing the stressor, and why reducing the stressor can fail when the athlete's resources were never adequate for the demand in the first place.

这也就是为什么增加资源可以在不改变压力源的情况下起作用，而当运动员的资源本来就不足以应对需求时，减少压力源会失败。

The stressors themselves are worth listing because they are not all the same kind of thing.

把压力源本身列出来是有用的，因为它们不是同一种东西。

Sport-specific ones are things like selection, injury, competition, and judgement by officials.

运动专项的有选拔、受伤、比赛、以及被裁判判罚；

Non-sport-specific ones are job, family, finances, health, and travel, and the second list is where most of the strain usually comes from.

非专项的有工作、家庭、经济、健康和出行，而通常大部分应激来自后面这张清单。

And the stressors interact: an athlete with a job change in the same month as pre-season is under a combined demand that no single stressor accounts for.

而且压力源之间会相互作用：一位在备赛月同时遇上换工作的运动员，承担的是一个没有任何单一压力源能解释的综合需求。

Appraisal is the step that connects the two lists.

评价是把两张清单连起来的那一步。

An event becomes stressful when the athlete appraises it as threatening to something they value, as exceeding their coping resources, or as something they cannot control - and each appraisal points at a different response.

当运动员把一个事件评价为“威胁到我看重的东西”、“超出我的应对资源”或“我无法控制”时，它才成为压力源--而每一种评价都指向不同的应对。

Threat appraisal points at resources, "I cannot cope" points at skills or support, and uncontrollability appraisal is the one most associated with the worst outcomes, because an event that cannot be influenced can still be worried about indefinitely.

威胁评价指向资源，“我应付不了”指向技能或支持，而“无法控制”这一种与最差结果关联最紧密，因为一个无法影响的事件仍然可以被

The practical consequence for a coach is that the assessment should begin with the athlete rather than with the calendar.

对教练的实际推论是：评估应当从运动员开始，而不是从赛历开始。

What is actually going on, when did it start, what has changed, what has been lost, who is available, and how are they sleeping and eating.

实际在发生什么、什么时候开始的、变了什么、失去了什么、谁是可以依靠的、以及他睡眠和进食如何。

Most cases of unexplained under-performance in sport resolve into a stress or strain problem that was never asked about, and the cost of asking is a conversation while the cost of not asking is a season.

运动中大多数无法解释的状态下滑，最终都会归结为一个从未被问起的压力或应激问题；

而问的代价是一次谈话，不问的代价是一个赛季。

The numbers / 算一算

Numbers to keep. The practical numbers here are all in the assessment rather than the theory. How long since the change. How many of the non-sport stressors are present at once, which is the count that most reliably predicts strain. How many people the athlete can actually talk to, which in elite sport is routinely smaller than the number of people they are surrounded by. And sleep, which is the cheapest single indicator available to a coach and the one most often abnormal in exactly the situations where the athlete is reporting no problem.

需要记住的数字。这里实用的数字全都在评估里而不在理论里。变化发生至今多久；非运动类压力源同时存在几个--这是最能可靠预测应

Common mistakes / 易错点

- "The stressor is the problem, so reduce the stressor." Strain is demand minus resources. Sometimes the stressor cannot be reduced and the resources can be found.
"压力源就是问题所在，所以去减少压力源。"应激是需求减去资源。有时压力源无法减少，而资源是可以找到的。
- "If he has not mentioned it, it is not bothering him." Strain is frequently unreported, particularly in athletes who have learned that reporting a problem is a weakness.
"如果他没提，那就不是困扰他。"应激常常不被报告，尤其是在那些学会了“报告问题是一种软弱”的运动员身上。
- "Only sport counts." Most of the strain usually comes from outside sport, and an athlete whose job or family situation has deteriorated is being asked to train on top of it.
"只算运动内的。"大部分应激通常来自运动之外，而一位工作或家庭状况恶化的运动员是被要求在这一切之上继续训练的。
- "Telling them not to worry will settle it." An event appraised as uncontrollable can be worried about indefinitely, and instruction to stop worrying does not change its controllability.
"告诉他们别担心就解决了。"一个被评价为无法控制的事件可以被无限期地担忧，而“别担心”这条指令并不改变它的可控性。

Why it matters / 为什么重要

Stress is the section that connects everything else in the theme to the rest of an athlete's life, and it is the one most often treated as either a character issue or an excuse. The demand-and-resource framing removes both of those readings: strain is a measurable relationship, it varies between people for identifiable reasons, and it can be altered from either side. For a coach the practical value is diagnostic - most under-performance that has no technical explanation is here - and for a student it is the section that makes the case that mental skills are part of performance preparation rather than an optional extra.

压力是这一节把主题中其他所有内容 with 运动员其余生活连起来的地方，而它最常被当作性格问题或者借口来对待。需求与资源的框架去

Positive strain can energise focus and performance when manageable; negative strain can reduce concentration and confidence and contribute to burnout or injury risk.

可管理的积极压力可以激活注意和表现；消极压力会降低注意和自信，并可能导致倦怠或受伤风险。

Key terms / 核心术语

Stressor 压力源

A demand or challenge that taxes resources.

消耗资源的需要或挑战。

Distress 消极困扰

Negative strain that overwhelms or threatens.

压倒或威胁性的消极压力反应。

Formulas / 公式

A final can be eustress when the athlete sees it as a manageable challenge and has resources; it can become distress if demands exceed perceived control.

运动员把决赛看作可管理的挑战且拥有资源时，压力可能具有激励作用；当需求超过知觉控制时，可能转为消极困扰。

Worked examples / 实例

>

Define stressor and strain.

定义压力源和压力反应。

Exam focus / 考核要点

Differentiate eustress and distress.

区分积极压力和消极困扰。

Three coping categories · 三类应对方式

First, picture it / 先这样想

Coping is not a personality type, it is what you do with a moment, and the moment matters more than the person. The same athlete copes well with one thing and badly with another, and often the difference is not in their character but in whether they have ever had to do this particular thing before.

应对不是一种人格类型，而是你在某一个时刻所做的事，而这个时刻比这个人更重要。同一位运动员对一件事应对得好、对另一件事应对得差，而这种差别往往不在他的性格里，而在于他以前是否遇到过这一类事情。

The detail / 详细讲解

The three categories are problem-focused, emotion-focused and avoidant.

三个类别是问题聚焦、情绪聚焦和回避型。

Problem-focused coping aims to change the situation: seek information, plan, take the direct action, solve it.

问题聚焦应对旨在改变情境：寻找信息、做计划、采取直接行动、把它解决。

Emotion-focused coping aims to manage the feeling: accept what cannot be changed, adjust expectations, find meaning in it, regulate the emotion.

情绪聚焦应对旨在管理感受：接受无法改变的、调整预期、从中找到意义、调节情绪。

Avoidant coping aims to move away from the feeling rather than handle either the problem or the emotion - denying, withdrawing, drinking, or simply not thinking about it.

回避型应对旨在从感受上移开，而不是处理问题或情绪--否认、退缩、饮酒，或者干脆不去想它。

The first two are generally adaptive; the third is generally not, and it is frequently the most pleasant in the moment.

前两种通常是有适应性的；第三种通常不是，而且它经常是当下最舒服的一种。

Which one is used depends on whether the stressor is perceived as changeable.

使用哪一种，取决于压力源被感知为可改变还是不可改变。

If it is, problem-focused coping is available and useful.

如果可改变，问题聚焦应对就是可用的、有用的。

If it is not, the only options are emotion-focused or avoidant - and here the distinction matters enormously, because emotion-focused coping processes the feeling while avoidant coping removes the athlete from it, and one of those is a skill and the other is a way of ensuring the problem and the feeling both stay.

如果不可改变，那么选择只有情绪聚焦或回避型--而这里的区分极其重要，因为情绪聚焦应对处理那种感受，而回避型应对把运动员从感受中移开，其中一个技能，另一个是确保问题和感受都留下来的办法。

The flexibility point is the one coaches most often get wrong, and it is the key to using the framework well.

灵活性这一点是教练最常搞错的，而它也是用好这个框架的关键。

The strategy that worked last time is not automatically the strategy that works this time, because the controllability of the situation may have changed while the athlete's habit has not.

上一次有效的策略，不会自动成为这一次有效的策略，因为情境的可控性可能已经变了，而运动员的习惯没有变。

An athlete who solved a previous setback by fighting hard will often try to fight a situation that cannot be fought, and the coach who recognises the change can redirect them to grief, acceptance or re-planning instead - which feels like giving up and is actually the competent response.

一位曾经靠“硬拼”解决过挫折的运动员，往往会去硬拼一个根本拼不了的情境；

而看出这一变化的教练，可以把他们转向哀伤、接纳或重新规划--这感觉像放弃，而实际上是胜任的回应。

And the practical coaching task is mostly about noticing which one is in use, because athletes rarely label it themselves.

而教练的实际任务主要是辨认正在使用的是哪一种，因为运动员很少会自己给它贴标签。

The signs differ: problem-focused coping is visible in preparation and planning, emotion-focused in talking about and accepting the situation, and avoidant in the absence of both - a flatness, a refusal to discuss, a sudden improvement that nobody believes.

它们的迹象不同：问题聚焦可见于准备和计划，情绪聚焦可见于谈论与接纳，而回避型则表现为两者都缺席--一种平淡、一种拒绝讨论。

The avoidant presentation is the one that is easiest to miss and most expensive to miss, because the athlete looks fine and the problem continues.

回避型的表现最容易漏掉、而漏掉的代价最高，因为这位运动员看起来没问题，而问题仍在继续。

The numbers / 算一算

Numbers to keep. The useful measurement is a short scale covering the three categories, administered weekly rather than after a crisis, and the figure that matters is the balance between them over time rather than any single score. A second practical habit is to record what the athlete actually did in a difficult week - the observable coping behaviour is more reliable than what they report about it. And the third is the clearest single indicator of avoidant coping: a sudden improvement in mood or performance with no change in circumstances, which almost always means something is being avoided rather than resolved.

需要记住的数字。有用的测量是一份涵盖三个类别的短量表，按周施测而不是在危机之后，而重要的数字是它们随时间的平衡，而不是任何单一分数。

Common mistakes / 易错点

- "Coping style is a personality trait." It is what a person does in a moment, and the same person does different things with different stressors.
"应对风格是一种人格特质。"它是一个人在某个时刻所做的事，而同一个人对不同压力源会做不同的事。
- "If you cannot change it, do not worry about it." If it cannot be changed, the options are emotion-focused coping or avoidance. Doing nothing about the feeling is the second, not the first.
"如果改变不了，就别为它担心。"如果改变不了，选择就是情绪聚焦应对或回避。对感受不做处理是后者，不是前者。
- "Use what worked last time." The controllability of the situation may have changed while the habit did not, and the coach who notices the change can redirect the strategy.
"用上次有效的办法。"情境的可控性可能已经变了而习惯没有变，而注意到这一变化的教练可以把策略转向别处。
- "They seem fine now, so it is resolved." A sudden improvement with no change in circumstances is the clearest sign of avoidance, and it is the presentation most often missed.

"他们现在看起来不错，所以已经解决了。"在情境没有变化的情况下突然好转，是回避最清晰的迹象，也是最常被漏掉的表现。

Why it matters / 为什么重要

The three categories are worth teaching because they convert "how are you coping" - a question that invites reassurance - into three observable patterns with different responses. That is the practical value: an athlete avoiding and an athlete processing grief from the same setback need opposite things from a coach, and they present similarly enough that the difference is easy to miss. The framework also gives a coach a licence to be flexible, which matters because the strategy that built an athlete's resilience in one situation can be the wrong one in the next, and the alternative to flexibility is a coach who keeps prescribing the same response and reads the athlete's flatness as ingratitude.

这三个类别值得教，是因为它们把"你是怎么应对的"--一个只会引出安慰的问题--转换成了三种可观察的模式，而每一种需要不同的回应。

-

No single strategy works for all stressors; assess controllability and combine problem- and emotion-focused strategies when needed.

没有一种策略适合所有压力源；应评估可控性，必要时结合问题聚焦和情绪聚焦策略。

Key terms / 核心术语

Problem-focused coping 问题聚焦应对

Changing the stressor or situation.

改变压力源或情境。

Emotion-focused coping 情绪聚焦应对

Managing the emotional and physiological response.

管理情绪和生理反应。

Formulas / 公式

For a controllable communication problem, plan a conversation; for weather or an opponent's ability, use acceptance, routine and self-talk.

可控的沟通问题可安排谈话；天气或对手能力等不可控因素可使用接纳、流程和自我对话。

Worked examples / 实例

>

Which strategies lower heart rate and reduce arousal?

哪些策略可以降低心率和唤醒？

Exam focus / 考核要点

Name the three coping categories and examples.

说出三类应对及例子。

Decision flow and maladaptive responses · 决策流程与适应不良反应

First, picture it / 先这样想

There is a shape to how a bad moment turns into a bad month, and it is worth learning because most of it is avoidable. The path usually runs: something happens, the athlete appraises it, the appraisal produces a feeling, the feeling produces an impulse, the impulse produces a behaviour, and the behaviour either solves the problem or makes it bigger. Every maladaptive response is a short-circuit somewhere on that path, and each one has a different short-circuit.

一个糟糕的时刻如何变成糟糕的一个月，是有形状的，而值得学会它是因为其中大部分是可以避免的。这条路径通常是：发生了某件事、运动员对它做出评价、评价产生感受、感受产生冲动、冲动产生行为，而行为要么解决了问题、要么让问题更大。每一种适应不良反应都是这条路径上某处的一次短路，而每一种的短路位置都不同。

The detail / 详细讲解

The adaptive route runs the whole path and finishes with a behaviour that addresses the problem.

适应性路径走完整条流程，并以一个针对问题的行为结束。

The maladaptive routes branch off early.

适应不良的路径会在早期分叉。

One branch avoids the feeling, which prevents the appraisal from being worked through and leaves the problem where it was.

一条分支回避感受，这使那个评价无法被处理，于是问题留在原地。

One branch avoids the problem, which is usually subtler and more expensive: the athlete solves the wrong problem, or solves a version of the problem that is easier.

另一条分支回避问题，这通常更隐蔽、代价也更高：运动员解决了错误的问题，或者解决了一个更容易的版本的问题。

A third branch substitutes a different activity entirely, and it is the branch most easily mistaken for a coping strategy.

第三条分支则用一个完全不同的活动替代它，而这条分支最容易被误认为是一种应对策略。

The clinical categories are worth naming because they are the vocabulary a coach will need when the situation is beyond ordinary stress.

这些临床类别值得点名，因为它们是教练在情况超出普通压力时需要的词汇。

Generalised anxiety is persistent and unfocused worry across domains.

广泛性焦虑是跨领域的、持续而不聚焦的担忧。

Specific phobia is an intense response to one object or situation with avoidance of it.

特定恐惧是对某一个对象或情境的强烈反应并回避它。

Panic is a sudden surge with a strong bodily component.

惊恐发作是突然涌起并带有强烈躯体成分的。

Social anxiety is fear of being observed and judged.

社交焦虑是对被观察和被评判的恐惧。

Each has a distinct signature, and the useful coaching response is mostly to recognise which one is present, because the referral threshold differs for each.

每一种都有独特的特征，而有用的教练反应主要是辨认出存在的是哪一种，因为每一种的转介阈值是不同的。

The maladaptive responses most worth knowing about in sport are the ones that look like dedication.

运动中最值得知道的适应不良反应，是那些看起来像投入的那些。

Training through injury because resting feels like letting the team down is avoidant emotion-focused coping with a moral story attached.

因为"休息感觉像在让团队失望"而带伤训练，是带着一个道德故事外衣的回避型情绪聚焦应对。

Continuing to train an athlete who is not sleeping is problem-focused coping that solved the wrong problem - the behaviour addresses the coach's need, not the athlete's.

继续训练一位没有睡眠的运动员，是解决了错误问题的问题聚焦应对--这个行为满足的是教练的需求，而不是运动员的需求。

And an athlete who becomes unshakeably upbeat about a serious injury has often made the same trade in the other direction.

而一位对严重损伤变得坚定不移地乐观的运动员，往往在另一个方向上做了同样的交易。

And the decision flow has one more feature that is worth drawing out, which is that early decisions constrain later ones more than they seem to.

而这条决策流程还有一个值得点出的特征：早期的决定对后续的约束比看上去更大。

The first thing an athlete does after a setback determines what information they then have access to, and the most common maladaptive sequence is: avoid the feeling, therefore avoid the information, therefore confirm the belief that there was nothing to know, therefore avoid again.

挫折之后运动员做的第一件事，决定了他之后能接触到哪些信息；

It is self-sealing, and the only way in is usually a change made before the athlete is asked to feel anything.

而最常见的适应不良序列是：回避感受，因此回避信息，因此确认了"本来就没有什么可知道"这个信念，因此继续回避。

它是自我封闭的，而打破它的入口通常是在要求运动员"先感受一下"之前所做的某个改变。

The numbers / 算一算

Numbers to keep. Two figures carry most of the practical weight. The first is how long the response lasts: a reaction that resolves within days is stress, one that persists for weeks is a problem that needs a different approach, and the transition is roughly the point at which an athlete stops talking about it and starts avoiding it. The second is the number of domains the worry appears in - worry confined to sport is one thing, and worry appearing

across training, sleep, relationships and mood at once is the pattern that most reliably predicts a clinical threshold.

需要记住的数字。有两个数字承担了大部分实际分量。第一个是反应的持续时间：在数天内缓解的是压力，持续数周的是需要用不同方式

Common mistakes / 易错点

- "They are training through it, so they are coping well." Training through injury, sleep deprivation or an unresolved loss is often avoidance with a story attached, and it is usually more expensive than it looks.
"他们带着这个在训练，所以应对得很好。"带着损伤、睡眠不足或未解决的失落继续训练，往往是带了一个说法的回避，而且代价更高。
-
"Stay positive and it will pass." The unshakeably upbeat athlete has often made the same trade in the other direction, and the appearance of wellbeing is the thing that delays the help.
"保持积极就会过去。"坚定不移地乐观的运动员往往在另一个方向上做了同样的交易，而看起来良好的状态正是延误求助的原因。
-
"The worry is about the competition." Check the domains. Worry appearing across training, sleep, relationships and mood is a different problem from worry about one event.
"这份担忧是关于比赛的。"检查一下领域。在训练、睡眠、人际关系和情绪上同时出现的担忧，是与"担心某一场比赛"不同的问题。
-
"Once the mood improves, the issue is resolved." A sudden improvement with no change in circumstances is the classic signature of avoidance rather than resolution.
"情绪一好转，问题就解决了。"在情境没有变化的情况下突然好转，是回避而不是解决的经典标志。

Why it matters / 为什么重要

This section is the one that tells a coach when to stop coaching and refer, which makes it the most responsible content in the theme. The categories matter less as labels than as thresholds: recognising that a presentation is beyond what a session plan can address is a professional judgement, and making it late costs an athlete months. The second thing worth taking from it is the sequence - avoid, therefore do not learn, therefore believe there was nothing to learn - because that sequence is the reason the first conversation is worth having early and gently, before there is anything to be gained from not having it.

这一节告诉教练什么时候该停止教练工作、去做转介，因此它是这个主题中最有责任心的内容。这些类别作为标签的意义，小于它们作为

- Self-talk should be short, positive, present-tense and realistic, such as "one point at a time" or "I control my effort."
自我对话应简短、积极、现在时且现实，例如"一次一分"或"我能控制自己的努力"。

Key terms / 核心术语

Avoidance 回避

Escaping the stressor rather than addressing or managing it.

逃避而非处理或管理压力源。

Formulas / 公式

Practise coping in training, review what worked and build a personal toolkit.

在训练中练习应对，复习有效方法并建立个人工具箱。

Worked examples / 实例

- >
How do you match coping to controllable versus uncontrollable stressors?
如何根据可控和不可控压力源匹配应对？

Exam focus / 考核要点

Name four maladaptive coping strategies.

说出四种适应不良应对策略。

Goal setting

目标设定

Guiding question / 引导问题

How can goal choice and adjustment direct attention and motivation?

目标选择和调整如何引导注意与动机？

Goals direct attention, organise effort, provide standards, enable evaluation and influence persistence. IB SEHS groups goals as outcome, performance and process goals. Goals should be adjusted when performance, injury, time, competition or environment changes.

目标引导注意、组织努力、提供标准、支持评价并影响坚持。IB

SEHS将目标分为结果、表现和过程目标。表现、伤病、时间、比赛或环境变化时，目标应调整。

Three goal types · 三种目标类型

First, picture it / 先这样想

A goal is a destination, and there are three different kinds of destination in sport, and they are not equally good for the person. The mistake is treating "has a goal" as the goal. An athlete with a clear destination can be heading somewhere that quietly makes them worse to be, and a coach who has only taught them to set goals has not asked where.

目标是目的地，而运动中有三种不同的目的地，而它们对这个人并不是同样好的。错误在于把“有目标”本身当成了目标。一位有着清晰目的地的运动员，可能正在前往一个会悄悄让他变得更不好相处的地方；而一位只教过他们设定目标的教练，从未问过那是哪里。

The detail / 详细讲解

The three types are performance, learning and health.

三种类型是表现目标、学习目标和健康目标。

Performance goals are about doing better than a standard or than others - a personal best, a placing, a selection.

表现目标是关于做得比某个标准或比别人更好--个人最好成绩、一个名次、一次入选。

Learning goals are about improving a specific skill or understanding something.

学习目标是关于改进某项具体技术或理解某件事。

Health goals are about the body staying healthy and the career lasting.

健康目标是关于身体保持健康、职业生涯得以延续。

All three are legitimate, all three can be set well, and the difference between them is what happens to the athlete when the goal is achieved - because every goal has a life beyond the moment it is met.

三者都正当、都可以设定好，而它们之间的差别在于目标达成之后这位运动员会怎样--因为每一个目标都有一个超出“被达成那一刻”的

A performance goal that is met produces a new question: what now?

一个被达成的表现目标会产生一个新问题：接下来怎么办？

The athlete who wanted a personal best and got it needs a new standard, and if the coach has not provided one, the athlete either drifts or finds one that is harder and more dangerous.

一位想要个人最好成绩并得到了它的运动员需要一个新的标准；

A learning goal that is met produces a natural progression, because there is always another level of the skill.

而如果教练没有提供，这位运动员要么漂移，要么自己找一个更难、更危险的标准。

A health goal that is met produces permission, which is the only one of the three that reliably produces a rest rather than an increase.

一个被达成的学习目标会产生自然的进阶，因为这项技术总有下一个层级。

而一个被达成的健康目标会产生许可，这是三者中唯一可靠地产生“休息”而不是“加量”的。

The evidence base favours process goals and mastery goals, and the reason is mechanical rather than moral.

证据基础更支持过程目标与掌握目标，而原因是机制性的而不是道德性的。

A process goal - "get the elbow in", "hit the first target" - is controllable, which means success is available on most days, which means the athlete experiences competence.

一个过程目标

--"把肘抬到位""先打到第一个靶"--是可控的，这意味着成功在大多数日子里都是可得的，这意味着这位运动员会体验到胜任。

An outcome goal is controllable only partly, which means most days contain information about the athlete rather than about the performance.

而一个结果目标只被部分可控，这意味着大多数日子提供的是关于这位运动员的信息，而不是关于这份表现的信息。

And a person who is receiving daily information about their own inadequacy will find a way to avoid the sport, whatever the goal says.

而一个每天都在接收关于自己不足的信息的人，会找到某种方式远离这项运动，不管目标是怎么写的。

The practical design is not to abandon outcome goals - a medal is a perfectly good reason to get up in the morning - but to pair them.

实际的设计不是放弃结果目标--一块奖牌完全是一个值得早晨起床的理由--而是把它们配对。

The pairing is: an outcome goal to give the season a direction, a process or learning goal to give each session a reason, and a health goal to give the athlete a permission that no sporting result can revoke.

配对方式是：一个结果目标

给整个赛季一个方向，一个过程或学习目标给每一堂训练课一个理由，一个健康目标给这位运动员一份任何比赛结果都无法撤销的许可。

Athletes who hold all three are more robust to a bad result precisely because the result was never the only thing in the frame.

同时持有三者的运动员，对一次糟糕结果的韧性更强，恰恰因为结果从来不是画面里唯一的东西。

The numbers / 算一算

Numbers to keep. The useful measure is the proportion of a session's goals that are process or learning rather than outcome, and for most teams it is lower than they would guess. The second figure is the time between an outcome goal being met and the next goal being set, which is the window in which drift happens and which is easiest to control simply by pre-agreeing what comes next. The third is the athlete's own ranking of their three goals when asked, which is usually different from what the coach assumed, and asking is cheap.

需要记住的数字。有用的度量是一堂训练课的目标中过程或学习类相对于结果类的比例，而对多数团队来说，它比他们自己猜的要低。

Common mistakes / 易错点

-

"Just set a goal." What kind matters as much as having one, and an outcome goal alone supplies motivation only while it is achievable.

"设个目标就行。"目标的类型与是否设有目标同样重要，而一个单独的结果目标只在其可达成期间提供动力。

-

"Process goals are less ambitious." They are more specific. A process goal is a controllable target for a specific moment, which is exactly what a session needs.

"过程目标不够有野心。"它们更具体。过程目标是针对某一具体时刻的可控靶标，而这正是训练课所需要的。

-

"Once they have won, motivation will be higher." It may be lower. The goal has been met, and without a next one the athlete either drifts or escalates to something riskier.

"一旦赢了，动力就会更高。"可能更低。目标已经达成，而没有下一个目标，这位运动员要么漂移、要么升级到更冒险的东西。

-

"Health goals slow athletes down." A health goal is the only one of the three that produces permission rather than demand, which is what makes it protective rather than limiting.

"健康目标会让运动员变慢。"健康目标是三者中唯一产生许可而不是要求的，这正是它具有保护性而不是限制性的原因。

Why it matters / 为什么重要

Goal setting is the most taught and least examined technique in applied sport psychology, and this section is where the examination happens. The value is in the asymmetry: a met outcome goal is a moment of satisfaction followed by an empty frame, and most of the "lost motivation" coaches complain about is actually the arithmetic of that moment rather than a change in the athlete. A coach who pairs an outcome goal with a process goal and a health goal has solved most of it in advance, and who can name which of the three is currently doing the work has a much better handle on a difficult conversation.

目标设定是应用运动心理学里教得最多、而最少被审视的技术，而这一节就是那个审视发生的地方。它的价值在于那个不对称：一个被达

Process goals build habits, regulate arousal and reduce worry because the athlete can directly control them.
过程目标因为可控，可以建立习惯、调节唤醒并减少担忧。

Key terms / 核心术语

Outcome goal 结果目标

A result compared with others or a required standard.
与他人或要求标准比较的结果。

Process goal 过程目标

A controllable technique or strategy target.
可控的技术或策略目标。

Formulas / 公式

Running: finish in the top three; run 5 km under 18:30; maintain cadence and relaxed shoulders.
跑步：进入前三；5公里用时低于18:30；保持步频并放松肩膀。

Worked examples / 实例

>
Convert an outcome goal into performance and process goals.
把结果目标转换为表现目标和过程目标。

Exam focus / 考核要点

What is the key difference between outcome and performance goals?
结果目标和表现目标的关键区别是什么？

Goal purpose and paradox · 目标的作用与悖论

First, picture it / 先这样想

A goal has a purpose beyond the thing it names, and a coach who only hears the thing named will build the wrong programme. Ask a sprinter why they want a season best and the answer may be about love of the sport; ask why they need a season best and the answer is usually about belonging. Both are real, and they lead to opposite plans.

一个目标除了它所指的那件事之外还有一个目的，而一位只听到所指之事的教练会构建错误的方案。问一位短跑运动员为什么想要一个赛季最好成绩，答案可能是关于对这项运动的热爱；问他为什么“需要”一个赛季最好成绩，答案通常关于归属感。两者都是真的，而它们导向相反的计划。

The detail / 详细讲解

Purpose and content are different questions.
目的与内容是两个不同的问题。

Content is what the goal says - run 11 seconds, win the district final, return to full training.
内容是目标所说的东西--跑进11秒、赢下地区决赛、完全恢复训练。

Purpose is what the goal is for - to feel competent, to belong to a group, to prove something to a parent, to avoid disappointing a coach, to enjoy the process.
目的是这个目标为了什么--感到有胜任感、属于一个群体、向父母证明什么、不让教练失望、享受过程。

The same content can serve very different purposes in different athletes, and the purposes are the ones that survive a losing run, because a goal of "run 11 seconds" gives an athlete nothing to hold on the day they run 11.4.
相同的内容可以在不同运动员身上服务于非常不同的目的；
而活下来的正是那些目的，因为“跑进11秒”这个目标在运动员跑出11.4的那天什么也留不住他。

And then there is the paradox, which is that some goals undermine the very thing they aim at.
然后是那个悖论：有些目标会破坏它们本身想要达成的东西。

A goal of proving you are not weak can prevent an athlete admitting fatigue, which increases injury risk.
“证明自己不是弱者”这个目标会阻止一位运动员承认疲劳，而那会增加受伤风险。

A goal of not disappointing a coach can sustain training through pain.
“不让教练失望”这个目标会让训练在疼痛中继续。

A goal of being the best can convert a good performance into a perceived failure, and the athlete then approaches the next one as a threat rather than an opportunity.

"成为最强的"这个目标会把一次好的表现转变成一次被感知到的失败，于是这位运动员把下一次当作威胁而不是机会来面对。

None of this is hypocrisy; it is what a badly chosen goal does to the person holding it.

这里没有虚伪；这是一个选错了的目标会对持有它的人所做的事。

The way this is applied is by asking the second question.

应用它的方式是问第二个问题。

Not "what do you want to achieve" but "what would it mean if you did" and "what would it mean if you did not", because those two answers reveal the purpose, and a purpose of proving a point and a purpose of enjoying the sport lead to different responses to the same bad day.

不是"你想取得什么"，而是"如果你做到了，那意味着什么"和"如果你没做到，那意味着什么"，因为这两个答案揭示了目的，而"证明某件事"是另一个目的。

It is also a conversation that an athlete will find strange, because most sporting culture treats wanting something as unremarkable and wanting it for a reason as an over-complication.

这也是一场运动员会觉得奇怪的对话，因为大多数运动文化把"想要某样东西"视为理所当然，而把"因为某个理由想要它"视为把事情想复杂了。

The practical benefit is that once a purpose is named, it can be protected, and that is what actually holds a difficult period together.

实际好处是：一旦目的被命名，它就可以被保护，而这才是把一个困难时期撑住的东西。

A coach who knows the athlete's purpose can point at it when the results are poor; can suggest that a rest serves the purpose rather than threatening it; and can tell the difference between an athlete who is coasting because the purpose has been met and one who is coasting because the purpose has stopped meaning anything.

一位知道运动员目的的教练，可以在成绩不好时指着它；可以建议休息服务于目的而不是威胁目的；

Those two athletes need opposite things and look identical from the field.

并且可以分辨一位运动员是在目的已达成而收工，还是在目的已失去意义而收工。

这两位运动员需要相反的东西，而从场边看一模一样。

The numbers / 算一算

Numbers to keep. The honest position is that there are no useful frequencies here, and the numbers that get invented are the problem. What is countable is whether the conversation happened: how many of the squad have been asked what a result would mean rather than what they want to achieve, which in most programmes is a small number, and whether the answer changed anything in the plan, which in most programmes is also a small number. The one time-based figure worth having is the point at which a season goal is met, because that is when purpose becomes load-bearing.

需要记住的数字。诚实的立场是：这里没有有用的频率，而那些被发明出来的频率恰恰是问题所在。可以计数的是这场对话是否发生过：

Common mistakes / 易错点

-

"Ask them what they want to achieve." That asks about content. Asking what it would mean is what reveals the purpose, and the purpose is what survives a bad day.

"问问他们想取得什么。"那问的是内容。问"那意味着什么"才揭示目的，而目的是在糟糕日子里留下来的东西。

-

"Purpose and content are the same thing." Run 11 seconds can be about love of the sport, about belonging, about proving a parent wrong, or about fear of a coach's reaction, and the four produce different programmes.

"目的和内容是同一回事。"跑进11秒可以是关于对这项运动的热爱、关于归属、关于向父母证明他们错了，或者关于害怕教练的报复。

-

"Strong goals never backfire." Goals of proving you are not weak, or of not disappointing someone, are the ones that most reliably produce the behaviour they were meant to prevent.

"强目标从不会适得其反。"那些"证明自己不是弱者"或"不让某人失望"的目标，恰恰是最可靠地产生它们本想防止的那种行为的目标。

-

"Just reinforce confidence." Confidence is downstream. If the purpose is "do not disappoint", confidence in the skill is not what is holding the athlete in the programme, and the coach is reinforcing the wrong variable.

"只要强化自信就行。"自信是下游的。如果目的是"不要让人失望"，那么对技术的自信并不是把这个运动员留在计划里的那个东西，而是对教练的信任。

Why it matters / 为什么重要

This section is the most conceptually sophisticated part of the goal-setting material and the least likely to be examined mechanically, which is exactly why it is worth teaching. The distinction between content and purpose turns goal setting from a technique into a conversation, and the conversation is the part that produces information no technique can. It also explains the most stubborn coaching problem in the section - an athlete whose effort has collapsed for reasons that have nothing to do with their ability or their goals - and it does so with a single question that costs nothing to ask.

这一节是目标设定材料中概念上最精细的部分，也是最不可能被机械考查的部分，而这正是它值得被教的原因。内容与目的的区分，把目

External rewards can replace internal motivation after success, so purpose and reasons for participation should be revisited.

成功后外部奖励可能替代内在动机，因此需要重新思考参与目的和理由。

Key terms / 核心术语

Goal adjustment 目标调整

Maintaining, modifying or replacing a goal as context changes.

随着情境变化保持、修改或替换目标。

Formulas / 公式

After achieving a qualification goal, set a process goal for the next phase and a new performance target.

达到资格目标后，为下一阶段设定过程目标和新表现目标。

Worked examples / 实例

>

Explain the goal-setting paradox.

解释目标设定悖论。

Exam focus / 考核要点

Why can success reduce drive?

为什么成功可能降低动力？

Adjustment and flexible goals · 目标调整与弹性目标

First, picture it / 先这样想

A plan that cannot bend is not a plan, it is a promise, and promises break at exactly the moment they are needed. Adjustment in sport is the ordinary business of changing the goal rather than the athlete: keeping the direction, moving the target, changing the timeframe. It is a skill because it is not spontaneous, and because a good adjustment is invisible to anyone who is not looking for it.

一份不能弯折的计划不是计划，而是一个承诺，而承诺恰恰在需要它的那一刻崩断。运动中的调整就是改变目标而不是改变运动员这件日常工作：保持方向、移动靶标、改变时限。它是一项技能，因为它不会自发出现，而且因为一次好的调整对任何不刻意去找它的人都是看不见的。

The detail / 详细讲解

Adjustment has a hierarchy, and it works because the levels are arranged by cost.

调整有一个层级结构，而它之所以有效是因为这些层级是按代价排列的。

Abandoning the direction - quitting the event - is the most expensive and should always be last.

放弃方向--退出这个项目--代价最高，而且永远应该放在最后。

Reducing the ambition while keeping the direction - training for a placing rather than a title, or for a season rather than a career - is usually available and almost always under-used.

保持方向但降低志向--为名次而不是为冠军训练、为一个赛季而不是为整个职业生涯--通常是可用的，而且几乎总是被用得太少。

Keeping the direction and the target but changing the timeframe is the most common and the least costly.

保持方向和目标但改变时限，是最常见也代价最低的。

A coach who moves up this list too fast gives up the sport; one who never moves up it produces athletes who train for a version of their life that no longer exists.

一位在这个列表上爬得太快的教练，会让运动员退出这项运动；

而一位从不往上爬的教练，会培养出一批在为一个已经不存在生命版本而训练的运动员。

The flexibility has to come from the goals themselves, which is where the pairing in the earlier section pays off.

这种弹性必须来自目标本身，而前面那一节的配对正是在这里产生回报。

A paired athlete who meets an outcome goal still has a process goal and a health goal to hold, so there is somewhere to redirect the energy on the day the result goes wrong.

一位配对过的运动员在达成结果目标之后，仍然握着一个过程目标和一个健康目标可以抓住，所以在结果糟糕的那天有地方可以把能量转向。

A single-goal athlete has nothing, and either the season ends abruptly or the athlete chases something bigger.

一位只有单一目标的运动员什么也没有，于是要么赛季戛然而止、要么他去追一个更大的东西。

The goals did not change; the room the athlete has to move in did.

目标没有变；变的是这位运动员可以活动的空间。

And adjustment is a social process, not only a personal one, which is the part that makes it hard.

而调整是一个社会过程，而不只是个人过程，这正是它难的地方。

An athlete who adjusts well is often read as lacking commitment, and the people who are loudest about that are frequently the ones whose own plans have not moved in years.

一位调整得好的运动员常常被读作投入不足，而对此最吵的人，往往是那些自己的计划多年没动过的人。

Making adjustment visible and normalising it - talking openly about a changed target, treating a reduced goal as a competent decision rather than a smaller ambition - is a cultural task, and it is the only way the individual skill becomes sustainable.

让调整变得可见并使它正常化--公开谈论一个被调整过的靶标、把一个降低的目标当作一个胜任的决定而不是一个更小的野心--是一项文化任务，也是唯一能让个人技能变得可持续的方式。

The practical toolkit is short.

实用工具箱很简短。

Keep a written record of the direction and the current target, so the adjustment is a change to a document rather than a crisis.

保留一份关于方向和当前靶标的书面记录，让调整变成对一份文件的修改而不是一场危机。

Set the next target at the moment the previous one is met, rather than after a gap.

在上一个目标被达成的当下就设定下一个，而不是留一段空档。

Review the timeframe deliberately at the midpoint of a block rather than at the end.

在一个训练周期的中点而不是终点有意识地回顾时限。

And use a health goal as the place to put any increase in difficulty, because it is the only goal whose achievement means stopping rather than continuing.

并把任何难度的增加放到健康目标里，因为它是唯一一个“达成即意味着停下而不是继续”的目标。

The numbers / 算一算

Numbers to keep. The one figure worth tracking is the gap between the planned target and the actual target across a season, expressed as a proportion. A programme that ends the season at 80 percent of its stated target has either a planning problem or an adjustment problem, and the two are distinguished by whether the direction was kept. The second figure is the interval between a target being met and the next being set, and the useful benchmark is that it should be close to zero - a gap of even a few weeks is where a season quietly loses its purpose.

需要记住的数字。唯一值得追踪的数字是整个赛季中计划靶标与实际靶标之间的差距，用比例表示。一个最终只完成既定目标80%的计划。

Common mistakes / 易错点

-

"Changing the goal means giving up." It means the target moved. Direction and timeframe are the parts that should be held, and the target is the part that is allowed to move.

"改变目标意味着放弃。"它意味着靶标移动了。方向和时限是应当保持的部分，而靶标是允许移动的部分。

-

"Flexibility is a weakness." It is a skill, and it is not spontaneous - which is why it has to be practised, in the same way a routine has to be rehearsed.

"弹性是一种软弱。"它是一项技能，而且它不会自发出现--这就是为什么它必须像例程必须被演练那样被练习。

- "A good coach keeps the plan." A plan that cannot bend produces athletes who train for a version of their life that no longer exists, and the coach is usually the last to notice.

"好教练会坚持计划。"一份不能弯折的计划会培养出一批在为一个已经不存在生命版本而训练的运动员，而教练通常是最后一个。

- "Set the next goal after a break to enjoy it." The gap is where seasons lose their purpose. The next target is set at the moment the previous one is met.

"休息一下再定下一个目标，享受一下。"这个空档正是赛季失去目的的地方。下一个靶标是在上一个被达成的当下就设定的。

Why it matters / 为什么重要

Adjustment is the section that separates a coach who has had one good season from a coach who has had ten, and it is the one most often left out of goal-setting instruction because it does not fit a workshop. It is also the most humane thing in the theme, because it takes seriously the possibility that an athlete's life has changed since the plan was written - that a job appeared, a child arrived, a body hurt - and it treats the resulting adjustment as evidence of a working plan rather than as evidence of a weak athlete. A student who can describe the hierarchy and give one example of each level has a genuinely useful tool for the conversation that every long-term project eventually requires.

调整是这一节把"有过一个好的赛季"的教练与"有过十个赛季"的教练区分开来的东西，而它最常被排除在目标设定教学之外，因为它装不

- A goal may be adjusted because of performance, injury, illness, time constraints, competition changes or environment.

表现、伤病、疾病、时间限制、比赛变化或环境都可能导致目标调整。

Key terms / 核心术语

Open goal 开放目标

A flexible goal focused on experience or participation.

关注体验或参与的灵活目标。

Formulas / 公式

An injured athlete can shift from winning to maintaining quality movement, completing safe rehabilitation and returning gradually.

受伤运动员可从获胜目标转向保持动作质量、完成安全康复并逐步回归。

Worked examples / 实例

>

Why are goal adjustment and open goals important?

目标调整和开放目标为何重要？

Exam focus / 考核要点

Describe the goal-adjustment steps.

描述目标调整步骤。

Imagery

运动意象

Guiding question / 引导问题

How can mental simulation improve sporting performance?

心理模拟如何提高运动表现?

Imagery is mental simulation of a performance or situation using multiple sensory modalities. It supports cognitive rehearsal and motivational preparation, works best when vivid, controllable, specific and regular, and should be integrated with physical practice and reflection.

意象是使用多种感觉模态对表现或情境的心理模拟。它支持认知演练和动机准备，效果在生动、可控、具体且规律时最好，并应与身

Sensory imagery and functions · 感觉意象与其功能

First, picture it / 先这样想

Imagery is rehearsing without the equipment. The brain does not have a clean divider between "imagined" and "real", so what you picture is wired in by much of the same machinery that handles what you see. That is why a skier who visualises a course has genuinely skied it, several times, before the first run - and also why visualisation of the wrong thing is not a neutral act.

意象是不带器材的排练。大脑并没有在“想象的”与“真实的”之间划一道干净的界线，所以你所描绘的东西，正是由处理你所看到的東西的那套机制中很大一部分接线的。这就是为什么一位在头脑中走过一遍线路的滑雪运动员，在第一次滑行之前已经真正滑过好几次--也是为什么对错误东西的意象不是一个中性的行为。

The detail / 详细讲解

Imagery is defined by the senses used.

意象是按所使用的感官来定义的。

Visual is the most studied and often the default: seeing the course, the target, the opponent.

视觉是被研究得最多的，也常常是默认的：看见线路、靶子、对手。

Kinaesthetic is imagining the feeling of the movement, the weight of the implement, the stretch in the backswing.

动觉是想象这个动作的感觉、器械的重量、引拍时的拉伸。

Auditory and olfactory imagery are less studied and there is reasonable evidence they can be effective.

听觉和嗅觉意象研究得较少，而有合理的证据表明它们可以是有效的。

Olfactory imagery in particular has produced some of the most striking results in the literature, and it appears to work through an emotional pathway rather than a motor one.

嗅觉意象在文献中产生了一些最惊人的结果，而且它似乎是通过一条情绪通路而不是运动通路起作用的。

The functions are usually grouped into three.

这些功能通常归为三类。

Cognitive functions are about understanding - rehearsing a plan, seeing the options before you get there, learning a skill conceptually.

认知功能关乎理解--排练一份方案、在到达之前看见选项、在概念上学会一项技术。

Affective functions are about regulation - reducing anxiety, managing arousal, building confidence, recovering after a bad performance.

情绪功能关乎调节--减轻焦虑、管理唤醒、建立信心、在表现糟糕之后恢复。

Performance-enhancement functions are about the skill itself, generating a stronger motor pattern.

表现增强功能关乎技术本身，生成一个更强的运动模式。

The three are separable, they have different evidence behind them, and an athlete can be helped by one and not by another.

这三类是可分的、背后证据各不相同，而一位运动员可能受益于其中一类而不受益于另一类。

The two broad modes are external and internal.

两种主要模式是外部与内部。

External imagery is seeing the scene from the performer's eyes, which is closer to a rehearsal and is what most athletes do by default.

外部意象是从表演者的眼睛看见场景，它更接近一次排练，也是多数运动员默认的做法。

Internal imagery is seeing the action from inside the body - from the feet, from the hands, from the trunk - which is closer to what the motor system actually needs and is associated with better movement quality.

内部意象是从身体内部--从脚、从手、从躯干--看见这个动作，它更接近运动系统真正需要的那个，并与更好的动作质量相关。

The finding is reasonably consistent, and the practical advice is that if an athlete imagines from the outside and their motor quality is the problem, they should try inside.

这个发现是相当一致的，而实用的建议是：如果一位运动员从外部想象、而问题出在动作质量上，那么他应当试试从内部。

And the limitation that matters most is about what happens when an athlete has never succeeded.

而最重要的局限与“运动员从未成功过时会怎样”有关。

Imagery works by using the motor system, and the motor system is best at reproducing what it has already produced.

意象是靠使用运动系统起作用的，而运动系统最擅长复制它已经产生过的东西。

For an athlete attempting something they have never done, kinaesthetic and external imagery have no existing pattern to draw on, and the honest options are either to build the skill physically first, or to use cognitive imagery about the plan and the environment instead of trying to imagine an unfamiliar movement feeling.

对于一位正在尝试自己从未做过的事的运动员，动觉和外部意象没有可供调用的既有模式，而诚实的选择要么是先在身体上把这项技

The numbers / 算一算

Numbers to keep. The figure worth knowing is the frequency: daily short sessions, commonly ten to twenty minutes, are associated with better outcomes than occasional long ones, and the short-and-often pattern is the one most consistently reported. The second is the proportion of sessions in which imagery is combined with physical practice rather than replacing it, because the combination is what produces transfer and the replacement is what produces a pleasant ten minutes. The third is the proportion of the movement imagined from an internal perspective, which is the easiest single thing to change in a session and the one most likely to be changed by accident.

需要记住的数字。值得知道的数字是频率：每天短时间、通常十到二十分钟，与更好的结果相关联，而“短而频繁”这个模式是报告最一致

Common mistakes / 易错点

- "See yourself winning." Outcome imagery is the least effective kind. Process imagery about a specific moment is what changes the motor pattern.
"看见自己赢。"结果意象是效果最差的一类。关于某一具体时刻的过程意象，才是改变运动模式的东西。
- "Imagine it from the outside like a TV camera." That is external imagery. Internal imagery, from inside the body, is more closely tied to the motor system and to movement quality.
"像电视摄像机那样从外面想象。"那是外部意象。内部意象--从身体内部--与运动系统和动作质量的关系更紧密。
- "Imagery replaces training." Imagery alone is effective for maintaining a skill and for rehearsing a plan, and it does not replace the physical practice that builds one.
"意象可以替代训练。"单独的意象对保持一项技术和排练一份方案是有效的，而它不能替代建立这项技术所需的身体练习。
- "Visualisation works the same for everyone." Kinaesthetic, auditory and olfactory imagery are all usable, and the sensory system that is most effective for one athlete is not necessarily the one for another.
"意象对每个人都一样有效。"动觉、听觉和嗅觉意象都可以使用，而对一位运动员最有效的感官系统，对另一位未必是。

Why it matters / 为什么重要

Imagery is the technique with the widest evidence base in the psychology theme, which makes it an unusual candidate for teaching honestly rather than as a mystical extra. The value to a coach is concrete: it is free, it needs no equipment, it is available at times when physical practice is impossible, and it can be prescribed in a session without changing anything else. The most useful thing a student can take from it is the mechanism - the brain does not separate imagined and real motor experience, which explains both why it works and why imagining the wrong

thing is not harmless.

意象是心理学主题中证据基础最广的技术，这使它成为一个不常被当作神秘附加物、而可以被诚实对待的候选。对教练而言它的价值是具

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Imagery is not visual-only and is not limited to perfect outcomes. It can include effort, mistakes, recovery, positive and negative scenarios.

意象不只是视觉，也不限于完美结果；可以包含努力、错误、恢复以及积极和消极情境。

Key terms / 核心术语

Skill rehearsal 技能演练

Mentally practising the physical execution of a skill.

心理练习技能的物理执行。

Strategy rehearsal 策略演练

Mentally rehearsing tactics, plans and problem solving.

心理演练战术、计划和解决问题。

Formulas / 公式

A skier imagines turns, edge control, pole-plant timing, the race line, contingency plans, speed and balance.

滑雪者想象转弯、刃控、撑杖时机、线路、备选方案、速度和平衡。

Worked examples / 实例

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Define imagery and explain how it mimics experience.

定义意象并解释它如何模拟体验。

Exam focus / 考核要点

Name two sensory modalities in imagery.

说出意象中的两种感觉模态。

PETTLEP and imagery quality · PETTLEP与意象质量

First, picture it / 先这样想

PETTLEP is a checklist that tells you which part of an imagined action to fix, and the reason it exists is that most people rehearse a movement in the one way it is not supposed to be done. Knowing that the difference is a single letter in a six-letter acronym is not trivial - it is the difference between ten minutes of pleasant scenery and a technique that changes.

PETTLEP 是一份清单，它告诉你该修正想象动作的哪一部分，而它之所以存在，是因为多数人是用唯一不该采用的那种方式来排练一个动作的。知道差别在于一个六字母缩写里的一个字母，并不是小事--它就是"十分钟愉快风景"与"一项会改变的技术"之间的差别。

The detail / 详细讲解

The letters are physical, environment, task, timing, learning, emotion and personality.

这些字母是身体、环境、任务、时间、学习、情绪和人格。

Physical means the felt position of the body and the movement of the segments, and it is the component most consistently associated with improved outcomes.

身体指身体各处能感觉到的位置和各节段的运动，它是与更好结果关联最一致的那个成分。

Environment means imagining the actual venue, including crowd and light.

环境指想象真实的场地，包括观众和灯光。

Task means imagining the specific technical and tactical problem you will face.

任务指想象你将要面对的具体技术与战术问题。

Timing means the right moment of the action.

时间指动作中正确的那个时刻。

Learning means the process of practising, and it is used for acquiring rather than performing.

学习指练习的过程，它用于习得而不是用于表现。

Emotion means imagining your own feelings, and personality means the part of you that performs - a confident performer imagining from a confident perspective.

情绪指想象你自己的感受，而人格指“做动作的那个自我”--一位自信的表演者从一个自信的视角去想象。

The most important use of it is diagnosis.

它最重要的用途是诊断。

When an athlete reports a plate of technical difficulty and their imagery has been consistently visual and external, the letter to change is usually P - the physical component.

当一位运动员报告某个技术难点、而他的意象一贯是视觉的和外部的，那么该改的字母通常是P--身体成分。

If the difficulty is with reading a tactical situation, it is T for task.

如果难点在于阅读战术局面，那是任务这个字母。

If the difficulty appears only under pressure, it is E for emotion, and the athlete needs to imagine the feeling of being nervous, not the absence of it.

如果难点只在压力下出现，那是情绪这个字母，而这位运动员需要想象“正在紧张”的感觉，而不是“没有紧张”。

Imagery quality is a diagnostic instrument, not just a prescription.

意象质量是一份诊断工具，而不只是处方。

The standards for good quality are concrete, and an athlete who has never been taught them will typically fail several.

好质量的判据是具体的，而一位从未被教过这些的运动员通常会在好几条上不合格。

The movement should be imaged from an internal perspective, kinaesthetically and specifically.

动作应当从内部视角去想象，带着动觉、并且具体。

It should be felt as achievable - not as a fantasy of perfection, which produces an unrepresentable comparison.

它应当被感觉为可达到的--而不是一个完美的幻想，那会制造出一个无法复现的比较。

It should be controlled and adjustable, so the athlete can slow it down and hold a position.

它应当是可控可调的，以便运动员能把它放慢、停住某个姿势。

And it should be emotionally congruent with how the athlete will actually feel, including the nerves.

而它应当与运动员真实的感受在情绪上一致，包括那份紧张。

And there is an external-to-external trap worth naming, because it is the commonest failure.

而有一个“外部对外部”的陷阱值得点名，因为那是最常见的失败。

An athlete imagines themselves from the outside - a camera view, watching themselves succeed, seeing themselves being praised - and it feels excellent and produces almost nothing, because the motor system is not being given the proprioceptive picture it needs.

一位运动员从外部想象自己--一个摄像机视角

、看着自己成功、看着自己被表扬--感觉极好，却几乎产生不了任何东西，因为运动系统没有拿到它所需的那张本体感觉图像。

The internal view feels markedly worse and is the one that does the work.

内部视角感觉明显更难受，而它才是干活的那个。

The discomfort is the honest signal that a session is doing something.

这种不适正是一堂训练课确实在起作用的诚实信号。

The numbers / 算一算

Numbers to keep. The numbers here are the length and the fidelity of the check. Imagery sessions of about ten to fifteen minutes with a short physical warm-up first are the pattern most commonly used, and the kinaesthetic component is the one worth the most of the session. The second figure is the proportion of the movement that should be internal, which is a number a coach can ask for directly: "how much of what you just imagined did you feel from inside?" The third is the frequency of re-checking quality, because an athlete will drift back to external visual imagery within a few weeks unless someone keeps asking.

需要记住的数字。这里的数字是检查的长度与保真度。约十到十五分钟的意象课、之前先做一次短的身体热身，是最常用的模式，而动觉

Common mistakes / 易错点

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"Internal imagery means looking down at yourself." It means viewing the action from inside the body, from the feet or hands or trunk, feeling the segments move.

"内部意象意味着低头看自己。"它的意思是从身体内部、从脚或手或躯干去看这个动作，感觉各节段的运动。

"Imagery quality is about detailed scenery." Detail matters, but internal perspective, kinaesthetic specificity, achievability and controllability are the criteria that predict effect.

"意象质量是关于细节丰富的场景。"细节确实重要，但内部视角、动觉具体性、可达到性和可控性才是能预测效果的判据。

"Imagine feeling calm, because that is the goal." Congruent emotion means imagining the nerves you will actually feel, because that is what needs rehearsing.

"想象自己感到平静，因为那就是目标。"情绪上一致意味着想象你真正会感到的那份紧张，因为那才是需要被排练的。

"If it feels good, it is working." The external fantasy view feels excellent and produces almost nothing. The internal view feels worse and is the one that works.

"如果感觉不错，那就在起作用。"外部的幻想视角感觉极好，却几乎产生不了任何东西。内部视角感觉更差，而它才是有效的那一个。

Why it matters / 为什么重要

PETTLEP earns its place by being diagnostic rather than merely descriptive, which turns it into something a coach can use in the middle of a session with an athlete who is stuck. Its letters are a way of asking a specific question about a specific problem, and the questions are short enough to be worth asking. It is also the section where a student has to be most careful about the difference between evidence and enthusiasm: several of the components have weaker support than the physical one, and the honest account is that the internal kinaesthetic standard is the part that carries the technique.

PETTLEP

值得占一个位置，是因为它是诊断性的而不只是描述性的，这使它成为教练在一堂训练课中间、对一个卡住的运动员就能用的东西。它的

Vivid, controllable, specific and regular practice is more likely to transfer to performance.

生动、可控、具体且规律的练习更可能迁移到表现。

Key terms / 核心术语

PETTLEP PETTLEP模型

A seven-part structure for designing effective imagery.

设计有效意象的七部分结构。

Perspective 视角

Internal or external viewpoint used in the mental rehearsal.

心理演练中的内部或外部视角。

Formulas / 公式

Redesign a serve by imagining the actual court, racquet contact sound, physical effort, timing, emotion and a balanced follow-through.

通过想象真实场地、击球声音、身体努力、时间、情绪和平衡随动来重新设计发球意象。

Worked examples / 实例

>

Use PETTLEP to redesign an athlete's serve imagery.

用PETTLEP重新设计运动员的发球意象。

Exam focus / 考核要点

Describe the PETTLEP model.

描述PETTLEP模型。

Paivio, specificity and applications · Paivio理论、特异性与应用

First, picture it / 先这样想

Two questions finish the story, and both are blunt. Can the athlete picture it? Some can and some cannot, and the reasons are about how the brain represents information. And should they picture this particular thing? The second question is the one most often skipped, because the answer is usually no, and a technique that is not specific to a task can be as wrong as one that is specific to the wrong task.

两个问题为这个故事收尾，而两个都很直接。这位运动员能想象得出来吗？有些人能、有些人不能，而原因在于大脑如何表征信息。以及，他们该不该想象这特定的东西？第二个问题是最常被跳过的，因为答案通常是否定的--而一个不针对任务的技巧，可能和“针对错误任务的技巧”一样错。

The detail / 详细讲解

Paivio's mental model is the organising idea: cognition and imagery are separate but interconnected systems that act as a large pool of imagery which is drawn on depending on the task and the person.

帕维奥的心理模型是组织性的概念：认知与意象是两个既分离又相互连接的系统，它们共同作为一个巨大的意象池，而意象池的调用依赖于任务。认知任务更多地调用言语编码；

A cognitive task draws more on verbal coding; a physical task draws more on kinaesthetic coding; some people have a larger visual pool and some have a larger kinaesthetic pool; and where the imagery pool is small, the athlete has to add words and images consciously to get imagery to run at all.

身体任务更多地调用动觉编码；有些人的视觉池更大，有些人的动觉池更大；而在意象池较小的人身上，运动员必须有意识地把言语和图像加进去，意象才会运转。

The practical consequences are about who imagery works for and how to teach it.

实际推论是关于“意象对谁有效”以及“如何教”。

It is easier for an athlete with a large imagery pool and harder for one with a small pool, and the second group often need the imagery to be cued rather than left to intention - a trigger, a routine, a phrase.

意象对于意象池大的运动员更容易、对于意象池小的更难，而第二组人通常需要意象被“提示”而不是仅靠意图--一个触发点、一个例行程序。

It is also more effective when the task is physical and the athlete is kinaesthetic, and less obviously effective when the task is verbal or when the athlete thinks in words, which is a real limitation on the technique's universality.

而且在任务是身体性的、而运动员是动觉型时它更有效；在任务是言语性的、或运动员是用词语思考时效果不那么明显--这是这项技术普适性的一个真实局限。

Specificity is the second question, and it works in both directions.

专门性是第二个问题，而它在两个方向上都成立。

Training a physical skill benefits from imagery that is specific to that skill, in that movement, at that moment, with that feeling.

训练一项身体技术，受益于对该技术、那个动作、那个时刻、那种感觉具体的意象；

Training a tactical decision benefits from imagery of the situation, the options and the cues to read.

训练一个战术决策，受益于对局面、可选项和要读的线索的意象。

Imagery that is general - seeing yourself being successful, seeing yourself as a champion - produces a pleasant state and no transfer, because the motor system and the decision-making system are both being left without the specific information they need.

笼统的意象--看见自己成功、看见自己是个冠军--产生一个愉快的状态和零迁移，因为运动系统和决策系统都没有拿到它们各自需要的信息。

And the applications follow from the two answers.

而各种应用正是从这两个答案推导出来的。

For an athlete with a large pool who needs a physical skill: internal kinaesthetic imagery, short, frequent, alongside physical practice, with the emotion matched to what they will feel.

对一位意象池大、需要一项身体技术的运动员：内部的动觉意象，短而频繁，与身体练习并行，情绪要与真实感受一致。

For an athlete with a small pool: the same imagery, but cued by a trigger and built up over weeks, and with more verbal scaffolding to compensate.

对一位意象池小的运动员：同样的意象，但由一个触发点提示、并在数周内逐步建立，同时用更多言语脚手架来补偿。

For a tactical problem: imagery of the situation and the options, which is a cognitive use of the same tool and is often the more valuable one in a sport where a mistake of judgement costs more than a mistimed swing.

对一个战术问题：关于局面和可选项的意象，这是同一工具的认知性用法，而且在一门“判断失误比时机失误代价更高”的项目里，它往往更有价值。

The numbers / 算一算

Numbers to keep. The useful figures here are all about matching effort to the athlete rather than to the protocol. Time spent per session on imagery should be small - commonly ten to fifteen minutes - and the honest reading of a negative result is that the imagery was too short to run, or cued too weakly, or replaced rather than combined. The second figure is the proportion of a cognitive application, which in a tactical sport is often larger than coaches assume: the time spent imagining situations and options is frequently comparable to or greater than the time spent imagining movements. The third is the number of weeks before a cued routine attaches to its trigger, which is typically several and should be budgeted for rather than assumed away.

需要记住的数字。这里有用的数字全都是关于“让努力与运动员匹配”、而不是与方案匹配的。每次训练课花在意象上的时间应当很少--通常

Common mistakes / 易错点

- "Imagery ability is a gift some athletes have." It varies, and the athletes with a smaller imagery pool are usually the ones who need the most explicit teaching and the most consistent cueing.
"意象能力是某些运动员才有的天赋。"它是有差异的，而意象池较小的运动员通常恰恰是最需要明确教学和最持续提示的那批人。
- "If you cannot feel it, you cannot do it." Imagery ability varies and is not a selection criterion. A cognitively strong athlete with a small imagery pool can still be an excellent performer.
"如果你感觉不到它，你就做不到。"意象能力有差异，而且不是一项选拔标准。一位意象池小、而认知能力强的运动员仍然可以是
- "Generic imagery builds confidence." It builds a good feeling and no transfer. Specificity to the task, the movement and the moment is what makes it work.
"笼统的意象能建立信心。"它建立的是一个好感觉和零迁移。对任务、对动作、对时刻的专门性才是让它起作用的东西。
- "Imagery is only for physical skills." The cognitive applications - imagining a situation, the options and the cues - are frequently the higher-value use in sport.
"意象只用于身体技术。"认知性应用--想象局面、可选项和线索--在运动中经常是价值更高的那个用法。

Why it matters / 为什么重要

Paivio closes the theme by explaining who the technique works for, and that is the part that makes it more than a demonstration. It is the reason a coach who has one athlete who "does imagery" successfully and another for whom nothing happens should not conclude that the second athlete is doing it wrong - the second athlete has a smaller pool and needs cuing, verbal scaffolding and a longer runway, and that is a teaching problem rather than a motivation problem. The specificity argument is the other thing worth keeping, because it explains the disappointing result that a pleasant, successful, general piece of imagery produces nothing measurable.

帕维奥为此主题收尾，因为它解释了这项技术对谁有效，而这正是它不只是一个演示的原因。这就是为什么一位教练，有一位运动员

- Before training, use imagery to plan and focus; before competition, prepare and regulate; after performance, review and learn.
训练前用意象计划和集中；比赛前准备和调节；表现后复查和学习。

Key terms / 核心术语

Specific imagery 具体意象

Detailed, concrete and sport-specific rehearsal.

详细、具体且专项的心理演练。

General imagery 一般意象

Broad, abstract and outcome-focused rehearsal.

宽泛、抽象且以结果为重点的演练。

Formulas / 公式

A skier can see a clean run, hear the wind, feel balance and also imagine correcting a turn after a mistake.

滑雪者可以想象干净完成一圈、听到风声、感到平衡，也可以想象犯错后纠正转弯。

Worked examples / 实例

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Use Paivio to explain cognitive, motivational, specific and general imagery.

用Paivio解释认知、动机、具体和一般意象。

Exam focus / 考核要点

Give an example of imagery improving confidence and arousal control.

举出意象提高自信和调节唤醒的例子。
